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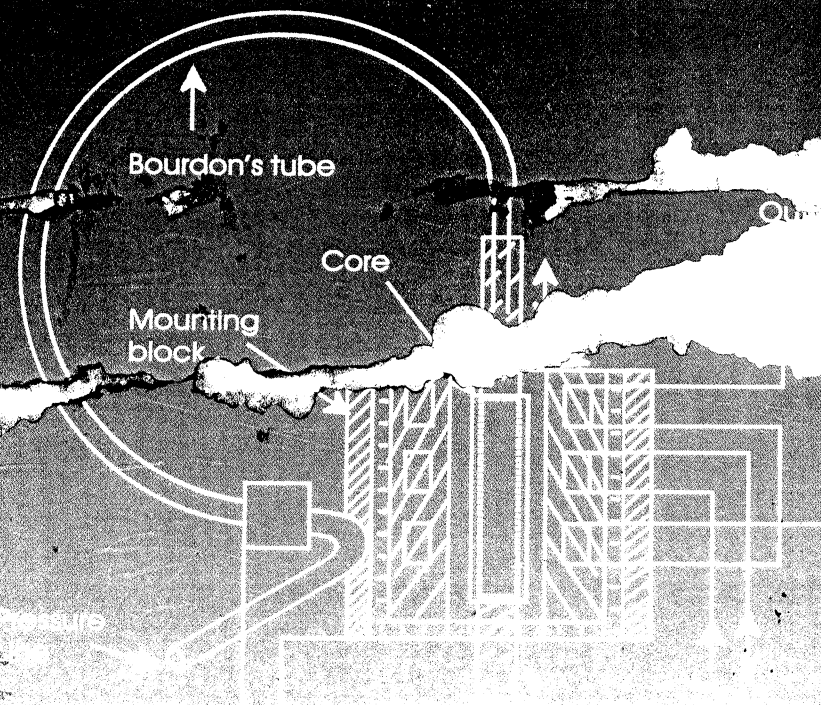
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TRANSDUCER ENGINEERING



Dr. S. Renganathan

TRANSDUCER ENGINEERING

by

S. Renganathan

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Preface

Instrumentation is a multidisciplinary subject. It needs inputs from various disciplines like Electrical, Electronics, Mechanical and Chemical Engineering. Instrumentation today comprises of measurement, control, computing and networking. Systematic study of this subject has to be started with transducers, signal-conditioners, controllers, final control elements, storage devices and display devices. The front-end of any Instrumentation system is Transducers. Transducer includes sensors as a sub-set. This sub-set sometimes becomes the full set. These transducers are used to convert any non-electrical quantity into an electrical quantity or a pneumatic quantity, which can be further processed for monitoring purposes, control purposes, transmission or display purposes. Now-a-days as computers are used for signal-processing and control functions, the electrical nature of the output is more useful than pneumatic one. This book is written to give basic understanding of the Transducers with electrical output used in Instrumentation systems. It is titled as Transducer Engineering because it is aimed at giving the theory of the various transducers before the other details. A separate chapter is set aside to discuss the generalized performance of the Transducers

There are basically two ways of categorizing the transducers. One is based on the application. The other is based on the principle of operation. In this book the second classification is used.

This book is written at the UG level, giving a coverage of the syllabus of the first level course in Transducers.

Preface

Students of Instrumentation and control, and Instrumentation and Electronics branches will find this very useful for their UG and PG programs. Students of the other branches like Electrical and Electronics and Communication, Mechanical and Chemical Engineering will find this book as a source of information for their courses and project work.

Simple style and language are used so that it is easy to follow by students from Tamil medium schools and other than English medium schools.

S. RENGANATHAN

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SCIENCE OF MEASUREMENT

1.1 IMPORTANCE OF MEASUREMENT

THE index of civilization of a country is better understood by the amount of use of the developments in science and technology. The easiest way to assess the amount of use of science and technology is to examine the number of measurements that are being made and how they are being used. This is because, as the human knowledge grows, new phenomena are observed which requires new types of measurements for better understanding of the event. Once an event or occurrence of an event is understood, it can be better harnessed for the welfare of the society.

A measurement is made by comparing the unknown quantity with a predefined standard. For example, the length of a wall is expressed as 5 m, after comparing it with pre-marked tape kept at standard environmental conditions. Similarly the weight of an object is said to be 2.4 kg after comparing it with standard weights. Measurements can be defined as a process of empirical but objective assignment of numbers to properties of objects or events of the real world in such a way as to describe them. In short measurement is nothing but quantifying an event or a thing.

1.2 PURPOSE OF MEASUREMENT

Measurements are generally made

- to understand an event or an operation
- to monitor an event or an operation
- to control an event or an operation
- to collect data for future analysis and
- to validate an engineering design.

Hence measurements are indispensable for identifying the parameters of a system, evaluating the performance of a system or a process, proving a discovery, formulating a statistical law and regulating or controlling a process or an event.

1.3 METHODS OF MEASUREMENT

The methods of measurement can be broadly classified as

- direct comparison method and
- indirect comparison method.

1.3.1 Direct Comparison Method

In direct comparison method the unknown quantity is compared directly with a primary or secondary standard.

Primary standards are maintained by national standard laboratories in different parts of the world for various quantities like length, mass, time, ampere, voltage, resistance, capacitance, inductance, temperature and luminous intensity. Details of the various standards are given in section 1.7.3.

Secondary standards are the basic reference standards used in industrial measurement laboratories. The accuracy of the primary standards are of the order of several parts in 10^8 whereas that of the secondary standards are of the order of several parts in 10^6 .

For example, the primary standard for voltage is a saturated Weston cell. If the unknown voltage is compared with this cell then the method of measurement is termed as direct comparison method.

1.3.2 Indirect Comparison Method

In indirect comparison method the unknown quantity is compared with a standard or a meter which is compared and checked with a primary or secondary standards which are kept at national laboratories periodically.

For example an unknown voltage is measured using a voltmeter which is periodically calibrated with primary or secondary standards.

As it is difficult to compare the unknown quantity directly with primary or secondary standards, most of the measurements are done by the indirect method.

1.3.3 Deflection and Null Methods

Another way of classifying the methods of measurement is based on the principle of operation. They are deflection and null-balance methods.

In deflection method of measurement, the unknown quantity causes a deflection or a mechanical displacement on a calibrated scale. For example, in a Bourdon pressure gauge, the pointer deflects proportional to the pressure on a calibrated dial as shown in Fig. 1.1.

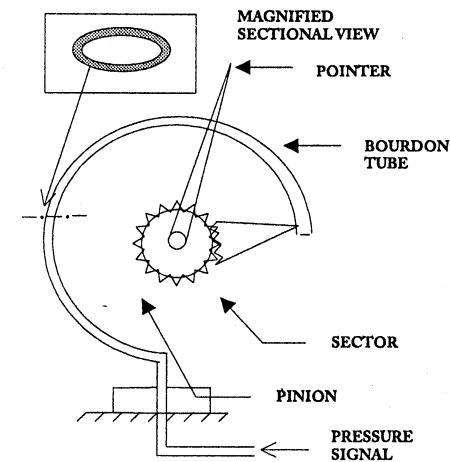


Fig 1.1

Deflectional method of measurement of pressure

In null method of measurement, the effect of unknown quantity is balanced by the application of another effect so as to maintain the deflection caused by the difference between the effects, at zero. The magnitude of opposing effect which is quantifiable, gives a measure of unknown quantity. For example, in a voltage balancing potentiometer which is employed to measure the output of a thermocouple, the thermo emf is balanced with a

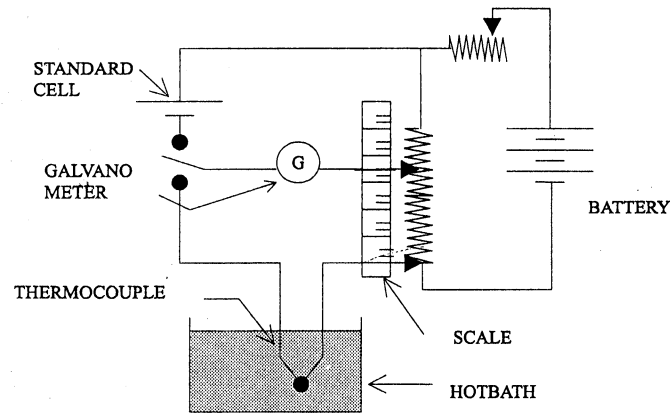


Fig 1.2

Null method of measurement of thermo e.m.f.

known voltage, obtained from a variable voltage source as shown in Fig. 1.2. A Galvanometer is used to detect the balance condition.

The null method offers higher accuracy of measurement than that by a deflection method because of the following features of the former method-

- The unknown quantity is compared with a source which is calibrated with a standard directly.
- The sensitivity of the detector used is very high as it has to measure only a small range around zero.
- The detector need not be calibrated.
- The conditions of the measuring quantity is not altered leading to zero loading effect. A detailed explanation of the loading effect is given in Chapter 3.

However, null method requires appreciable time for balancing and hence it is not suitable if the measuring quantity changes faster than the time required for balancing.

1.4 FUNCTIONAL BLOCKS OF A MEASUREMENT SYSTEM

The various functions of a measurement system can be represented by a block schematic as shown in Fig. 1.3.

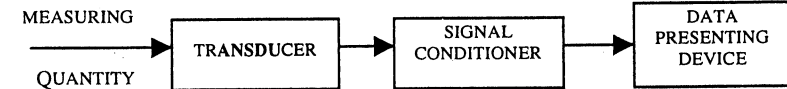


Fig1.3

Functional block diagram of a measurement system

A transducer produces an output signal which is a function of the measured quantity. The input signal quantity may be temperature, flow, speed, vibration etc. The output signal may be a voltage, current or air pressure. For example a strain gauge connected in a wheatstone bridge converts strain into voltage. Here the strain gauge is called a sensor and the strain gauge along with wheatstone bridge is called a transducer. A sensor is an element which detects a change. It simply changes its character when an input quantity undergoes a variation. For example in a strain gauge, a change in strain causes a change in resistance. In order to convert this change into a voltage variation, a bridge circuit and or supply voltage source are needed. Strain gauge, bridge circuit and the voltage source put together is a transducer. There transducer is something more than a sensor. Transducer is a bigger set in which sensor is a part or subset. Sometimes sensor and transducer are one and the same as one set. For example a thermocouple is a sensor and also it is a transducer. It might be noted here that some authors use these two words namely sensor and transducers interchangeably. A flapper-nozzle converts displacement into air pressure, which is a transducer.

The second block namely the signal conditioner modifies the output of the transducer into a convenient form so that it can be further processed. The conditioning of the transducer output signal may include one or more of the following-

- Amplification
- Demodulation
- Filtering
- Impedance matching
- Linearisation
- Analog to digital conversion

For example, a low pass filter allows only the low frequency content of the signal.

The last block namely the data presenting device presents the measured quantity to the observer. The data can be presented by any one of the following ways:

- Analog indication by means of deflection of a pointer
- Digital indication using displays like LCD, LED etc.,
- Recording the variation of signal as a function of time
- Totalizing the signal over a period of time
- Interfacing with a computer or controller for further processing.

1.5 TYPES OF ERRORS

A measurement cannot be made without errors. These errors can be only minimized but not eliminated completely. It is essential to know the different errors that can possibly enter into the measurement.

The error can be classified into three categories:

- Gross errors
- Systematic errors
- Random errors

1.5.1 Gross Errors

Gross errors are largely due to human factors such as misreading of instruments, incorrect adjustment and improper application of instruments. The computational errors are also grouped under this type errors.

When human beings are involved in measurement, gross errors will inevitably be committed. Complete elimination of gross errors is probably impossible in analog meters. This error is almost eliminated in autoranging digital meters. One common gross error frequently encountered in measurement work involves the improper selection of the instrument. When a voltmeter is used to measure the potential difference across two points in a circuit, the input impedance of the voltmeter chosen should be atleast 10 times

greater than the output impedance of the measuring circuit. As the output impedance of a circuit is normally not known before hand, the selection of the voltmeter may not be made correctly, leading to a gross error. The error caused by the improper selection of a voltmeter is shown by the following example.

EXAMPLE 1.1

A voltmeter reads 40V in its 50V scale when connected across an unknown resistor as shown in Fig. 1.4. The resistance of the voltmeter coil is 2000 ohms/volt. If the milli ammeter reads 2.4 mA, calculate (i) apparent value of the unknown resistor (ii) actual value of the unknown resistor (iii) gross error.

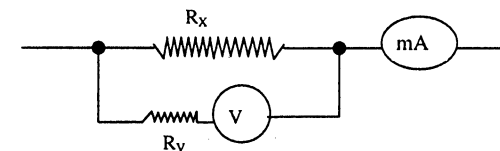


Fig 1.4
For example 1.1

SOLUTION

(i) Apparent value of resistance

$$R_A = \frac{V}{I} = \frac{40}{2.4} = 16.7K \Omega$$

(ii) Voltmeter resistance,

$$R_v = 2000 \times 50 = 100 k \Omega$$

Since the voltmeter is connected in parallel with the unknown resistor.

$$R_A = \frac{R_x \cdot R_v}{R_x + R_v}$$

where R_x is the actual value of the unknown resistor, i.e.

$$R_x = \frac{R_A \cdot R_v}{R_v - R_A}$$

$$= \frac{16.7 \times 10^3 \times 100 \times 10^3}{10^3 (100 - 16.67)} = 20 \text{ K } \Omega$$

$$\begin{aligned} \text{(iii) \% error} &= \frac{\text{Apparent} - \text{Actual}}{\text{Actual}} \times 100 \\ &= \frac{16.67 - 20}{20} \times 100 \\ &= -16.67 \% \end{aligned}$$

This error is due to the appreciable current drawn by the voltmeter which is commonly known as loading effect. Electronic voltmeters have in general, very high input impedance and hence error due to loading effect is less.

A large number of gross errors is due to the carelessness of the observer. For example, failure to set zero before measurement is made, failure to note the reading from the scale corresponding to the selected range and improper positioning of the instrument cause gross errors. These errors cannot be treated mathematically. These errors can be eliminated by taking a number of readings of the same quantity and by a number of observers. It is recommended to take atleast three separate readings at different times.

1.5.2 Systematic Errors

Systematic errors are due to shortcomings of the instrument and changes in external conditions affecting the measurement. These are classified into two categories-

- Instrumental errors
- Environmental errors

Instrumental errors arises out of the changes in the properties of the components used in the instrument. This can be avoided by calibrating the instrument frequently.

Environmental errors are due to the changes in the environmental conditions such as temperature, humidity, pressure and electrostatic and magnetic fields. For example, the resistance of a strain gauge changes with variation in temperature. These errors can be minimised by controlling the environmental

conditions in the laboratory. However, it is difficult to have a controlled environment in an industrial atmosphere.

Systematic errors can also be divided into static and dynamic errors. Static errors are caused by the limitations of the measuring device or the assumption in the physical laws governing its behaviour. Dynamic errors are caused by the instruments slow response in following the changes in the measured variable.

Systematic errors have definite magnitude and direction. These are usually more troublesome as repeated measurements may not reveal them.

1.5.3 Random Errors

Random errors are unpredictable errors and occur even when all systematic errors are accounted for. Although the instrument is used under controlled environmental and accurately pre-calibrated before measurement. It will be found that the readings vary slightly over a period of observation. This change cannot be corrected by any method and it cannot be explained without detailed investigation. However, these errors can be reduced by taking more number of readings and using statistical methods to obtain the best approximation of the true value.

1.6 ERROR ANALYSIS

The analysis of the measurement data is necessary to obtain the probable true value of the measured quantity. Any measurement is associated with a certain amount of uncertainty. The systematic way of specifying this uncertainty is the statistical method. A large number of measurements is usually required to make statistical methods and interpretations meaningful. Also the systematic errors should be small compared with random errors, because statistical treatment of data cannot remove a fixed bias contained in all the measurements.

When fluid pressure in a container is to be measured, 10 readings are taken over a period of time by means of a Bourdon gauge. Each of these 10 readings may be different from the others.

One will be wondering which one of these is the correct reading. The statistical methods will give the most probable true value of the pressure. The statistical methods use terms like arithmetic mean, deviation, mode, median, which are explained below.

1.6.1 Arithmetic Mean

The most probable value of a measured variable is the arithmetic mean of the number of readings taken. Arithmetic mean is defined as the average value of the readings which is given by

$$\begin{aligned} \bar{X} &= \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \\ &= \frac{\sum_{\alpha=1}^n x_{\alpha}}{n} \end{aligned}$$

where x_{α} is the individual reading and n is the total number of readings. When n is very large, best results are obtained.

1.6.2 Deviation

Deviation is the departure of a given reading from the arithmetic mean of a group of readings. For a reading x_1 , the deviation is given by

$$d_1 = x_1 - \bar{x}$$

Average deviation is defined as the average of the modulus of the individual deviations and is given by

$$\begin{aligned} D &= \frac{|d_1| + |d_2| + \dots + |d_n|}{n} \\ &= \frac{\sum_{\alpha=1}^n |d_{\alpha}|}{n} \\ &= \frac{\sum_{\alpha=1}^n |(x_{\alpha} - \bar{X})|}{n} \end{aligned}$$

The average deviation gives an indication of the repeatability of the instrument.

1.6.3 Standard deviation

Another deviation which is commonly used in statistical analysis is standard deviation which is defined as

$$\sigma = \sqrt{\frac{\sum_{\alpha=1}^n d_{\alpha}^2}{n-1}} = \sqrt{\frac{\sum_{\alpha=1}^n (x_{\alpha} - \bar{x})^2}{n-1}}$$

when the number of readings are less than or equal to 20.

When the number of reading is more than 20, standard deviation is defined as

$$\sigma = \sqrt{\frac{\sum_{\alpha=1}^n d_{\alpha}^2}{n}}$$

1.6.4 Variance

Variance is another term which is sometimes used in statistical analysis. This is the square of the standard deviation and is given by

$$\begin{aligned} V = \sigma^2 &= \frac{\sum_{\alpha=1}^n d_{\alpha}^2}{n-1} && \text{for } n \leq 20 \\ &= \frac{\sum_{\alpha=1}^n d_{\alpha}^2}{n} && \text{for } n > 20 \end{aligned}$$

EXAMPLE 1.2

In an experiment, ten observations of pressure are made which are given below:

Trial no.	1	2	3	4	5
Scale reading (K P _a)	10.02	10.20	10.26	10.20	10.22
Trial no.	6	7	8	9	10
Scale reading (K P _a)	10.13	9.97	10.12	10.09	9.9

Calculate (i) arithmetic mean (ii) average deviation (iii) standard deviation and (iv) variance.

SOLUTION

(i) Arithmetic mean

$$\bar{x} = \frac{10.02+10.02+10.26+10.20+10.22+10.13+9.97+10.12+10.09+9.99}{10}$$

$$= 10.111 \text{ KP}_a$$

(ii) Average deviation absolute

$$d = |d_1| + |d_2| + \dots + |d_n|$$

$$d = \frac{0.091+0.089+0.149+0.089+0.109+0.019+0.141+0.009+0.021+0.211}{10}$$

$$= 0.0928 \text{ KP}_a$$

(iii) [Standard deviation]² = σ^2

$$= \frac{10^{-3}[8.281+7.921+22.201+7.921+11.881+0.361+19.881+0.081+0.441+44.521]}{9}$$

$$= -0.12349 / 9 = 0.0137$$

$$\sigma = \text{standard deviation} = 0.117 \text{ KP}_a$$

(iv) Variance $V = \sigma^2 = 0.0137$

1.6.5 Median

Median is also used to indicate the most probable value of the measured quantity when a set of readings are taken. When the readings are arranged in ascending or descending order of magnitude, the middle value of the set is taken as the median. For example, the temperature of a bath is noted by eleven observers as follows:

66.5°C, 63.8°C, 65.7°C, 66.1°C, 64.8°C, 67.0°C, 65.3°C,
63.9°C, 64.4°C, 65.9°C, 66.5°C

It is rearranged in ascending order as follows:

63.8°C, 63.9°C, 64.4°C, 64.8°C, 65.3°C, 65.7°C, 65.9°C,
66.1°C, 66.5°C, 66.5°C, 67.0°C

Now the median is the sixth reading, i.e. 65.7°C.

1.6.6 Mode

Mode is the value which occurs most frequently in a set of observations and around which other items of the set cluster densely.

For example, the frequency distributions of a set of 101 observations is given as follows

Pressure readings KP _a	50	51	52	53	54	55	56	57
No. of readings	4	9	16	25	22	15	7	3

The value of pressure reading corresponding to maximum number of occurrences is 53 KP_a. hence mode is 53 KP_a.

1.6.7 Probability Of Errors

By the very nature of the random errors, the uncertainty associated with any measurement cannot be predetermined. Only the probable error can be specified using statistical error analysis. Some of the statistical methods of analysing the errors are discussed in the following sections.

1.6.7.1 Normal distribution of errors

The measurement data can be pictorially represented by a histogram for a better visual appeal and quick understanding of information. Histogram is a bar graph display of the number of occurrences of a particular observed value in a measurement. For example, 60 temperature readings were taken over a small period of time and tabulated as shown in table 1.1.

The histogram is plotted taking temperature reading on the X-axis, and number of occurrences in the Y-axis, as shown in Fig. 1.5.

Fig. 1.5 shows that the largest number of readings (22) occurs at the central value of 80.0°C, while the other readings are distributed more or less symmetrically on either side of the middle. If more readings were taken, the distribution of observation would have remained symmetrical about the central value. With increased number of observations taken with high resolution meters, the contour of the histogram would become a smooth curve as shown

by the broken line in Fig. 1.5. This bell shaped broken line curve is known as the *Gaussian curve*. If the curve is sharp, then one may state with confidence that the most probable true value is the central value.

Table 1.1: Tabulation of temperature readings

Temperature reading (deg. C)	Number of readings
78.5	2
79.0	5
79.5	12
80.0	22
80.5	14
81.0	4
81.5	1
	60

If a precise method is used for measurement, the Gaussian

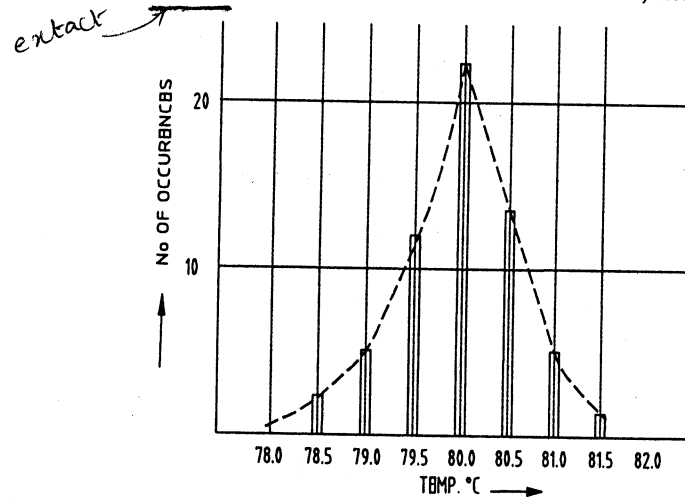


Fig 1.5

Fig. 1.5 : Histogram showing the frequency of occurrence

curve will be sharper (A) and the curve will be flattened when a crude method is used (B) as shown in Fig. 1.6.

unfinished (or) not refined

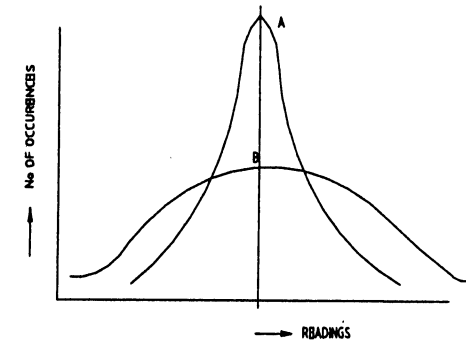


Fig 1.6

Gaussian curves for precise and crude methods of measurements

1.6.7.2 Probable error

It is quite often useful to specify the probable error in a measurement due to random error.

If the central value of a Gaussian curve is assumed as the true value, then error V_s number of occurrences in the measurement can be plotted as shown in Fig. 1.7.

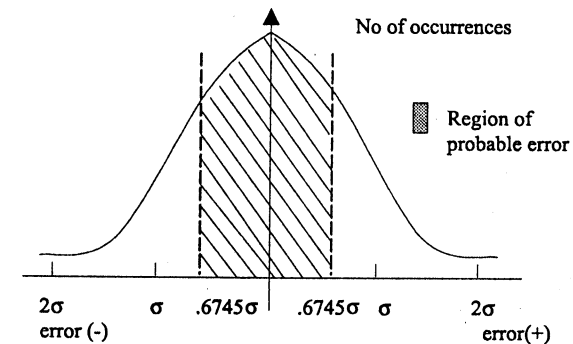


Fig 1.7

Error distribution

For a Gaussian distribution of data, it is found that about 68% of the total number of observations have errors lying within $\pm \sigma$. It is also found that about 50% of the total number of observations have errors lying within $\pm 0.6745 \sigma$ and this is taken as the probable value of error because there is an even chance for any one

observation to have a random error more than this value. Hence probable error, $r = \pm 0.6745 \sigma$.

1.6.7.3 Limiting error

The accuracy of a measuring instrument is usually specified by its manufacturer as a percentage of the full scale reading. However, the percentage error for various values of readings taken will be more than the percentage specified. The limiting error is defined as the maximum deviation in the reading. For example, the accuracy of a 0–150°C thermometer is specified as $\pm 1\%$ of full scale reading. The limiting error of this thermometer is $\pm 1.5^\circ\text{C}$. Hence, when the thermometer reads 60°C , the maximum possible deviation is $\pm 1.5^\circ\text{C}$ and the percentage error at this reading is $\pm 1.5/60 \times 100 = \pm 2.5\%$. Limiting error is also called as guarantee error.

1.6.7.4 Odds in specification

The specification of limiting error is in itself uncertain because the manufacturer himself is not sure about the accuracy because of the presence of random errors. To add a further specification of the uncertainty in a measurement Klins and McClintoch have proposed to specify certain odds for the uncertainty. For example, a temperature reading may be expressed as

$$T = 110^\circ\text{C} \pm 1.5^\circ\text{C} \text{ (20 to 1)}$$

This means that, if 21 readings are taken, one reading is likely to have an error more than $\pm 1.5^\circ\text{C}$.

The relation between probability of occurrence and odds is given by

$$\text{odds}/(\text{odds}+1) = \text{probability of occurrence}$$

Table 1.2 gives the relation between deviation, probability and odds.

σ is the standard deviation.

Table 1.2:

Deviation	Probability of occurrence within the deviation	Odds
$\pm 0.6745\sigma$	0.5000	1 : 1
$\pm \sigma$	0.6828	2.15 : 1
$\pm 2\sigma$	0.9546	21 : 1
$\pm 3\sigma$	0.9974	256 : 1

1.6.8 Propagation of Errors

When the readings from a number of instruments are used to compute some quantity, the overall limiting error should be computed from the individual limiting errors of the instruments.

Consider the problem of computing a quantity α from n independent measurements $u_1, u_2, \dots, u_n$. Now α can be expressed as

$$\alpha = f(u_1, u_2, \dots, u_n) \quad (1.1)$$

Let the limiting errors in $u_1, u_2, \dots, u_n$ be $\pm \Delta u_1, \pm \Delta u_2, \dots, \pm \Delta u_n$ respectively. These errors lead to an error of $\pm \Delta \alpha$ in the computed value of α . Therefore,

$$\alpha \pm \Delta \alpha = f(u_1 \pm \Delta u_1, u_2 \pm \Delta u_2, \dots, u_n \pm \Delta u_n) \quad (1.2)$$

Expanding the function f in equation (1.2) using Taylor's series, we get

$$\begin{aligned} & f(u_1 \pm \Delta u_1, u_2 \pm \Delta u_2, \dots, u_n \pm \Delta u_n) \\ &= f(u_1, u_2, \dots, u_n) + \Delta u_1 \frac{\partial f}{\partial u_1} + \Delta u_2 \frac{\partial f}{\partial u_2} + \dots + \Delta u_n \frac{\partial f}{\partial u_n} \\ & \quad + \frac{1}{2} \left[\Delta u_1^2 \frac{\partial^2 f}{\partial u_1^2} + \Delta u_2^2 \frac{\partial^2 f}{\partial u_2^2} + \dots + \Delta u_n^2 \frac{\partial^2 f}{\partial u_n^2} \right] \quad (1.3) \end{aligned}$$

In actual practice, the limiting errors $\Delta u_1, \Delta u_2, \dots$ etc. are small quantities and hence the higher powers of Δu 's are negligible. Hence equation 1.3 becomes

$$f(u_1 + \Delta u_1, u_2 + \Delta u_2, \dots, u_n + \Delta u_n) \\ = f(u_1, u_2, \dots, u_n) + \Delta u_1 \frac{\partial f}{\partial u_1} + \Delta u_2 \frac{\partial f}{\partial u_2} + \dots + \Delta u_n \frac{\partial f}{\partial u_n} \quad (1.4)$$

Hence the absolute error Δ is given by

$$\Delta = \left| \Delta u_1 \frac{\partial f}{\partial u_1} \right| + \left| \Delta u_2 \frac{\partial f}{\partial u_2} \right| + \dots + \left| \Delta u_n \frac{\partial f}{\partial u_n} \right| \quad (1.5)$$

The absolute value signs are used because some of the partial derivatives might be negative and for a positive Δu such a term would reduce the total error.

When the Δu 's are considered not as absolute limits of error, but rather as statistical bounds such as $\pm 3\sigma$ limits, probable errors, or uncertainties, the formulae for computing overall errors must be modified. It can be shown that the proper method of combining such errors is according to the root-sum square (r.s.s) formula.

$$E_{a_{r.s.s}} = \sqrt{\left(\Delta u_1 \frac{\partial f}{\partial u_1} \right)^2 + \left(\Delta u_2 \frac{\partial f}{\partial u_2} \right)^2 + \dots + \left(\Delta u_n \frac{\partial f}{\partial u_n} \right)^2}$$

EXAMPLE 1.3

The power dissipated in a resistor of $50 \pm 1 \Omega$ is computed by measuring the current through the resistor using the formula $P = I^2 R$. The ammeter used has an accuracy of $\pm 1\%$ and its full scale reading is 5A. Calculate the limiting error in the computed value of power, when the ammeter reads 3A.

SOLUTION

$$\text{Limiting error for ammeter} = \frac{\pm 1 \times 5}{100} = \pm 0.05A$$

$$\Delta = \left| \Delta u_1 \frac{\partial f}{\partial u_1} \right| + \left| \Delta u_2 \frac{\partial f}{\partial u_2} \right| = \left| \Delta I \frac{\partial I^2 R}{\partial I} \right| + \left| \Delta R \frac{\partial I^2 R}{\partial R} \right| \\ = |0.05 \times 2IR| + |1 \times I^2|$$

$$= |0.05 \times 2 \times 3 \times 50| + |1 \times 3^2| = 15 + 9$$

$$= 24 \text{ watts}$$

$$P = I^2 R = 3 \times 3 \times 50 = 450 \text{ W}$$

Percentage error in computed value of power

$$= \frac{\Delta}{P} \times 100 = \frac{24 \times 100}{450} = 5.33\%$$

1.7 UNITS AND STANDARDS

1.7.1 Fundamental And Derived Units

It is necessary to define a physical quantity both in kind and magnitude in order to use this information for further proceedings. The standard measure of each kind of physical quantity is named as the unit. This measure has been accepted by many of the users world over. Units can be broadly classified as *fundamental units* and *derived units*. Units which are fundamental to most other physical quantities are called fundamental units. They are length, mass, time, electric current, thermodynamic temperature and luminous intensity. All other units which can be expressed in terms of the fundamental units are called derived units. For example, the velocity of a moving vehicle is expressed as distance traveled divided by time.

1.7.2 System of Units

A number of systems of units are in use since 16th century. The important system of units are

- FPS System (foot, pound, second)
- CGS System (centimeter, gram, second)
- MKS System (meter, kilogram, second)
- Rationalised MKSA System (Meter, kilogram, second, ampere)

The international MKSA system of units was adopted in 1968 by the Eleventh general conference of weights and measure under the name Systems International d'Units (SI).

The SI system is replacing all other systems in the metric countries including India and it has been accepted internationally.

There are six fundamental units, two supplementary units and about twenty seven derived units which are listed in table 1.3.

Table 1.3: International System of units (SI units)

Sl. No	Quantity	Symbol	Unit	Unit symbol
FUNDAMENTAL				
1	Length	L	Meter	m
2	Mass	m	Kilogram	kg
3	Time	t	Second	s
4	Electric current	I	Ampere	A
5	Thermodynamic temperature	T	Kelvin	K
6	Luminous intensity	Z	candela	cd
SUPPLEMENTARY				
1	Plane angle	α, β	Radian	rad
2	Solid angle	Ω	steradian	sr
DERIVED				
1	Area	A	Square meter	m ²
2	Volume	V	Cubic meter	m ³
3	Frequency	f	Hertz	Hz
4	Density	ρ	Kilogram per cubic meter	Kg/m ³
5	Velocity	v	Meter per second	m/s
6	Angular velocity	ω	Radian per second	rad/s
7	Acceleration	a	Meter per second squared	m/s ²

Sl. No	Quantity	Symbol	Unit	Unit symbol
8	Angular acceleration	α	Radian per second squared	rad/s ²
9	Force	F	Newton	N
10	Pressure (stress)	P	Newton per square meter	N/m ²
11	Dynamic viscosity	μ	Newton-second per square meter	Ns/m ²
12	Kinematic viscosity	v	Square meter per second	m ² /s
13	Work, energy	W	Joule	J(Nm)
14	Power	P	Watt	W(J/s)
15	Quantity of electricity	Q	Coulomb	C(As)
16	Potential difference	V	volt	V(W/A)
17	Electric field strength	E	Volt per meter	V/m
18	Resistance	R	ohm	Ω (V/A)
19	Capacitance	C	farad	F(As/V)
20	Magnetic flux	ϕ	weber	Wb(Vs)
21	Magnetic field strength	H	Ampere per metre	A/m
22	Magnetic flux density	B	tesla	T(wb/m ²)
23	Inductance	L	henry	H(Vs/A)
24	Magnetic force	U	ampere	A
25	Luminous flux	F	Lumen	lm(cd/sr)

Sl. No	Quantity	Symbol	Unit	Unit symbol
26	Luminance	B	Candela per square metre	Cd/m ²
27	Illumination	E	lux	Lm/m ²

Even though SI units have been accepted by many, other system of units are also in vogue.

1.7.3 Standards of Measurement

The physical embodiment of a unit of measurement is a standard. For example, the fundamental unit of mass in the international system is the kilogram and defined as the mass of a cubic decimetre of water at its temperature of maximum density of 4°C. The physical representation of kilogram is the platinum-iridium alloy cylinder which is preserved at the International Bureau of Weights and Measures at Sevres near Paris. Standards are available for fundamental units of measurement and for some of the derived units.

1.7.3.1 Classification of standards

Standards can be classified by their function and application as follows:

- International standards
- Primary standards
- Secondary standards
- Working standards

The international standards are standards accepted internationally. They represent certain units of measurements which are very close to the maximum possible accuracy attainable by the current technology of production and measurement. These standards are maintained at the International Bureau of Weights and Measures at Sevres near Paris. They are not available for the common user of measuring instruments for the purpose of calibration or comparison.

Primary standards are maintained by National standards laboratory in different parts of the world. In India, National Physical laboratory at New Delhi is responsible for maintaining primary standards for various fundamental units. Primary standards are not available for use outside the national laboratories. One of the main functions of the primary standards is the verification and calibration of secondary standard.

Secondary standards are the basic reference standards available in different laboratories. These standards are maintained by a particular laboratory and are easily accessible to the measurement engineer.

1.7.3.2 Fundamental standards

The four fundamental quantities of the International measuring system, for which independent standards have been defined are mass, length, time and temperature.

The international standard for mass is standard kilogram, which is the mass of the platinum-iridium cylinder preserved at the International Bureau of Weights and Measures in France.

The international standard for length is standard meter, which is materially represented by the distance between two lines engraved on a platinum-iridium bar preserved again in France.

The fundamental unit of time, the second, has been defined in the past as 1/86, 400 of a mean solar day. The solar day is measured as the time interval between the two successive transits of the sun across a meridian of the earth. Subsequently it was discovered that the duration of mean solar day is subjected to irregular variations. The current definition of standard second is the interval of time corresponding to 9, 192, 631, 770 cycles of the atomic resonant frequency of cesium 133.

The SI unit for thermodynamic temperature is Kelvin. The magnitude of the Kelvin has been fixed by defining the thermodynamic temperature of triple point of water at exactly 273.16°K. The triple point of water is the temperature at which the solid, liquid and vapour phases of water are in equilibrium. The temperature measurements on the thermodynamic scale are

inherently difficult because it depends on an ideal Carnot cycle. Hence a practical scale has been adopted in 1927 which has been modified several times and now is called as the International Practical Temperature Scale. The temperature on this scale is designated as degree Celsius ($^{\circ}\text{C}$). The Celsius scale has two fundamental fixed points, viz., the boiling point of water of 100°C and triple point of water as 0.01°C .

A number of primary fixed points has been established above and below the two fundamental points. The primary fixed points are given in table 1.4.

Various secondary fixed points also are established with the lowest being the triple point hydrogen (-259.34°C). some of the secondary fixed points are given in table 1.5.

Table 1.4: Primary points for the International Temperature scale

S.No.	Primary fixed point at atmospheric pressure	Temperature $^{\circ}\text{C}$
1	Boiling point of oxygen	-182.962
2	Triple point of water	0.01
3	Boiling point of water	100.00
4	Freezing point of zinc	419.58
5	Freezing point of silver	961.93
6	Freezing point of gold	1064.43

Table 1.5: Secondary fixed points for the International Temperature Scale

S.No.	Secondary fixed point	Temperature $^{\circ}\text{C}$
1	Freezing point of mercury	-38.87
2	Freezing point of tin	231.9
3	Freezing point of aluminium	660.1
4	Freezing point of nickel	453.0
5	Freezing point of platinum	1769.0
6	Freezing point of iridium	2443.0

1.7.3.3 Electrical standards

All electrical measurements are based upon three fundamental quantities viz. Voltage, current and impedance. The SI units for these quantities are volt, ampere and ohm respectively.

The International Ampere is defined as the current which when passed through a solution of silver nitrate, causes deposition of silver at the rate of 1.118 mg/sec . In 1948 the International Ampere was superseded by the Absolute Ampere which is defined as the constant current which, if maintained in two straight parallel conductors of infinite length and negligible cross section placed 1m apart in vacuum, will produce between these conductors a force equal to 2×10^{-7} newton per meter length. The absolute ampere is now the fundamental unit of electric current in the SI. The following relation exists between International Ampere and Absolute Ampere.

$$\text{International Ampere} = 0.99985 \text{ Absolute Ampere}$$

The standard volt was based on an electrochemical cell called the saturated standard cell for many years. The standard cell suffers from the drawback of temperature dependence of about $-40 \mu\text{V}/^{\circ}\text{C}$ from the nominal value of 1.01858 volt . In 1962, a new standard for voltage was developed by Brain Josephson. This

is based on the voltage developed across a thin film junction which is cooled to nearly absolute zero and irradiated with microwave energy. The voltage developed is related to the irradiating frequency by the following relationship

$$v = \frac{hf}{2e}$$

where h is Planck's constant
 f is the frequency of the microwave irradiation.
 e is the electron volt

The accuracy of standard volt based on this technique is one part in 10^8 as the standard volt is related to the standard of time/frequency.

The impedance in an electrical circuit is due to resistance, inductance and capacitance. The SI unit for resistance is ohm, which is defined in terms of fundamental units of mass, length and time. As the international standards for current and voltage have been defined, the absolute measurement of resistance can be made using ohm's law. The national laboratories usually preserve a group of primary resistance standards which are periodically verified by absolute measurements. The standard resistor is a cell of wire of some alloy like manganin which has a high electrical resistivity and a low temperature coefficient of resistance. The secondary standards for resistance are available from instrument manufacturers over a wide range of values.

The SI unit for capacitance is farad. The absolute measurement of capacitance can be carried out using a Maxwell d.c. commutated bridge which is described below.

Fig. 1.8 shows the Maxwell d.c. commutated bridge. The capacitor C is alternately charged and discharged through the commutating contact and resistance R . The frequency of the commutation is decided by the frequency of a.c. excitation.

The bridge balance is obtained by adjusting R_3 . The capacitance value is determined in terms of the bridge arm constants and the frequency of commutation. The primary standards for capacitors are usually constructed from interleaved

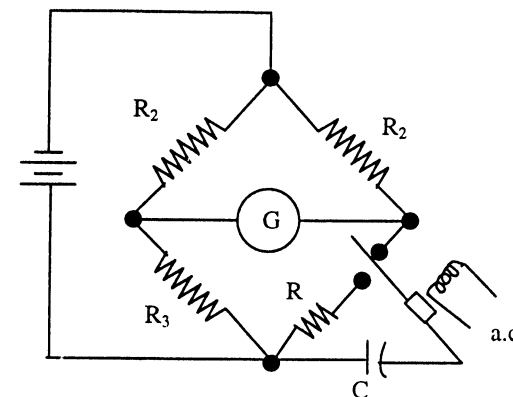


Fig 1.8

Maxwell d.c. commutated bridge to measure capacitance

metal plates with air as the dielectric medium. The area of the plate and the distance between them are measured very accurately and the capacitance is calculated from the basic dimensions. These capacitors are preserved in the specified environmental conditions such as humidity, temperature and pressure. A bank of standard air capacitors are preserved in National Physical laboratory.

The SI unit for inductance is henry. The inductance standard can be derived from resistance and capacitance standards. The campbell standard of mutual inductance is considered as the primary standard for both mutual and self-inductance. (Ref. 1)

1.7.3.4 Standards for luminous intensity

The primary standard of luminous intensity is derived from full radiator (black body or planckian radiator) at the temperature of solidification of platinum (2042 °K approx). The candela is then defined as one-sixtieth (1/60) of the luminous intensity per square centimetre of the full radiator. Secondary standards of luminous intensity are obtained from special tungsten filament lamps, operated at a temperature whereby their spectral power distribution in the visible region matches that of the basic standard. These secondary standards are recalibrated against the basic standard at periodic intervals.

1.7.3.5 Standards for pressure

Even though pressure is a derived quantity from force and area, pressure standards in the form of very accurate instruments are available. The basic standards for pressures ranging from 10^{-1} mm Hg to several hundred thousand kilopascals, are in the form of precision mercury column manometers and dead weight piston gauges. For pressures in the range 10^{-1} to 10^{-3} mm Hg, the McLeod vacuum gauge is considered as the standard.

1.8 WORKED OUT EXAMPLES

EXAMPLE 1.8.1

One hundred temperature readings were taken at small intervals of time and recorded to the nearest 0.5°C . The frequency of occurrences of the readings is given below:

Temperature readings $^{\circ}\text{C}$	98.5	99	99.5	100	100.5	101.0	101.5
Frequency	4	13	19	35	17	10	2

Calculate (i) arithmetic mean, (ii) average deviation, (iii) standard deviation, (iv) variance and (v) probable error

SOLUTION

(i) Arithmetic mean,

$$\bar{x} = \frac{x_1 f_1 + x_2 f_2 + \dots + x_7 f_7}{f_1 + f_2 + \dots + f_7}$$

$$= \frac{98.5 \times 4 + 99 \times 13 + 99.5 \times 19 + 100 \times 35}{100}$$

$$= \frac{100.5 \times 17 + 101 \times 10 + 101.5 \times 2}{100}$$

$$= \frac{9993}{100}$$

$$= 99.93^{\circ}\text{C}$$

(ii) Deviation from the mean,

$$d_1 = 98.5 - 99.93 = -1.43$$

$$d_2 = 99.0 - 99.93 = -0.93$$

$$d_3 = 99.5 - 99.93 = -0.43$$

$$d_4 = 100.0 - 99.93 = +0.07$$

$$d_5 = 100.5 - 99.93 = +0.57$$

$$d_6 = 101.0 - 99.93 = +1.07$$

$$d_7 = 101.5 - 99.93 = +1.57$$

Average deviation, absolute

$$D = \frac{|d_1| f_1 + \dots + |d_7| f_7}{f_1 + f_2 + \dots + f_7}$$

$$= \frac{(1.43 \times 4) + (0.93 \times 13) + (0.43 \times 19) + (0.07 \times 35)}{100}$$

$$= \frac{+ (0.57 \times 17) + (1.07 \times 10) + (1.57 \times 2)}{100}$$

$$= \frac{51.96}{100} = 0.5196^{\circ}\text{C}$$

(iii) The standard deviation

$$\sigma = \sqrt{\frac{d_1^2 f_1 + d_2^2 f_2 + \dots + d_7^2 f_7}{f_1 + f_2 + \dots + f_7}}$$

$$= \frac{(-1.43)^2 \times 4 + (0.93)^2 \times 13 + (0.43)^2 \times 19 + (0.07)^2 \times 35}{100}$$

$$= \frac{+ (0.57)^2 \times 17 + (1.07)^2 \times 10 + (1.57)^2 \times 2}{100}$$

$$= \sqrt{\frac{45.01}{100}} = 0.671^{\circ}\text{C}$$

(iv) Variance $\sigma^2 = 0.4501$

$$\begin{aligned} \text{(v) Probable error} &= 0.6745 \times \sigma \\ &= 0.6745 \times 0.671 \\ &= 0.4526^{\circ}\text{C} \end{aligned}$$

EXAMPLE 1.8.2

Two resistors of $470\ \Omega \pm 10\%$ and $330\ \Omega \pm 5\%$ are connected in parallel. Calculate (i) effective resistance neglecting errors and (ii) effective resistance taking errors into account.

SOLUTION

(i) Effective resistance neglecting errors

$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{470 \times 330}{470 + 330} = 193.9\ \Omega$$

(ii) Error in R

$$\Delta R = \left| R_1 \frac{\partial R}{\partial R_1} \right| + \left| R_2 \frac{\partial R}{\partial R_2} \right|$$

$$= R_1 \frac{R_2^2}{(R_1 + R_2)^2} + R_2 \frac{R_1^2}{(R_1 + R_2)^2}$$

$$= \frac{470 \times 330^2 + 330 \times 470^2}{800^2} = 13.69\ \Omega$$

Effective resistance taking errors into account = $193.9 \pm 13.69\ \Omega$
 $= 193.9\ \Omega \pm 7.06\%$

EXAMPLE 1.8.3

Five observers had measured a voltage and recorded as 117.02, 117.11, 117.02, 117.03, 117.08. Calculate (i) the average voltage, (ii) max. deviation and (iii) average deviation.

SOLUTION

(i) Average voltage

$$= \frac{117.02 + 117.11 + 117.02 + 117.03 + 117.08}{5}$$

$$= 117.052$$

(ii) Maximum deviation = $117.11 - 117.052 = +0.058$

(iii) $5 \times$ Ave. deviation

$$= |117.02 - 117.052| + |117.11 - 117.052|$$

$$+ |117.02 - 117.052| + |117.03 - 117.052|$$

$$+ |117.08 - 117.052|$$

Ave. deviation

$$= \frac{0.032 + 0.0058 + 0.032 + 0.022 + 0.028}{5}$$

$$= \frac{|0.172|}{5} = 0.0344$$

EXAMPLE 1.8.4

The value of a resistor is specified as $500\ \Omega \pm 10\%$ by a manufacturer. Find the limits of resistance between which the value is guaranteed.

SOLUTION

The limiting value of resistance = $(10/100) \times 500 = 50\ \Omega$

$\therefore$ The limits of the resistance is = $450\ \Omega$ to $550\ \Omega$.

EXAMPLE 1.8.5

Temperature of a metal bath is measured 100 times with variations in apparatus, procedures and persons. The readings are tabulated below.

Temp.°C	397	398	399	400	401	402	403	404	405
Freq.	1	3	12	23	37	16	4	2	2

Find (i) Mean, (ii) Mode, (iii) Mean deviation, (iv) Standard deviation, (v) probable error of one reading.

SOLUTION

Temp. T_i	Freq f_i	$T_i \times f_i$	$d_i = T_i - \bar{T}$	$ f_i \times d_i $	d_i^2	$f_i d_i^2$
397	1	397	-3.78	3.78	14.2884	14.288
398	3	1194	-2.78	8.34	7.7284	23.185
399	12	4788	-1.78	21.36	3.1684	38.020
400	23	9200	-0.78	17.94	0.6084	13.993
401	37	14837	+0.22	8.14	0.0484	1.790
402	16	6432	+1.22	19.52	1.4884	23.814
403	4	1612	+2.22	8.88	4.9284	19.214
404	2	808	+3.22	6.44	10.3684	20.734
405	2	810	+4.22	8.44	17.8084	35.618
40078	100	40078		102.84		190.656

- (i) Mean temperature $\bar{T} = \frac{400780}{100} = 400.78^\circ\text{C}$
(ii) Mode $= 401^\circ\text{C}$
(iii) Mean deviation $= 102.8/100 = 1.028^\circ\text{C}$
(iv) Standard deviation (σ) $= \sqrt{(190.65/100)} = 1.38^\circ\text{C}$
(v) Probable error of any one reading $= 0.6745\sigma = 0.93^\circ\text{C}$.

EXAMPLE 1.8.6

The following 10 observations were recorded when measuring a voltage 41.7, 42.0, 41.8, 42.0, 42.1, 41.9, 42.5, 42.0, 41.9, 41.8. Find (i) Mean, (ii) S.D, (iii) Probable error of one reading.

SOLUTION

x_i	d	d^2
41.7	-02.7	0.0729
42.0	+0.03	0.0009
41.8	-0.17	0.0289
42.0	+0.03	0.0009
42.1	+0.13	0.0169
41.9	-0.07	0.0049
42.0	+0.03	0.0009
41.9	-0.07	0.0049
42.5	+0.53	0.2809
41.8	-0.17	0.0289
419.7		0.441

$$\text{Mean} = \frac{\sum x_i}{N} = \frac{419.7}{10} = 41.97$$

$$\text{S.D.} = \sqrt{\frac{\sum d^2}{N-1}} = \sqrt{\frac{0.441}{9}} = 0.221$$

$$\text{Probable error of one reading } r_i = 0.6745 (\sigma) \\ = 0.1490 \text{ v}$$

EXAMPLE 1.8.7

A batch of colour coded resistors of value 5.6 k Ω were measured and were found to have the following values. Determine the mean and standard deviation. Can any resistor be discarded on the basis of $\pm 3\sigma$ limits.

5.75, 5.60, 5.65, 5.50, 5.70, 5.55, 5.80 and 5.55 k Ω

SOLUTION

x_i	$ d_i = x_i - \bar{x} $	d_i^2
5.75	0.1125	0.0127
5.60	0.0375	1.406×10^{-3}
5.65	0.0125	1.5625×10^{-4}
5.50	0.1375	0.0189
5.70	0.0625	3.906×10^{-3}
5.55	0.0875	7.656×10^{-3}
5.80	0.1625	0.0264
5.55	0.0875	7.656×10^{-3}
45.10	0.7	0.0788

$$\bar{x} \text{ Mean} = \frac{45.10}{8} = 5.6375 \text{ k}\Omega$$

$$\text{S.D. } \sigma = \sqrt{\frac{\sum d_i^2}{n-1}} = 0.106 \text{ k}\Omega$$

$$\pm 3\sigma = 0.3182 \text{ k}\Omega$$

No resistor can be discarded.

EXAMPLE 1.8.8

Six measurements are made on a quantity as given below. Determine the mean, median, average deviation, standard deviation and variance.

12.35, 12.71, 12.48, 10.24, 12.63, 12.58.

SOLUTION

x_i	$d_i = x_i - \bar{x} $	d_i^2
12.35	0.185	0.034
12.71	0.545	0.297
12.48	0.315	0.099
10.24	1.925	3.706
12.63	0.465	0.216
12.58	0.415	0.172
72.99	3.850	4.524

$$\text{Median} = \frac{12.48 + 12.58}{2} = 12.53$$

$$\text{Mean} = \frac{\sum x_i}{n} = \frac{72.99}{6} = 12.165$$

$$\text{Variance } (\sigma^2) = \frac{\sum d_i^2}{n-1} = \frac{4.524}{5} = 0.905$$

$$\text{Std. deviation } \sigma = 0.951$$

$$\text{Ave. deviation} = \frac{\sum |d_i|}{n} = \frac{3.85}{6} = 0.6417$$

EXAMPLE 1.8.9

For the following readings, find the mean, median, standard deviation and variance.

24.2, 24.3, 23.9, 24.6, 24.1

SOLUTION

x_i	$d_i = x_i - \bar{x} $	d_i^2
24.2	0.02	0.0004
24.3	0.08	0.0064
23.9	0.32	0.1024
24.6	0.38	0.1444
24.1	0.12	0.0144
121.1	0.92	0.2680

$$\text{Median} = 24.2$$

$$\text{Mean} = \bar{x} = \frac{\sum x_i}{n} = \frac{121.1}{5} = 24.22$$

$$\text{Variance } (\sigma^2) = \frac{\sum d_i^2}{n-1} = \frac{0.268}{5-1} = 0.067$$

$$\text{Std. deviation } \sigma = \sqrt{\text{variance}} = 0.2588$$

$$\text{Ave. deviation} = \frac{\sum |d_i|}{n} = \frac{0.92}{5} = 0.184$$

EXAMPLE 1.8.10

A series circuit is having three resistances whose values are given by $R_1 = 37 \Omega \pm 5\%$, $R_2 = 75 \Omega \pm 5\%$, $R_3 = 50 \Omega \pm 5\%$. Determine the total resistance and the limiting value.

SOLUTION

Let R_T the total resistance be $= R_1 + R_2 + R_3$
 $= 162$

Total resistance $R = R_1 + R_2 + R_3$

Limiting error

$$\begin{aligned}\Delta R_T &= \left| \Delta R_1 \frac{\partial}{\partial R_1} R_T \right| + \left| \Delta R_2 \frac{\partial}{\partial R_2} R_T \right| + \left| \Delta R_3 \frac{\partial}{\partial R_3} R_T \right| \\ &= |\Delta R_1| + |\Delta R_2| + |\Delta R_3| \\ &= 1.85 + 3.75 + 2.5 = 8.1\end{aligned}$$

The limiting value of resistance $= 162 \pm 8.1 \Omega$
 $= 162 \pm 5\%$

EXAMPLE 1.8.11

A resistance is measured by voltmeter - ammeter method. Ammeter shows 3.22A on a 10A scale and the voltmeter shows 96.6v on 300v range. Both ammeter and voltmeter are guaranteed to be accurate within $\pm .5\%$ of full scale reading. Find the indicated value of resistance, and the limits within which you can guarantee the result.

SOLUTION

Indicated value of resistance $= 96.6/3.22 = 30 \Omega$

Max. possible error in ammeter $= \pm (0.5/100) \times 10 = \pm 0.05 \text{ A}$

Max. possible error in voltmeter $= \pm (0.5/100) \times 300 = \pm 1.5 \text{ V}$

$$\therefore \text{ the Limiting error } |\Delta R| = \left| \Delta V \frac{\partial}{\partial V} \left(\frac{V}{I} \right) \right| + \left| \Delta I \frac{\partial}{\partial I} \left(\frac{V}{I} \right) \right|$$

$$\begin{aligned}&= \left| \Delta V \left(\frac{1}{I} \right) \right| + \left| \Delta I \left(-\frac{V}{I^2} \right) \right| \\ &= 1.5 (1/3.22) + 0.05 (96.6/3.22^2) = 0.932\end{aligned}$$

The actual resistance may be $30 \pm 0.932 \Omega$

EXAMPLE 1.8.12

The power factor in a circuit having sinusoidal voltage and current are determined by measuring current, voltage and power.

The voltage is 125.0 v on a voltmeter of 150 v scale.

The current is 3.0 A on a ammeter of 5A scale.

The power is 225W on a 500W wattmeter.

Ammeter and voltmeter are guaranteed to be accurate within $\pm 0.5\%$ of full scale and wattmeter $\pm 0.25\%$ of full scale.

- to what % accuracy can you guarantee the power factor obtained from the readings.
- For the possible error in (a) show the amount of contribution by each instrument

SOLUTION

(a) voltmeter reading $= 125\text{V}$

Limiting error $= \pm 0.75\text{V}$

Ammeter reading $= 3\text{A}$

Limiting error $= \pm 0.025\text{I}$

Power $= 225\text{W}$

Limiting error $= \pm 1.25\text{W}$

Power factor as per the reading $= P/V1 = 225/(125 \times 3) = 0.6$

Error in power factor reading is

$$\begin{aligned}|\Delta \text{PF}| &= \left| \Delta P \frac{\partial}{\partial P} \left(\frac{P}{VI} \right) \right| + \left| \Delta V \frac{\partial}{\partial V} \left(\frac{P}{VI} \right) \right| + \left| \Delta I \frac{\partial}{\partial I} \left(\frac{P}{VI} \right) \right| \\ &= \left| 1.25 \times \frac{1}{125 \times 3} \right| + \left| 0.75 \frac{-225}{3 \times 125^2} \right| + \left| 0.025 \frac{-225}{125 \times 3^2} \right|\end{aligned}$$

$$= 0.003 + 0.005 + 0.0036 = 0.0116$$

The actual power factor = 0.6 ± 0.0116

Accuracy of the power factor is $\pm 1.93\%$

(b) contribution by each instruments

$$\text{Voltmeter contributes } \frac{0.0036}{0.6} \times 100 = 0.6\%$$

$$\text{Ammeter contributes } \frac{0.005}{0.6} \times 100 = 0.833\%$$

$$\text{Wattmeter contributes } \frac{0.003}{0.6} \times 100 = 0.5\%$$

$$\underline{\underline{1.933\%}}$$

EXAMPLE 1.8.13

A wheatstone bridge has ratio arms which are accurate within $\pm 0.025\%$, and a variable arm within $\pm 0.05\%$. What is the possible error in a resistance measurement on this bridge?

SOLUTION

Let the ratio arms be P & Q and the variable arm be S.

The unknown resistance $R = (P/Q) \times S$

$$|\Delta R| = \left| \Delta P \frac{\partial}{\partial P} \left(\frac{PS}{Q} \right) \right| + \left| \Delta S \frac{\partial}{\partial S} \left(\frac{PS}{Q} \right) \right| + \left| \Delta Q \frac{\partial}{\partial Q} \left(\frac{PS}{Q} \right) \right|$$

$$= \left| \Delta P \times \frac{S}{Q} \right| + \left| \Delta S \frac{P}{Q} \right| + \left| \Delta Q (-) \frac{PS}{Q^2} \right|$$

For equal values of P & Q

$$|\Delta R| = \left| \Delta P \times \frac{S}{Q} \right| + |\Delta S| + \left| \Delta Q \frac{S}{Q} \right|$$

$$\therefore \Delta R = 2 \times 0.025 \frac{S}{Q} + 0.05 = \pm 0.05 \left(1 + \frac{S}{Q} \right)$$

EXAMPLE 1.8.14

A liquid flows through a pipe having a diameter of 100 mm with a velocity of 1 m/s. Calculate the flow rate. If the diameter is

guaranteed within $\pm 1\%$ and the velocity is given to be within $\pm 3\%$ error of the measured value, find the limits within which rate of flow can be specified.

SOLUTION

$$\text{Diameter of the pipe} = 100 \text{ mm}$$

$$= 0.1 \text{ m}$$

$$\text{Area} = 7.854 \times 10^{-3} \text{ m}^2$$

Error in flow rate

$$\Delta Q = \left| \Delta d \frac{\partial}{\partial d} \left(\frac{\pi d^2 v}{4} \right) \right| + \left| \Delta v \frac{\partial}{\partial v} \left(\frac{\pi d^2 v}{4} \right) \right|$$

$$= \Delta d \left(\frac{\pi dv}{2} \right) + \Delta v \left(\frac{\pi d^2}{4} \right)$$

$$= \frac{1}{100} \times \frac{\pi}{2} \times 0.1 \times 1 + \frac{3}{100} \times \frac{\pi}{4} \times (0.1)^2$$

$$= 0.01 \times \frac{\pi \times 0.1 \times 1}{2} + 0.03 \times \frac{1 \times \pi \times (0.1)^2}{4}$$

$$= 1.8055 \times 10^{-3}$$

The limits of the measured flow rate

$$= 7.854 \times 10^{-3} \pm 1.8055 \times 10^{-3}$$

$$= 7.854 \times 10^{-3} + 22.98\%$$

CHAPTER 2

CLASSIFICATION OF TRANSDUCERS

2.1 DEFINITION OF TRANSDUCER

A number of definitions are available for a transducer in the literature. The ISA – the international society for measurement & control – USA, defines a transducer as a sensing device that converts physical phenomenon into electrical, pneumatic or hydraulic output signal. A transducer is defined as a device that draws energy from the measured medium and produces an output depending in some way on the measured quantity. The transducer is also defined as a device which, when actuated by one form of energy, is capable of converting it to another form of energy. However the efficiency of the energy conversion is not important. Hence transducers can be defined as devices which convert one form of signal into another form preferably electrical form.

In literature an instrument transducer is defined as a device which for the purpose of measurement turn physical input quantities into electrical output signal, its output-input and output-time relationships being predictable to a known degree of accuracy at specified environmental conditions.

Because of the recent technological advancement in electronics and computers, the instrumentation systems are getting oriented towards more and more of electronics. Hence the transducer technology is also concentrating on the devices having electrical outputs. Therefore, the above definition assumes more significance in the present day context. This book deals with transducers having electrical outputs only.

For example, a tachnogenerator is a transducer as it converts speed signal into voltage signal for measurement purpose.

2.2 CLASSIFICATION OF TRANSDUCERS

The transducers can be classified in three different ways.

- Based on the physical effect employed
- Based on the physical quantity they convert
- Based on source of energy for their output

2.2.1 Classification based on Physical Effect Employed

The basic principle of operation of a transducer is the physical effect or effects employed to convert the physical quantity into electrical quantity. For example, a change in temperature causes variation of resistance in a copper wire and this effect could be used to convert temperature into an electrical output.

The physical effects commonly employed in the design of transducers are given below:

2.2.1.1 Variation of resistance

The resistance of a length of metallic wire is given by

$$R = \frac{\rho l}{a}$$

where R is the resistance in ohm

ρ is the specific resistance of the material in ohm-m

l is the length of wire in m

and a is the area of cross-section in m².

In other words, $R = f(\rho, l, a)$ when any one of the physical quantities ρ , l or a changes due to a physical effect, a variable resistance transducer can be designed to convert that physical quantity.

Some of the transducers based on this principle are potentiometer, strain gauge, resistance thermometer, carbon microphone and photoconductive cell.

The resistance thermometer is based upon thermo resistive effect which is the change in electrical resistivity of a metal or

semiconductor due to change in temperature coefficient of resistivity. Carbon microphone works on the principle of change in contact resistance due to applied pressure. Photoconductive cell is based upon photoconductive effect which is the change in electrical conductivity due to incident light.

2.2.1.2 Variation of inductance

The inductance of a coil is given by

$$L = N \frac{d\phi}{dt}$$

$$= \frac{N^2 \mu_0 \mu_r A}{l}$$

where L is the inductance
 N is the no. of turns
 μ_0 is the absolute permeability
 μ_r is the relative permeability
 A is the area of cross section of the core
 l is the length of magnetic path
 and $d\phi/dt$ is the rate of change of magnetic flux.

Hence $L = f(N, \mu_r, A, l)$

When any one of the quantities within the parenthesis changes, the inductance changes, which leads to the design of a variable inductance transducer.

Some of the transducers of this category are induction potentiometer, linear variable differential transformer and synchros.

2.2.1.3 Variation of capacitance

The capacitance between two conductor plates is given by

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

where C is the capacitance
 ϵ_0 is the absolute permittivity

ϵ_r is the relative permittivity of the separating medium

A is the area of cross-section of the plates

d is the distance between the plates

Hence $C = f(\epsilon_r, A, d)$

A variable capacitance transducer can be made for any physical quantity that causes a change in ϵ_0 , A or d .

An example of variable capacitance transducer is a capacitor microphone which converts sound pressure into an electrical signal.

2.2.1.4 Piezoelectric effect

Whenever a piezoelectric crystal like quartz or Rochelle salt is subjected to a mechanical stress, an electric charge is generated. This is known as piezoelectric effect.

2.2.1.5 Magnetostrictive effect

The permeability of a magnetic material changes when it is subjected to a mechanical stress. This is called villari effect. Another physical effect which is also used in magnetostrictive transducers is known as wertheim effect, which is the development of a transient voltage between the ends of a wire twisted in a longitudinal magnetic field.

2.2.1.6 Elastic effect

When an elastic member is subjected to mechanical stress it is deformed. This deformation can be used to construct a primary sensor.

2.2.1.7 Hall effect

The development of a transverse electric potential gradient in a current-carrying conductor, upon the application of a magnetic field at right angles to the direction of current is known as Hall effect.

The physical effects mentioned above are not complete. Transducers making use of other effects are available in plenty.

2.2.2 Classification based on Quantity Converted

This classification is based on the end use of the transducers. For example, transducers used to convert temperature into electrical signal are termed as temperature transducer - for example, a thermocouple is a temperature transducer.

The classification is as follows:

- Temperature transducers
- Pressure transducers
- Flow transducers
- Liquid level transducers
- Humidity transducers
- Displacement transducers
- Velocity / speed transducers
- Acceleration / vibration transducers
- Force / torque transducers

Transducers under each of the above categories may make use of more than one physical effect.

2.2.3 Classification based on Source of Energy for the Output

A third way of classifying transducers is based upon the source of energy for their output. Accordingly there are two types namely active and passive transducers.

A passive transducer is one which absorbs energy from the input medium and converts it directly into the output signal. An example of a passive transducer is a thermocouple which extracts heat energy from the input medium and converts it into an electrical voltage. In this process, a disturbance will be caused to the input medium as some energy is absorbed by the medium. These transducers are of self-generating type.

An active transducer on the other hand, uses the energy it absorbs from the input medium as a control signal to transfer energy from the power supply to produce a proportional output. An example of an active transducer is a strain gauge. Strain in a member is converted into a resistance variation in a strain gauge. The energy extracted from the strained member is very small. The

energy for the output signal is supplied by an external power source.

The active and passive transducers are illustrated in fig. 2.1.

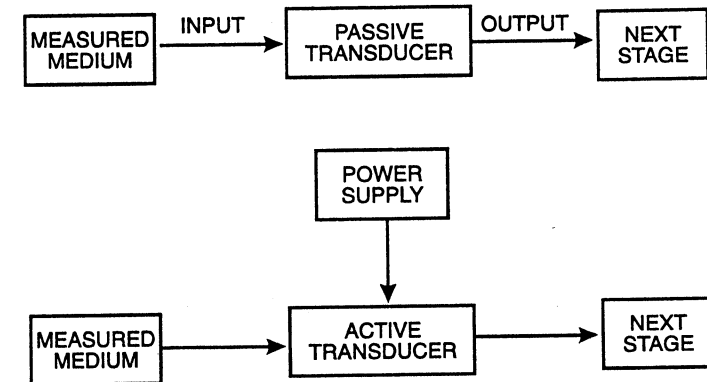


Fig 2.1

Active and passive transducers

Some authors, however, have defined active and passive transducers exactly in the opposite way.

GENERALISED CHARACTERISTICS OF TRANSDUCERS

3.1 INTRODUCTION

The most important step in designing an Instrumentation System is the selection of the transducer. The knowledge of the performance characteristics of the transducer is essential for the proper choice of the transducer. The performance characteristics can be broadly classified into two categories namely Static Characteristics and Dynamic Characteristics.

Static characteristics is a set of performance criteria that give a meaningful description of the quality of measurement while the measured quantities are either constant or vary slowly.

On the other hand, dynamic characteristics describe the quality of measurement when the measured quantities vary rapidly with time. The dynamic characteristics do not play any role in deciding the static behavior of the transducer and the static characteristics do not affect the dynamic behavior of the transducer. The dynamic characteristics are generally described by linear differential equations. When static characteristics are also taken into account, these differential equations become non-linear and unmanageable to deal with. Hence, the conventional approach is to treat the two aspects of the problem separately.

3.2 STATIC CHARACTERISTICS

The important static characteristics of a transducer or instrument are

- static sensitivity
- linearity

- precision / Accuracy
- threshold
- resolution
- hysteresis
- range and span
- input impedance and loading effect.

All these characteristics are obtained in one form or other through static calibration. The desired input is one of the many inputs to the transducer which is of interest in the particular application. A transducer may respond to a number of inputs out of which one may be the desired quantity i.e. the measured and the others are called interfering or the modifying inputs. Static calibration refers to a situation in which all inputs except the desired one are kept constant at some particular value.

It may be noted that some of the inputs are determinable and measurable whereas some of them may be random and unmeasurable. The desired input is varied over some range in steps and the output values are noted. The input - output relationship thus developed is called the static calibration valid under the stated constant conditions of all other inputs.

3.2.1 Static Sensitivity

Static sensitivity of a transducer can be defined as the slope of the static calibration curve. If the calibration curve is not a straight line then the sensitivity will vary with input value as shown in fig. 3.1. hence the sensitivity should be taken depending on the operating point. The sensitivity is expressed in output unit / input unit. For example, the sensitivity of a mercury-in-glass thermometer is expressed in $\text{mm}/^{\circ}\text{C}$ and that of a pressure gauge is expressed in angular degrees/kilopascal (KPa).

While the sensitivity of a transducer to the desired input is important, the sensitivity to other inputs namely interfering and modifying inputs are also necessary to select transducer for a particular application.

Sensitivity at X_1 , $S_1 = \Delta Y_1 / \Delta X$

Sensitivity at X_2 , $S_2 = \Delta Y_2 / \Delta X$

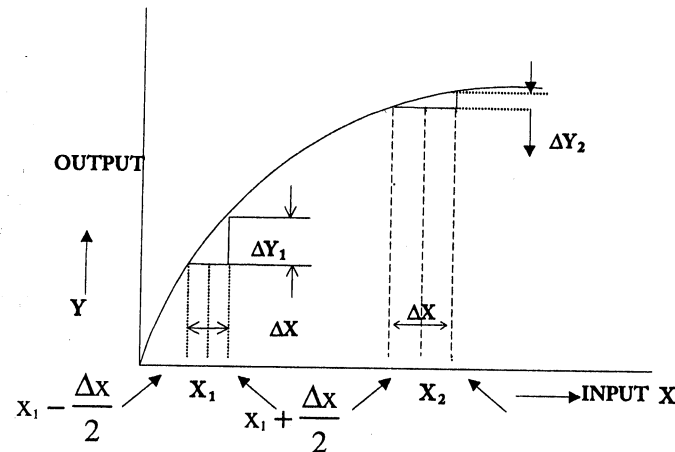


Fig 3.1
Calibration curve

3.2.2 Linearity

In many cases the calibration curve of a transducer may not be linear. However a linear relationship is most desirable for many applications. For example, when a transducer is part of a control system, linear behavior of the transducer simplifies the analysis and design of the system. Also the conversion from the scale reading to the measured value of the input quantity becomes easier, if it is to be multiplied by a fixed constant rather than looking into a calibration chart or a graph.

Linearity is a measure of the maximum deviation of the plotted transducer response from a specified straight line. To specify or to select a straight line for a plotted calibration curve there are a number of ways. Some of these are

- the straight line connecting the calibration point at zero input to that at full-scale input.
- the straight line may be drawn through as many calibration points as possible.

- the straight line may be determined by the least squares fit method mathematically. The input-output relationship of a transducer is generally given by the equation

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots + a_n x^n \quad (3.1)$$

where x is the input quantity, y is the output quantity and $a_0, a_1, \dots, a_n$ are the calibration factors. The best-fit straight line is mathematically determined by evaluating the deviation of the response curve from the straight line at a number of calibration points and choosing the one that gives the minimum of the sum of the squares of the deviations. This procedure is described as least squares fit.

Method of least squares

Assume that the input to a transducer ' x ' is varied over its full range and output ' y ' is measured. Let the total number of measurements be n . The linearised relation between x and y can be expressed as

$$Y = ax + b \quad (3.2)$$

Where a and b are constants. The constants ' a ' and ' b ' are determined using least-square fit.

The deviation of the i th reading from the straight line specified by $y = ax + b$

$$= y_i - (ax_i + b) \quad (3.3)$$

Sum of the squares of the deviation

$$S = \sum_{i=1}^n [y_i - (ax_i + b)]^2 \quad (3.4)$$

S would be minimised by setting the following derivatives equal to zero.

$$\frac{\partial S}{\partial a} = 0 = \sum_{i=1}^n (bx_i + ax_i^2 - x_i y_i) \quad (3.5)$$

$$\frac{\partial S}{\partial b} = 0 = \sum_{i=1}^n (b + ax_i - y_i) \quad (3.6)$$

Solving the above two equations, we get

$$a = \frac{n \sum x_i y_i - \sum (x_i) \sum (y_i)}{n \sum x_i^2 - (\sum x_i)^2} \quad (3.7)$$

$$b = \frac{(\sum y_i) (\sum x_i^2) - \sum (x_i y_i) \sum (x_i)}{n \sum x_i^2 - (\sum x_i)^2} \quad (3.8)$$

This method of least squares can also be used for determining higher-order polynomial for a data set.

Linearity can be expressed as a percentage of the actual reading or a percentage of full-scale reading or a combination of the two. The most realistic method of expressing linearity is the combination of both actual and full-scale reading, which is known as the independent linearity.

Independent linearity = $\pm A\%$ of reading or
 $\pm B\%$ of full-scale

whichever is greater.

The specification of independent linearity is illustrated in fig. 3.2.

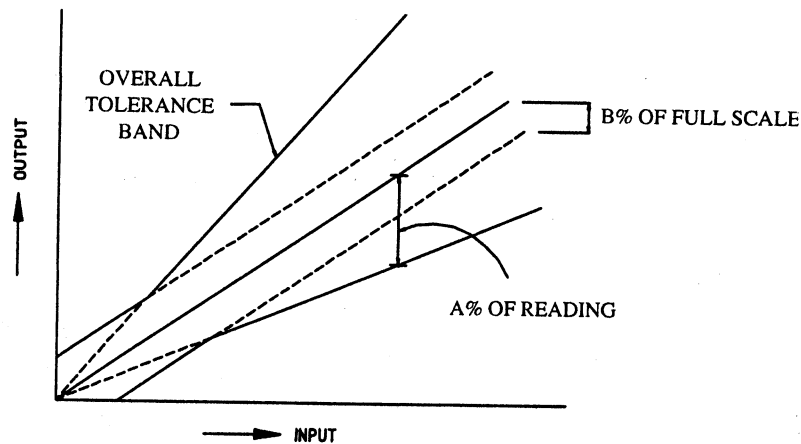


Fig 3.2
Linearity Specification

In commercial transducers however, linearity is specified as percentage of full-scale reading only. In such cases, the transducer gives more accurate result only for readings above 50% of the full-scale value.

For transducers which are considered linear, the specification of linearity is the specification of overall accuracy. Hence if only linearity specification is given by the manufacturer it may be taken as the accuracy specification.

3.2.3 Precision / Accuracy

Precision is the degree of closeness with which a given value may be repeatedly measured. When a transducer is used to measure the same input at different instances, the output may not be same. The deviation from the nominal output in absolute units or a fraction of full-scale is called the precision error or repeatability error. The input is varied from zero to full-scale value and the repeatability at different input values are calculated (Ref. Fig. 3.3a). The maximum of these will give the repeatability error of the transducer.

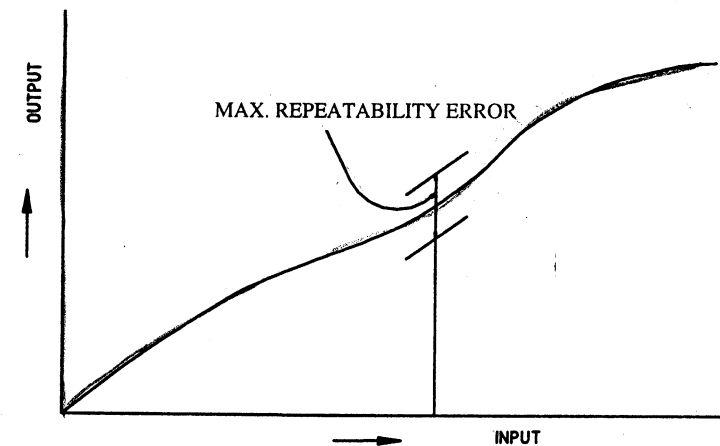


Fig 3.3a
Repeatability Error

Accuracy can never be better than half of precision.

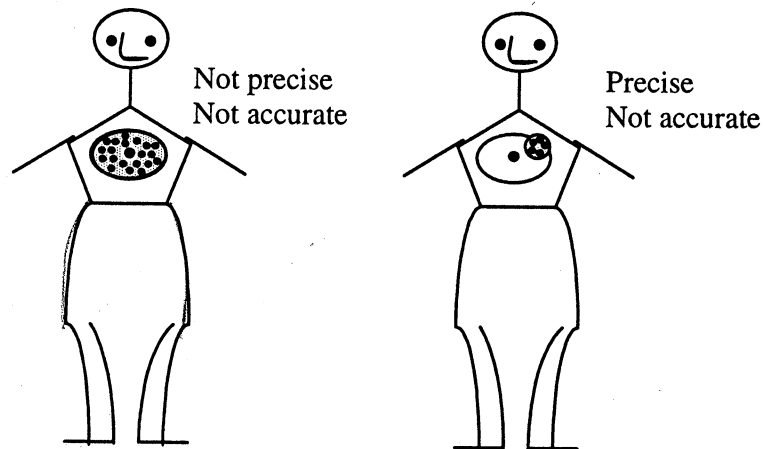


Fig 3.3b

Difference between accuracy and precision

A precise instrument does not mean that it is an accurate instrument. However, a precise instrument can be calibrated to give accurate readings. Accuracy is the closeness to true value whereas precision is the closeness amongst the readings. Fig. 3.3b explains the difference between accuracy and precision.

3.2.4 Threshold

When the input to a transducer is increased from zero, there is a minimum value below which no output can be detected. This minimum value of the input is defined as the threshold of the transducer (fig. 3.4). However this threshold depends on the output sensing device and the observer. Because of the above vagueness in specifying the threshold, it may be preferable to state a definite numerical value for output change and find the corresponding input change. This input change can be called the threshold.

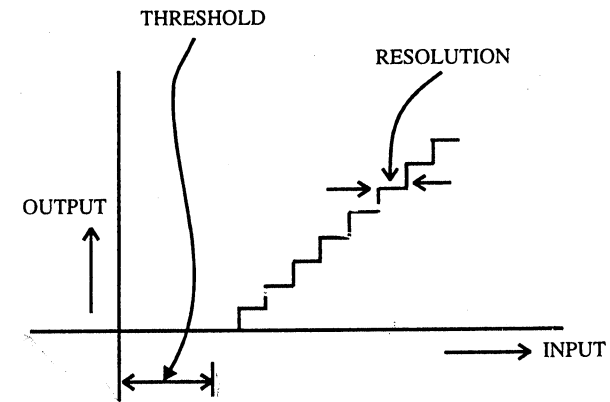


Fig 3.4

Threshold of an instrument

3.2.5 Resolution

When the input to a transducer is increased slowly from some non-zero arbitrary value, the change in output is not detected at all until a certain input increment is exceeded. This increment is defined as the resolution. To reduce the vagueness, it is desirable to define resolution as the input increment that gives a small but definite numerical change in output. In short, the threshold defines the smallest measurable input at the beginning whereas the resolution defines the smallest measurable input change. Both threshold and resolution may be given either in absolute terms or as a percentage of full-scale reading.

It may be noted that the resolution of digital instruments is decided by the number of digits used for display. For example, the resolution of a four-digit voltmeter with a range of 999.9 volts is 0.1 volt. Whereas for a five-digit voltmeter of the same range, the resolution would be 0.01 volt.

3.2.6 Hysteresis

When the input to a transducer which is initially at rest is increased from zero to full-scale and then decreased back to zero, there may be two output values for the same input. This mismatching of the

input-output curves is mainly due to internal friction, free play and change in damping of the spring elements in the transducer. Hysteresis effects can be minimised by taking readings corresponding to ascending and descending values of the input and then taking their arithmetic average.

It may be noted that if the input is increased after the first experiment immediately, the curves obtained are different for the ascending values of input. For ascending and descending value a loop is observed as shown in fig. 3.5. This loop will be different for different starting and ending points. Almost all the elastic members such as springs, bellows, diaphragms exhibit considerable amount of hysteresis effect.

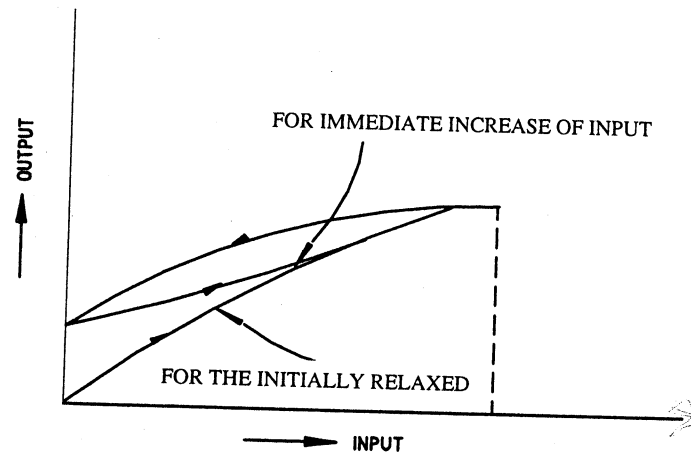


Fig 3.5
Hysteresis effect

3.2.7 Range and Span

Generally a transducer is recommended to be used between a higher and a lower values of input. The range of the transducer is specified as from the lower value of input to higher value of input.

The span of the transducer is specified as the difference between the higher and lower limits of recommended input values.

For example, if a thermocouple is recommended to be used between 100 and 600 degree C, its range is specified as 100 to 600

degree C whereas its span is 500 degree C. When a voltmeter is specified to be used between 0 to 250 V, its range is 0 to 250 V and its span is 250 V.

3.2.8 Input Impedance and Loading Effect

A transducer used in any measurement normally extracts some energy from the measuring medium and thereby disturbs the value of the measured quantity. This property is known as the loading effect of the transducer. An ideal transducer is one which does not absorb any energy and hence does not disturb the prevailing state of the measured quantity. The loading effect of a transducer gives a measure of its disturbance on the measuring quantity. The loading effect is usually expressed in terms of input impedance and stiffness.

For example, the input impedance of a voltmeter is nothing but the impedance of the deflecting coil plus any resistance connected in series. When the input impedance of the voltmeter is very large it draws less current from the circuit and hence does not affect the conditions of the circuit very much. The definition of input impedance and stiffness can be accomplished more generally with the help of the following two terms namely, effort variable and flow variable.

$$\text{Input impedance} = \frac{\text{effort variable}}{\text{flow variable}}$$

Consider, for example, a voltmeter connected across two nodes in an electrical circuit as shown in fig. 3.6. the current drawn by the voltmeter, i_m is the flow variable and the voltage across the meter, e_m is the effort variable. The input impedance of the voltmeter is given by

$$Z = e_m / i_m$$

The effort variable and flow variable are also called 'across variable' and 'through variable' respectively.

When an ammeter is connected between source and load as shown in fig. 3.7, its input admittance Y is given by

$$Y = i_m / e_m$$

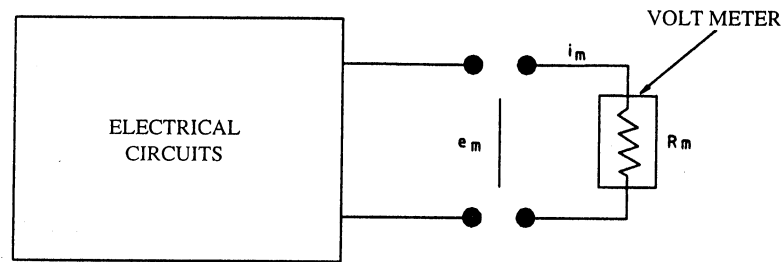


Fig 3.6
Input impedance of voltmeter

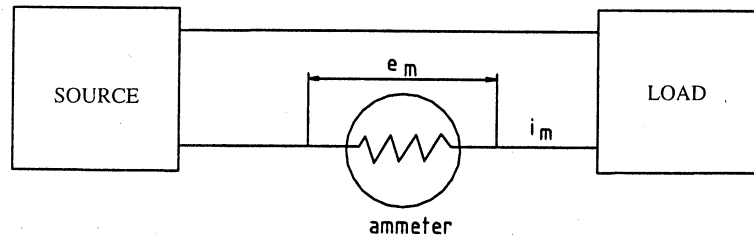


Fig 3.7
Input admittance of ammeter

The input admittance of the ammeter should be very large so that the conditions of the circuit are not altered considerably. This is true for all the instruments measuring 'through variable' or 'flow variable'.

The concepts of input impedance and input admittance are not used for mechanical systems where a number of situations exist with infinite input impedance and admittance. Hence the concepts of stiffness and compliance are introduced. Static stiffness is defined as

$$S = \frac{\text{effort variable}}{(\text{flow variable}) dt}$$

Consider a force measuring instrument, namely spring scale shown in fig. 3.8. Here force is the effort variable and velocity is the flow variable. If we try to calculate the mechanical impedance

of the instrument, it becomes infinity as the velocity of motion becomes zero at steady-state. The static stiffness is a definite quantity which is given by

$$S = \frac{\text{force}}{\text{velocity} \times dt} = \frac{\text{force}}{\text{displacement}}$$

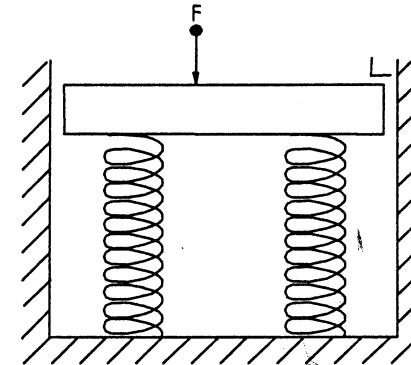


Fig 3.8
Stiffness of a spring scale

3.3 DYNAMIC CHARACTERISTICS OF TRANSDUCERS

The dynamic characteristics of a transducer refers to the performance of the transducer when it is subjected to time-varying input. The number of parameters required to define the dynamic behavior of a transducer is decided by the group to which the transducer belongs. The transducers can be categorized into

- zero-order transducers
- first-order transducers
- second-order transducers
- higher-order transducers

The order of a transducer is the highest derivative of the differential equation which describes the dynamic behavior of a transducer for a specified input. For example, let the input output relationship of a transducer be described by

$$\frac{d^3y(t)}{dt^3} + 3\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} + 4y(t) = r(t) \quad (3.9)$$

where $y(t)$ is the output and $r(t)$ is the input. The order of this transducer is same as the highest derivative of the output, that is 3.

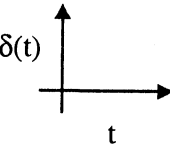
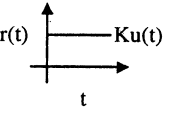
3.4 TEST INPUT

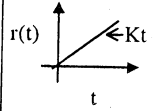
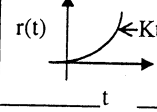
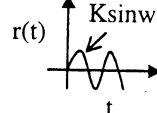
Normally, transducers are subjected to inputs which are random in nature. As it is not possible to predict the random input, the following test inputs are used to determine the dynamic behavior of the transducers:

- impulse input
- step input
- ramp input
- parabolic input
- sinusoidal input

The time function, laplace function and its pictorial representation of the test inputs are given in table 3.1.

Table 3.1: Test Inputs

	Name of the input	Time function	Laplace function	Pictorial representation
1	Impulse input If the area of the impulse is 1 then it is called unit impulse	$r(t) = \delta(t)$ $= 1$ for $t = 0$ $= 0$ for $t \neq 0$	1	
2	Step input	$r(t) = K$ for $t > 0$ $= 0$ for $t < 0$ If $K=1$ $r(t) = u(t) = \text{unit step.}$	K/s	

	Name of the input	Time function	Laplace function	Pictorial representation
3	Ramp input	$r(t) = Kt$ for $t \geq 0$ $= 0$ for $t \leq 0$	K/s^2	
4	Parabolic input	$r(t) = Kt^2$ for $t \geq 0$ $= 0$ for $t \leq 0$	$2K/s^3$	
5	Sinusoidal input	$r(t) = K \sin wt$ for $t > 0$ $= 0$ for $t \leq 0$	$\frac{kw}{s^2 + w^2}$	

3.5 ZERO-ORDER TRANSDUCER

The input-output relationship of a zero-order transducer is given by

$$y(t) = K r(t) \quad (3.10)$$

where $r(t)$ is the input, $y(t)$ is the output and K is the static-sensitivity of the transducer. The laplace transfer function of the zero-order transducer is given by

$$Y(s) / R(s) = K \quad (3.11)$$

The relationship clearly shows that the output varies exactly the same way as the input. Hence, a zero-order transducer response, represents ideal dynamic performance.

A potentiometer used for displacement measurements is an example for zero-order transducer. The output of a potentiometer is given by

$$e_o = E \cdot x/l$$

$$= K \cdot x$$

where x is the displacement of the slider, l is the total length of the potentiometer, E is the excitation and e_o is the output in volts. The static sensitivity of the potentiometer is $K = E/l$ volts/cm. The

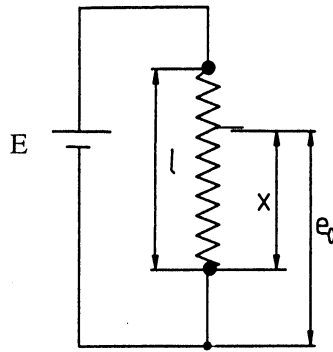


Fig 3.9
Potentiometer

potentiometer will behave as a zero-order instrument only when it is a pure resistance. It may be noted that displacement is considered as the input and not the force causing the displacement.

The response of zero-order transducers for some of the inputs are shown in fig. 3.10a & 3.10b

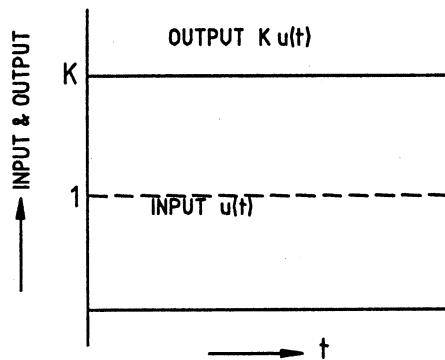


Fig 3.10a
Step response of zero-order transducer

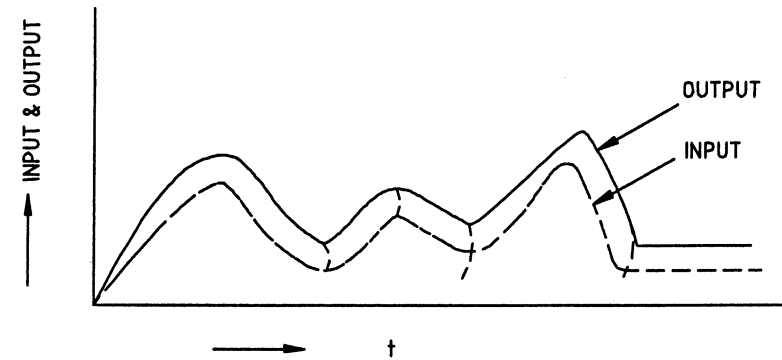


Fig 3.10b
Response of zero-order transducer for random input

3.6 FIRST-ORDER TRANSDUCER

The input-output relationship of a first-order transducer is given by

$$a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_0 u(t)$$

where a_1, a_0 and b_0 are the parameters of the transducer. The Laplace transfer function of the first-order transducer is given by

$$\frac{y(s)}{u(s)} = \frac{b_0/a_0}{\left(\frac{a_1}{a_0}s + 1\right)} = \frac{K}{(\tau s + 1)} \quad (3.13)$$

where $K = b_0 / a_0 =$ static sensitivity

and $\tau = a_1 / a_0 =$ time constant

The two parameters namely static sensitivity and time constant characterize the first-order transducer.

A thermocouple used for temperature measurements is a first-order transducer. Consider a thermocouple immersed in a fluid medium as shown in fig. 3.11. The heat balance equation is given by

$$UA(T_2 - T_1) = MS \frac{dT_1}{dt} \quad (3.14)$$

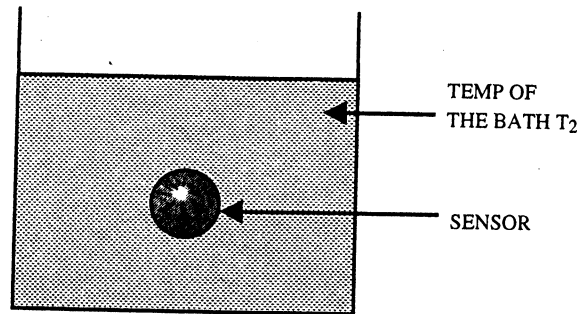


Fig 3.11

Example - I: order transducer

where U is the overall heat-transfer coefficient. A is the heat transfer area, T_2 is the temperature of the medium, T_1 is the temperature indicated by the thermocouple, M is the mass of the sensing portion of the thermocouple and S is the specific heat of the sensing bead.

The Laplace transfer function is given by

$$\frac{T_1(s)}{T_2(s)} = \frac{1}{\tau s + 1} \quad (3.15)$$

where $\tau = MS / UA$

The voltage output of a thermocouple is proportional to the temperature of the bead at the hot junction while the cold junction is kept constant at 0°C . Hence

$$V = K T_1$$

Where v is the thermocouple output in volts and K is proportionality constant.

The overall transfer function of the thermocouple is given by

$$\frac{V(s)}{T_2(s)} = \frac{V(s)}{T_1(s)} \frac{T_1(s)}{T_2(s)} \quad (3.16)$$

$$= \frac{K}{\tau s + 1} \quad (3.17)$$

which is a first order system.

It may be noted that when the hot junction of a thermocouple is kept inside a thermal wall in order to protect it from abrasive and corrosive effects of the surroundings, the transducer becomes a second order one.

3.6.1 Impulse Response of First Order Transducer

When a I order transducer is subjected to an Impulse input, the output of the transducer is known as the impulse response of the I order transducer.

This can be obtained as given below.

Let $y(s)$ be the Laplace transform of the response. Then $y(s)$, in general, is given as

$$y(s) = \frac{K}{1 + s\tau} u(s) \quad (3.18)$$

where $u(s)$ is the Laplace transform of the input.

If the input is assumed to be an unit impulse, then $u(s) = 1$. Therefore

$$y(s) = \frac{K}{1 + s\tau} \quad (3.19)$$

The time response can be determined by taking the Laplace inverse of $y(s)$

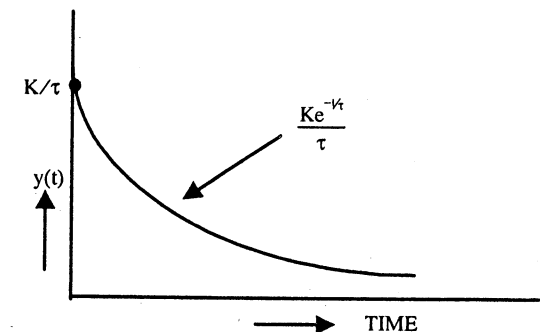


Fig 3.12a

Ideal impulse response of I order transducer

$$L^{-1} y(s) = y(t) = L^{-1} \frac{K}{1+s\tau} = \frac{K}{\tau} e^{-t/\tau} \quad (3.20)$$

The variation of $y(t)$ with respect to time is plotted in fig. 3.12a. The actual output, however, cannot rise from zero to K/τ in zero time but will take a small but negligible time to reach the peak. The practical input and output curves of impulse response of I order instrument / transducer is shown in fig. 3.12b.

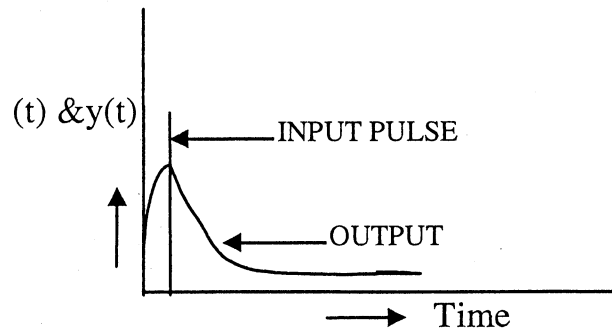


Fig 3.12b

Practical impulse response of I order transducer

The impulse response of three first order transducers with different time-constants are portrayed in fig. 3.13.

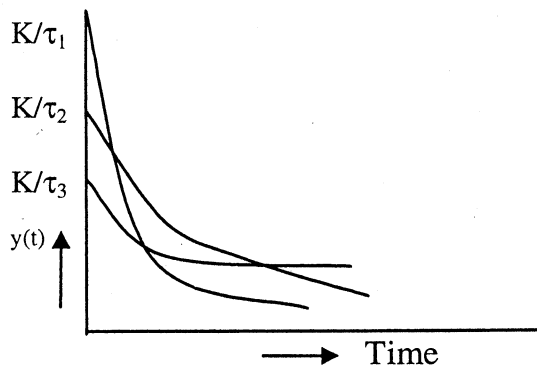


Fig 3.13

Impulse response for different time constants - I order transducer

As the time-constant becomes larger, the response becomes flatter. When the time-constant is very small the system approaches zero-order and the output is similar to the input except the change in magnitude.

The impulse response is also useful in determining the transfer-function of the transducer in situations where the transfer function is not already known. It may be noted that when the transducer is excited with a unit impulse, laplace transform of the system output is same as the transfer function of the transducer itself.

3.6.2 Step-Response Of First-Order Transducer

When a first-order transducer is excited by a unit-step input,

Now $\alpha(s) = 1/s$ $y(s) = \frac{K}{(1+s\tau)} u(s)$

$$y(s) = \frac{K}{s(1+s\tau)}$$

$$\begin{aligned} y(t) &= L^{-1} y(s) \\ &= L^{-1} \frac{K}{s(1+s\tau)} \end{aligned}$$

From the Laplace transform table,

$$y(t) = K(1 - e^{-t/\tau}) \quad (3.21)$$

The variation of $y(t)$ with respect to time is plotted in fig. 3.14a. The output rises at a faster rate initially and slowly afterwards.

The transducer takes theoretically infinite time to reach the steady-state value of the output, K . however, in practice the final value is assumed to have reached in four or five times the time-constant of the transducer. Actually the output would have reached 95.0% of the final value in three times the time-constant, 98.2% in 4 times the time-constant and 99.3% in five times the time-constant. The output will reach 63.2% of the final value when $t = \tau$ i.e. in one time constant. It is the normal practice to define a

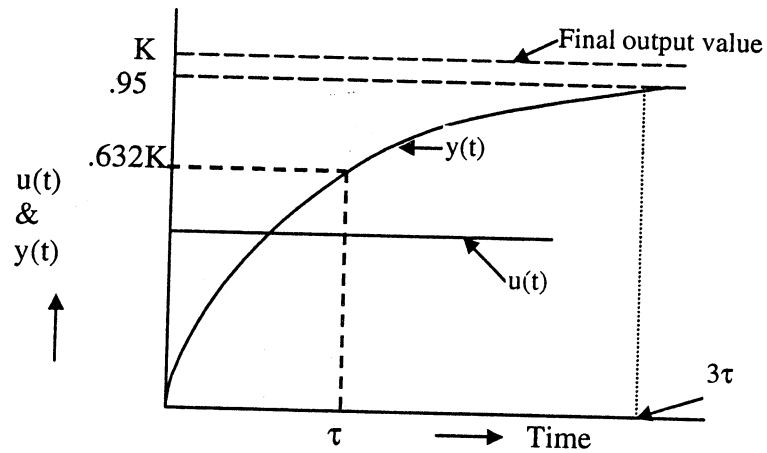


Fig 3.14a

Step response of 1 order transducer

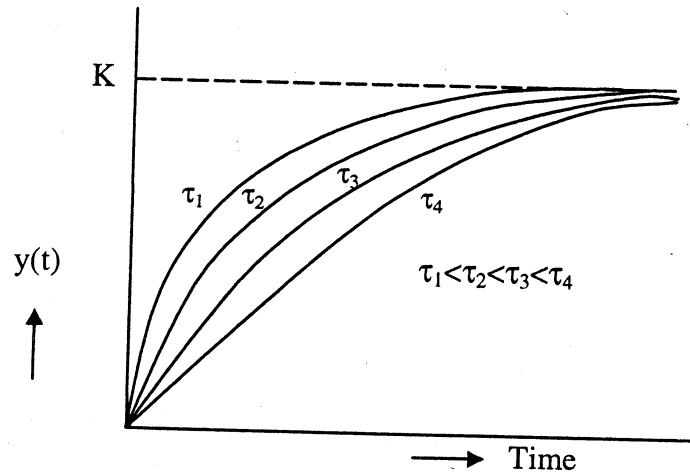


Fig 3.14b

Step response for different time constants

reading time for a first-order transducer as 4 times the time constant.

Reading time of transducer is $= 4\tau$.

In the expressions derived thus far in this section, it is conveniently assumed that the initial conditions are zero. However,

when the initial conditions are non-zero, the expression for the step-response will get modified as

$$y(t) = y(0) + (K - y(0))(1 - e^{-t/\tau}) \quad (3.22)$$

for example, assume a simple RC network, fig. 3.14b, whose transfer function is a first-order one. i.e.

$$\frac{V_o}{V_i} = \frac{1}{1 + RCs}$$

When the capacitor is initially charged to 5V the expression for the voltage across the capacitor is given by,

$$V_o = 5 + (20 - 5)(1 - e^{-t/\tau}) \text{ volts}$$

The variation of V_o with respect to time is given in fig. 3.15.

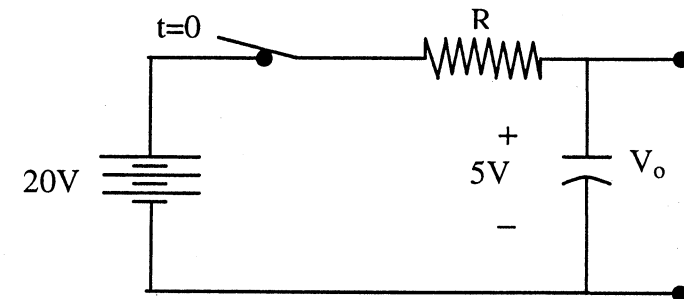


Fig 3.15a

First-order system with non-zero initial condition

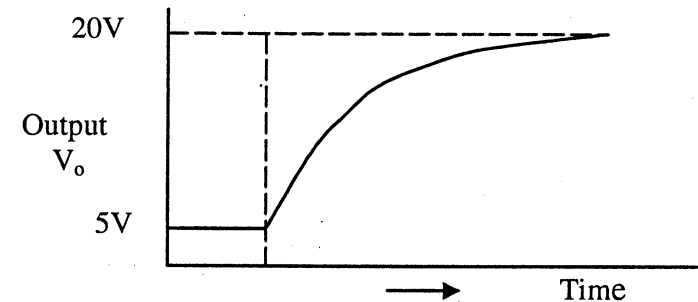


Fig 3.15b

Step response of 1-order system with non-zero initial condition

3.6.3 Experimental Determination of Time-Constant

The time-constant of the first-order transducer can be determined experimentally from the step-response in three ways. The transducer is selected to a step change in input and its output is recorded. To do this, the following steps are taken the transducer is initially relaxed. An unit-step input, is applied. Simultaneously a stop watch is started. For every second the output value is noted. A graph of the output V_s time is plotted. from the output curve the following procedure is adopted to determine the time constant.

First method

The equation for the step output is given by

$$y(t) = K(1 - e^{-t/\tau})$$

assuming zero initial conditions.

When $t = \tau$, $y(t) = K(1 - e^{-1}) = 0.632 K$

Hence the time-constant is the time taken for the output of the transducer to reach 63.2% of the final value for a step-input. The time taken for the output to reach 63.2% of the final value is determined by the graph which gives the time constant of the system.

Second method

The output, $y(t) = K(1 - e^{-t/\tau})$

The slope of the output curve, (ref. fig 3.16)

$$\frac{dy(t)}{dt} = -Ke^{-t/\tau} \left(-\frac{1}{\tau}\right) = \frac{K}{\tau} e^{-t/\tau}$$

$$\frac{dy(t)}{dt} = \frac{K}{\tau} e^{-0} = \frac{K}{\tau}$$

If the transducer output increases with the same slope it has at $t = 0$, then the final value of the output will be reached in t . This fact can also be used to determine the time constant from the step-response. A tangent to the experimentally obtained output curve at $t = 0$ is drawn. This line cuts the final value of the output at τ .

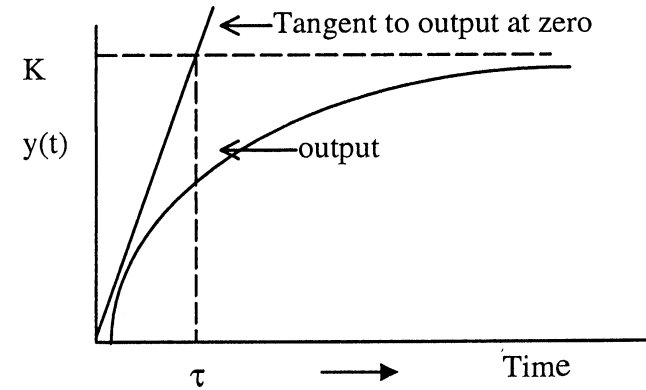


Fig 3.16
Time constant by initial slope method

Third method

When the complete plot of the step-response is not available, the time-constant can be determined if the output at two different instances are known. (ref. fig 3.16)

Assume $y(t) = y_1$ at $t = t_1$

and $y(t) = y_2$ at $t = t_2$

now $y_1 = K(1 - e^{-t_1/\tau})$

$$y_2 = K(1 - e^{-t_2/\tau})$$

$$\frac{y_1}{y_2} = \frac{1 - e^{-t_1/\tau}}{1 - e^{-t_2/\tau}}$$

If t_2 is chosen as $2t_1$, where t_1 can be arbitrarily chosen, then

$$\frac{y_1}{y_2} = \frac{1 - e^{-t_1/\tau}}{1 - e^{-2t_1/\tau}} = \frac{1}{1 + e^{-t_1/\tau}}$$

$$y_1(1 + e^{-t_1/\tau}) = y_2$$

$$e^{-t_1/\tau} = \frac{y_2 - y_1}{y_1} = \left(\frac{y_2}{y_1} - 1\right)$$

$$\tau = \frac{t_1}{\ln \left(\frac{y_1}{y_2 - y_1} \right)} \quad (3.23)$$

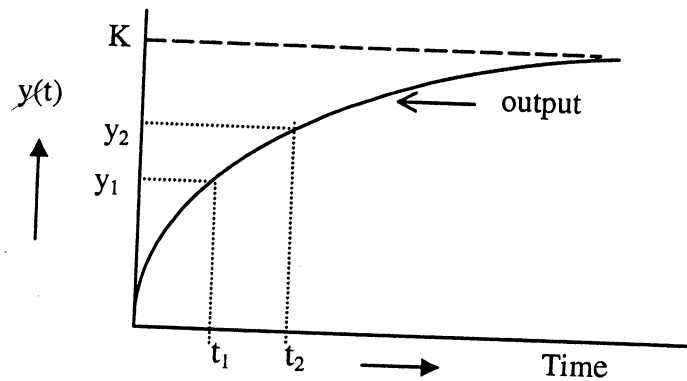


Fig 3.17

Time constant by two point method

3.6.4 Ramp Response Of First-Order Transducer

Consider a first-order transducer subjected to a ramp-input given by

$$\begin{aligned} u(t) &= R t & \text{for } t \geq 0 \\ &= 0 & \text{for } t < 0 \end{aligned}$$

where R is the slope of the ramp.

$$u(s) = \frac{R}{s^2}$$

$$y(s) = \frac{K}{1 + s\tau} \cdot \frac{R}{s^2}$$

$$\begin{aligned} y(t) &= L^{-1} \frac{KR}{s^2(1 + s\tau)} \\ &= KR(\tau e^{-t/\tau} + t - \tau) \end{aligned} \quad (3.24)$$

for an unit ramp, $R=1$ and $y(t) = (\tau e^{-t/\tau} + t - \tau)$
The plot of $y(t)$ Vs time is shown in fig. 3.18.

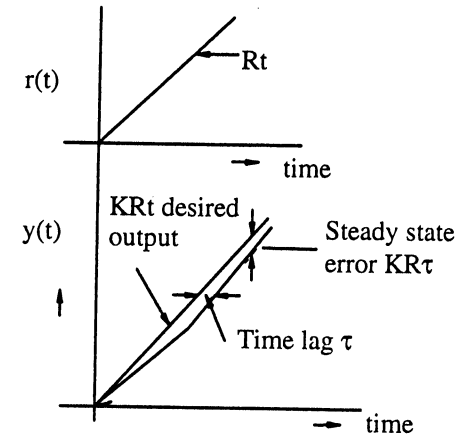


Fig 3.18

Ramp response of first-order transducer

By examining the equation (3.24), it is observed that the first-order system always has a measurement error for a ramp input which is given by

$$e(t) = K u(t) - y(t)$$

where $K u(t)$ is the desired output and $y(t)$ is the actual output.

$$e(t) = \tau KRt - KR(\tau e^{-t/\tau} + t - \tau)$$

$$e(t) = \underbrace{-KR\tau e^{-t/\tau}}_{\text{transient error}} + \underbrace{KR\tau}_{\text{steady-state error}} \quad (3.25)$$

The transient- error is the error which dies down as time becomes large. The steady-state error persists even if time becomes infinity. When the time-constant of the transducer is very small, the transient error disappears quickly and the magnitude of the steady-state error is also very less. When the transducer is subjected to a fast changing input (large R), the steady-state error also increases as the transducer finds it difficult to follow rapid change in input. During steady-state, the horizontal displacement between desired output and actual output curves is observed to be τ . This can be interpreted as that the transducer is reading the output which was there τ seconds ago.

3.6.5 Terminated Ramp-Response

In practice, a perfect step input cannot be given to any system since any physical quantity cannot change in zero time. The more realistic input is the terminated ramp rather than the step.

Also it is very difficult to excite a system or a transducer with an ideal ramp for a long time. The input function gets saturated or terminated after a finite time giving rise to a terminated ramp input. Such an input function is graphically represented in fig. 3.19.

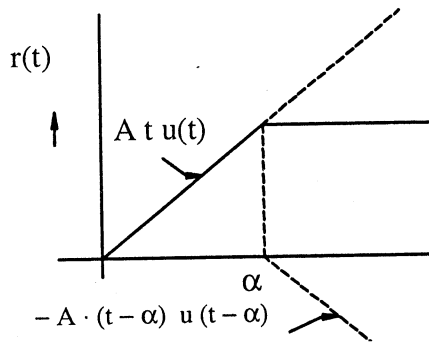


Fig 3.19
Terminated ramp function

The response of the transducer for this type of input can be obtained by considering the input to be composed of a ramp and a delayed negative ramp. That is

$$r(t) = A t u(t) - A \cdot (t - \alpha) u(t - \alpha) \quad (3.26)$$

$$\text{where } u(\theta) = 1 \quad \text{for all } \theta > 0 \\ = 0 \quad \text{for all } \theta < 0$$

It may be noted that θ is the variable within parenthesis.

Here $\theta = t$ or $(t - \alpha)$ as the case may be. Hence

$$R(s) = \frac{A}{s^2} - \frac{A}{s^2} e^{-\alpha s} \\ y(s) = \frac{A}{s^2} \frac{K}{1 + s\tau} - \frac{A}{s^2} e^{-\alpha s} \frac{K}{1 + s\tau}$$

$$\therefore y(t) = KA(t - \tau + \tau e^{-t/\tau}) - KA(t - \alpha - \tau + \tau e^{-(t-\alpha)/\tau}) u(t - \alpha) \\ = KA\alpha - KA\tau e^{-\alpha/\tau} (e^{-t/\tau} - 1) \quad (3.27)$$

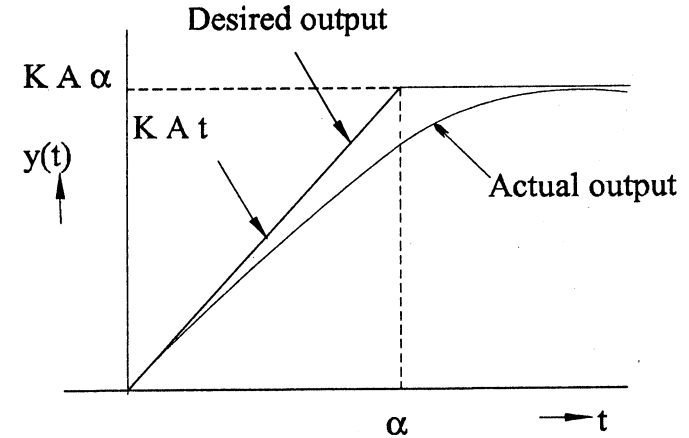


Fig 3.20
Terminated Response

The response $y(t)$ is plotted in fig. 3.20.

3.6.6 Frequency Response

When a linear transducer is excited with a sinusoidal input the output will also be a sinusoid after the transient has died down. The amplitude and phase of the output may be different from that of the input. This is illustrated in fig. 3.21. the output comprises of two parts namely transient and steady-state. It is normal practice to consider only the steady-state portion of the response.

Frequency response is thus defined as the steady-state output of a transducer when it is excited with sinusoidal input. This is now one of the most important characteristics of a transducer. The frequency response is represented with the help of two plots namely amplitude ratio versus frequency and phase angle shift versus frequency. These are shown in fig. 3.22.

The frequency response of any system can be obtained from the frequency transfer function. Frequency transfer function is obtained by replacing s by $j\omega$, in the laplace transfer function.

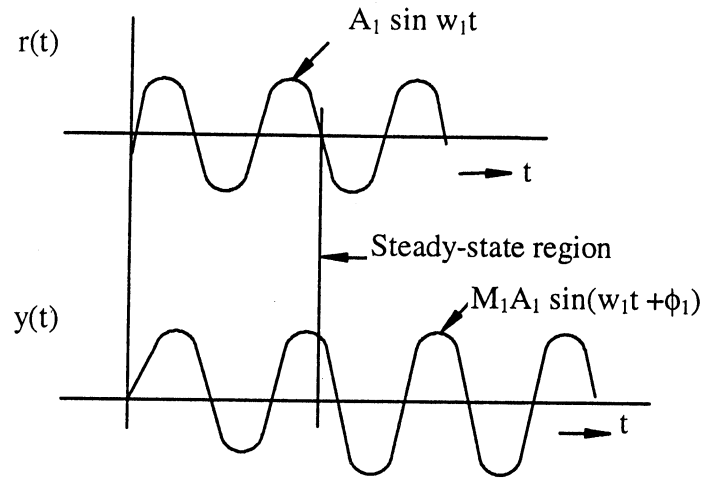


Fig 3.21

Sinusoidal input and output of linear system

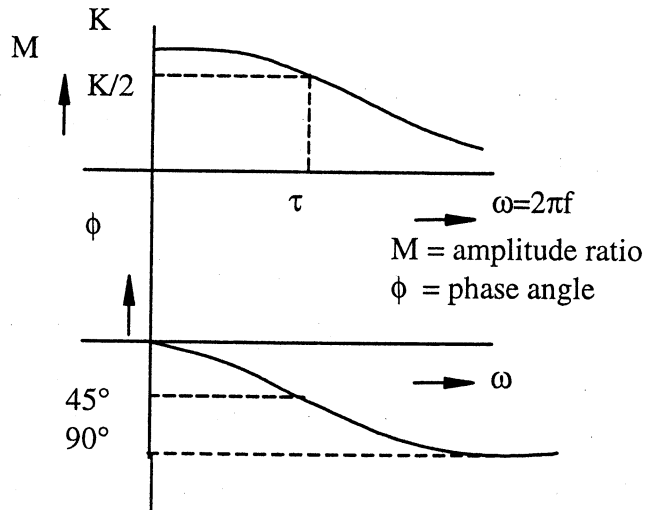


Fig 3.22

Frequency response of first-order transducer

3.6.7 Frequency Response of First-Order Transducer

The frequency transfer function of a first-order transducer is given by

$$\frac{Y(j\omega)}{R(j\omega)} = \frac{K}{1 + j\omega\tau} \quad (3.28)$$

The amplitude ratio is given by

$$\begin{aligned} \left| \frac{Y(j\omega)}{R(j\omega)} \right| &= \left| \frac{K}{1 + j\omega\tau} \right| \\ &= \frac{K}{\sqrt{1 + \omega^2\tau^2}} \end{aligned} \quad (3.29)$$

The phase shift between output and input is given by

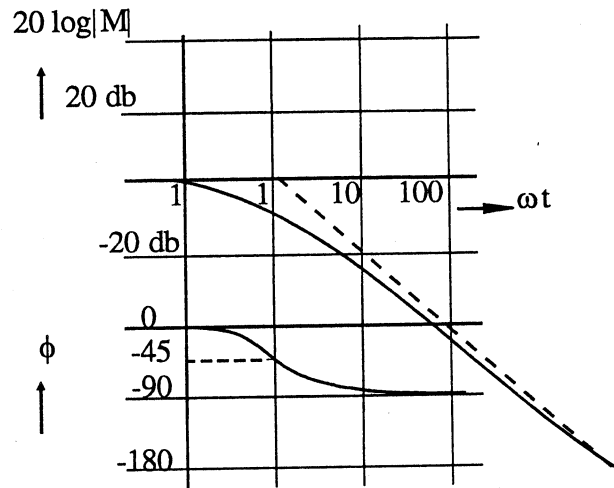
$$\begin{aligned} \phi &= \tan^{-1} \frac{0}{K} - \tan^{-1} \frac{\omega\tau}{1} \\ &= 0 - \tan^{-1} \omega\tau \\ \phi &= -\tan^{-1} \omega\tau \end{aligned} \quad (3.30)$$

These are plotted in fig. 3.23. As the time constant τ becomes smaller the amplitude ratio plot tends to become flatter, approaching an ideal situation. It may be noted that an ideal transducer is one whose amplitude ratio is constant for all frequencies. If the frequency in the x axis is taken in the logarithmic units it is possible to cover a wide range of frequencies. This is what is done in Bode plots.

x - axis in the logarithmic scale

y - axis in decibel gain and phase angle in degrees.

For any value of τ , the time-constant, there will be some frequency of input below which the transducer can be considered to give accurate measurements. It is the normal practice to take $\omega = 1/\tau$ as the limiting value for acceptable measurements since at this point of frequency the amplitude ratio becomes $\frac{K}{\sqrt{2}}$



x - axis in the logarithmic scale
y - axis in decibel gain and phase angle in degrees.

Fig 3.23

Frequency Response of 1-Order Transducer

3.7 SECOND-ORDER TRANSDUCER

The input-output relationship of a second-order transducer is given by

$$a_2 \frac{d^2 y(t)}{dt^2} + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_0 u(t)$$

the Laplace transfer function of the transducer is given by

$$\frac{Y(s)}{R(s)} = \frac{b_0/a_0}{\frac{a_2}{a_0} s^2 + \frac{a_1}{a_0} s + 1} \quad (3.31)$$

when all the initial conditions are zero.

Equations (3.31) can be rewritten so that it confirms with the general form as

$$\frac{Y(s)}{R(s)} = \frac{K}{\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n} s + 1} \quad (3.32)$$

where

$K = b_0/a_0$	= static sensitivity
$\omega_n = a_0/a_2$	= undamped natural frequency
$\zeta = \frac{a_1}{\sqrt{a_0 a_2}}$	= damping ratio

Hence, three parameters namely, K , ω_n and ζ characterise the second-order transducer.

The damping ratio ζ is an important parameter which decides the nature of oscillation in the transducer output. When $\zeta = 0$, the second-order system is said to be undamped and the system behaves like an oscillator. Its output oscillates continuously, the magnitude of outputs decided by others factors. When $0 < \zeta < 1$, the system is said to be under damped and the oscillations will die down as time elapses. This is known as damped oscillation. When $\zeta = 1$, the system is said to be critically damped and the response just becomes non-oscillatory. When $\zeta > 1$, the system is said to be over damped and the response is similar to first-order system without any oscillation. The output of a second-order transducer for a sudden change in input for various damping conditions are shown in fig. 3.27.

3.7.1 Examples of Second Order Instrument

A vibration galvanometer is an example of second-order instrument, which is shown in fig. 3.24. This consists of a suspended coil placed in a magnetic field between a pole pair.

When a current i flows through the coil, the coil is subjected to a deflecting torque equal to Gi where G is the displacement constant of the galvanometer. This results in a deflection of the coil by an angle θ . The three opposing torque to this motion are due to moment of inertia, damping effect and elasticity of suspension, which are given by

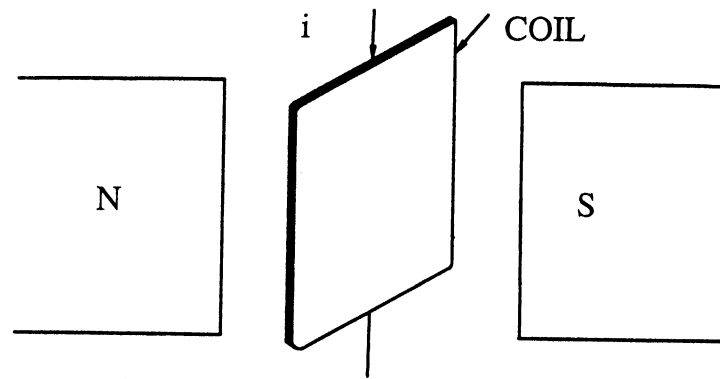


Fig 3.24
Vibration galvanometer

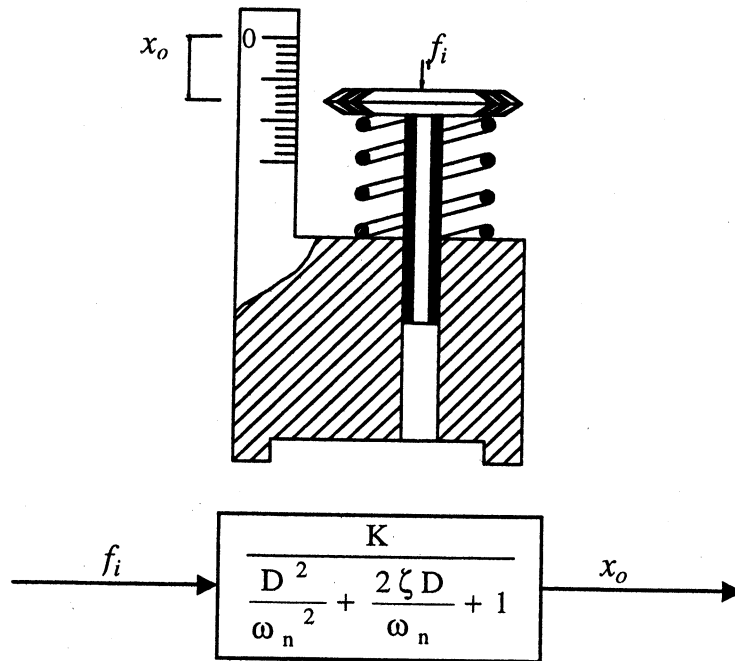


Fig 3.25
Second-order instrument - Spring scale system.

$$a \frac{d^2 \theta}{dt^2}, b \frac{d \theta}{dt} \text{ and } c \theta$$

respectively, where a , b and c are constants. According to Newton's laws of motion, the deflecting torque is equal to the opposing torque at every instant of time. Hence

$$a \frac{d^2 \theta}{dt^2} + b \frac{d \theta}{dt} + c \theta = Gi(t) \quad (3.33)$$

The Laplace transfer function of the galvanometer dynamics is

$$\frac{\theta(s)}{I(s)} = \frac{G}{as^2 + bs + c} \text{ which is a second order system.} \quad (3.34)$$

Spring scale system is shown in fig. 3.25 is another example of a second order system. The applied force is opposed by the mass, damping force and the spring force manometer, a thermometer in a thermowell are some of the other examples of second-order transducers.

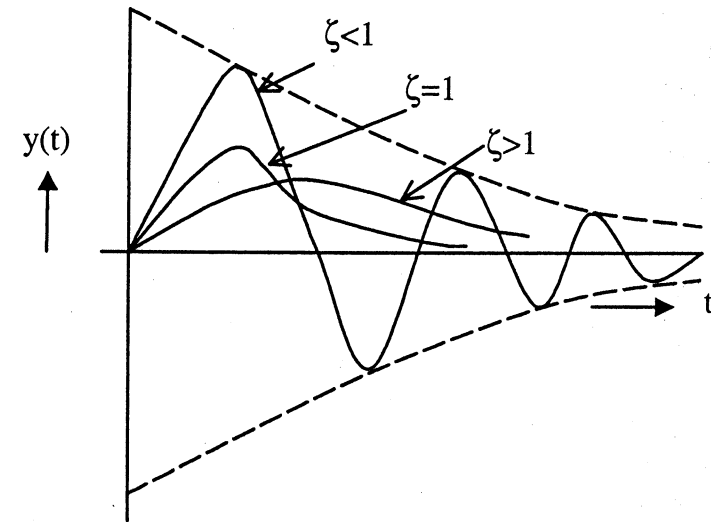


Fig 3.26
Impulse-response of second-order transducer

3.7.2 Impulse-Response of Second-Order Transducers

When a second-order transducer is subjected to an unit impulse, the laplace transform of the output is given by

$$Y(s) = \frac{K}{\omega_n^2 + \frac{2\zeta}{\omega_n}s + 1} R(s) \quad (3.35)$$

$$= \frac{K}{\omega_n^2 + \frac{2\zeta}{\omega_n}s + 1} \quad \because R(s) = 1 \quad (3.36)$$

$$\begin{aligned} y(t) &= L^{-1} Y(s) \\ &= L^{-1} \frac{K \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2} \\ &= L^{-1} \frac{K \omega_n^2}{(s + \zeta \omega_n)^2 + \omega_n^2 (1 - \zeta^2)} \end{aligned}$$

(see the derivation for step response)

$y(t)$ can be obtained from Laplace transform table as given below.
For under damped conditions ($\zeta < 1$),

$$y(t) = \frac{K \omega_n}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \sin(\omega_n \sqrt{1 - \zeta^2} t) \quad (3.37)$$

For critically damped condition ($\zeta = 1$),

$$y(t) = K \omega_n^2 t e^{-\omega_n t} \quad (3.38)$$

For over damped conditions ($\zeta > 1$),

$$y(t) = \frac{K \omega_n}{2\sqrt{\zeta^2 - 1}} e^{(-\zeta + \sqrt{\zeta^2 - 1}) \omega_n t} - e^{(-\zeta - \sqrt{\zeta^2 - 1}) \omega_n t} \quad (3.39)$$

The response of the second-order transducer for unit impulse is shown in fig. 3.26 for different damping conditions.

3.7.3 Step-Response of Second-Order Transducer

When a second-order transducer is subjected to an unit step input,

$$R(s) = 1/s$$

The laplace transform of the output is given by

$$Y(s) = \frac{K \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2} \cdot \frac{1}{s} \quad (3.40)$$

This can be split into partial fractions as given below

$$Y(s) = K \left[\frac{1}{s} - \frac{s + 2\zeta \omega_n}{s^2 + 2\zeta \omega_n s + \omega_n^2} \right] \quad (3.41)$$

which can be written by adding and subtracting $\zeta^2 \omega_n^2$ to the denominator of the second term as

$$= K \left[\frac{1}{s} - \frac{s + \zeta \omega_n}{(s + \zeta \omega_n)^2 + \omega_n^2 (1 - \zeta^2)} - \frac{\zeta \omega_n}{(s + \zeta \omega_n)^2 + \omega_n^2 (1 - \zeta^2)} \right]$$

By using the laplace transform table, for $\zeta < 1$,

$$y(t) = K \left[1 - \frac{\zeta}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \cos(\omega_n \sqrt{1 - \zeta^2} t) - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \sin(\omega_n \sqrt{1 - \zeta^2} t) \right] \quad (3.42)$$

This can be reduced to the following form

$$y(t) = K \left[1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1 - \zeta^2}} \sin(\omega_n \sqrt{1 - \zeta^2} t + \phi) \right] \quad (3.43a)$$

$$\text{where } \phi = \tan^{-1} \frac{\sqrt{1 - \zeta^2}}{\zeta} \quad (3.43b)$$

In a similar way, for $\zeta = 1$, $y(t)$ can be obtained as

$$y(t) = K [1 - (1 + \omega_n t) e^{-\omega_n t}] \quad (3.44)$$

For $\zeta > 1$,

$$y(t) = K \begin{bmatrix} 1 + \frac{\zeta - \sqrt{\zeta^2 - 1}}{2\sqrt{\zeta^2 - 1}} e^{(-\zeta - \sqrt{\zeta^2 - 1})\omega_n t} \\ - \frac{\zeta + \sqrt{\zeta^2 - 1}}{2\sqrt{\zeta^2 - 1}} e^{(-\zeta + \sqrt{\zeta^2 - 1})\omega_n t} \end{bmatrix} \quad (3.45)$$

The step response of second-order transducer for various values of damping ratios are shown in fig. 3.27.

Whenever a second-order transducer is suddenly connected to an input, it is equivalent to the application of step input. To have a quick indication of the measured values, the time taken for the transducer response to reach the steady-state value should be minimum. As the second-order system subjected to step-input takes infinite time (theoretically) to reach the steady-state value, it is customary to define settling time for such systems. The settling time is the time taken for the output to reach and stay within a specified percentage of steady-state value. For example, 5% settling time means, the time taken for the system output to reach and stay within 95% to 105% of the steady-state value.

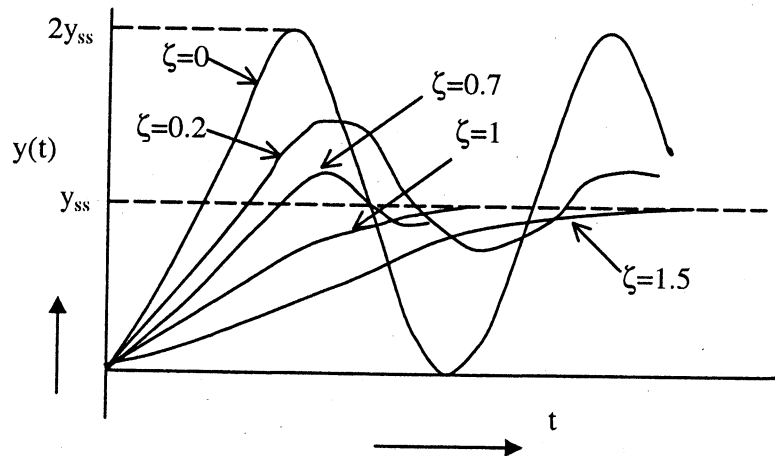


Fig 3.27

Step response of a II-order transducer for $\zeta = 0, 0.2, 0.7, 1, 1.5$

When the second-order transducer is undamped ($\zeta = 0$), the settling time is infinity. As the damping ratio is increased, the settling time decreases and reaches an optimum value and again increases for over damped conditions. If we are interested in 10% settling time, then $\zeta = 0.6$ gives optimum value whereas $\zeta = 0.7$ to 0.8 gives the optimum value for 5% settling time. Many commercial transducer systems are designed to have a damping ratio of 0.6 to 0.7.

The two parameters of the second-order transducer, ζ and ω_n , can be determined from the step-response. This is illustrated in section 3.9.

3.7.4 Ramp Response of Second-Order Transducer

Consider a second-order transducer subjected to ramp input given by

$$\begin{aligned} r(t) &= A t \\ R(s) &= A/s^2 \end{aligned} \quad (3.46)$$

The laplace transform of the output is

$$y(s) = \frac{K \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2} \cdot \frac{A}{s^2} \quad \text{Pr} \quad (3.47)$$

This can be split into its partial fraction as follows

$$y(s) = \frac{B_1}{s} + \frac{B_2}{s^2} + \frac{B_3 s + B_4}{s^2 + 2\zeta \omega_n s + \omega_n^2} \quad (3.48)$$

B_1, B_2, B_3, B_4 are determined comparing the coefficients of s .

$$y(s) = KA \left[\frac{-2\zeta/\omega_n}{s} + \frac{1}{s^2} + \frac{2\zeta/\omega_n s + 4\zeta^2 - 1}{s^2 + 2\zeta \omega_n s + \omega_n^2} \right] \quad (3.49)$$

The output $y(t)$ is found to be

$$\begin{aligned} & \text{(i) for } \zeta < 1 \\ y(t) &= KA t - \frac{KA 2\zeta}{\omega_n} \left[1 - \frac{e^{-\zeta \omega_n t}}{2\zeta \sqrt{1-\zeta^2}} \sin(\omega_n \sqrt{1-\zeta^2} t + \phi) \right] \end{aligned} \quad (3.50)$$

where $\phi = \tan^{-1} \frac{2\zeta\sqrt{1-\zeta^2}}{2\zeta^2-1}$ (3.51)

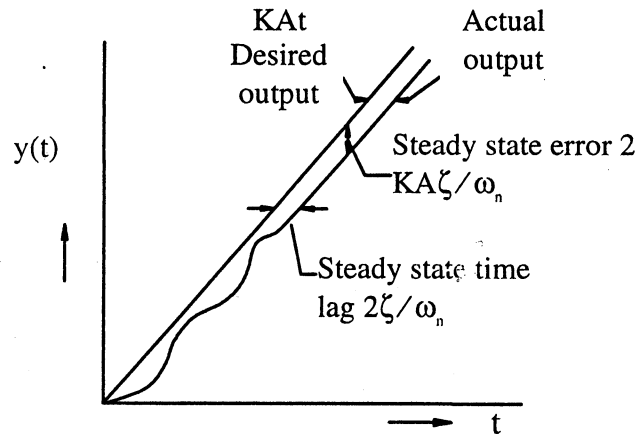


Fig 3.28
Ramp response of II order transducer

(ii) for $\zeta = 1$

$$y(t) = KA t - \frac{2KA}{\omega_n} \left[1 - e^{-\omega_n t} \left(1 + \frac{\omega_n t}{2} \right) \right] \quad (3.52)$$

(iii) for $\zeta > 1$

$$y(t) = KA t - \frac{2\zeta KA}{\omega_n} \left[\frac{1 + \frac{2\zeta^2 - 1 - 2\zeta\sqrt{\zeta^2 - 1}}{4\zeta\sqrt{\zeta^2 - 1}} e^{(-\zeta + \sqrt{\zeta^2 - 1})\omega_n t}}{+ \frac{-2\zeta^2 + 1 - 2\zeta\sqrt{\zeta^2 - 1}}{4\zeta\sqrt{\zeta^2 - 1}} e^{(-\zeta - \sqrt{\zeta^2 - 1})\omega_n t}} \right] \quad (3.53)$$

steady-state error = $KA t - y(t)$ for $t \neq \infty$ but large

$$= KA t - \left[KA t - \frac{KA 2\zeta}{\omega_n} \right] \quad (3.54)$$

for all damping conditions.

$$\text{Steady-state error} = KA (2\zeta/\omega_n) \quad (3.55)$$

Hence the steady-state error decreases as ζ is decreased and ω_n is increased.

The ramp response of the second-order transducer is shown in fig. 3.28.

3.7.5 Terminated Ramp Response of Second Order Transducer

As discussed in section 3.6.5, the more realistic input to any transducer is a terminated ramp rather than a pure step or a pure ramp. The practical example of a second-order transducer subjected to terminated ramp input is a piezo-electric accelerometer (explained in later chapters). The damping ratio, ζ , of such a transducer is around 0.01. Therefore the response of the transducer to terminated ramp input is discussed for critically damped and over damped conditions can be obtained following the same procedure.

As explained in section 3.6.5, a terminated-ramp input is expressed as

$$r(t) = At u(t) - A \times (t - \alpha) u(t - \alpha) \text{ where } u(\theta) = 1 \text{ for all } \theta > 0 \\ = 0 \text{ for all } \theta < 0$$

$$R(s) = \frac{A}{s^2} - \frac{A}{s^2} e^{-\alpha s} \quad (3.57)$$

The laplace transform of the output is

$$Y(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \frac{A}{s^2} (1 - e^{-\alpha s}) \quad (3.58)$$

The output $y(t)$ can be obtained using partial fraction method (as discussed in section 3.7.4).

For under damped condition

$$y(t) = KA t - \frac{KA 2\zeta}{\omega_n} \left[1 - \frac{e^{-\zeta\omega_n t}}{2\zeta\sqrt{1-\zeta^2}} \sin(\omega_n \sqrt{1-\zeta^2} t + \phi) \right]$$

for $t < \alpha$

$$= KAt - \frac{KA 2 \zeta}{\omega_n} \left[1 - \frac{e^{-\zeta \omega_n t}}{2 \zeta \sqrt{1 - \zeta^2}} \sin(\omega_n \sqrt{1 - \zeta^2} t + \phi) \right] \quad (3.59)$$

$$- \left\{ KA(t - \alpha) - \frac{KA 2 \zeta}{\omega_n} \left[1 - \frac{e^{-\zeta \omega_n (t - \alpha)}}{2 \zeta \sqrt{1 - \zeta^2}} \sin(\omega_n \sqrt{1 - \zeta^2} (t - \alpha) + \phi) \right] \right\}$$

for $t > \alpha$

In a similar way, the expression for $y(t)$ for critically damped and over damped conditions can be obtained.

The terminated ramp response for under damped condition is plotted in fig. 3.29.

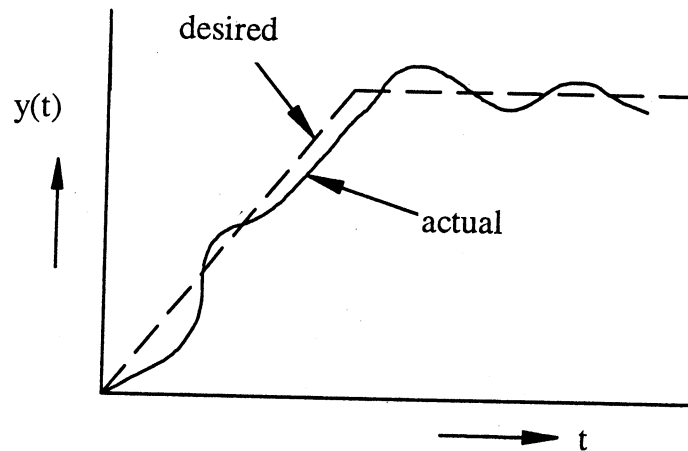


Fig 3.29

Terminated ramp response of II-order damped transducer

3.7.6 Frequency Response of Second-Order Transducer

The frequency transfer function of a second-order transducer is given by replacing s by $j\omega$

$$\frac{Y(j\omega)}{R(j\omega)} = \frac{K}{\left(\frac{j\omega}{\omega_n}\right)^2 + 2\zeta \frac{j\omega}{\omega_n} + 1} \quad (3.61)$$

This can be put in the form

$$\frac{Y(j\omega)}{R(j\omega)} = \frac{K}{\sqrt{\left(1 - \left(\frac{\omega}{\omega_n}\right)^2\right)^2 + \frac{4\zeta^2 \omega^2}{\omega_n^2}}} \quad \phi \quad (3.62)$$

$$\text{where } \phi = -\tan^{-1} \left(\frac{2\zeta}{\frac{\omega_n}{\omega} - \frac{\omega}{\omega_n}} \right) \quad (3.63)$$

The various frequency response plots namely, Bode plot, Polar plot and Phase angle Vs Magnitude plot can be obtained for the above function.

As the normalised frequency response plot will be more useful for a generalised analysis. Fig. 3.30 shows the normalised frequency response plot. This can be obtained by plotting

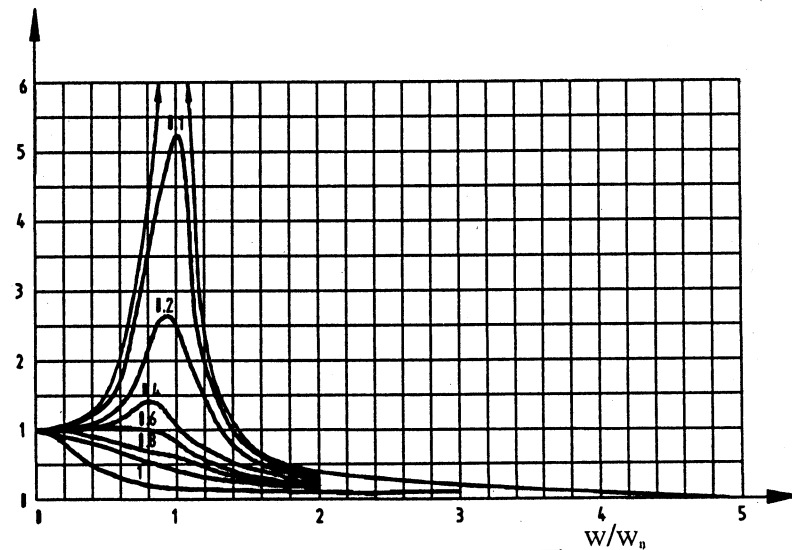


Fig 3.30a

Amplitude plot - Second order system.

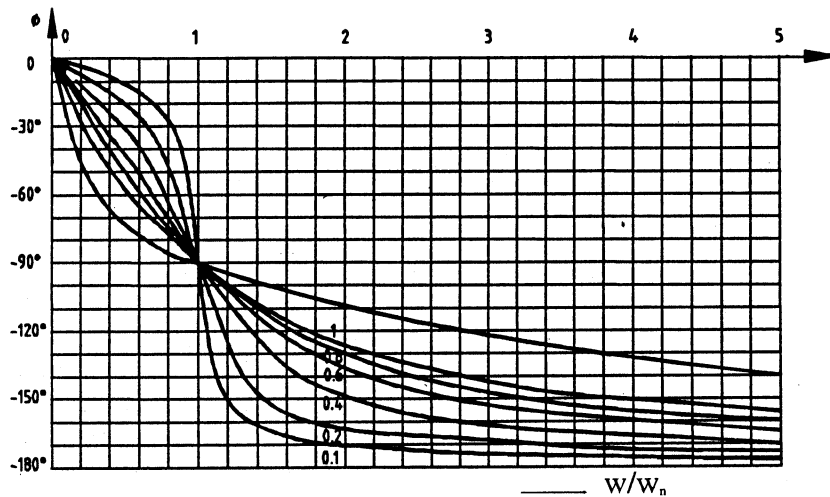


Fig 3.30b

Phase angle plot - Second order system

$$\frac{Y(j\omega)}{KR(j\omega)} = V_s \frac{\omega}{\omega_n}$$

let
$$M = \left| \frac{Y(j\omega)}{KR(j\omega)} \right| \quad (3.64)$$

A transducer is considered to be good if it has a flat amplitude plot and zero phase shift for all frequencies. In a practical situation, it is difficult to get a transducer with zero phase shift for all frequencies. In that case, a linear variation of phase shift with frequency is acceptable. It may be observed from fig. 3.30, the amplitude plot is flat for the largest frequency range when damping ratio, ζ lies between 0.6 to 0.7. The phase shift, θ also varies linearly with frequency ω for the same range 0.6 to 0.7. Hence the designers of transducers normally choose this range of damping ratio.

3.8 HIGHER ORDER TRANSDUCERS

In practice, many transducers have higher order dynamics which can be described by a higher order differential equations. For engineering analysis purposes, they can be represented by either first-order and second-order differential equations with certain linearising assumptions. However, when an accurate analysis is required for research purposes, the higher-order equations can be taken as it is and solved. It may be noted that the response of the higher-order transducers would be similar to that of second-order transducers with a sluggish rise in the initial period.

A third-order transducer namely piezo-electric accelerometer and its frequency response are discussed in a later section.

3.9 SIGNIFICANCE OF FREQUENCY RESPONSE

The response of a transducer to a frequency input is an important characteristic since most of the signals in the real world can be considered to be a combination of signals of different frequencies. If a transducer is to represent a variation of a quantity realistically, it should treat all the frequency components of the signal equally. That is, the sensitivity should be same for all frequencies and phase shift should be either zero or it should increase linearly with frequency.

In other words, the amplitude plot of the frequency response should be flat for all frequencies. In general, this plot droops at higher frequencies. A term bandwidth is used to quantize the flat useful region of the amplitude plot of the frequency response. The bandwidth is defined as the frequency range in which the amplitude ratio is more than 0.707 of the final value. This is illustrated in fig. 3.31.

AR = Amplitude ratio, f_1 & f_2 are -3db or $1/\sqrt{2}$ AR frequencies.

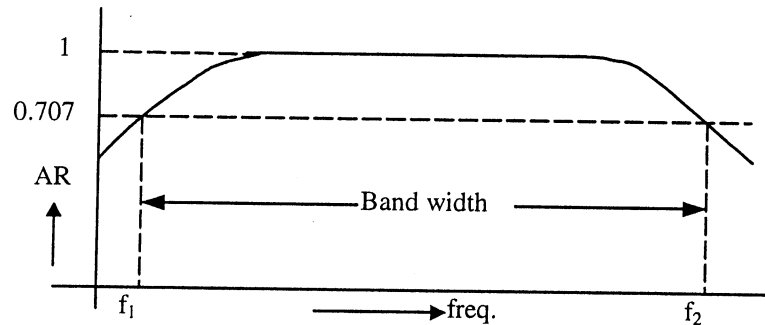


Fig 3.31

Band width of a transducer - frequency response.

A transducer will faithfully reproduce the input if all the frequency components of the input lie within the bandwidth of the transducer. When the input signal contains frequency component outside the bandwidth, the output will be distorted. If important information is in the frequencies outside the bandwidth, then this information might be missed.

3.10 IDENTIFICATION OF TRANSDUCER MODELS

Transducer mathematical models are nothing but the differential equations that describe the dynamics of transducers. These models can be derived from the knowledge of the components, their interconnection and the physical laws governing their functioning. Ofcourse a number of assumptions are mandatory to derive the equations representing model. In many practical situations, the components used, their numerical values, their behavior, their interconnections and the physical laws followed by them may not be precisely known. Therefore, the model cannot be obtained using the above conventional method.

In such situations, the transducer can be assumed to be a black box, whose inputs and outputs are accessible for measurements. Number of methods are available to identify the transducer model by measuring the inputs and outputs of the transducer. If the order

of the model is known apriori, the method of identification becomes simple.

3.10.1 Identification From Impulse Response

Let the transfer function of the transducer be of the form $K/(1+s\tau)$ where K and τ are the only unknown parameters. When this transducer is excited with a unit impulse, the output transform, $Y(s) = K/(1+s\tau)$ as $R(s) = 1$.

$$\begin{aligned} \text{Therefore } y(t) &= L^{-1} Y(s) \\ &= K e^{-t/\tau} \end{aligned} \quad (3.65)$$

The output of the transducer will be as shown in fig. 3.32.

From the experimentally obtained outputs of the transducer $y(t_1)$ and $y(t_2)$ at two different times t_1 and t_2 , the two unknown parameters K , τ of the transducer can be estimated.

$$y(t_1) = \frac{K}{\tau} e^{-t_1/\tau} \quad (3.66)$$

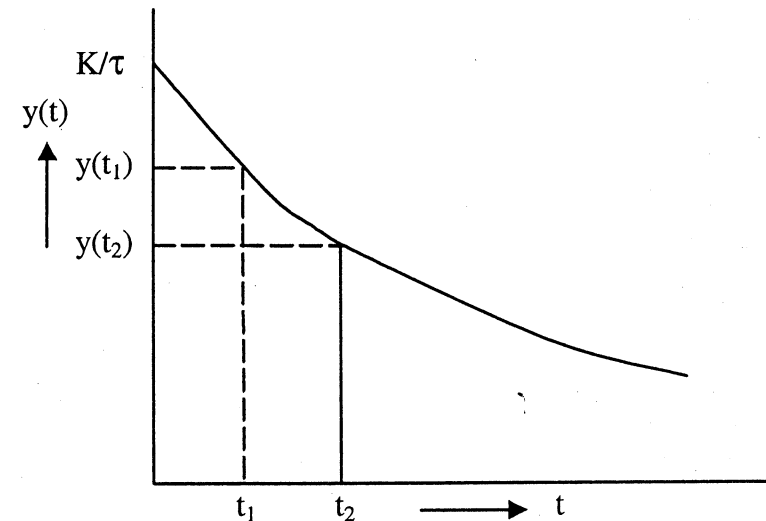


Fig 3.32

First-order transducer output record for impulse excitation

$$y(t_2) = \frac{K}{\tau} e^{-t_2/\tau} \quad (3.67)$$

$$\frac{y(t_1)}{y(t_2)} = e^{(t_2 - t_1)/\tau} \quad (3.68)$$

$$\tau = \frac{(t_2 - t_1)}{\ln \frac{y(t_1)}{y(t_2)}} \quad (3.69)$$

K can be calculated by substituting τ in one of the above equations. If the transfer function is of the form

$$\frac{K}{\omega_n^2 + \frac{2\zeta}{\omega_n} s + 1}$$

where ζ , ω_n and K are the unknown parameters. When such a system is subjected to an unit impulse, the response for the under damped case will be as shown in fig. 3.33. As shown earlier in section 3.7.1.

$$y(t) = \frac{K \omega_n}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \sin \omega_n [\sqrt{1 - \zeta^2} t] \quad (3.70)$$

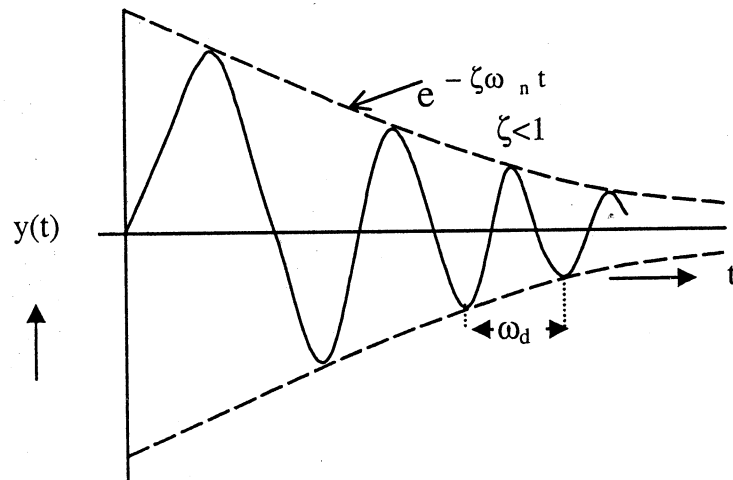


Fig 3.33

II-order transducer output record for impulse excitation.

From the experimental output curve, ζ , ω_n is calculated taking the envelope (dotted line) only. As the envelope is a decaying exponential curve $1/\zeta \omega_n$ is the time constant of the exponential curve.

The time between two successive peaks T_d is determined which is equal to

$$\omega_d = \frac{2\pi}{T_d} = \omega_n \sqrt{1 - \zeta^2} \quad (3.71)$$

Now the values of ζ , and ω_n are determined from the values of $1/\zeta \omega_n$ and the values of the envelope at $t = 0$ is

$$\frac{K \omega_n}{\sqrt{1 - \zeta^2}} \quad (3.72)$$

which can be determined from the experimental response. As ζ and ω_n have already been evaluated, K can be calculated.

3.10.2 Identification From Step Response

When the transfer function of the transducer is of the form $K/(1+s\tau)$, the parameters K and τ have to be determined from the step response. Three methods of determination of time-constant τ have already been discussed in section 3.6.3. The static sensitivity K is calculated as

$$K = \frac{\text{Steady state output change}}{\text{Input change}} \quad (3.73)$$

In situations where it takes a long time for the output to reach steady state, the two-point method discussed in section 3.6.3 can be used to determine K. In that method, the output at two different instants are noted and the time-constant τ is first calculated. By substituting τ in one of the two equations, the static sensitivity K can be calculated.

For a second-order transducer, the parameters, K, ζ , and ω_n can be determined from the step-response. The response of an under damped transducer for an unit step input is shown in fig. 3.34.

The expression for the output $y(t)$ of the transducer is given by

$$y(t) = K \left(1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \right) \sin(\omega_n \sqrt{1-\zeta^2} t + \phi) \quad (3.74)$$

which is derived in section 3.7.2. The time instances at which the maximum and minimum values of the response curve occur can be found out by differentiating $y(t)$ with respect to time and equating to zero.

As shown below

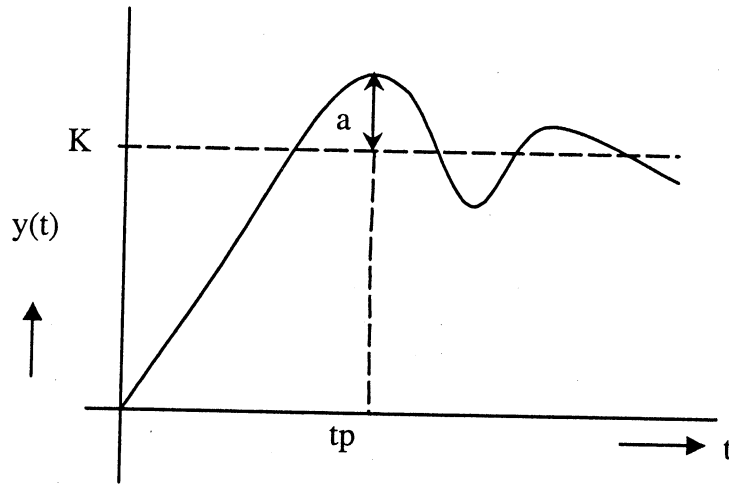


Fig 3.34

Second-order transducer output record for step excitation

$$\begin{aligned} \frac{dy(t)}{dt} = K \left\{ + \zeta \omega_n \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_n \sqrt{1-\zeta^2} t + \phi) \right. \\ \left. - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \cdot \cos(\omega_n \sqrt{1-\zeta^2} t + \phi) \cdot \omega_n \sqrt{1-\zeta^2} \right\} \quad (3.75) \end{aligned}$$

when this expression is equated to zero, one gets

$$\tan(\omega_n \sqrt{1-\zeta^2} t + \phi) = \frac{\sqrt{1-\zeta^2}}{\zeta} \quad (3.76)$$

Recall the definition (eqn 3.43b) of $\phi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$

$$\text{Therefore, } \tan(\omega_n \sqrt{1-\zeta^2} t + \phi) = \tan \phi \quad (3.77)$$

This equation is true for all values of

$$t = 0, \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}, \frac{2\pi}{\omega_n \sqrt{1-\zeta^2}}, \frac{3\pi}{\omega_n \sqrt{1-\zeta^2}} \text{ etc.} \quad (3.78)$$

when $t = 0$, $y(t)$ is 0, minimum value

$$t = t_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} \quad (3.79)$$

$y(t)$ is first maximum and t_p is the peak time.

$$t = t_v = \frac{2\pi}{\omega_n \sqrt{1-\zeta^2}} \quad (3.79)$$

$y(t)$ is second minimum and t_v is the valley time

Because the oscillation is a damped one, the time at which first maximum occurs will be maximum overshoot.

∴ This overshoot shown as 'a' in fig. 3.34 can be obtained as

$$a = y(t) |_{\max} - y(t) |_{\text{steady state}} \quad (3.80)$$

$y(t)_{\max}$ is obtained by substituting $t_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$ in eqn 3.74.

i.e.,

$$y(t) \Big|_{\max} = K \left\{ 1 - \frac{e^{-\zeta \omega_n \pi / \omega_n \sqrt{1-\zeta^2}}}{\sqrt{1-\zeta^2}} \sin \left(\omega_n \sqrt{1-\zeta^2} \cdot \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} + \phi \right) \right\} \quad (3.81)$$

$$= K \left\{ 1 + \frac{e^{-\pi \zeta \sqrt{1-\zeta^2}}}{\sqrt{1-\zeta^2}} \sin \phi \right\} \quad (3.82)$$

$$= K \left(1 + e^{-\pi \zeta \sqrt{1-\zeta^2}} \right) \text{ as } \sin \phi = \sqrt{1-\zeta^2} \quad (3.83)$$

$$\begin{aligned} y(t) / \text{steady state} &= \lim_{t \rightarrow \infty} y(t) \quad (3.84) \\ &= K(1 - 0) \\ &= K \end{aligned}$$

$$\therefore \text{Overshoot } a = K e^{-\pi \zeta \sqrt{1-\zeta^2}} \quad (3.85)$$

From the step-response plotted from experimental results, t_p , a and K can be obtained from eqn. 3.73 ζ can be evaluated from eq.(3.85) as a and K are already known. Substituting this value of ζ in eq.(3.79), ω_n can be determined.

For an over damped second-order transducer the output $y(t)$ is given by

$$y(t) = K \left[\begin{aligned} &1 + \frac{\zeta - \sqrt{\zeta^2 - 1}}{2\sqrt{\zeta^2 - 1}} e^{((- \zeta - \sqrt{\zeta^2 - 1}) \omega_n t)} \\ &- \frac{\zeta + \sqrt{\zeta^2 - 1}}{2\sqrt{\zeta^2 - 1}} e^{((- \zeta + \sqrt{\zeta^2 - 1}) \omega_n t)} \end{aligned} \right] \quad (3.86)$$

It is very difficult to determine ζ , ω_n for a over damped system.

It is easier to express the response in terms of two time-constants τ_1 and τ_2 rather than ζ and ω_n and determine T_1 & T_2 .

$$\frac{y(t)}{K} = 1 + \frac{\tau_1}{\tau_2 - \tau_1} e^{-t/\tau_1} - \frac{\tau_2}{\tau_2 - \tau_1} e^{-t/\tau_2} \quad (3.87)$$

$$\text{where } \tau_1 = \frac{1}{(\zeta - \sqrt{\zeta^2 - 1}) \omega_n} \quad (3.88)$$

$$\text{and } \tau_2 = \frac{1}{(\zeta + \sqrt{\zeta^2 - 1}) \omega_n} \quad (3.89)$$

τ_1 and τ_2 can be evaluated using the followin procedure.

The unit step response of an overdamped system is shown in fig. 3.35.

Let us define the percent incomplete response R_{pi} as

$$R_{pi} \triangleq \left(1 - \frac{y}{k} \right) 100 \quad (3.90)$$

This R_{pi} is plotted on a semi logarithmic paper versus time as shown is fig. 3.36.

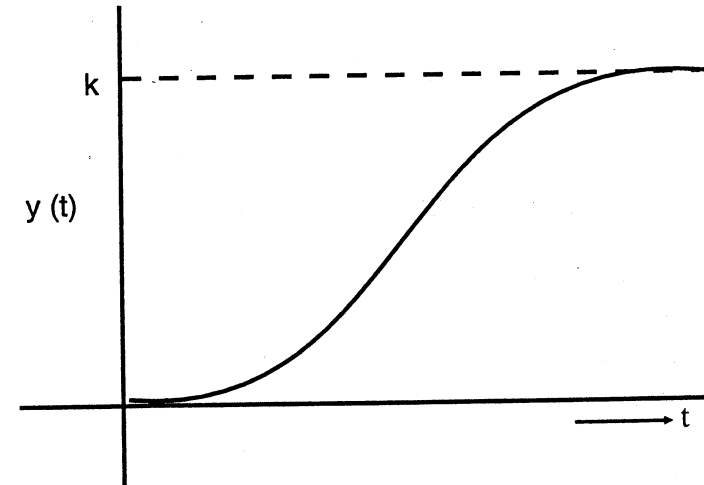


Fig 3.35
II-order system step respond (over damped)

This curve will approach a straight line for large values of time 't' for a second order system. Draw an asymptote for this curve. This asymptote will intersect the vertical axis at P_1 as shown

in fig. 3.36. The value of τ_1 can be taken to be the time at which R_{pi} is $.368P_1$.

Now another curve is plotted taking the difference between the straight line asymptote and the R_{pi} . This will be again a straight line for a second order system. The time at which this straight line has $.368(P_1 - 100)$ is equal to τ_2 .

3.10.3 Identification from Frequency Response

When a first-order transducer is subjected to sinusoidal input with different frequencies, the input-output amplitude ratio at steady-state is given by

$$\frac{Y(j\omega)}{R(j\omega)} = \frac{K}{\sqrt{1 + \omega^2 \tau^2}}$$

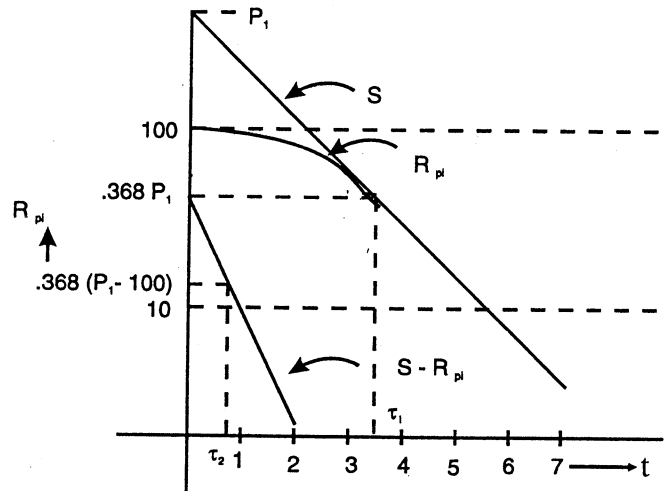


Fig 3.36
Log plot of R_{pi}

At

$$\omega = \frac{1}{\tau}, \quad \frac{Y(j\omega)}{R(j\omega)} = \frac{K}{\sqrt{2}} = 0.707 K \quad (3.92)$$

By noting the frequency at which the amplitude-ratio is 0.707 times the maximum value, the time-constant can be determined.

The amplitude-ratio curve is plotted as shown in fig. 3.37 by conducting a frequency response experiment. At very low values of ω , say $\omega = 0.01$ rad/sec, the amplitude-ratio is almost equal to K . After finding out K , the value of ω for which the amplitude-ratio is $0.707 K$, say ω_c is determined. The time-constant of the transducer is given by $\tau = 1/\omega_c$.

When a second-order transducer is subjected to sinusoidal input with different frequencies, the input-output amplitude ratio at steady-state is given by

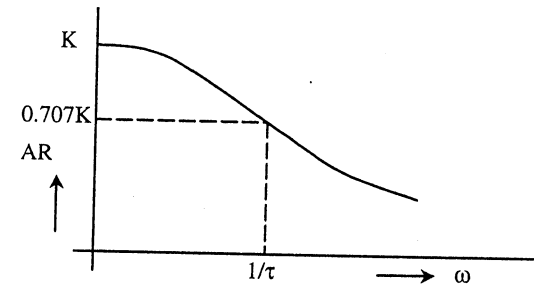


Fig 3.37
Determination of time-constant from frequency response
1-order system

$$\frac{Y(j\omega)}{R(j\omega)} = \frac{K}{\sqrt{1 - \left(\frac{\omega}{\omega_n}\right)^2 + \frac{4\zeta^2\omega^2}{\omega_n^2}}} \quad (3.93)$$

If the transducer is under damped, the experimentally obtained amplitude-ratio versus log ω curve will be as shown in fig. 3.38.

For large values of ω , the amplitude-ratio curve becomes asymptotic to a straight line with a slope of -40 decibels/decade. The gain in decibel is defined as $20 \log [Y(j\omega)/R(j\omega)]$. The number of decades between two frequencies f_1 and f_2 is defined by

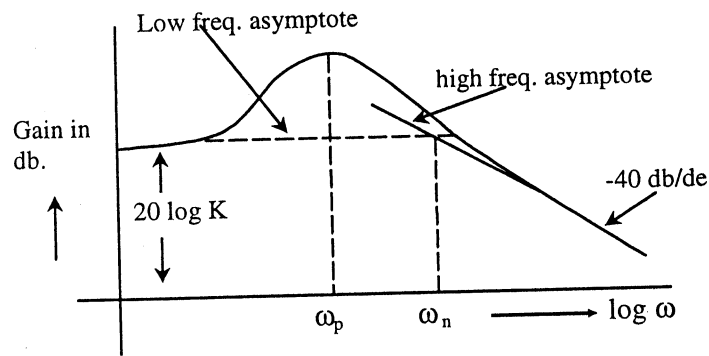


Fig 3.38

Frequency response test of second-order transducer - under damped.

$$\frac{\log f_2/f_1}{\log 10} = \log \frac{f_2}{f_1} \quad (3.94)$$

The low frequency asymptote and the high frequency asymptote meet at a point which corresponds to natural frequency of the transducer. Thus ω_n is determined. The peak amplitude occurs at a frequency ω_p which is given by

$$\omega_p = \omega_n \sqrt{1 - 2\zeta^2} \quad (3.95)$$

The damping ratio ζ can be evaluated by substituting ω_n in eqn 3.95. The low-frequency asymptote has a magnitude of $20 \log K$ at $\omega = 1$. By noting down this magnitude, K can be calculated.

When the second-order transducer is over damped, the amplitude-ratio versus $\log \omega$ curve will be as shown in fig. 3.39.

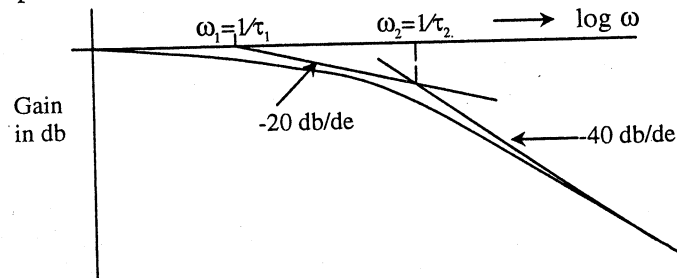


Fig 3.39

Frequency response test of II-order over damped transducer

The amplitude-ratio curve starts with zero slope and the slope increases gradually and finally reaches a value of -40 db/de for large values of ω .

If the order of the transducer is known to be second-order but the actual values of K , ζ , and ω_n are not known, then an experiment can be conducted to obtain the frequency response plots and from the plots the values of K , ζ , and ω_n can be calculated as described in the next section.

3.10.4. Procedure to Determine the Constants and Transfer Functions of a System

The frequency response curve of a transducer is given in fig. 3.40. The transfer function of the system is determined using the following steps.

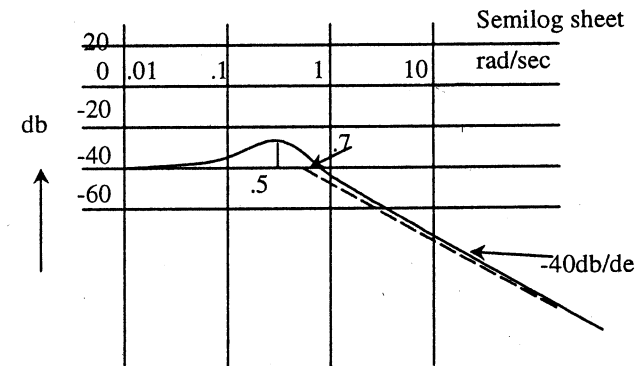


Fig 3.40

Frequency response plot of lightly damped system.

Inference : As this is similar to a II-order under damped one, the following steps are followed.

Step - 1 : Asymptotes at low frequency and high frequency are drawn. They meet at (0.7, 40) point.

$$\therefore \omega_n = 0.7 \text{ rad/sec}$$

Step - 2 : The frequency at which the peak value of the response occurs is found to be 0.5 rad/sec

$$\therefore \omega_p = 0.5 = \omega_n \sqrt{1 - 2\zeta^2}$$

$$1 - 2\zeta^2 = \left(\frac{.5}{.7}\right)^2 = \frac{.25}{.49} \therefore \zeta = .494$$

Step - 3 : The initial slope is zero db/decade.
The decibel value of this horizontal line is 40db.
 $\therefore 20 \log K = 40$
 $K = 100.$

$\therefore$ The transfer function of the transducer is

$$\begin{aligned} & \frac{100}{\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n}s + 1} \\ &= \frac{100}{\frac{s^2}{.49} + \frac{2 \times .494 s}{.7} + 1} \\ &= \frac{100}{1 + 1.41 s + \frac{s^2}{.49}} \end{aligned}$$

In case of higher order system, the procedure is similar except that the straight line approximations has to be done in as many places as necessary and the corner frequencies are to be determined accordingly.

3.11 WORKED OUT EXAMPLES:

EXAMPLE 3.11.1

A pressure transducer gives voltage output as shown below during a static calibration experiment. Obtain least squares fit equation for the input output relationship of the data

Pr. in KPa P	0	5	10	15	
Output in volts V	-0.07	1.08	2.05	3.27	
Pr. in KPa P	20	25	30	35	40
Output in volts V	4.28	5.41	6.43	7.57	8.66

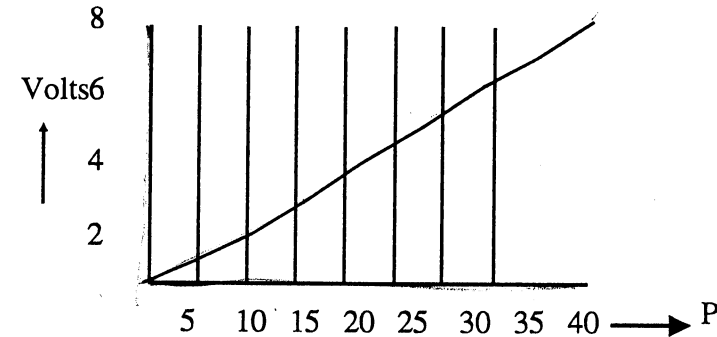


Fig 3.41
Volts Vs Pressure plot for example 3.11.1

SOLUTION

A table can be formed to make the calculations simpler

	x_i	y_i	x_i^2	$x_i y_i$
	0	-0.07	0	0
	5	1.08	25	5.40
	10	2.05	100	20.50
	15	3.27	225	49.05
	20	4.28	400	85.60
	25	5.41	625	135.25
	30	6.43	900	192.90
	35	7.57	1225	264.95
	40	8.66	1600	346.40
Sum	180	38.68	5100	1100.05

$n = 9$

A plot of the input output characteristic fig. 3.41 shows a curve close to straight line. Let us assume the equation of the input output equation be

$$y = ax + b$$

where y is the output in volts and x is the pressure, a and b can be calculated using the least squares fit. The required equations are given by eqn. 3.7 and eqn. 3.8 of section 3.2.2

$$a = \frac{n \sum (x_i y_i) - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

$$b = \frac{(\sum y_i) (\sum x_i^2) - \sum (x_i y_i) \sum x_i}{n \sum x_i^2 - (\sum x_i)^2}$$

Substituting these values in the equations we get

$$a = \frac{9 \times 1100.05 - 180 \times 38.68}{9 \times 5100 - 180 \times 180} = \frac{2938.05}{13500} = 0.2176$$

$$b = \frac{38.68 \times 5100 - 1100.05 \times 180}{9 \times 5100 - 180 \times 180} = \frac{-741}{13500} = -0.055$$

The least squares fit equation for the transducer is

$$v = 0.2176 p - 0.055$$

EXAMPLE 3.11.2

An industrial mercury in glass thermometer is described by the equation $\alpha dh/dt + h = \beta$.

where 'h' is the height of mercury in capillary and 'T' is the temperature at the thermometer bulb. When the temperature is 25°C the height of mercury in the capillary is 10 mm.

The thermometer is immersed in a bath of boiling water suddenly. How does the height of mercury in the capillary rise? Take $\alpha = 7$ sec and $\beta = 1.4$

SOLUTION

The thermometer is described by a I order dynamics.

The Laplace transfer functions is

$$\frac{H(s)}{T(s)} = \frac{\beta}{1 + \alpha s}$$

This thermometer is subjected to a step change in temperature of

$$(100 - 25)^\circ\text{C} = 75^\circ\text{C}.$$

The step response equation of I order system is

$$h(t) = 75 \beta \left(1 - e^{-t/\alpha}\right) + h(o)$$

The initial height $h(o) = 10$ mm

Substituting the values of α , β and $h(o)$, we get

$$h(t) = 75 \times 1.4 \left(1 - e^{-t/7}\right) + 10 \quad (3.97)$$

$$= 115 - 105 e^{-t/7}$$

The height of mercury rises exponentially with time in the capillary as per the equation.

EXAMPLE 3.11.3

A temperature transducer with a time constant of 0.4 sec and a static sensitivity of 0.05 mv/°C is used to measure the temperature of a hot liquid medium which changes from 25°C to 65°C. The transducer is adjusted to read 0 at 25°C.

- Determine the time taken to read 80% of the final voltage value if the temperature changes as a step.
- Calculate the reading of the transducer at the end of 4 seconds if the temperature changes at a constant rate of 10° per sec from 25° to 65°C.

SOLUTION

The given transducer is a I order instrument with transfer function

$$\frac{V(s)}{T(s)} = \frac{0.05 \times 10^{-3}}{1 + 0.4s}$$

where $V(s)$ is the voltage output and $T(s)$ is the temperature of the medium.

- then for a step change of 40°C

$$r(t) = 40 \times u(t).$$

$$v(t) = 40 \times 0.05 \times 10^{-3} (1 - e^{-t/0.4}) \text{ volts}$$

The steady state voltage value of the transducer $V_{ss} = 2$ mV
 $(40 \times 0.05 \times 10^{-3})$

80% of this final value is 1.6 mV

Time taken to read 1.6 mV is given by the equation

$$1.6 \times 10^{-3} = 2 \times 10^{-3} (1 - e^{-t/0.4})$$

$$0.8 = 1 - e^{-t/0.4}$$

$$t = 0.64 \text{ sec}$$

- (b) The input is ramp of slope 10°C/sec . $r(t) = 10t u(t)$ up to 6 sec and then it is terminated. For $t < 6$

$$\begin{aligned} V(t) &= k \times 10 (t - \tau + \tau e^{-t/\tau}) \\ &= 0.05 \times 10 (t - 0.4 + 0.4e^{-t/0.4}) \\ &= 0.5 (t - 0.4 + 0.4e^{-t/0.4}) \end{aligned}$$

for $t = 4 \text{ sec}$,

$$\begin{aligned} V(t) &= 0.5 (4 - 0.4 + 0.4e^{-4/0.4}) \\ &= 0.5 (3.6 + 0.4e^{-10}) \text{ mV} = 1.8 \text{ mV} \end{aligned}$$

EXAMPLE 3.11.4

A thermometer was suddenly dipped in a water bath kept at 100°C , the following data was generated.

t sec	0	2	4	6	8	10	12	14	16	18
Temp. $^\circ\text{C}$	30	54	74.2	81	86	93	96	97	98.4	99

- Find the time constant of the thermometer.
- Find the steady state error if the thermometer is used to measure temperature of a liquid cooling at a constant rate of 2° for every 5 sec.

SOLUTION

From data generated from experiment a graph showing the response can be drawn as in fig. 3.42.

- The response is a typical I order transducer response 63.2% of the step change is
 $(100 - 30) \times 0.632 = 44.2$

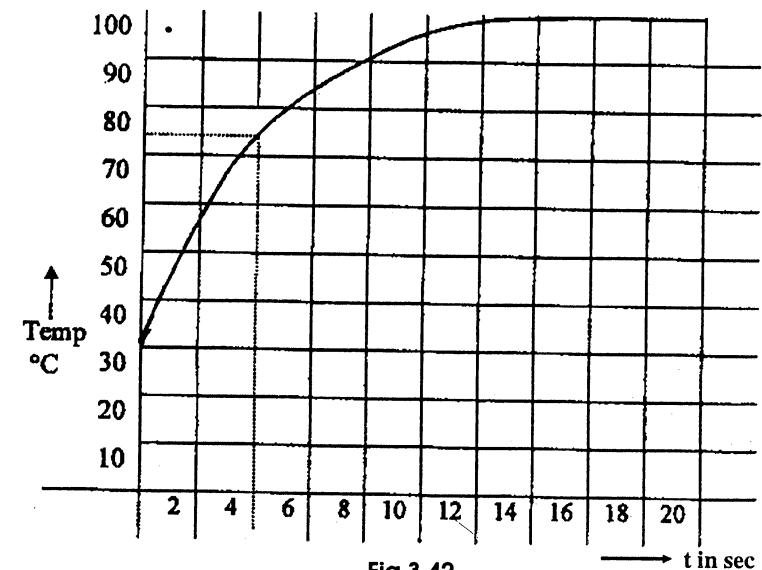


Fig 3.42
Output curve for ex. 3.11.4

The corresponding thermometer reading would be
 $30 + 44.2 = 74.2^\circ\text{C}$.

The time taken to reach this temperature value is 4 sec as per the output curve and the time constant of the thermometer is therefore 4 sec.

The transfer function of the thermometer is $1/(1+4s)$.

- The thermometer is subjected to a cooling at a ratio of 2° per 5 sec.

$$R = (2/5)^\circ\text{C/sec}$$

The input is $r(t) = -(2/5)t$

We know from the equation (3.25), the steady state error = $KR\tau$

Where K is the static gain
 τ is the time constant
 R the ramp input constant

For this problem $K = 1$, $\tau = 4$, $R = -2/5$

The steady state error = $-8/5^\circ\text{C}$.

EXAMPLE 3.11.5

A thermometer at room temperature of 28°C is suddenly immersed in a steaming water bath. Calculate the time constant of the thermometer if it takes 30 seconds to show a reading of 96.4°C .

SOLUTION

Steaming water bath temperature = 100°C .

Step input given to the thermometer is $100 - 28 = 72^\circ\text{C}$

Output temperature $T = 28 + 72(1 - e^{-t/\tau})$ where $t = 30$ sec, $\tau = 96.4$.

$$96.4 = 72(1 - e^{-30/\tau}) + 28$$

$$72(1 - e^{-30/\tau}) = 68.4$$

$$(1 - e^{-30/\tau}) = 68.4/72 = 0.95$$

τ can be calculated from this equation as 10 sec.

we can also infer that when $t = 3$ times the time constant, the output will be 0.95 or 95% of the step change.

$\therefore$ when $t = 30$ sec, τ should be 10 sec.

EXAMPLE 3.11.6

A balloon carrying a first order thermometer with a 15 seconds time constant rises through the atmosphere at 10 meters per second. Assume that the temperature varies at $0.15^\circ\text{C}/50$ meters. The balloon radios temperature and altitude reading back to the ground. At 5000 meter the balloon radios the temperature as 15°C . (i) What is the true temperature at 5000 meters? (ii) What is the true altitude at which 15°C occurs?

SOLUTION

The instrument is I order with $\tau = 15$ sec. It is subjected to a ramp input of 10 m/sec equivalent to $(0.15/50) \times 10 = 0.03^\circ\text{C}/\text{sec}$ as the temperature gradient is $0.15^\circ\text{C}/50\text{m}$.

We know the ramp response is

$$y(t) = KR(t - \tau e^{-t/\tau} - \tau)$$

At steady state $\tau e^{-t/\tau}$ term can be deleted

$$y(t)_{ss} = KR(t - \tau)$$

In this problem $K=1$, $R=0.03^\circ/\text{sec}$ and $\tau = 15$ sec. in the steady state the temperature reading will always lag the true value by $KR\tau$, which is here = 0.45°C .

The temperature is reducing as the balloon rises.

$\therefore$ When the instrument says the temperature is 15°C the actual temperature will be $15 - 0.45 = 14.55^\circ\text{C}$. At 5000 meters the true temperature is 14.55°C . 15°C would have reached at a lower height. The height reduction for an increase of 0.45°C is $0.45 \times (50/0.15)$ m = 150m.

The actual altitude at which 15°C occurs = $5000 - 150 = 4850$ m.

EXAMPLE 3.11.7

The transfer function of a I order instrument with dead time is given by $e^{-1.5s}/(1+0.5s)$. Calculate the output of this system after 2 seconds for a unit step input.

SOLUTION

This instrument has a dead time of 1.5 seconds and a time constant of 0.5 seconds.

The output can be calculated from the equation

$$\frac{y(s)}{u(s)} = \frac{e^{-1.5s}}{1 + 0.5s}$$

$$u(t) = \text{unit step}, \therefore u(s) = 1/s$$

$$y(t) = \frac{e^{-1.5s}}{1 + 0.5s} \times \frac{1}{s}$$

The solution for this equation is

$$y(t) = 1 - e^{-\frac{t-1.5}{0.5}}$$

∴ output after 2 seconds will be

$$y(t)_{t=2} = (1 - e^{-1}) = 0.632$$

EXAMPLE 3.11.8

A temperature measuring system with a time constant of 3 seconds is used to measure temperature of a heating medium which changes sinusoidally between 350°C and 300°C with a period 20 seconds. Find the maximum and minimum values of the temperature indicated by the measuring system and the time lag between the output and the input temperature variation.

SOLUTION

The system is a first order one with transfer function $\frac{1}{1 + 3s}$

The frequency T.F = $\frac{1}{1 + 3j\omega}$

The input temperature variation is $325 + 25 \sin(2\pi/T)t$
 $= 325 + 25 \sin 0.314 t$

For the frequency .314 rad/sec

the amp. ratio M $= \sqrt{\frac{1}{1 + (3 \times .314)^2}} = .727$

The maximum and minimum value of the reading
 $= 325 \pm 0.727 \times 25$
 $= 325 \pm 18.2$

The phase angle difference between input and output
 $= -\tan^{-1}(\omega T)$

$$\begin{aligned} &= -\tan^{-1}(0.314 \times 3) \\ &= -\tan^{-1}(.942) \\ &= -0.7555 \text{ rad.} \end{aligned}$$

EXAMPLE 3.11.9

A signal at 2 cpm has a noise content with an essential frequency content of 0.3 cps with a noise level 0.707 of signal level. What will be the ratio of signal to noise after its transmission through an instrument having a time constant of 6 seconds.

SOLUTION

Since only time constant is given it is a first order instrument whose transfer function is

$$\frac{y(s)}{u(s)} = \frac{K}{1 + 6s} = G(s)$$

The frequency response function is $G(j\omega) = \frac{K}{1 + 6j\omega}$

The frequency of the signal is $2/60 = \frac{1}{30}$ Hz and noise is .3 Hz.

Let us say signal is $A \sin \frac{2\pi}{30} \times t = A \sin \frac{\pi}{15} t$

Then noise is $= .707 A \sin 2\pi \times .3 t$

Signal output by the instrument = $\frac{AK}{\sqrt{1 + \left(6 \times \frac{\pi}{15}\right)^2}}$
 $= \frac{AK}{\sqrt{1 + \left(\frac{2\pi}{5}\right)^2}}$

$$\begin{aligned}
 \text{Noise output by the instrument} &= \frac{.707AK}{\sqrt{1 + (6 \times 2\pi \times .3)^2}} \\
 &= \frac{.707AK}{\sqrt{1 + (3.6\pi)^2}} \\
 \text{Signal/noise ratio after the signal} \\
 \text{is transmitted through the instrument} &= \frac{\sqrt{1 + (3.6\pi)^2}}{.707 \sqrt{1 + \left(\frac{2\pi}{5}\right)^2}} \\
 &= \frac{11.348}{1.135} = 9.998
 \end{aligned}$$

EXAMPLE 3.11.10

A first order instrument must measure signals with frequency content upto 200Hz with an amplitude inaccuracy of 2 percent. What is the maximum allowable time constant ?. What will be the phase shift at 100 and 200 Hz for the cohse time constant?

SOLUTION

T.F of a I order instrument is

$$G(s) = \frac{K}{1 + s\tau}$$

The frequency T.F is

$$G(j\omega) = \frac{K}{1 + j\omega\tau}$$

$$M = |G(j\omega)| = \left| \frac{K}{1 + j\omega\tau} \right| = \frac{K}{\sqrt{1 + \omega^2 \tau^2}}$$

Allowable accuracy at 200 Hz is 2% that means

$$.98K = \frac{K}{\sqrt{1 + (2\pi \times 200 \tau)^2}}$$

$$\begin{aligned}
 \therefore (.98)^2 [1 + (400\pi)^2 \tau^2] &= 1 \\
 (400\pi)^2 \tau^2 &= \frac{1 - .9604}{.9604}
 \end{aligned}$$

$$15.775 \times 10^5 \tau^2 = 0.412$$

$$\tau^2 = 0.26 \times 10^{-6}$$

$$\tau = 0.509 \times 10^{-3} \text{ sec.}$$

$$\text{Phase shift} = -\tan^{-1}(\omega\tau) \text{ radians}$$

$$\text{At 100 Hz / } \phi_{100} = -\tan^{-1} 2\pi \times 100 \times 0.509 \times 10^{-3} =$$

$$\text{At 200 Hz / } \phi_{200} = -\tan^{-1} 2\pi \times 200 \times 0.509 \times 10^{-3} =$$

EXAMPLE 3.11.11

The temperature of water in a deep sea varies as $T_x = T_o - 0.01 x$, where T_x the temperature at a depth of x meter and T_o is the temperature at the sea level. A submarine equipped with a depth and temperature measuring transducer is allowed to descend at a rate of 0.5m/sec. The depth gage is a zero order instrument and the temperature instrument is a I order instrument with a time constant of 50 secs. The surface water temperature is 26°C, what is the temperature reported by the thermometer and the depth gage after 400 secs of descend ?

SOLUTION:

The submarine is descending at a rate of 0.5m/sec.

$\therefore$ After 400 secs the submarine would have gone down to 200m.

The depth gage being a zero order instrument will read 200m.

The temp. measuring instrument being a I order one with time constant of 50 sec, the temperature reading at a distance of x meter is given by

$$T_x = 25 - KR \left[t - \tau (1 - e^{-t/\tau}) \right]$$

where $K = 1$ and $R = 0.01 \times 0.5$ (temp. decreases with 0.01°C/m and the submarine descends at 0.5m/sec. Hence the temp. decreases at a rate of $0.01 \times 0.5 = 0.005^\circ/\text{sec}$)

After = 400sec, reading of the thermometer,

$$T_x = 25 - 0.005 \left[400 - 50 (1 - e^{-400/50}) \right]$$

$$= 23.245^\circ$$

The actual temperature = $25 - 0.005 \times 400 = 23^\circ\text{C}$.

EXAMPLE 3.11.12

Determine the damping coefficient, natural frequency of oscillation for a second order transducer described by

$$\frac{d^2 y(t)}{dt^2} + 2.8 \frac{dy(t)}{dt} + 4y(t) = 11.7u(t)$$

where $y(t)$ is the output and $u(t)$ is the input.

SOLUTION

The generalised equation of a second order transducer is

$$\frac{d^2 y}{dt^2} + 2\xi\omega_n \frac{dy}{dt} + \omega_n^2 y = K\omega_n^2 u(t)$$

where ω_n = natural frequency of oscillation

ξ = damping coefficient or ratio

K = static gain

Comparing the given equation with the standard generalised equation

$$\omega_n^2 = 4, \text{ so } \omega_n = 2, 2\xi\omega_n = 2.4, \text{ so } \xi = 0.7$$

damping coefficient is 0.7

natural frequency of oscillation is 2.

EXAMPLE 3.11.13

A transducer is described by Laplace transfer function

$$\frac{y}{x}(s) = \frac{K}{(1 + 0.1s)(1 + 0.5s)}$$

Find the amplitude inaccuracy and phase shift when x is a signal with frequency $100/2\pi$ Hz.

SOLUTION

The frequency response of this transducer can be obtained by replacing s by $j\omega$.

$$\frac{y}{x}(j\omega) = \frac{K}{(1 + 0.1j\omega)(1 + 0.5j\omega)}$$

Amplitude ratio

$$M = \left| \frac{y}{x}(j\omega) \right| = \frac{K}{\sqrt{(1 + (0.1\omega)^2)} \sqrt{1 + (0.5\omega)^2}}$$

Phase angle between the output and the input signal is

$$\frac{y}{x}(j\omega) = \phi = -\tan^{-1} 0.1\omega - \tan^{-1} 0.5\omega$$

For a frequency of $100/2\pi$, $\omega = 100$ ($\omega = 2\pi f$)

For $\omega = 100$,

Amplitude ratio

$$M = \frac{K}{\sqrt{1 + 10^2} \sqrt{1 + 50^2}} = \frac{K}{\sqrt{101} \sqrt{2501}} = \frac{K}{500}$$

If input magnitude is K output magnitude is $K/500$.

$$\therefore \% \text{ error} = \frac{K - \frac{K}{500}}{K} \times 100 = \frac{499}{500} \times 100 = 99.8\%$$

The output is just 0.2% of the input. So much attenuated

$$\phi = -\tan^{-1}(10) - \tan^{-1}(50) = -173.14^\circ$$

EXAMPLE 3.11.14

A second order transducer has a damping ratio of 0.5 and a natural frequency of oscillation of 3 radians. This transducer transduces a step change in input. What is the maximum output of the transducer?

SOLUTION

The general transfer function of a transducer is

$$\frac{y}{r}(s) = \frac{K\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \text{ where } K = \text{gain, } \xi = \text{damping ratio and}$$

ω_n = natural frequency of oscillation

The step response of this transducer is given by equation 3.43

$$y(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_n \sqrt{1-\xi^2} t + \phi)$$

The time at which the maximum will occur can be found out by differentiating $y(t)$ and equating to zero.

$$\begin{aligned} \frac{dy(t)}{dt} &= \xi \omega_n \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_n \sqrt{1-\xi^2} t + \phi) \\ &\quad - \frac{e^{-\xi\omega_n t} \omega_n \sqrt{1-\xi^2}}{\sqrt{1-\xi^2}} \cos(\omega_n \sqrt{1-\xi^2} t + \phi) \\ &= 0 \end{aligned}$$

This gives after simplification

$$\begin{aligned} \xi \sin(\omega_n \sqrt{1-\xi^2} t + \phi) &= \sqrt{1-\xi^2} \cos(\omega_n \sqrt{1-\xi^2} t + \phi) \\ \tan(\omega_n \sqrt{1-\xi^2} t + \phi) &= \frac{\sqrt{1-\xi^2}}{\xi} = \tan \phi \end{aligned}$$

The maximum and minimum values of $y(t)$ occurs when

$$t = 0, \quad t = \frac{\pi}{\omega_n \sqrt{1-\xi^2}}, \quad t = \frac{2\pi}{\omega_n \sqrt{1-\xi^2}}$$

The first value of t is for minimum, the second value is for maximum etc. Because oscillation is damped one, the first maximum will be the global maximum.

Substituting this value of t in $y(t)$, we get

$$\begin{aligned} y(t)|_{\max} &= 1 + \frac{e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}} \sin \phi, \text{ as } \sin \phi = \sqrt{1-\xi^2} \\ y(t)|_{\max} &= 1 + e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}} \end{aligned}$$

For the problem on hand

$$\begin{aligned} y(t)|_{\max} &= 1 + e^{-\frac{0.5\pi}{0.866}} = 1 + e^{-0.577\pi} \\ &= 1.1632 \end{aligned}$$

RESISTANCE TRANSDUCERS

4.1 BASIC PRINCIPLE

Resistance variation type transducers is one of the important group of transducers that are quite popular, simple and versatile. Many system variables like displacement, acceleration, vibration, force, pressure, temperature, humidity, sound level, light intensity etc. can be transduced using this category of transducers.

The basic principle of this kind of transducer is very simple. Resistance of a conducting wire is given by the expression

$$R = \frac{\rho l}{a} \quad (4.1)$$

Where 's' is the specific resistance of the material of the wire, 'l' is the length of the wire and 'a' is the area of cross section of the wire.

Any stimulus or measurand or a variable which changes or affects any one of the quantities like l , a or s , the resistance of the wire is changed. This change in resistance can be suitably converted by a electrical circuitry to a change in voltage. Thus a transducer is obtained.

4.2 POTENTIOMETER.

A resistive potentiometer is a resistive wire wound on a former provided with a sliding contact and excited with a d.c. voltage source or an a.c. voltage source. The movement of the slider can be translational, rotational or a combination of these two, like, helical motion, permitting measurement of linear or rotational motion or a combination of both. It may be noted here that this term potentiometer has a slightly different description in Electrical measurement and electronic circuits. In electrical measurements,

standard potentiometers are used to measure the unknown voltage by comparing it with a standard known voltage. In electronic circuits the potentiometer is used to denote a resistance whose value can be changed by a slide wire, i.e. the voltage source is not included.

A schematic diagram of a resistive potentiometer is shown in fig. 4.1a and 4.1b.

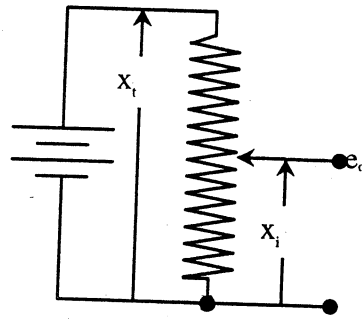


Fig. 4.1a
Linear potentiometer

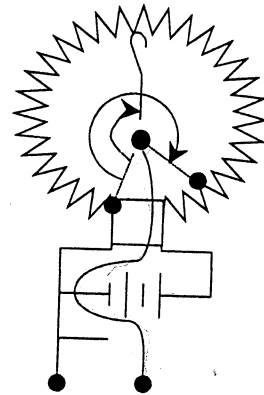


Fig. 4.1b
Rotary potentiometer

The output voltage

$$e_o = \frac{x_i}{x_t} e_{ex} \quad (4.2)$$

where e_{ex} is the excitation voltage, x_i is the input displacement and x_t is the total length of travel of the sliding contact. The total resistance of the potentiometer is taken as R_p . Therefore the resistance corresponding to x_i and x_t will be R_{xi} and R_p respectively. For the rotary potentiometer

$$e_o = \frac{\theta_i}{\theta_p} e_{ex} \quad (4.3)$$

In many rotary potentiometers θ_p is 350° or 1.95π radian. It can be 360° also.

The output voltage in potentiometers will be proportional to the displacement. If the excitation is sinusoidal then the output voltage will be a sinusoidal voltage with no phase shift. The magnitude will be proportional to the displacement.

4.2.1 Loading Effect on Potentiometers

The output of a potentiometric transducer is normally connected to an amplifier or a recorder or a meter which has a definite input impedance and hence a current will be drawn by this meter or amplifier or recorder. The practical situation is shown in fig. 4.2.

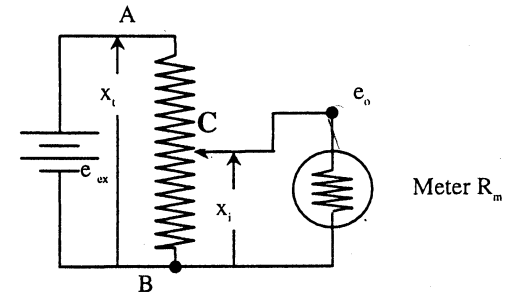


Fig. 4.2
Loading effect on potentiometer

When a meter of input resistance R_m is connected across the potentiometer then the total resistance across points C and B is given by

$$R_o = \frac{R_m R_p (x_i / x_t)}{R_m + R_p (x_i / x_t)} \quad (4.4)$$

where R_p is the total potentiometric resistance and R_m is the meter resistance, x_t total length of travel of the potentiometer and x_i is the input displacement.

$$\text{Let } \alpha = \frac{x_i}{x_t}, \text{ then} \quad (4.5)$$

$$R_o = \frac{R_m R_p \alpha}{R_m + R_p \alpha} \quad (4.6)$$

Then the output voltage across C and B

$$e_o = \frac{e_{ex} R_o}{(1 - \alpha) R_p + R_o} \quad (4.7)$$

where $(1 - \alpha) R_p$ is the resistance between A and C.

After substituting (4.6) into (4.7), one gets

$$e_o = \frac{e_{ex} \frac{R_m R_p \alpha}{R_m + R_p \alpha}}{(1 - \alpha) R_p + \frac{R_m R_p \alpha}{R_m + R_p \alpha}} \quad (4.8)$$

$$= \frac{e_{ex} R_m \alpha}{R_m + R_p \alpha - R_p \alpha^2} \quad (4.9)$$

Then

$$\frac{e_o}{e_{ex}} = \frac{\alpha}{1 + \frac{R_p \alpha}{R_m} (1 - \alpha)} \quad (4.10)$$

Now if R_p/R_m becomes very small, near to zero, the second term in the denominator can be omitted when compared to the first term, then

$$e_o / e_{ex} = \alpha = x_i / x_t$$

This is the condition when there is no loading. In reality R_m is not very large. Hence e_o becomes a nonlinear function of displacement x_i . If the actual plots of e_o/e_{ex} against x_i/x_t is obtained, this fact is well exhibited as shown in fig. 4.3.

Let us examine the situation when $R_p/R_m = 1.0$ and

$$\frac{e_o}{e_{ex}} = \frac{\alpha}{1 + \alpha (1 - \alpha)} \quad (4.11)$$

$$e_o = \frac{\alpha}{(1 + \alpha - \alpha^2)} e_{ex} \quad (4.12)$$

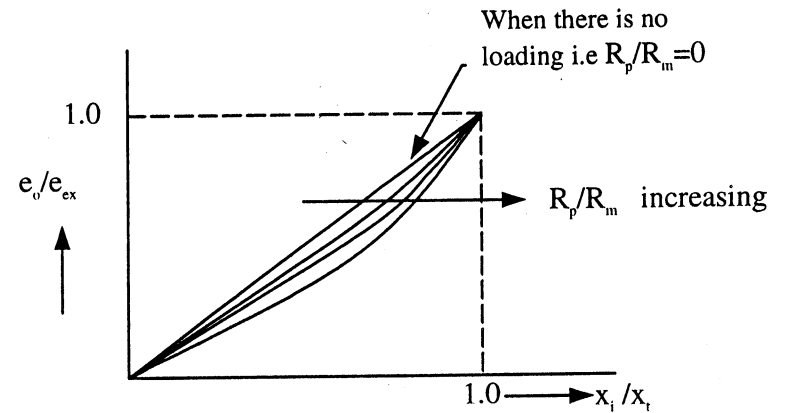


Fig. 4.3
Loading effect on potentiometer

If there were no loading then the output would have been

$$e_o = \alpha e_{ex}$$

$$\varepsilon = \% \text{ error (FS)} = \left(\frac{\alpha e_{ex} - \frac{\alpha e_{ex}}{1 + \alpha - \alpha^2}}{e_{ex}} \right) \times 100 \quad (4.13)$$

$$= 100 \left(\alpha - \frac{\alpha}{1 + \alpha - \alpha^2} \right) \quad (4.14)$$

$$= \frac{100 \alpha^2 (1 - \alpha)}{1 + \alpha - \alpha^2}$$

$$\text{Max. } \varepsilon \text{ will occur at } d\varepsilon / d\alpha = 0 \quad (4.15)$$

$$\begin{aligned} \frac{d\varepsilon}{d\alpha} &= \frac{100 (2\alpha - 3\alpha^2)}{1 + \alpha - \alpha^2} - \frac{100 \alpha^2 (1 - \alpha) (1 - 2\alpha)}{(1 + \alpha - \alpha^2)^2} \\ &= \frac{100 \alpha [(2 - 3\alpha) (1 + \alpha - \alpha^2) - \alpha (1 - \alpha) (1 - 2\alpha)]}{(1 + \alpha - \alpha^2)^2} \\ &= \frac{100 \alpha (2 - 2\alpha - 2\alpha^2 + \alpha^3)}{(1 + \alpha - \alpha^2)^2} \\ &= 0 \end{aligned} \quad (4.16)$$

$$\text{i.e., } 2 - 2\alpha - 2\alpha^2 + \alpha^3 = 0 \quad (4.17)$$

The above equation gives α a value of 0.7 approximately. That means that when x_i/x_t ratio becomes 0.7, the error is maximum. The maximum error works out to be

$$\text{Max. percent error} = \frac{100 (.7)^2 (1 - .7)}{1 + .7 - (.7)^2} = 12.14$$

Similarly the error when $R_p/R_m = 0.1$ can be shown to be 1.51%.

But for values of $R_p/R_m < 0.1$, the maximum error is approximately 15 R_p/R_m percent of full scale reading. This fact is proved below.

Equation 4.10 can be rewritten as

$$\frac{e_o}{e_{ex}} = \frac{1}{1/\alpha + R_p/R_m (1 - \alpha)} \quad (4.18)$$

$$\begin{aligned} \epsilon = \text{Percent error (fs)} &= 100 \left[\alpha - \frac{1}{1/\alpha + R_p/R_m (1 - \alpha)} \right] \quad (4.19) \\ &= 100 \left[\frac{\alpha (1/\alpha + R_p/R_m (1 - \alpha)) - 1}{1/\alpha + R_p/R_m (1 - \alpha)} \right] \\ &= 100 \frac{R_p/R_m \alpha^2 (1 - \alpha)}{1 + R_p/R_m (1 - \alpha) \alpha} \end{aligned}$$

Now α is always ≤ 1 and $R_p/R_m \leq 0.1$ as per the assumption, therefore the above equation can be written as, neglecting $R_p/R_m \alpha$ & $R_p/R_m \alpha^2$ in the denominator

$$\text{Percent error} = 10 R_p/R_m \alpha^2 (1 - \alpha) \quad (4.20)$$

$$\frac{d\epsilon}{d\alpha} = 100 R_p/R_m (2\alpha - 3\alpha^2) = 0 \quad (4.21)$$

$$\text{i.e., } \alpha = 2/3 \text{ or } 0 \quad (4.22)$$

$\alpha = 0$ corresponds to the minimum error point.

$\alpha = 2/3$ corresponds to the maximum error point.

$$\text{The maximum percent error} = 100 R_p/R_m (2/3)^2 (1 - 2/3)$$

$$\begin{aligned} &= (400/27) R_p/R_m \\ &= 15 R_p/R_m \end{aligned} \quad (4.23)$$

The above equations tell us that the error due to loading depends on the ratio of the potentiometer to meter resistance. For a given potentiometer, higher the meter resistance better will be linearity. On the other hand given a meter with a resistance R_m , lower the potentiometer resistance better will be linearity. But the potentiometer resistance can not be lowered much for it will reduce the sensitivity of measurement. It is shown below how this happens. Sensitivity is defined as the change in the output for a change in input. Here in the potentiometer

$$e_o = (e_{ex}/x_t) x_i$$

So sensitivity is e_{ex}/x_t . One may think that sensitivity can be increased by just increasing the excitation voltage. But this is not possible since potentiometers have definite power ratings. This power ratings depend on the size, material used and the configuration and not on the actual resistance value. Power ratings is actually the heat that can be dissipated away by the potentiometer. So the power that can be supplied to the potentiometer is fixed depending on the heat dissipation capacity. Normally manufacturers design a series of potentiometers with a total resistance R_p ranging from 100 to 100000 ohms. But all of them may be 5cm in dia and having same mechanical configuration and hence having the same heat transfer capability and thus having the same power rating say 5 watts at 25°C ambient. And this power rating will fix the maximum excitation voltage for a given R_p . Let P be the power rating of a potentiometer in watts and R_p be the total resistance of the potentiometer.

$$\begin{aligned} \text{Then } P &= (\text{Max. } e_{ex})^2 / R_p \\ \text{Max. } e_{ex} &= \sqrt{P R_p} \end{aligned} \quad (4.24)$$

Therefore for a given meter of resistance R_m , if low R_p is chosen to have a better linearity then the maximum excitation is also low and hence a lower sensitivity. This fact can be understood further by the following examples.

EXAMPLE 4.1

Determine the maximum sensitivity of 5 cm, 5 watts single turn 360° potentiometers of total resistance of 100 ohm, 1000 ohm and 10 kilo ohms.

Sensitivity = S

$$\begin{aligned}
 &= \frac{\text{Max. } e_{ex}}{x_t} = \frac{\text{Max. } e_{ex} \text{ volts}}{\pi D \text{ cm}} \\
 &= \frac{\sqrt{PR_p}}{\pi \times 5} = \frac{\sqrt{R_p}}{\sqrt{5} \pi} \\
 &= 0.142 \sqrt{R_p}
 \end{aligned}$$

for 100Ω S = 0.142√100 = 1.42 volts/cm

for 1000Ω S = 0.142√1000 = 4.5 volts/cm

for 10 KΩ S = 0.142√10000 = 14.2 volts/cm

The above arguments tell us that the choice of R_p is a trade off between loading error and sensitivity. The maximum sensitivity achievable varies considerably from type to type and also with size of potentiometers. If the size is increased then the heat dissipation can be increased but the stroke length is also increased. So the sensitivity can not be increased so easily by increasing size alone. But suitable design and proper choice of material will increase the sensitivity. In fact the shorter stroke devices generally have higher sensitivity.

Extreme values are of the magnitude of 15 volts/degree for sector potentiometers and 120 volts/cm for short stroke of about 0.6 cm translational pots.

EXAMPLE 4.2

The potentiometer output is to be connected to a recorder of 10 KΩ input resistance. Nonlinearity is to be held within 1 percent. A family of potentiometers having power rating of 5 watts and resistance ranging from 100 to 10000 ohms in 50 ohms step is available. Choose from this family of potentiometers that will give

the highest sensitivity. If the potentiometers are of a single turn 360° units what is the sensitivity?

Max. error in linearity = Nonlinearity = 1%

Therefore R_p/R_m should be less than 0.1. If $R_p/R_m \leq 0.1$,

the % error = 15 R_p/R_m

$$\begin{aligned}
 15 R_p/R_m &= 1 \\
 R_p/R_m &= 1/15 \\
 R_p &= 1/15 R_m \\
 &= 1/15 \times 10000 \\
 &= 666.6 \text{ ohms}
 \end{aligned}$$

We can choose a potentiometer with a total resistance $R_p = 666.6\Omega$ at the maximum. Any value of R_p less than 666.6Ω would be alright as far as the nonlinearity specification is concerned. But lower the value of R_p , lower will be the sensitivity. Therefore we choose 650Ω potentiometer from the family which will have max. sensitivity and at the same time having a nonlinearity less than 1%.

$$\text{Max. } e_{ex} = \sqrt{(5 \times 650)} = 57$$

$$\text{Sensitivity for } 360^\circ = 57/360 = 0.16 \text{ volts/degree}$$

4.2.2 Resolution

The resolution of a potentiometer is the smallest change in displacement that can be measured or identified. If the excitation is fixed then it is the smallest change in resistance that can be obtained by slider movement. This factor strongly depends on the construction of the resistance element. To get a high resolution, a single slide-wire can be used as the resistance element of the potentiometer. This will give a continuous stepless resistance variation and hence very fine resolution. Here the accuracy is limited by other components of the system. This type of potentiometers are available but limited to small resistance values because the length of the wire depends on the desired stroke in a translational device and on the space restrictions in rotational devices. Resistance of a given length of wire can be increased by

decreasing the diameter of the wire. But lesser the diameter less will be the strength and will get worn out quickly.

To get fairly high resistance values in small space the wire wound resistance element is made. The resistance wire is wound on a mandrel or on a card which is then formed into a circle or helix, as shown in fig. 4.4.

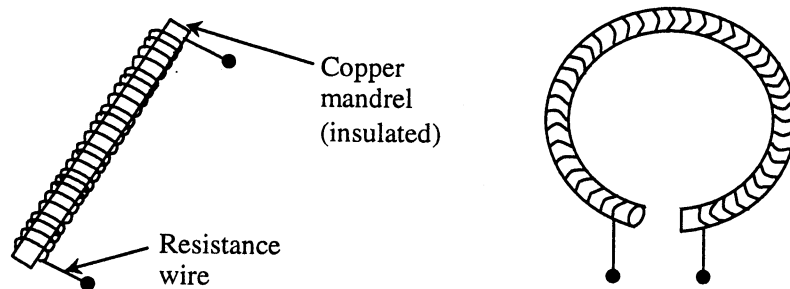


Fig. 4.4
Wire wound resistance

The resistance variation for such potentiometers is linear but continuous step change as shown in fig. 4.5.

That means that resolution is limited depending on the resistance wire size. For example, if a translational device has 100 turns of resistance wire on a card of 1 cm. long, displacement motion changes smaller than 0.01 cm. can not be detected. The practical limit for wire spacing at present times is between 75 to 150

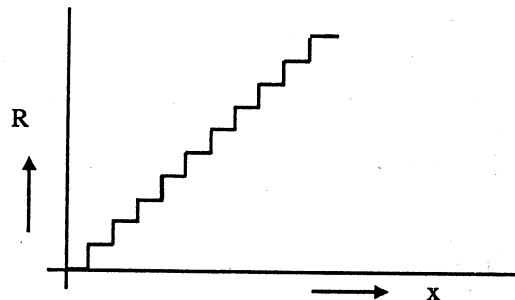


Fig. 4.5
Resolution of potentiometer

turns per cm. So far translational devices, the resolution is limited to 0.001 to 0.002 cm. while single turn rotational device has, on the basis of 150 turns per cm., best angular resolution

$$= \frac{360}{150 \pi D} = \frac{0.764}{D} \quad (4.25)$$

where D is in cm.

The resolution is related to total resistance since the fine wire is required to get close wire spacing and fine wire will have high resistance. A carbon film or conductive plastic resistance element can be used if fine resolution and low resistance are needed. This carbon film element may have resolution as fine as 12×10^{-6} cm. but the over all resolution is little bit poor because of mechanical defects in bearings and wiper springs. In carbon film devices, the wiper is having relatively high resistance. Therefore the amount of current drawn from the potentiometer must be kept low.

Another way to improve the resolution is to use multi-turn potentiometers. Here the resistance element is in the form of a helix and the wiper travels along a lead screw. But still the number of wire per cm. is limited. The increase in resolution can be obtained with the help of gears. The gears make the potentiometer shaft to go through a number of rotations when the shaft of the measurand goes through one rotation. For instance one rotation of the measured shaft can cause the potentiometer shaft to rotate 10 times. In fact multi turn potentiometers are available up to about 60 rotations. In a similar way motion amplifying mechanisms can be used for translational devices.

4.2.3 Linearity of Potentiometers

We have already seen that there can be error in linearity due to loading effect. But there are other factors which give rise to nonlinearity. One of them is the non uniformity in the wire area. Other one is the non uniform winding. Unless specified the potentiometers supplied are nonlinear in nature.

Errors in linearity can be corrected by adding fixed resistance in series and or parallel at proper locations on the winding.

The best linear potentiometers available in the market have nonlinearities of the order of 1% of full scale for 1/2" dia single turn pots and 0.02% for 2" dia. Multiturn. Whatever it might be, the accuracy can be no better than 1/2 resolution in potentiometers.

4.2.4 Nonlinear Potentiometers

All potentiometers are nonlinear in nature. But to generate a specific nonlinear functions we have to design it. If a function $f(d) = kd^2$ is to be generated then a potentiometer as shown in fig. 4.6 is suitable.

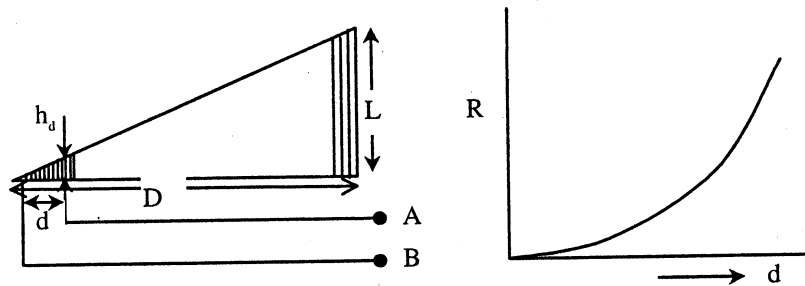


Fig. 4.6

Nonlinear potentiometers and its characteristics

In the potentiometer shown in fig. 4.6, using similar triangle principles it can be written as

$$\frac{L}{D} = \frac{h_d}{d} \quad \text{i.e.,} \quad h_d = \frac{dL}{D} \quad (4.26)$$

The resistance R of this potentiometer upto the slider is proportional to twice the area. This is because the length of the wire upto to this point is proportional to twice the area.

$$l \propto 2 \text{ area} = d h_d = d^2 L/D$$

$$R_d \text{ Resistance upto } d, = kd^2 \quad (4.27)$$

Where $k = \frac{\rho L}{a D}$, where ρ is the specific resistance of the wire and a is the area of the wire which are constants.

To generate sine wave, a potentiometer given in fig. 4.7 is suitable.

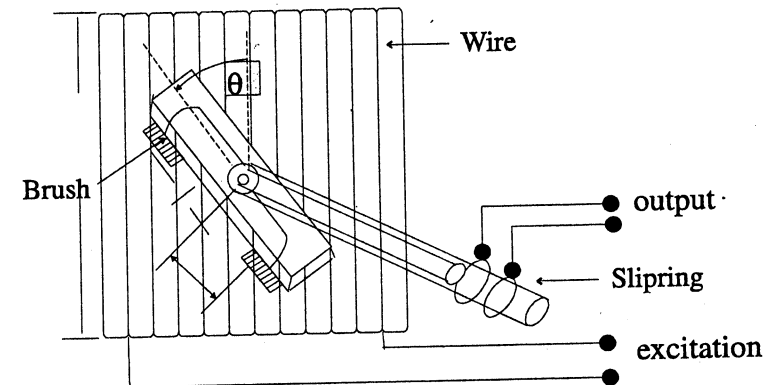


Fig. 4.7

Sine wave potentiometer

It can be shown that the length of the wire between brushes is

$$l = 4 \times L \times r \sin \theta$$

Hence resistance between brushes $\propto \sin \theta$, where θ is the angle through which the potentiometer shaft is rotated.

In general any nonlinear function can be approximated to a number of linear segments and each linear segment can be generated between the tappings of the potentiometer as shown in fig. 4.8.

4.2.5 Noise in Potentiometer

In potentiometers noise is the spurious output voltage fluctuation during the movement of the slider. This spurious output voltage is produced by poor resolution and various mechanical and electrical defects. One of the main sources of noise is vibration of the slider. The motion of the slider over the resistance wire of the wire wound

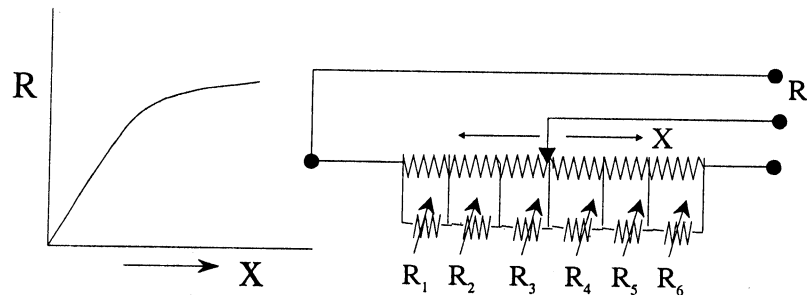


Fig 4.8

General nonlinear potentiometer.

potentiometers, may cause vibration of the contact at certain speed, thus making only intermittent contact. The magnitude of vibration is significant if the frequency of vibration is near the resonant frequency of the spring loaded contact. To overcome this effect, two wipers are used with different resonant frequency as shown in fig. 4.9.

Here if one wiper resonates at a certain speed of wiper the other will not and hence making good contact with the pot wire. Another way of reducing this intermittent contact is by filling the potentiometer with a damping fluid. This damping also increases the tolerance of the unit to shock and vibration.

Another source of noise is the dirt and the wear out products of the potentiometer which come between the wiper and the wire.

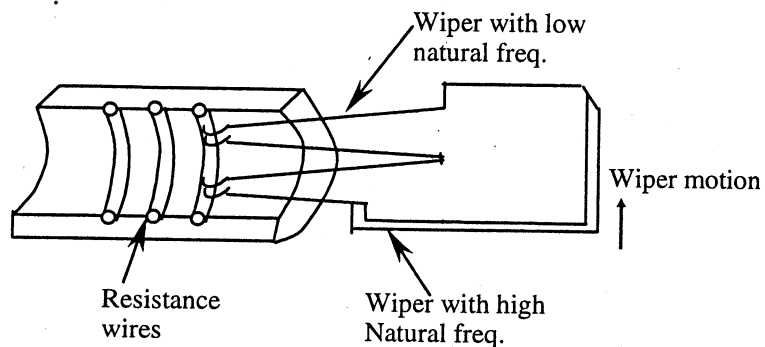


Fig. 4.9

Anti vibration wiper contact.

This makes the contact resistance different at different positions. In addition, the contact resistance varies during the movement of wiper due to unevenness of the resistive wire. This random variation of contact resistance cause a random iR drop when a load current of ' i ' is flowing through.

Noise voltage can also appear due to a current i_c flowing from one turn of the resistive wire to the next through the contact, fig. 4.10.

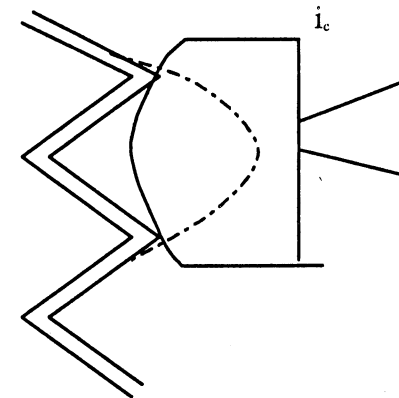


Fig. 4.10

Short circuit of one turn of wire by the wiper

The dynamic response of potentiometer is essentially that of a zero order instrument if the displacement is considered as the input and the voltage is considered as the output. The order of an instrument or system refers to the order of the differential equation describing the mathematical model of the instrument or system. Therefore any change in the input will immediately change the output correspondingly.

The Force or Torque required to start the movement of the brush and maintain a motion depends on the mechanical loading of the instrument. The mass or moment of inertia and the friction between the wiper and the wire are to be considered to find out the mechanical input impedance of the system. These values vary over a wide range, depending on the construction of the potentiometer. Special low friction rotary pots are available with a starting torque

of as small as 200 dyne-cm (0.003 oz.in) but commercial pots may have 6600 to 33000 dyne-cm (approximately 1oz. in to 0.5 oz in). Translational pots may have friction values 10 to 100 times more than rotary devices and inertia value vary widely with size.

A typical 22 mm diameter single turn pot has a moment of inertia of 0.12 g-cm² while a 50 mm diameter 10 turn pot has about 18 g-cm².

If the input variable to the pot, that is the measured variable, is force, the order of the system becomes second order. If friction is neglected as it can be made small by suitable methods, the potentiometer system can be analysed as if it contains only inertia. Let us consider system shown in fig. 4.11.

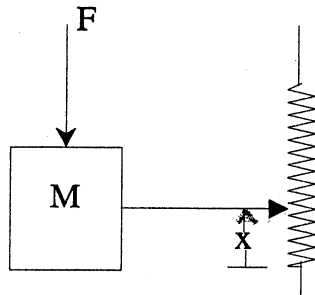


Fig. 4.11

Potentiometer with force input

The force and the displacement x are related by

$$f = M \frac{d^2 x}{dt^2} = MD^2 x$$

where D is the differential operator.

Then the generalised input admittance or compliance

$$C_i(D) \triangleq \frac{\text{displacement}}{\text{force}} = \frac{x}{f}(D) = \frac{1}{MD^2} \quad (4.28)$$

If we draw the block diagram of the system with the Laplace transfer function, we get, the system shown in fig. 4.12, where k_p is the potentiometer constant in k_p volts / displacement

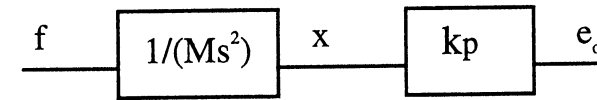


Fig. 4.12

Block diagram of potentiometer with force as input

The frequency response of the admittance set of this system is given by

$$C_i(j\omega) = \frac{1}{M(j\omega)^2} = \frac{1}{M\omega^2} = \frac{1}{|M\omega^2|} \quad |180| \quad (4.29)$$

So, for lower frequencies, the $C_i(j\omega)$ is very high and therefore the loading is negligible even if the output impedance of the previous stage is not very small. If the mass is reduced, the low frequency range also can be reduced further.

The performance of the instrument may be degraded if the environmental factor like temperature, shock, humidity and altitude are extreme. Special precautions must be taken in selecting the potentiometer specially the materials of the potentiometers, for specific application. Under specified environment, the life of the potentiometer may be more than 20 million full strokes or rotations.

The main advantages of potentiometric transducers are their accuracy and simplicity. Potentiometers are hardly influenced by acceleration and vibration. But they require precision machining and their resolutions are limited.

EXAMPLE 4.3

A potentiometer whose moving part weighs 1 gm measures the displacement of a spring mass system shown in fig. 4.13, subjected to a step input. What is the measured natural frequency ?

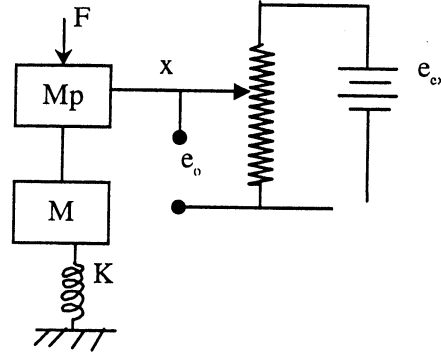


Fig. 4.13
For example 4.3

$$M_p = 1$$

e_{ex} The equation of motion for this system is

$$(M + 1) \ddot{x} + kx = F$$

Comparing this with the general second order system of chapter 3, natural freq. $= \sqrt{k/(M + 1)}$

EXAMPLE 4.4:

A potentiometer whose moving part weighs 1.5 gm is used to measure a spring mass system shown in fig. 4.13 subjected to a step input. The measured natural frequency is 30 cps. (i) If the spring constant and mass M of the system are unknown, can the true natural frequency be deduced from the above data. (ii) If an additional weight of 1.5 gm is attached to the mass of the system and then the measured natural frequency is 25 cps. Calculate the true natural frequency of the system.

(i) No.

$$(ii) (M + M_p) \ddot{x} + kx = u(t) \\ = 1$$

Taking the Laplace transform, we get,

$$\frac{x(s)}{u(s)} = \frac{1}{(M + M_p)s^2 + k} \quad (4.30)$$

Compare this with the general transfer function of a second order system, the natural frequency ω_n is

$$\omega^2 = \frac{k}{(M + M_p)}$$

w_n initially without adding extra weight is $2\pi \times 30$

$$(2\pi \times 30)^2 = \frac{k}{(M + 1.5 \times 10^{-3})} \quad (4.31)$$

when additional weight is added, $w_n = 2\pi \times 25$, we get

$$\{(2\pi \times 25)^2 = \frac{k}{[M + (1.5 + 1.5) \times 10^{-3}]}\} \quad (4.32)$$

There are two equations and two unknowns. Therefore k and M can be calculated as given below:

Dividing eqn. 4.31 by eqn. 4.32

$$\frac{900}{625} = \frac{(M + 3 \times 10^{-3})}{M + 1.5 \times 10^{-3}} \quad (4.33)$$

$$\therefore 900M - 625M = (3 \times 625 - 1.5 \times 900) \times 10^{-3} \quad (4.34)$$

$$275M = (1875 - 1350) \times 10^{-3}$$

$$M = (525/275) \times 10^{-3} = 1.91 \times 10^{-3} \text{ kg} = 1.9 \text{ gm.}$$

using equation 4.3, we get $(2\pi \times 30)^2 (1.91 + 1.15) \times 10^{-3} = k$

$$\therefore k = 121.16 \text{ Newton/m.}$$

The natural frequency of the system without the potentiometer is

$$= \sqrt{\frac{121.6}{1.91 \times 10^{-3}}} = 251.86 \text{ rad} \approx 40.08 \text{ cps.}$$

4.3 RESISTANCE STRAIN GAGES

The next important transducer making use of resistance variation as a fundamental property is resistance strain gage. We know the resistance of a conductor of uniform cross-sectional area 'a', length L and specific resistivity ρ is given by

$$R = \rho L/a \quad (4.35)$$

Now if this conductor is subjected to a stress, this resistance R will change because of dimensional changes and also because of the

change in the fundamental property of materials called piezo resistance. If the equation (4.35) is differentiated partially with respect to ρ , L and a , we get the following equation as the total change in R .

$$dR = \frac{a(\rho dL + Ld\rho) - \rho Lda}{a^2} \quad (4.36)$$

Dividing the LHS by R and RHS by $\rho L/a$, we get

$$\frac{dR}{R} = \frac{dL}{L} + \frac{d\rho}{\rho} - \frac{da}{a} \quad (4.37)$$

$$\text{But } da = a \left(1 - \nu \frac{dL}{L} \right)^2 - a \quad (4.38)$$

Where ν is the poisson's ratio.

$$\begin{aligned} \frac{da}{a} &= -2\nu \frac{dL}{L} + \nu^2 \left(\frac{dL}{L} \right)^2 \\ &\approx -2\nu \frac{dL}{L} \end{aligned} \quad (4.39)$$

neglecting the second term as it is very small. Substituting this value of equation (4.39) into equation (4.37) we get

$$\frac{dR}{R} = \frac{dL}{L} + \frac{d\rho}{\rho} + 2\nu \frac{dL}{L} \quad (4.40)$$

Dividing throughout by dL/L we get,

$$\frac{dR/R}{dL/L} = 1 + 2\nu + \frac{d\rho/\rho}{dL/L} \quad (4.41)$$

The left hand side of the above equation is called 'gauge factor'.

The gauge factor is unit resistance change per unit strain, which is due to three factors as revealed by the above equation. The first term is the resistance change due to length change, the second term is the resistance change due to area change and the third term is the resistance change due to piezo resistance change.

If the gauge factor for a strain gage is known, then a measurement of resistance change allows one to determine strain to which the gage is subjected to. The gauge factor for a strain gage is normally supplied by the manufacturer. This gauge factor can be computed and experimentally verified.

$$\text{Gage factor} = 1 + 2\nu + (d\rho/\rho)/(dL/L)$$

' ν ' the poisons ratio which is always between 0 and 0.5 for all metals.

$$\frac{d\rho/\rho}{dL/L} = \Pi_1 E \quad (4.42)$$

where Π_1 = longitudinal piezo resistance coefficient

E = modulus of elasticity.

Some of the most common type of strain gage are made of Advance or Isoelastic material.

Advance is an alloy of 55% copper and 45% nickel. The gages factor is about 2.

Isoelastic is an alloy of 36% nickel, 8% chromium, 4% manganese, silicon and molybdenum and 52% iron. This has a gage factor of about 3.5.

Semiconductor materials can also be used as strain gage material. These semiconductor strain gages have very high gage factor of about 125. This high gage factor is due to the large change in resistance of the strain gage material due to piezo resistive effect. This effect is not linear with strain as the longitudinal piezo resistance coefficient varies significantly with stress. But in metallic gages the resistance change under strain is mostly due to dimensional change and the gage factor is fairly constant. Table 4.1 gives a list of different gage materials and their gage factors.

Table 4.1: Gauge factor for different material

Grid material	composition	G F	App. Resistivity Micro ohm-cm	App. Temp. coefficient ohm/ohm°F
Nichrome	80% Ni, 20% Cr	+2.0	112	0.00022
advance	45% Ni, 55% Cu	+2.0	49	0.000006

Grid material	composition	G F	App. Resistivity Micro ohm-cm	App. Temp. coefficient ohm/ohm°F
Isoelastic	36% Ni, 8% Cr, 0.5% Mo, remaining Manganese, Fe Si	+3.5	112	0.00026
Manganin	4% Ni, 12% Mn, 84% Cu	0.47	48	0.000006
Iridium-Platinum	5% Ir, 95% Pt	5.10	24	0.0007
MoveI	67% Ni, 33% Cu	1.9	42	0.0011
Nickel	—	-12	7.8	0.0024

In metallic strain gages, there are essentially two varieties namely, (i) unbonded type and (ii) bonded type.

4.3.1 Unbonded Type Strain gage.

In unbonded type, the resistance wires of about 0.025 mm diameter (0.001 inch) are fixed with some initial tension between two frames which can move relative to each other. This initial tension or preload is necessary to avoid buckling under compression or negative displacement and this preloading should be greater than any expected compression or negative displacement. A simplified figure is shown in fig. 4.14.

Unbonded type strain gage for rotational motion is shown in fig. 4.15.

The angular motion given to the inner member which is pivoted to the outer stationary member, increases the tension on the wires and reduces the preload on the other two wires. For example, clockwise twist given to the centre beam increases the tension on wires A and C and reduces the preloaded tension on wires B and D. If they are connected in a bridge as shown then the output voltage available is four times the voltage that would have

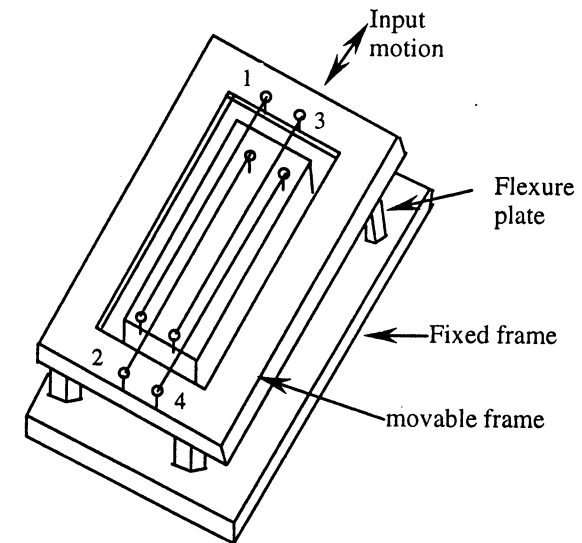


Fig. 4.14a
Unbonded type strain gage

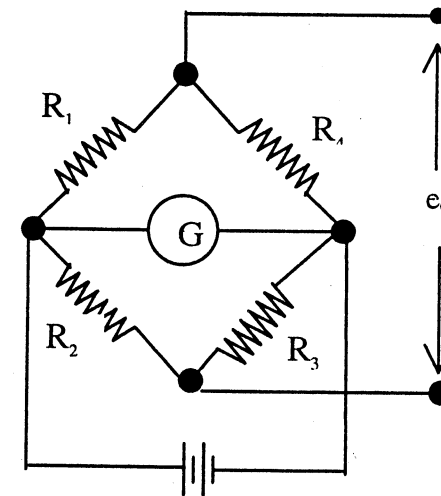


Fig. 4.14b
Circuit Connection

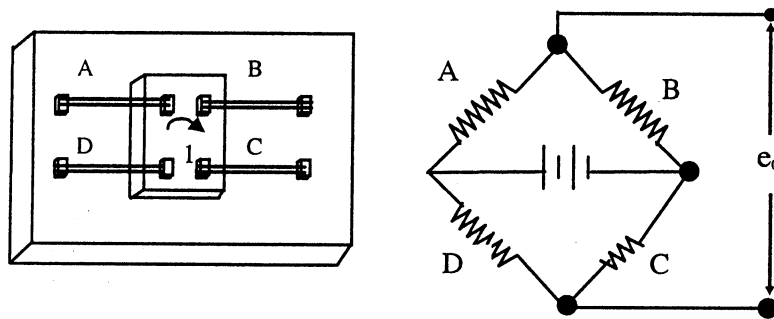


Fig. 4.15

Unbonded type strain gage for rotational stress

been obtained due to a single wire. This arrangement is useful for measurement of Torsional Strain and angular displacement. This type of gages can be used to measure only very small displacements of the order of .004 cm full scale. Normally these gages are used as sensors for force, pressure and acceleration. In these cases the strain wires serve as the necessary spring elements to transduce force to displacement and this displacement is sensed as a resistance variation. The range of force and deflection values are decided by the size, length of wires and the number of wires used.

The sensitivity for a bridge excitation of 5 volts is 40mv full scale output for .006 cm full scale displacement. The nominal value of resistance of the bridge arms is 350 ohms. The thermal sensitivity shift is 0.02 %/°C between -18°C and 120°C.

The order of the instrument is zero order if displacement is considered as the input and the bridge voltage as output. If the input is force then the order is two.

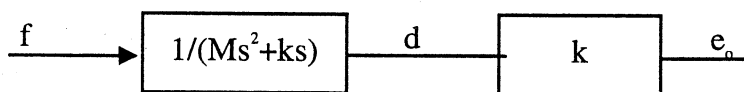


Fig. 4.16

Block diagram of a strain gage with force as input

Here the moving mass is M and the spring constant K is determined by the resistance wires and the flexure.

4.3.2 Bonded Type

In case of bonded-filament strain gages, the resistance wire is made into a form of a grid and cemented between two pieces of thin paper. The factors to be considered to have a detailed description are

- Filament construction
- Material of the filament wire
- Base carrier material or backing material
- Cement used to bond the filament to the carrier
- Lead wire connections

4.3.2.1 Filament construction

The filament wire in bonded type is made in the form of a flat grid or flattened helix or a thin foil etched to give a flat grid pattern as shown in fig. 4.17.

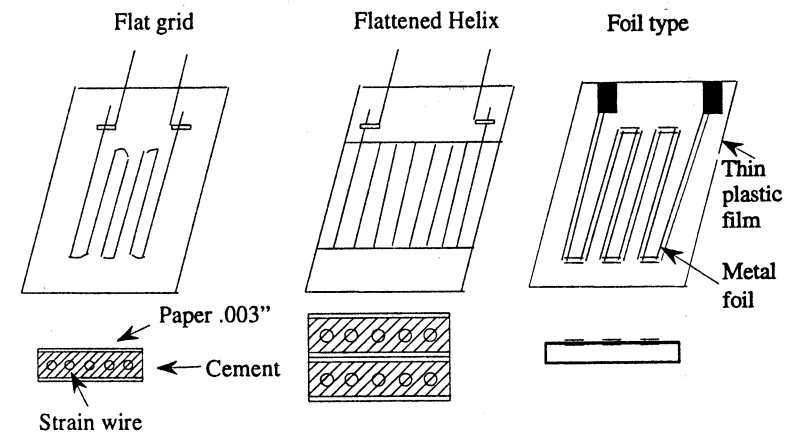


Fig. 4.17

Filament construction details

In the flat grid type the gage lengths were around 2.5 cm and wire size 0.025 mm. for shorter gage lengths up to .15 cm helical grid type is suitable. In this type the resistive wire is wound round a thin walled cylinder in the form of a helix. This cylinder is then flattened and bonded between two sheets of insulating material.

For still shorter gages up to 1mm, foil type is suitable. In this type a metal foil of about 0.0004 mm thickness is formed on plastic film of 0.025 mm giving a total thickness of 0.025 mm. This foil is etched by suitable process so that a metallic grid is formed.

These strain gages are mainly sensitive to the component of strain along their longitudinal axis. However they are also sensitive to a small extent to strain in the transverse direction. This is because of the loops at the end of each turn of wire. In construction this end loop should be made as small as possible. For this reason in foil type the end loops are made thicker to reduce the resistance of this portion of the gage. The nominal value of resistance for these gages range from 40 to 2000 ohms, but 120, 350 and 1000 being the common values.

4.3.2.2 *Material of the filament wire*

The material chosen should have high gage factor. Common materials are Advance, Isoelastic and Nichrome – V (Nickel-chrome alloy with a G.F of 2.2). the choice of material depend not only upon gage factor but also its temperature sensitivity. The metal alloys used for strain gages are temp. sensitive. For example, if strains are to be measured over long period and the temperature is likely to change widely between each tests then low temperature sensitive materials like copper nickel alloy is to be used inspite of the fact it has low gage factor. For measurement of strain above 250°C Nichrome V is suitable. For strain measurements at still higher temperature platinum alloys are used.

As a matter of fact many more points are to be considered when the grid material is selected. Selection is actually a compromise between the following factors.

- High gage factor
- High resistivity
- Low temperature sensitivity
- High electrical stability
- High yield point
- High endurance limit

- Good workability
- Good solderability
- Low hysteresis
- Low thermal emf.
- Good corrosion resistance.

4.3.2.3 *Base carrier material*

The carrier material or the support material should have the following properties.

- Good adherence to cements used
- High dielectric strength
- High mechanical strength
- Minimum thickness consistent with other factors
- Minimum temperature restrictions.

For room temperature applications, the carrier material for the filament wire is a nitrocellulose – impregnated paper. For higher temperatures, phenolic plastic impregnated cellulose or glass fibres are used. For cryo-genic applications, Nichrome foil gages are available in the phenolic-glass carrier construction with integral welded leads.

4.3.2.4 *Strain gage cements*

The cement to be used for fixing the strain gage to the straining member should have the following properties.

- High mechanical strength
- High creep resistance
- High dielectric constant
- Good adherence
- Minimum moisture attraction
- Minimum temperature restriction
- Ease of application
- The capacity to dry fast

4.3.2.5 *Lead wire connections*

The vulnerable point for failure in the ordinary wire strain gage is at the discontinuity formed by the junction between the grid and the lead. A soldered junction between the fine filament and heavier

lead wire or ribbon may be alright for static or slowly varying loads. But for dynamic strain measurements, resistance welding is necessary to provide a fatigue free resistant joint. Lead wire materials should have low, stable resistivity, minimum temperature coefficient of resistance. These lead wires are to be insulated properly with materials of the same thermal classification as the gage carrier and bonding cements. The recommended lead wire insulation material for the temperature range is given below.

Below	–	75°C	–	nylon
Between	–	75°C and 65°C	–	vinyl
Between	–	75°C and 95°C	–	polyethylene
Between	–	75°C and 260°C	–	teflon

Above 260°C glass sleeving or glass impregnated silicone.

4.3.3 Temperature Effect on Strain Gages

Temperature effect is one of the most troublesome factors in the use of resistance strain gages. The strain gage changes its resistance due to temperature variation in two ways besides the actual strain of our interest.

- The strain produced by differential thermal expansion existing between the grid support and the gage.
- The change in resistance due to temperature variations.

Unless proper correction or compensation is done, the indicated strain will be very much different from the actual strain. This temperature effect can be cancelled in various ways. One way is to use a dummy gage identical to the active gage. This dummy gage is cemented to a piece of the same material which is unstrained and kept in the vicinity of the active gage so that the same temperature variation is experienced by the dummy. This dummy and the active are placed in the adjacent arms of a wheatstone bridge, so that resistance change due to differential thermal expansion and due to the temperature variation will have no effect on the bridge output voltage. Another way is to construct a special inherently temperature compensated gages.

The temperature may change the gage factor also. This effect is negligible in metallic gages but affects seriously in case of semiconductor gages. This temperature problem has been successfully solved in the past for the temperature range from liquid Helium temperature -70°C to 1100°C .

4.3.4 Measurement of Strain

The strain measured with the help of the strain gage is going to be some sort of average strain as strain gage spreads over an area. But normally strain is defined with respect to a point. Our measurement will be correct only if the strain gradient is constant and the strain is uniaxial. The strain gage is to be fixed so that its axis is aligned with the strain axis and its mid point (which is marked by the manufacturer on the gage) coincides with the point of interest. If the strain gradient is not constant over the area of gage and its form is not known then the strain read by the gage cannot be associated with a point. In such a situation, the smallest practical gage is very useful. There are situations where the direction and magnitude are both unknowns. In such a case, the strain measurement in three different directions at a point is necessary to determine the maximum strain and its direction. If three different gages are to be fixed at a point, then one over the other arrangement is to be done which may not be very satisfactory for obvious reasons. Fixing them side by side will cover more area and hence the uncertainty of the measurement will increase. To overcome all these

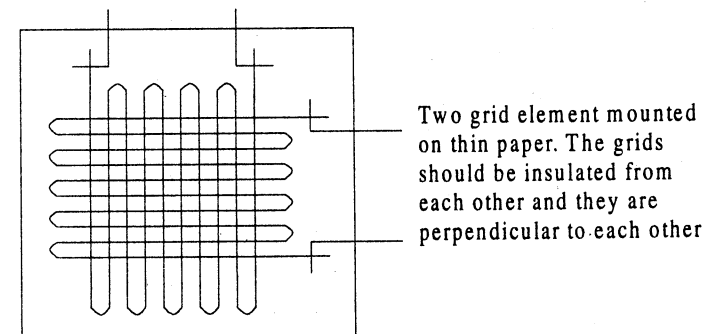


Fig. 4.18
Two element rosette

difficulties to certain extent, strain gage 'rosettes' have been developed. They are available both in wire type and also in foil type. Fig. 4.18 shows a two element rosette.

Often two strain measurements at right angle suffice to determine a stress both direction and magnitude in one quadrant of a plane. This rosette can also be used for measurement of strain in a known direction with temperature compensation. The grid perpendicular to the strain axis will compensate the resistance variation due to temperature change and resistance variation due to strain is negligible. Fig. 4.19a is delta rosette and fig. 4.19b rectangle rosette. These rosettes can be used for any strain measurements whose direction and magnitude is completely unknown and it can be in any quadrant of a plane.

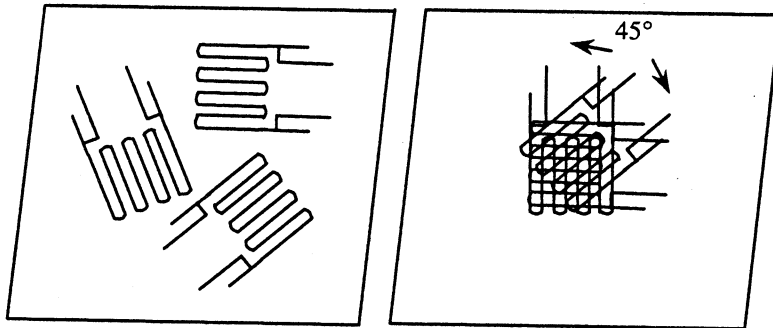


Fig. 4.19a

(a) Delta rosette (b) Rectangle rosette

4.3.5 Strain Gage Circuitry

The strain gage can transduce a strain into a resistance change only. If a voltage change corresponding to the strain is needed, then a potentiometer circuit or a bridge circuit with an excitation should be thought of. One may think that an ohmmeter can be used to find out the change in resistance which can be properly calibrated to give the corresponding strain. This is not possible because the sensitivity of the ohmmeter is far less than the sensitivity required for strain measurement. As a matter of fact strains of the order of one micro m/m are detectable with commercial equipment. For a

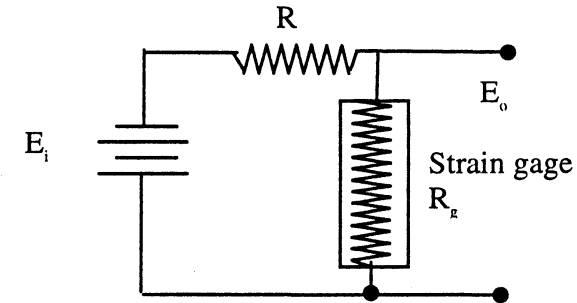
typical gage constants of gage factor $F = 2.0$ and $R_g = 120$ ohms, the change in resistance that is to be measured for a microstrain, is

$$\begin{aligned}\Delta R &= F \epsilon R_g \text{ where } \epsilon \text{ is the strain} \\ &= 2 \times 1 \times 10^{-6} \times 120 \\ &= 0.00024 \text{ ohms.}\end{aligned}\quad (4.43)$$

No ohmmeter will be able to measure this small resistance change reasonably accurately.

Therefore bridge and potentiometric methods are used.

4.3.5.1 The ballast circuit

Fig. 4.19b
Ballast circuit

$$E_o = \frac{R_g E_i}{R + R_g} \quad (4.44)$$

$$\begin{aligned}dE_o &= \frac{((R_g + R) dR_g - R_g dR_g) E_i}{(R_g + R)^2} = \frac{R dR_g E_i}{(R_g + R)^2} \\ &= \frac{R R_g E_i}{(R + R_g)^2} \frac{dR_g}{R_g}\end{aligned}$$

$$\text{But } G.F. = F = \frac{dR_g}{R_g} \cdot \frac{1}{\epsilon}$$

$$\therefore dE_o = \frac{R R_g E_i}{(R + R_g)^2} F \epsilon \quad (4.45)$$

The value of R is large relative to R_g so that it acts as a ballast resistance to keep the current I constant regardless of gage resistance change. The normal value of R is to fourteen times the magnitude of R_g .

4.3.6 The Wheatstone Bridge Circuits

The simplest form of wheatstone bridge for the measurement of strain is shown in fig. 4.20.

The bridge is balanced initially when there is no strain. When the gage is strained, the resistance changes and thereby causing an unbalance in the bridge. The voltage output is given by the following expression.

Considering the upper half of the bridge, potential of the centre point is

$$E_c = \frac{R_g}{R_g + R_1} E_i \quad (4.46)$$

The change in this voltage for a change in R_g is the output voltage as the potential of the reference point is E_c , as R_2 and R_3 do not change.

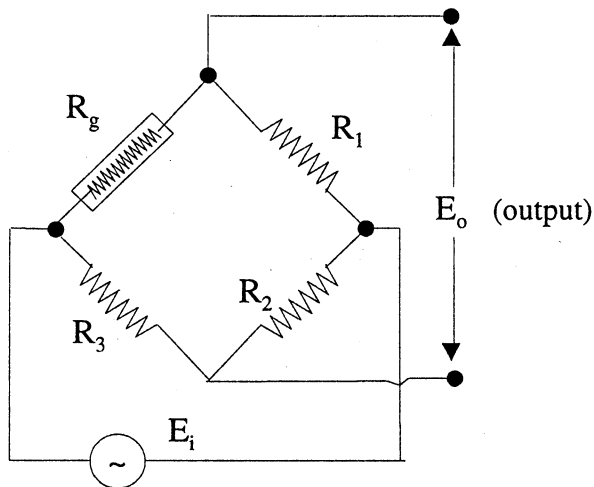


Fig. 4.20
Strain gage bridge circuits

$$E_o = \frac{dE_c}{dR_g}$$

As shown in the previous ballast circuit

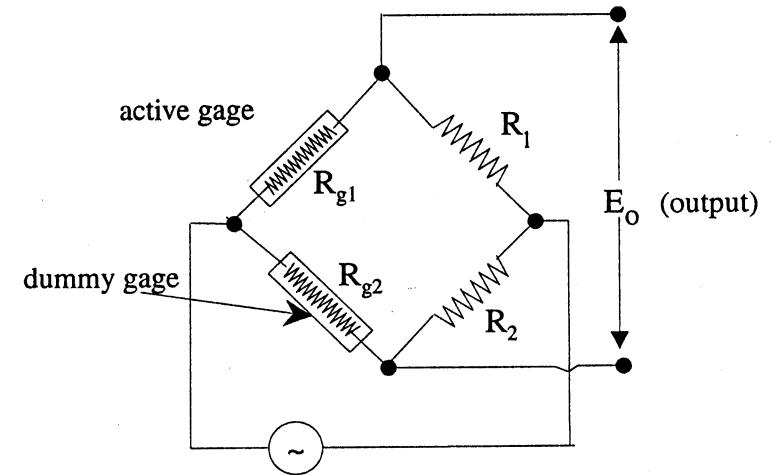


Fig. 4.21
Strain gage circuit with temperature compensation

$$E_o = \frac{R_1 R_g}{(R_g + R_1)^2} E_i F\epsilon$$

In this basic circuit there is no temperature compensation. To have temperature compensation, another dummy gage, fixed to an unstrained piece of the same material, which is kept in the neighborhood of the active gage is used as shown in fig. 4.21.

The active gage and the dummy gage are same in all respects and therefore have same temperature coefficient of resistance and temperature coefficient of expansion. Any change in R_{g1} due to temperature change is exactly compensated by a same change in R_{g2} .

The dummy gage can also be put in the other arm as shown in fig. 4.22.

In this bridge also temperature effect will change R_{g1} and R_{g2} to the same extent and thereby the potential of the centre point is unaltered.

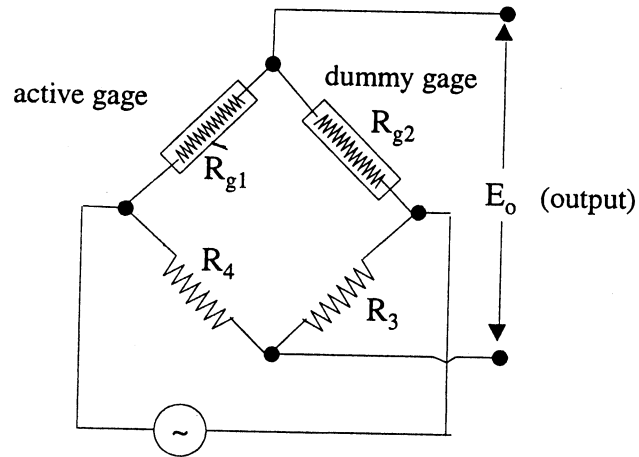


Fig. 4.22

Strain gage bridge circuit with temperature compensation - another way

If higher voltage output is required another set of active and dummy gages can be used as shown in fig. 4.23.

$$E_o = \frac{2 R_{g1} R_{g2}}{(R_{g1} + R_{g2})^2} E_i F \epsilon \quad (4.47)$$

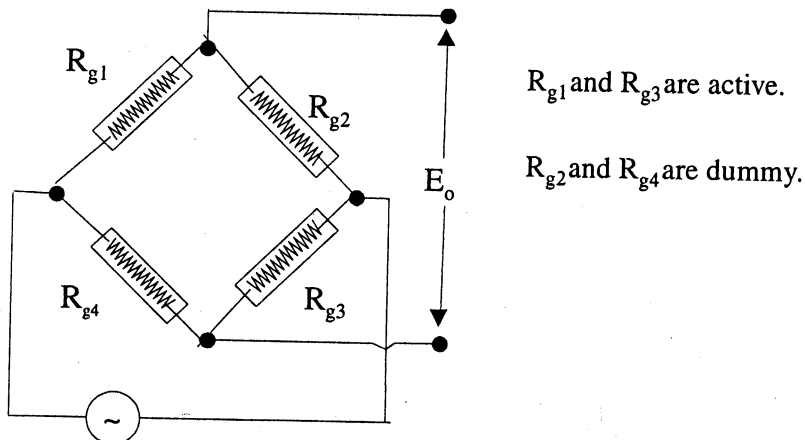


Fig. 4.23

Strain gage bridge with two active gages

When a bending stress is to be measured the circuit in fig. 4.23 can be used by just making the dummy gages R_{g2} and R_{g4} active by fixing them at the bottom side of the cantilever beam just beneath the R_{g1} & R_{g3} gages. They will be subjected to compression. The gages R_{g1} & R_{g3} fixed at the top side will be subjected to tension as shown in fig. 4.24. Here the output is

$$E_o = \frac{4 R_{g1} R_{g3}}{(R_{g1} + R_{g3})^2} E_i F \epsilon \quad (4.48)$$

When $R_{g1} = R_{g2} = R_{g3} = R_{g4} = R_g$

$$E_o = R_g E_i F \epsilon \quad (4.49)$$

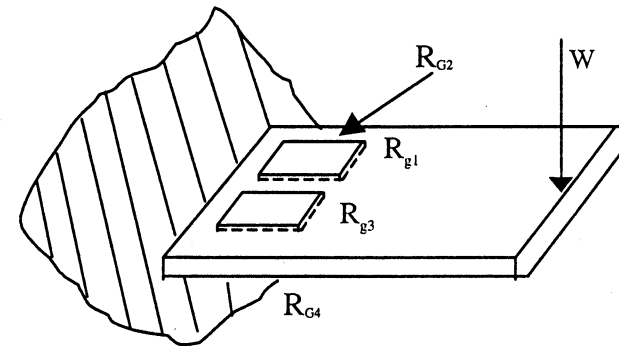


Fig. 4.24

Strain gage arrangement for bending stress

All the strain gage bridge output voltages considered earlier are open circuit voltages. When a load is connected across the terminals, the output voltage is calculated using Thevenin's network theorem as shown below fig. 4.25.

$$Z_o = \frac{R_1 R_2}{R_1 + R_2} + \frac{R_3 R_4}{R_3 + R_4} \quad (4.50)$$

When $R_1 = R_2 = R_3 = R_4 = R \quad \therefore \quad Z_o = R$

$$E_L = \frac{E_o R_L}{Z_o + R_L}$$

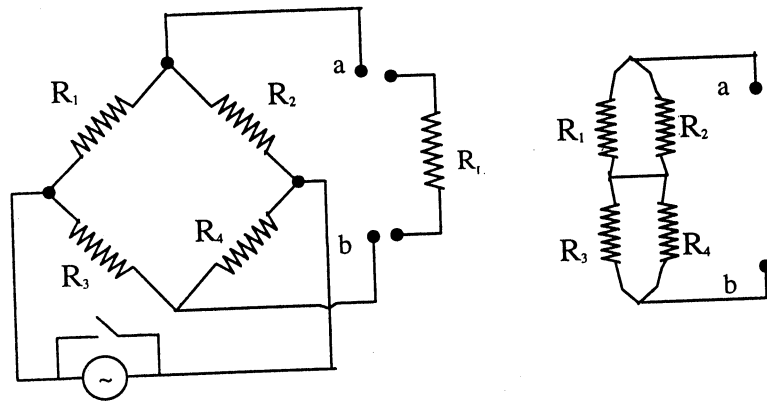


Fig. 4.25

Fig. 4.26 gives the equivalent circuit of the bridge when loaded.

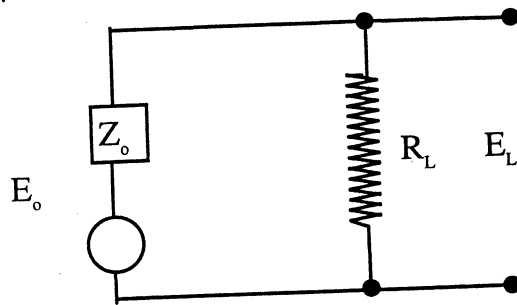


Fig. 4.26

Thevenin's equivalent circuit for strain gage bridges with loading condition.

EXAMPLE 4.3.6.1:

A bridge used for strain measurement is having a gage of 350 ohms nominal resistance in arm 1 and other arms are of 350 ohms resistors. A micro ammeter of coil of resistance 150 ohms is used to measure the output. The gage is having a gage factor of 2.04. Input excitation is 22.5 volts. Find the reading of the meter for a strain of 1200 $\mu\text{m/m}$.

Assume that there is no temperature change.

$$\begin{aligned}
 dE_o &= \frac{R_1 R_2}{(R_1 + R_2)^2} E_i \epsilon & (4.52) \\
 &= \frac{350 \times 350}{(2 \times 350)^2} \times 22.5 \times 2.04 \times 1200 \times 10^{-6} \text{ volts} \\
 &= 13.77 \times 10^{-3} \text{ volts} \\
 Z_o &= (350/2) + (350/2) = 350 \\
 I_L &= (13.77 \times 10^{-3})/500 \\
 &= 27.54 \times 10^{-6} \text{ Amps.}
 \end{aligned}$$

4.3.7 Circuit For Calibration

Normally calibration of any measuring system is done by applying an accurately known sample of the variable to be measured and observing the measuring systems response in a test bench. In case of bonded resistance strain gage this is not possible because once the gage is bonded to a strain situation, it cannot be transferred to the test bench for calibration. So some other method is to be thought of.

Resistance strain gages are manufactured under carefully controlled conditions and the gage factor for each group is provided by the manufacturer. Within indicated tolerance normally about $\pm 0.2\%$, the nominal resistance value is also known. Knowing these two, a simple method to calibrate is to introduce a known small resistance change at the gage and determine the systems response. This small resistance change can be obtained by connecting a known shunt resistance as shown in fig. 4.27. Then the equivalent strain which will introduce the same resistance change can be calculated and a calibration chart can be prepared.

For calculating the equivalent strain the following equation can be used. When a shunt resistance of R_s is connected across the strain gage of nominal resistance R_g , the change in resistance in arm 1,

$$\Delta R_g = R_g - \frac{R_g R_s}{R_g + R_s} \quad (4.53)$$

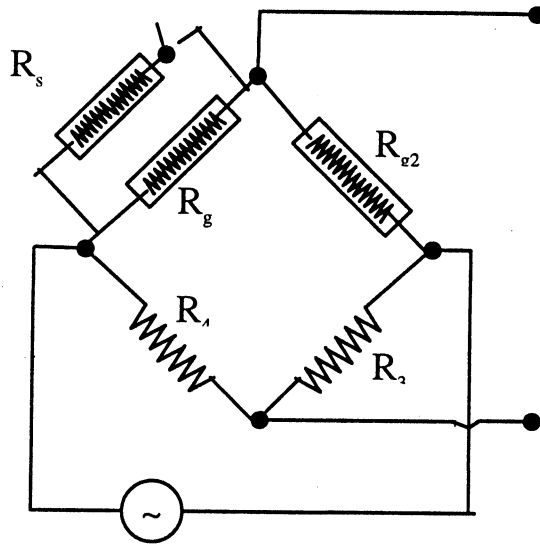


Fig. 4.27
Strain gage circuit for calibration.

$$= \frac{R_g^2}{R_g + R_s}$$

In terms of the change in resistance, the equivalent strain is

$$\begin{aligned} \epsilon &= \frac{1}{F} \frac{\Delta R_g}{R_g} \\ &= \frac{1}{F} \left(\frac{R_g}{R_g + R_s} \right) \end{aligned} \quad (4.54)$$

EXAMPLE 4.3.6.2:

Consider a strain gage of $R_g = 350$ ohms and gage factor of 2.82. Calculate the equivalent strain if a resistance of 100 Kilo ohms is shunted across the strain gage.

$$\text{equivalent strain} = \epsilon_e = \frac{1}{F} \left(\frac{R_g}{R_g + R_s} \right)$$

$$\begin{aligned} &= \frac{1}{2.82} \left(\frac{350}{350 + 100000} \right) = 0.001238 \text{ cm/cm} \\ &= 1238 \times 10^{-6} \text{ cm/cm} = 1236 \mu \text{ strain} \end{aligned}$$

EXAMPLE 4.3.6.3:

In the above example calculate the shunt resistor necessary to provide an equivalent strain ϵ_e of 100×10^{-6} cm/cm.

$$\begin{aligned} \text{equivalent strain} = \epsilon_e &= \frac{1}{F} \left(\frac{R_g}{R_g + R_s} \right) \\ R_g + R_s &= \frac{R_g}{F \epsilon_e} \\ R_s &= \frac{R_g}{F \epsilon_e} - R_g = R_g \left(\frac{1}{F \epsilon_e} - 1 \right) \\ &= 350 \left(\frac{1}{2.82 \times 100 \times 10^{-6}} - 1 \right) \\ &= 350 (3546 - 1) \\ &= 124.0 \text{ K } \Omega \end{aligned}$$

For dynamic calibration, an electrically driven switch or chopper is made use of to make and break the shunt resistance circuit and the response is observed on the CRO.

EXAMPLE 4.3.6.4:

In a wheatstone bridge, leg 1 is an active strain gage of Advance alloy with 120 ohms resistance, leg 4 is a similar dummy gage for temperature compensation, and legs 2 and 3 are fixed 120 ohm resistors. The maximum gage current is to be kept below 0.030 amp.

- What is the maximum permissible d-c bridge excitation voltage? (Use this value in the remaining parts of this problem)
- If the active gage is on a steel member, what is the bridge output voltage for 70 kg/cm^2 of stress?

- c) If temperature compensation were not used, what bridge output would be caused by the active gage by increasing temperature by 56°C if the gage is bonded to steel? What stress value would be represented by this voltage? Thermal-expansion coefficients of steel and Advance alloy are $11.7 \times 10^{-6} \text{ m/m-}^\circ\text{C}$ and $26.82 \times 10^{-6} \text{ m/m-}^\circ\text{C}$ respectively. The temperature coefficient of resistance of Advance is $10.8 \times 10^{-6} \text{ ohm/ohm-}^\circ\text{C}$.
- d) Compute the value of a shunt calibrating resistor that would give the same bridge output as 600 kg/cm^2 stress in a steel member.

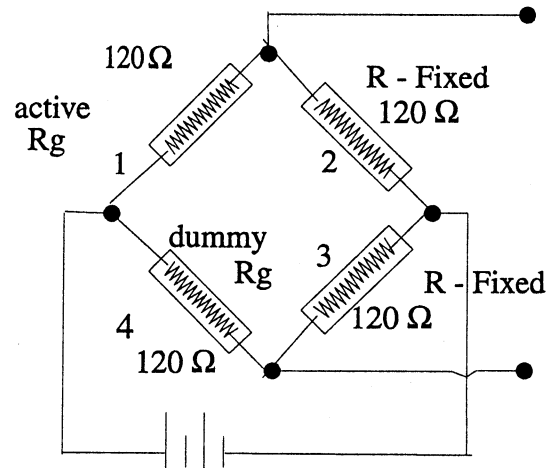


Fig. 4.28

Figure for example 4.3.6.4

- a) The maximum d-c excitation = $0.030 \times 240 = 7.2 \text{ volts}$.
- b) Young's Modulus for mild steel $E = 2.2 \times 10^{11} \text{ NM}^{-2} = 2.24 \times 10^6 \text{ kg/cm}^2$
- The strain for $70 \text{ kg/cm}^2 = 70/2.24 \times 10^6$
- $$= 3.125 \times 10^{-5} \text{ cm/cm}$$
- Gage factor for advance alloy = 2
- G.F. = $(\Delta R/R)/\epsilon$

$$\Delta R/R = 2 \times \epsilon$$

$$\Delta R = 2 \times \epsilon \times R$$

$$= 2 \times 3.125 \times 10^{-5} \times 120$$

$$= 7.524 \times 10^{-3} \text{ ohm}$$

$$dE_g = \frac{ERR_g}{(R + R_g)^2} F\epsilon$$

$$\text{when } R = R_g,$$

$$= 7.2 \times 7.524 \times 10^{-3} / (4 \times 120) \text{ volts}$$

$$= 0.11286 \text{ mv}$$

- c) Change in resistance of Advance due to temperature change.

$$\Delta R = 10.8 \times 10^{-6} \times 120 \times 56$$

$$= 7.2576 \times 10^{-2}$$

because of differential expansion, strain induced in the gage is compressive in nature; Left free Advance alloy would have expanded more than steel for any temperature rise.

$$\begin{aligned} \text{Compressive strain /}^\circ\text{C change} &= (26.82 - 11.7) \times 10^{-6} \text{ cm/cm-}^\circ\text{C} \\ &= 15.12 \times 10^{-6} \text{ cm/cm-}^\circ\text{C} \end{aligned}$$

$$\begin{aligned} \text{Compressive strain for } 56^\circ\text{C change} &= 15.12 \times 10^{-6} \times 56 \text{ cm/cm} \\ &= 8.467 \times 10^{-4} \text{ cm/cm} \end{aligned}$$

$$\text{Corresponding reduction in resistance} = F \epsilon R$$

$$= 2 \times 8.467 \times 10^{-4} \times 120$$

$$= 20.32 \times 10^{-2} \Omega$$

$$\begin{aligned} \text{net change in resistance } \Delta R_T &= (-20.32 + 7.257) \times 10^{-2} \\ &= -13.06 \times 10^{-2} \Omega \end{aligned}$$

$$\text{The output voltage due to } \Delta R_T, dE_o = \Delta R_T / (4R) E$$

$$\begin{aligned} &= (-13.06 \times 10^{-2} \times 7.2) / 4 \times 120 \\ &= 1.96 \text{ mv} \end{aligned}$$

$$\begin{aligned} \text{d) Strain } \epsilon \text{ for } 600 \text{ kg/cm}^2 &= 600 / (2.24 \times 10^6) \\ &= 267.85 \times 10^{-6} \text{ cm/cm} \end{aligned}$$

The corresponding change in resistance

$$\begin{aligned} \Delta R &= F \epsilon R \\ &= 2 \times 267.85 \times 10^{-6} \times 120 \\ &= 6.43 \times 10^{-2} \Omega \end{aligned}$$

Shunt calibrating resistance needed is R_s

$$R_g - \frac{R_g R_s}{R_g + R_s} = \Delta R$$

$$\Delta R = \frac{R_g^2}{R_g + R_s}$$

$$\therefore \Delta R (R_g + R_s) = R_g^2$$

$$R_s = \frac{R_g^2 - \Delta R R_g}{\Delta R} = \frac{R_g^2}{\Delta R} - R_g$$

$$= \frac{120^2}{6.43 \times 10^{-2}} - 120$$

$$= 223.83 \text{ k} \Omega$$

4.3.8 Circuits For Temperature Compensation

The effect of temperature on the measurement of strain using gages has been discussed earlier. Use of dummy gage for temperature compensation as seen earlier, is one method of reducing the error due to temperature variation. Some of the other methods are discussed in this section.

4.3.8.1 Compensation by thermocouple

When a bonded strain gage is used in circuit without temperature compensation, the output voltage changes as a function of temperature which can be expressed as millivolts per degree change in temperature. This change can be altered by changing the excitation voltage. If the bridge is excited with D.C., then the

change in bridge output V_s temperature is very similar to the output of a thermocouple for a temperature change. Therefore a thermocouple can be used to compensate the bridge output due to temperature variation. A typical network having a variable thermocouple compensation is shown in fig. 4.29

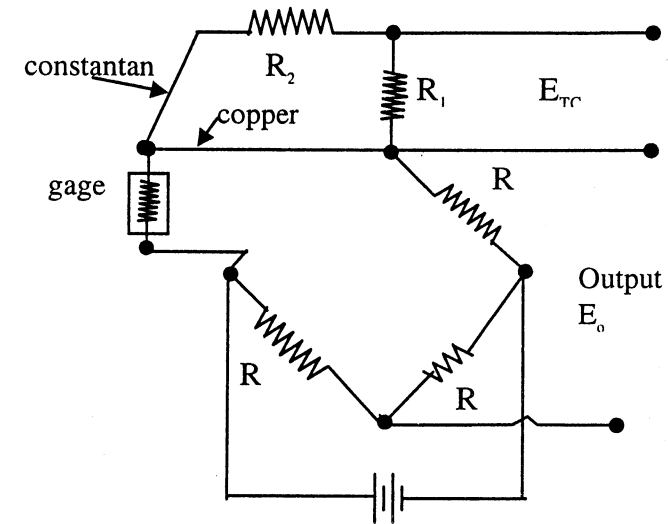


Fig. 4.29

Thermocouple circuit for temperature compensation

R_1 and R_2 are for varying the E_{TC} , the thermocouple output voltage. For compensation the bridge output E_o due to temperature change should be equal and opposite to the thermocouple output E_{TC} , across R_1 for some temperature change. That is why the gage and the thermocouple junction are kept as close as possible. Here the choice of the thermocouple and excitation voltage are dependent on each other.

4.3.8.2 Dual grid strain gage circuit

The gage is made of two elements so that a single gage form two halves of the wheatstone bridge and provide self temperature compensation. Here a thermometer grade platinum wire element is put within the boundaries of a nickel chromium alloy foil grid as shown in fig. 4.30a.

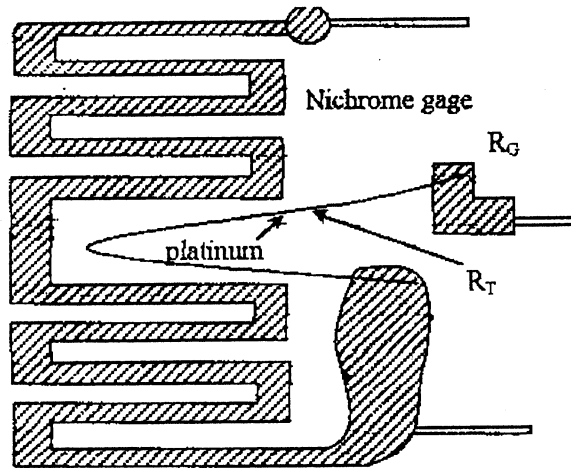


Fig. 4.30a

Dual grid gage for temperature compensation

The resistance R_T of the platinum wire is very small of the order of 3.5 ohms. The circuit for this dual grid gage is given in fig. 4.30b. In this circuit the resistor R_B is an adjustable ballast resistor, connected in series with the platinum element to provide initial balance at different ambient temperatures. The platinum element being so small, the change in resistance due to strain is negligible. But the change in resistance due to temperature change

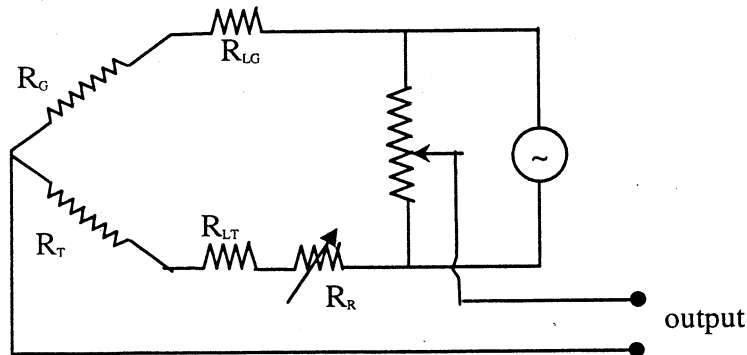


Fig. 4.30b

Circuit for dual grid gage

is equal to that of the gage. Thus this element being an integral part of the gage, provide temperature compensation very effectively. The value of R_B can be calculated as follows:

Let ΔR_g and ΔR_T be the change in resistance of the gage and the platinum for a temperature change. R_{LG} and R_{LT} are the lead wire resistances.

Then the percentage change in these two arms should be equal for compensation.

$$\frac{\Delta R_g}{R_g + R_{LG}} = \frac{\Delta R_T}{R_T + R_{LT} + R_B} \quad (4.55)$$

Let $\Delta R_g / R_g = \Delta G$ and $\Delta R_T / R_T = \Delta T$

Then

$$\frac{\Delta G R_g}{R_g + R_{LG}} = \frac{\Delta T R_T}{R_T + R_{LT} + R_B}$$

$$(R_T + R_{LT} + R_B) \Delta G R_g + \Delta T R_T (R_g + R_{LG})$$

$$R_B = \frac{\Delta T}{\Delta G} \left[R_T \left(1 + \frac{R_{LG}}{R_g} \right) \right] - R_T - R_{LT} \quad (4.56)$$

To compensate the unwanted temperature errors perfectly, the ratio $\Delta T / \Delta G$ should be constant over the entire operational temperature range.

4.3.8.3 Three wire strain gage for temperature compensation of lead wires

Fig. 4.31 is a three wire method for the temperature compensation of lead wires which introduce some error if it is not compensated.

The wires a and b are of same material and are subjected to same temperature change. Therefore any change in 'a' is cancelled by the same change in 'b' due to temperature variation. And the wire 'c' may be of any other material as this introduces a resistance in series with battery and hence no effect in the balance of the bridge. But if an output voltage from the bridge is required then the

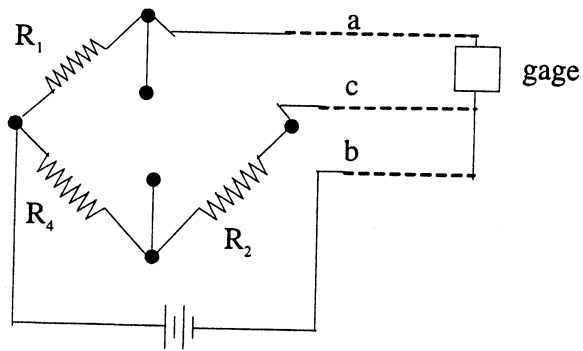


Fig. 4.31

Strain gage lead wire compensation

temperature coefficient of resistance of the wire 'c' should be small so that no error is introduced due to its resistance change which will alter the applied voltage to the bridge.

4.3.9 Mounting of Strain Gages

Mounting of strain gages is an important problem because improperly mounted gage can cause a large error, however much, one is careful in selection of gage and circuitry. For the purpose of mounting, the strain gages may be divided into three groups as follows:

- Ordinary paper backed gage
- The phenolic or bakelite backed gage
- The epoxy film - supported gage.

In general, the mounting area should be cleaned of all aberrations, corrosion, paint and dirt and the test material must be exposed. Emery cloth is a good material to use for this step. After the bare material is prepared, all traces of greasy film must be removed. For this step a suitable solvent may be utilised. Acetone is a satisfactory material. The surface should be swabbed with surgical cotton repeatedly until no trace of soil or oil is found on the cotton. The gage also must be cleaned from any grease or oil. Cleaning the gage requires special care as this may damage the gage itself.

The gage can be mounted slowly and carefully. There is no necessity for hurrying up as several hours are required for the bond to become usable.

4.3.9.1 Paper-backed gages

Paper gages can be fixed with nitrocellulose cement of the ordinary type – 'Duco' cement also can be used. Normally a generous quantity of cement is applied on the surface and the cleaned gage is fixed on it by pressing it gently against the base member. Care should be taken not to puncture the grid. If punctured, the grid will get shorted to the base metal. It is advisable to keep it pressed for some time either by weights or by clamps. If no such method is applicable, atleast pressing it with fingers for ten to fifteen minutes is necessary. The cement should be allowed to harden atleast for eight hours before use.

4.3.9.2 Bakelite baked gages

This type of gages are more difficult to mount as a minimum clamping pressure of 2.5kg/cm^2 is necessary when the curing of the cement takes place. Higher pressures of 6 to 12 kg/cm^2 are recommended for better results. In addition to this the following heating cycle is to be employed.

For 1 hour the bonding area is kept at 45°C , then for two hours at 56°C , then for two hours at 87°C , then it is heated to 97°C to 125°C after removing the clamps. After this the gage is heated several times to the temperature at which it is to be used.

4.3.9.3 Epoxy-resin backing gages

Foil grid gages are formed on very thin epoxy plastic film and these must be mounted with an epoxy cement of the thermosetting type. The procedure is as given below:

- The test surface and the gage are cleaned with trichloroethylene. It is better to use a pair of tongs for handling the gage.
- An epoxy cement is prepared from resin and hardener. Just then apply a generous coating to the test surface with a brush avoiding any air bubbles. The gage is positioned so that the

grid is facing upwards and gently pressed so that excess cement and air bubbles are removed from under the gage.

- A piece of teflon or polystyrene paper or any flat solid material which will not stick to the cement is placed on the gage and .4 to .7 kg/cm² pressure is applied while the cement cures.
- The gage is ready for use after 24 hours. If shorter curing time is required then the temperature may be raised to 79°C and kept for 3 hours.

The maximum temperature for good result is less than 60°C, and the time should be longer than 4 hrs.

4.3.10 Commercial Strain Measuring Systems

For static strain measurements a strain indicator is sufficient. In this type generally the resistance bridge is provided with a fine and coarse adjustments for balance. It is excited with an oscillator. Any unbalance is amplified and fed to a phase-sensitive demodulating circuit which supplies the balancing voltage to the meter. The bridge is balanced for null deflection and the balancing adjustments is calibrated in terms of strain in micro cm/cm.

For dynamic strain measurements, the indicator type is not suitable. A strain gage bridge with a CRO is suitable. Basically this instrument consists of a d-c excited resistance bridge, a chopper, a three stage amplifier and a cathode ray oscilloscope. If the strain is a fast varying one then the chopper is not necessary.

Multi channel strain-measuring systems with an oscillographic recorder is also available.

4.3.11 Gage Arrangements on a Stressed Structure

Fig. 4.32 shows the different arrangements possible on the stressed structure. In all these cases the circuit employed is shown in fig. 4.33.

In fig. 4.32a, the output is low and there is no temperature compensation. In fig. 4.32b, the output is again low and the temperature compensation may be improper because the dummy is also fixed on the straining member in the transverse direction. Fig. 4.32c gives a moderate output. Fig. 4.32d gives the highest output with temperature compensation, as all the four gages are active. The gages on the top surface are subjected to tension and

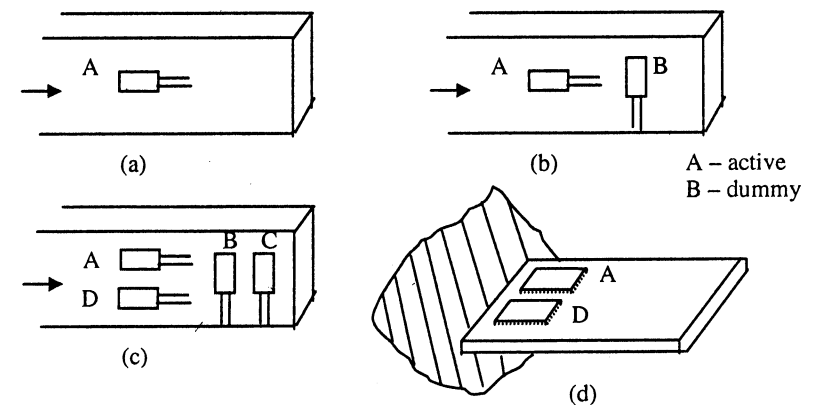


Fig. 4.32

Strain gage arrangements

the gages at the bottom are subjected to compression when the cantilever is loaded as shown in fig 4.32d.

4.3.12 Measurement of Torque

For the measurement of torsional stresses, two gages are fixed at 45° to the axis of the shaft as shown in fig. 4.34. The other two are exactly behind the given gages. All the four gages are active and temperature compensation is also achieved.

4.3.12.1 Stress measurement on rotating members

The real problem in using strain gages for the measurement of stresses in rotating members is to make connection with the rotating members. The common method used is to use slip rings and brushes. There are lots of problems in this method. For example, the wear and tear of brushes, the variation of contact pressure every now and then. All these troubles are not worth taking if there is a temporary method. Normally the set up with strain gages will be used for 10 to 20 minutes and discarded, as the necessary information can be obtained within that time. Of course, there are situations where the experiment set up will be retained for months and use it now and then. Majority of cases the set up is used for a

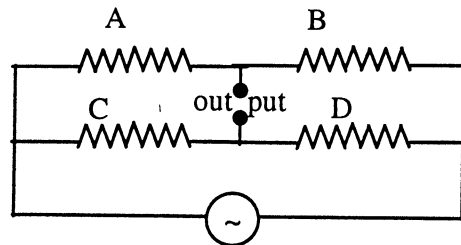


Fig. 4.33

Strain gage circuit

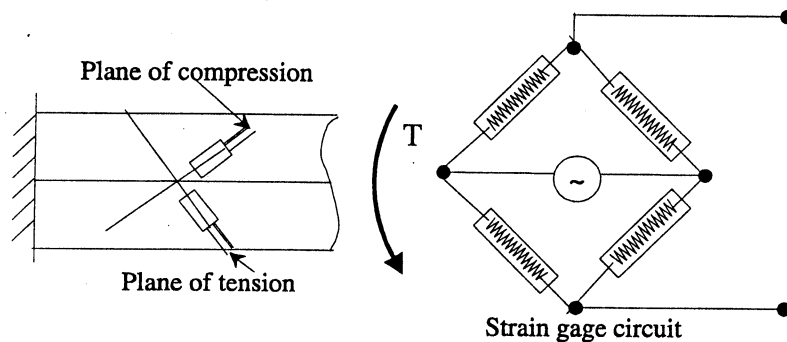


Fig. 4.34

Strain gage arrangement for Torque measurement

short time and dismantled. In such a case, it is advisable to think of simpler methods. If the rotating member is rotating slowly the leads may be allowed to wrap around and stop the rotation at the end of completion of the lead wire. In fact the available time for taking the reading may be doubled for the same lead wire length if it is first wrapped on the shaft in such a way that the shaft rotation causes it to unwrap and then to wrap itself on the shaft. This method is not suitable if the rotating member cannot be stopped quickly enough as the end of the cable is approached. A third method is to provide a fast disconnecting arrangement. This need be no more than soldered connections that can be quickly peeled off. Even though this method is somewhat limited, it is quite workable at slow speeds and avoids many of the inherent problems in the other methods.

A fourth method is to use radio frequency transmitter mounted on the shaft which transmits the strain gage information. A receiver placed nearby picks up the signal. This method is used successfully by many researchers. The equipment used is bulky and the cost is very high.

A fifth method would be to use painted rings. Here an insulating liquid say glyptal is painted on the shaft. Over this silver paint is brushed on, dried and polished to form the slip rings. But this method is suitable for gage installations which are dismantled after the tests for a short time.

4.3.13 Special problems in Strain Gage applications

- **Cross sensitivity:** The majority of the strain-sensitive grids of the bonded type strain gage are normally arranged along the sensitive axis of the gage. But a part of it is aligned transversely. This transverse portion of the grid senses the strain in that direction and this effect is superimposed upon the longitudinal output. This is known as cross sensitivity. The error is small, not exceeding 2 or 3 percent in the worst cases. For engineering applications this error can be tolerated. For some scientific and research activities this error has to be reduced further. Some of the methods used for this purpose are
 - a) The transverse portions of the grid is made much heavier in section than the longitudinal portions. This is easily done in the case of etched foil type gages.
 - b) The grid is made of two wire sizes. The transverse elements are of relatively large diameter wire while the longitudinal strain sensitive elements are of small section.

In both cases, the heavier elements are not appreciably strained because the cement is incapable of supplying the required force and also because the resistance of this heavier part is a small fraction of the total resistance of the gage.

- **Fatigue:** Repeated straining of the gage cause fatigue failure. This failure is mostly due to the fracture at the

discontinuity formed at the junction of the grid wires and the lead wires. The endurance can be improved by using a *dual lead* gages. The construction feature of this type of gage is shown in fig. 4.35.

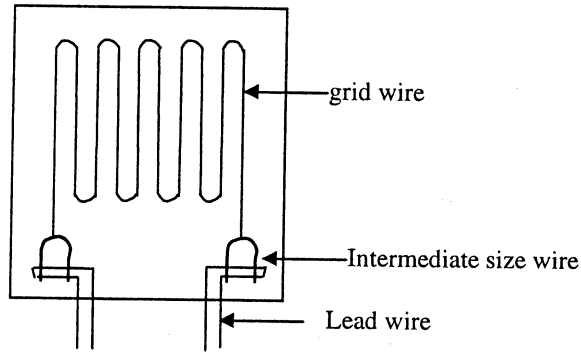


Fig. 4.35

Dual lead gage

A wire of intermediate size between the grid wire size and the lead wire size is interposed between the grid and the lead. This construction markedly increase the fatigue endurance of gage.

4.3.14 Semi-conductor Strain Gages

The gage factor for semi-conductor strain gages is mainly contributed by the piezo resistance effect. For simple tension or compression, when the current through the gage is along the stress axis, the per unit change in resistivity ρ is given by

$$\frac{d\rho}{\rho_0} = \alpha_L \sigma \quad (4.57)$$

where α_L is the longitudinal piezo resistive coefficient and σ is the stress. Then the gage factor is

$$\begin{aligned} \text{G.F.} &= 1 + 2\nu + \alpha_L \frac{\sigma}{\epsilon} \\ &= 1 + 2\nu + \alpha_L E \end{aligned} \quad (4.58)$$

where ν is the poisson's ratio and E the young's modulus of elasticity. The gage factors for semi-conductive materials can be as high as 200 and hence the usual strain gage amplifier is not needed.

Several ceramic type of semi-conductors have been tried out, but silicon is employed as the strain sensitive material mostly. The elastic nature of a single silicon crystal below 550°C is nearly perfect and is essentially free of hysteresis, drift or creep. The magnitude of the piezo resistance effect depends upon the directions of the applied stress and current flow with respect to the crystallographic axes. The strain sensitive element in these gages consist of a filament sliced from a single crystal. Typical dimensions of a semi-conductor strain gage are given in fig. 4.36a.

Semi-conductor gages are highly nonlinear. The gage factor decreases rapidly with increasing temperature due to increase in resistance. But at a given temperature the gage sensitivity is

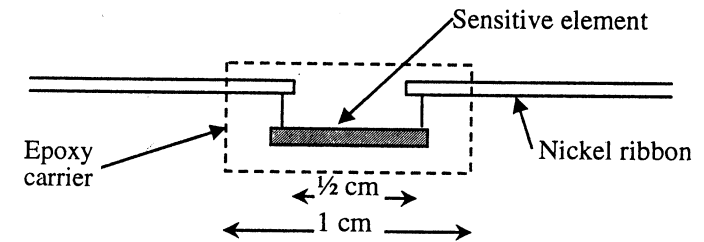


Fig. 4.36a

Semi-conductor strain gage

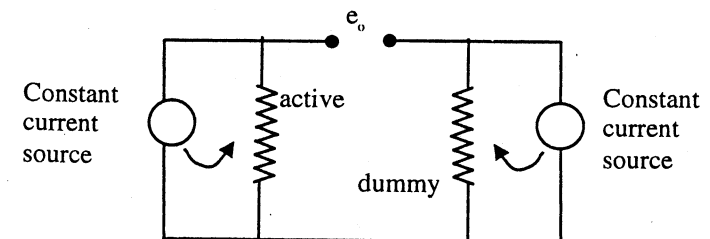


Fig. 4.36b

Constant current bridge circuit of semi-conductor strain gage

constant and better than 0.1 percent. Temperature compensation can be effected by proper choice of resistive elements or resistance thermometers.

The drastic drop in gage sensitivity with increasing temperature can be reduced by using a constant current bridge circuit as shown in fig. 4.36b. but the voltage output is nonlinear.

4.4 RESISTANCE THERMOMETERS

4.4.1 Introduction

A resistance thermometer consists of a resistive element which is exposed to the temperature to be measured.

The electrical resistance of various materials changes in a reproducible manner with temperature, thus forming the basis of a temperature sensing method.

Materials normally used are broadly classified as

- Conductors
- Semiconductors.

If the conductors or metals are used to measure the temperature, they are known as resistance thermometers and if semiconductors are used then they are known as thermistors.

4.4.2 Resistance Thermometers

Resistance thermometers use conductive elements like nickel and copper or tungsten and nickel/iron alloys. The variation of resistance R with temperature T for most metallic materials can be represented by an equation of the form

$$R_T = R_0 (1 + a_1 T + a_2 T^2 + \dots + a_n T^n) \quad (4.59)$$

where R_0 is the resistance at $T = 0^\circ\text{C}$.

The change in resistance for different metals are given in the form of a graph in fig. 4.37.

For engineering purposes and also if range of variation of temperature is narrow then

$$R_t = R_0 (1 + \alpha (t - t_0)) \quad (4.60)$$

$$R_t = R_0 (1 + \alpha \Delta t) \quad (4.61)$$

where α is the temperature coefficient at t_0 and R_0 is the resistance at t_0 .

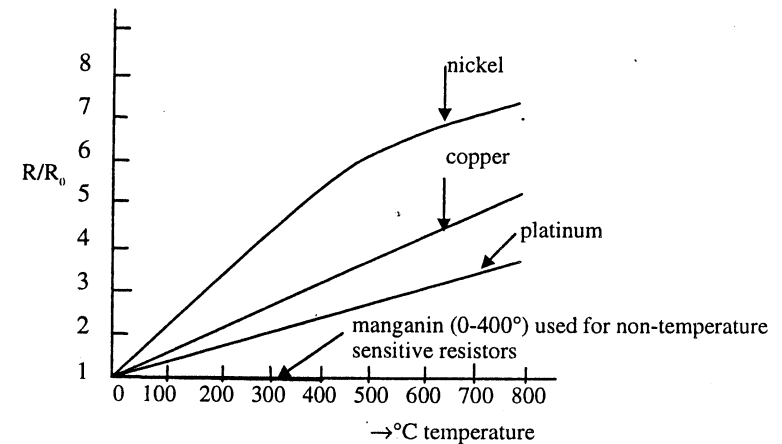


Fig. 4.37

Variation of resistance with temperature for different metals

4.4.3 Construction Details of Resistance Thermometer

Resistance thermometers are constructed using platinum wire or copper or copper alloys. These wires are wound on an aluminium support which is suitably insulated by paint or ceramic coating. Aluminium is used to provide the necessary strength and flexibility for the support.

The wire is wound like a double helical coil on the support to reduce the inductance effect. The entire assembly is housed in a metallic tube with suitable bushings. The terminals are brought out on the top of the pipe. The outer metallic tube acts as a good thermal conductor and also it provides the necessary mechanical strength. Fig. 4.38 gives the details of a platinum RTD.

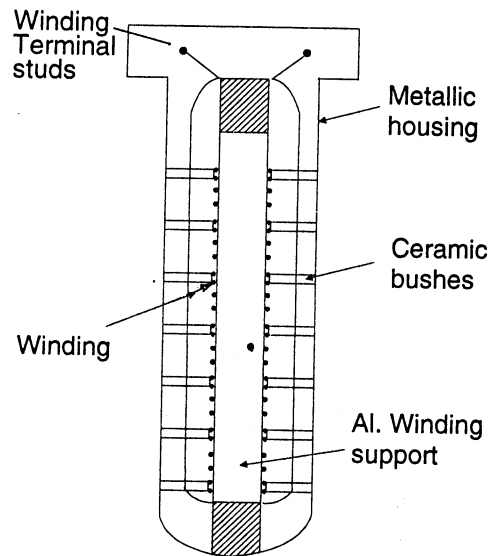


Fig. 4.38

Construction details of a platinum RTD

4.4.4 Wafer Type Resistance Bulb for Surface Temperature Measurement

The surface temperature measurement needs some kind of special construction for the thermometer. For this application wafer type resistance bulb is used, which is shown in fig. 4.39.

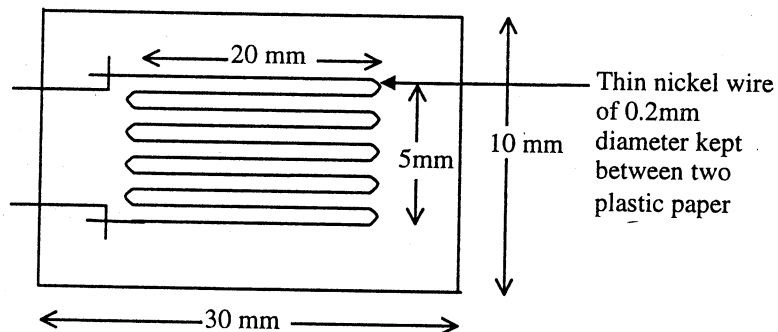


Fig. 4.39

Wafer type resistance bulb

For large temperature ranges the resistance variation follows the form

$$R_t = R_0 (1 + \alpha \Delta t + \beta \Delta t^2) \quad (4.62)$$

where α and β are temperature coefficients of the material at temperature t_0 .

4.4.5 RTD Circuits

The variation in resistance is measured and converted into a voltage signal with the help of a bridge circuit - Bridge circuits employ either deflection mode of operation or the null mode. (manually or automatically balanced). Fig. 4.40 is a bridge for null method of measurement.

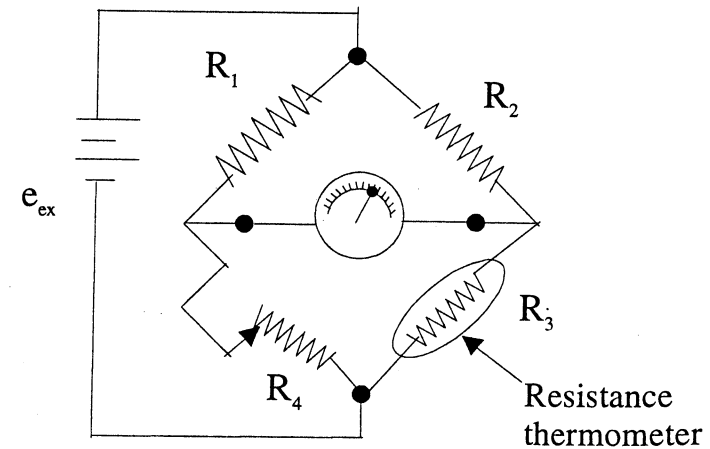


Fig. 4.40

Null balance bridge circuit of resistance thermometer

R_4 is varied until balance is achieved. When better accuracy is required the arrangement shown in fig. 4.41 is preferred.

In this circuit the contact resistance in the adjustable resistor has no influence on the resistance of the bridge legs.

If long lead wires subjected to temperature variations are unavoidable, then three wire resistance thermometers is used with the circuit configuration as shown in fig. 4.42.

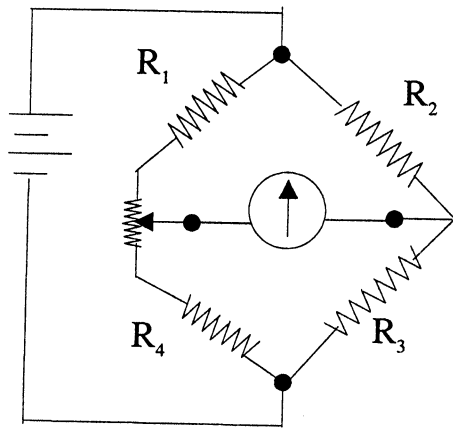


Fig. 4.41

Bridge balance circuit for better accuracy

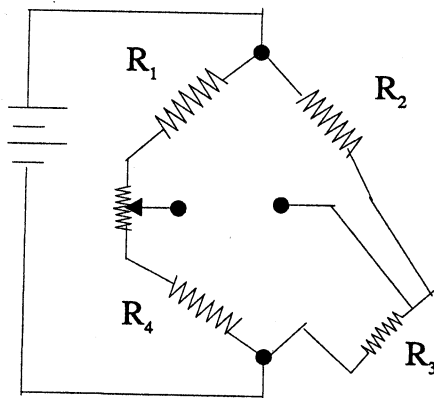


Fig. 4.42

Three wire resistance thermometer circuit

To get a fairly linear relationship between the output voltage and the temperature, the values of R_1 and R_2 of the above circuits are made at least 10 times greater than that of the thermometer.

4.4.6 Self Heating Error

Resistance thermometer bridges may be excited with either d.c. or a.c. The direct or rms alternating current through the thermometer is usually in the range of 2 to 20 mA. This current causes an I^2R heating which raises the temperature of the thermometer above its surrounding, causing the so called self-heating error. The magnitude of this error depends also on heat transfer conditions and is usually small.

A 450 ohm platinum element of open construction carrying 25 mA current has a self heating error of 0.1°C .

To reduce the error due to self heating and at the same time get higher output voltage we can excite the bridge with a pulse whose rms value is small compared with its peak value.

Such pulse excitation voltage can be obtained by commutating a d.c source as shown in fig. 4.43a and this also allows time-sharing of the bridge among several other resistance sensors.

As much as 5 volts full scale bridge output signal can be obtained from resistance sources used this way.

Resistance thermometer elements range in resistance from about 10 ohms to as high as 25000 ohms.

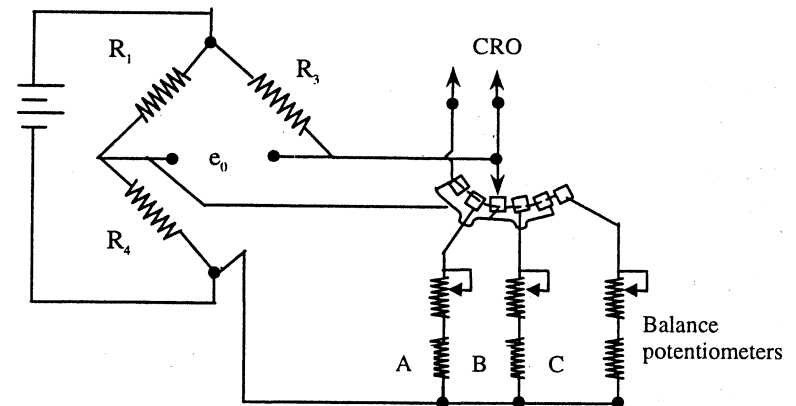


Fig. 4.43a

Pulse excitation circuits for RTD

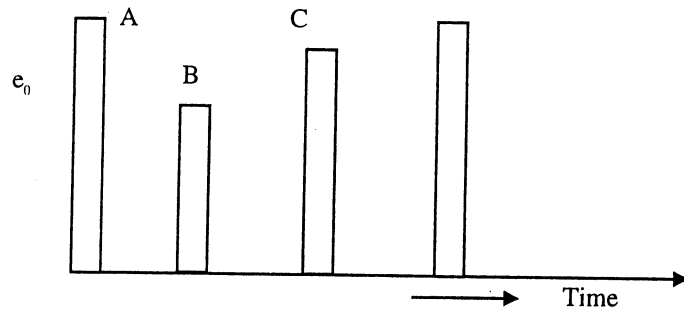


Fig. 4.43b

Pulse excitation for RTD

The choice of elements for the resistance thermometer depends on the temperature range of operation. Table 1 gives the temperature range for different elements.

Table 1:

Platinum	From 250 to 1000°C
Copper	From 160 to 260°C
Nickel	From 160 to 400°C
Tungston	From 250 to 1100°C

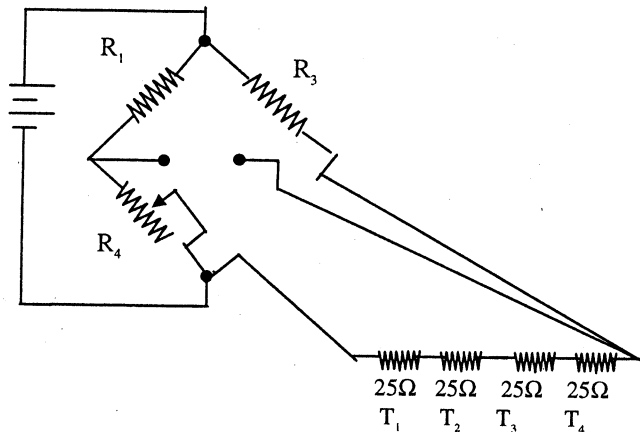


Fig. 4.44

Average temperature measurement circuit

To measure average temperature we may use the circuit shown in fig. 4.44.

To measure differential temperature, circuit in fig. 4.45 is suitable.

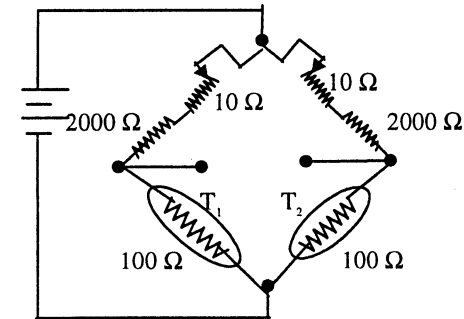


Fig. 4.45

Differential temperature measuring circuits

4.4.7 Advantages of wire resistance thermometers

- They are suitable for measuring large temperature differences and high temperatures.
- They are very accurate which makes them suitable for small temperature measurement.
- Well designed resistance thermometers have excellent stability.
- Unlike thermocouples, they do not need a reference junction and this favors them in many aerospace and some industrial applications.

4.4.7.1 Disadvantages

- Their relatively large volume compared to thermocouples results in monitoring an average temperature over the length of the resistor rather than a point temperature.
- They need auxiliary apparatus and power supply.
- The resistance element is usually more expensive than a thermocouple.

- There are errors due to self heating and thermoelectric effect of the resistive element and connecting leads (dissimilar metal junctions).

4.4.8 Semiconducting Resistance Temperature Transducers (Thermistors)

Thermistors are thermal resistors with a high negative temperature coefficient of resistance.

They are made of manganese, nickel, copper, iron, uranium and cobalt oxides which were milled, mixed in proper proportions with binders pressed into the desired shape and sintered.

Standard forms now available are leads, discs, probes washers and rods as shown in fig. 4.46.

The resistivity may be from 10^{-1} to 10^9 ohm - cm.

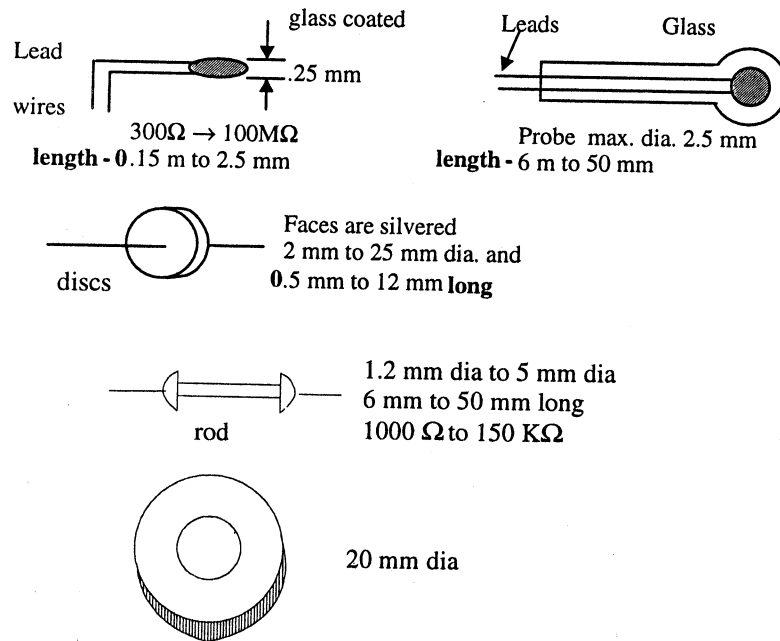


Fig. 4.46
Thermistors

Thermistors have a large negative temperature coefficient and it is highly nonlinear. The resistance at different temperature can be found out using the following equation,

$$R_T = R_0 e^{\beta(1/T - 1/T_0)} \quad (4.63)$$

where R_T = resistance at temperature T .

R_0 = resistance at temperature T_0

β = const. characteristic of material

e = base of natural log,

and T, T_0 = absolute temperature °K

The value of β for the semiconductor made of the above materials is 4000.

The temperature coefficient α for thermistor is expressed as

$$\alpha = \frac{1}{R_T} \frac{dR_T}{dT} \bigg|_{T=T_0} \quad (4.64)$$

$$R_T = R_0 e^{\beta(1/T - 1/T_0)}$$

$$\frac{dR_T}{dT} = -R_0 \beta e^{\beta(1/T - 1/T_0)} \frac{1}{T^2}$$

$$\frac{1}{R_T} \frac{dR_T}{dT} \bigg|_{T=T_0} = -\frac{\beta}{T_0^2} \quad (4.65)$$

when $T_0 = 25^\circ\text{C} = 298^\circ\text{K}$

$$\alpha = -(4000/298^2) = -0.045$$

The resistivity versus temperature graphs are shown in fig. 4.47.

The voltage to current characteristics of thermistors is as shown in fig. 4.48.

The current through the semiconductor element is time dependent for a constant voltage as the resistance varies due to self heating as shown in fig. 4.49.

Due to self heating the resistance decreases and the current increases. As the current is more the heating is also more and hence

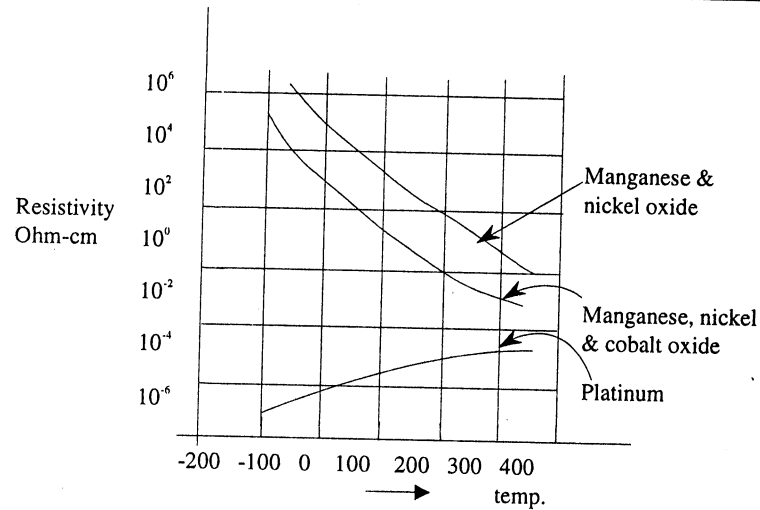


Fig. 4.47

Resistive curves for thermistors

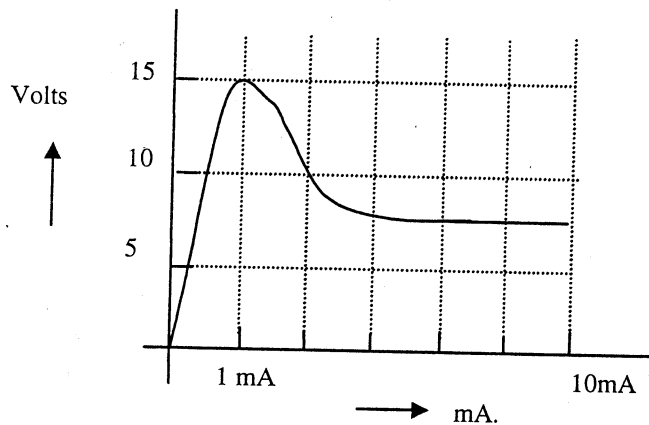


Fig. 4.48

V-I characteristics of thermistors

resistance will decrease. Some kind of chain action takes place here. This process will continue until the thermistor reaches the maximum temperature possible for the amount of power available at which time a steady state will exist.

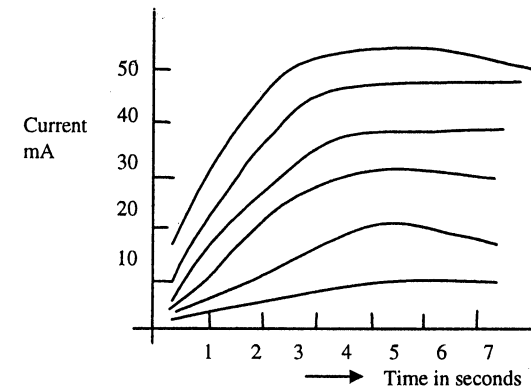


Fig. 4.49

Current variation due to self heating in thermistor

Fig. 4.49 shows a typical current time characteristic curves for a semiconductor material. The thermal dissipation constant for typical thermistor ranges from $0.1 \text{ mW}/^\circ\text{C}$ for glass covered beads to $7 \text{ mW}/^\circ\text{C}$ for relatively large discs. All are measured in still air. Other semiconductor temperature sensors include carbon resistors, silicon and germanium devices.

Carbon resistors are merely the commercial carbon - composition elements commonly used as resistance elements in electronic circuitry. The normal power rating is from 0.1 to 1 watt and the resistance value varies from 2 to 150 ohm. They are also used for cryogenic temperature measurements in the range 1 to 20°K . From about 20°K downward these elements exhibit a large increase in resistance with decrease in temperature given by the relation

$$\log_{10} R = \frac{K}{\log_{10} T} = A + \frac{B}{T} \quad (4.66)$$

R is the resistance T is the temperature in degree kelvin and A , B and K are constants determined by calibration of the individual resistors. Reproducibility of the order of 0.2% is obtained in the range of 0 to 20°C .

Silicon with boron impurities can be designed to have either a positive or negative temperature coefficient over a particular temperature range. A typical element shows from the normal value at 25°C, a change of 80% at -150°C to +180% at 200°C.

Germanium doped with arsenic and gallium is used for cryogenic temperatures where it exhibits a large decrease in resistance with increase in temperature.

4.4.9 Circuitry Employed for Thermistors

The circuit normally used is a wheatstone bridge with thermistor in one arm. Individual calibration is necessary for each circuit, employing thermistor.

A 3 ohm resistance is connected in parallel with the thermistor as shown in fig. 4.50, to have linearised relationship as shown in fig. 4.51.

4.4.10 Advantages of Thermistors

- Semiconductor resistance thermometers have extremely high temperature sensitivity (10 times higher than that of metallic resistance thermometer).
- It can be manufactured in almost any size or shape.
- If fast response time is desired flake thermistors can be made with thermal relaxation time in the order of a few milli seconds.
- Since resistance at normal operating temperature is sufficiently high to make lead resistances appear negligible, thermistors can be used for remote indications.
- Stability at normal temperature is good for they hold their original calibration for long periods of time.

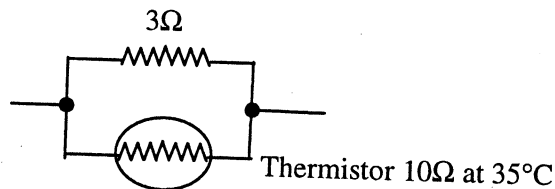


Fig. 4.50

Linearising circuit for thermistor

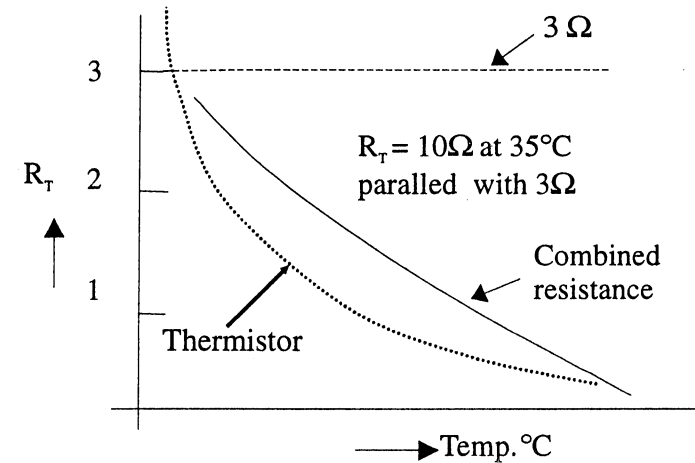


Fig. 4.51

Characteristics of linearising circuit

4.4.10.1 Disadvantages

- It is highly nonlinear.
- Its low temperature limit is set by insensitivity.
- Its upper limit is set by instability.

Thermistors are particularly useful for narrow ranges of measurements at extreme temperature such as those of liquid Oxygen which normally vary from -170°C to -185°C. They are also useful in fuel and air-conditioning temperature measurements.

4.4.11.1 Example

A resistance thermometer circuit shown in fig. 4.52 has $R_1 = R_2 = 10000$ ohms and is to cover the temperature range 0 to 400°C. The thermometer element is platinum with a resistance of 1000 ohms at 200°C. Plot the curve of bridge output versus temperature if the bridge excitation is 20 volts. Temperature coefficient of resistance of platinum is 0.0036 at temperature T_0

$$e_o = \frac{20}{11000} \times 1000 - \frac{20 \times 1000 (1 + \alpha (T - 200))}{10000 + 1000 (1 + \alpha (T - 200))}$$

$$= \frac{20000 [(11000 + 1000 \alpha (T - 200) - 11000 (1 + \alpha (T - 200)))]}{11000 (11000 + 1000 \alpha (T - 200))}$$

$$e_o = \frac{-20000 [10000 \alpha (T - 200)]}{11000 [11000 + 1000 \alpha (T - 200)]}$$

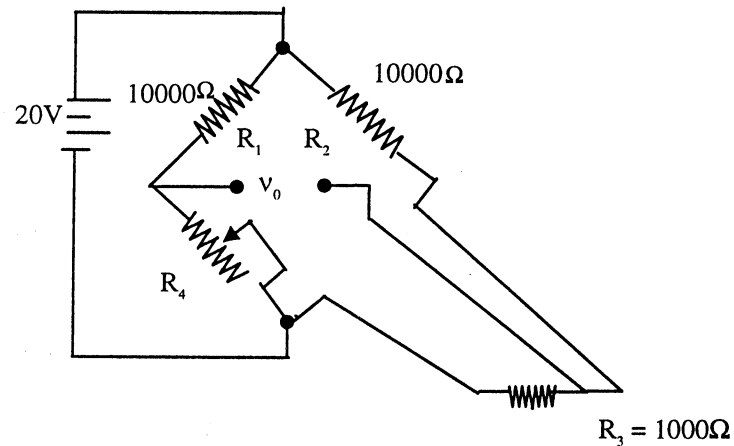


Fig. 4.52

For the example

neglecting $1000\alpha (T-200)$ compared to 11000, we get

$$e_o \approx \frac{-200 \alpha (T - 200)}{11 \times 11} = \frac{-0.72 (T - 200)}{121}$$

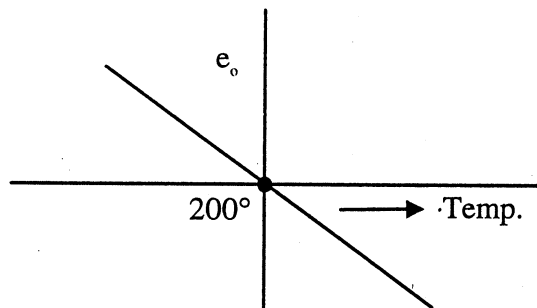


Fig. 4.53

Bridge output voltage for temperature

4.4.11.2 Example

A pulse excited resistance thermometer has an excitation voltage in the form of a rectangular pulse of 100 volts height and 0.1 sec duration. The pulse is on for 0.1 sec and off for 0.9 sec in a repetitive cycle. Compute the ratio of peak/rms voltage for this pulse. What average heating power would this voltage pulse produce in a 500 ohms resistor? If the power is continuously on, what is the heating power?

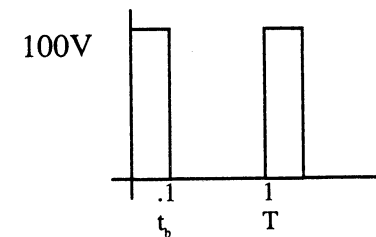


Fig. 4.54

For example

Peak value = 100V

$$\text{RMS value} = \sqrt{(100^2 \times 0.1)/1} = \sqrt{10^3} = \sqrt{1000} = 31.6$$

Ratio of peak/rms = $100/31.6 = 3.16$

Average heating power for the pulse

$$\begin{aligned} &= (V^2/R) (t_b/T) \\ &= (100^2/500)(.1/1) = 2 \text{ watts} \end{aligned}$$

heating power for continuous

$$\text{supply} = 100^2/500 = 20 \text{ watts.}$$

4.5 HOT WIRE ANEMOMETER

4.5.1 Introduction

Another resistance variation type transducers is hot wire anemometer. In general anemometers are devices used for measurement of velocity of flow. In hot wire anemometers, resistive wire is used as a basic sensor, which is heated initially by

passing an electric current. This heated resistive wire mounted on a probe is exposed to air/flow or wind, which is cooled because of fanning effect. The amount of cooling depends on the velocity of air flow. The resistance of the probe when it is hot is different from that when it is cooled. This difference in resistance, or this variation in resistance is converted into a voltage variation and thereby the flow velocity is transduced into a voltage variation. Broadly hot wire anemometers can be classified into two categories namely (i) constant current type and (ii) constant temperature type.

4.5.2 Constant Current Type (CCT)

In this type a fine resistance wire carrying a fixed current is exposed to the flow velocity.

The wire attains an equilibrium temperature when the $I^2 R$ heat generated in it is just balanced by the convective heat loss from

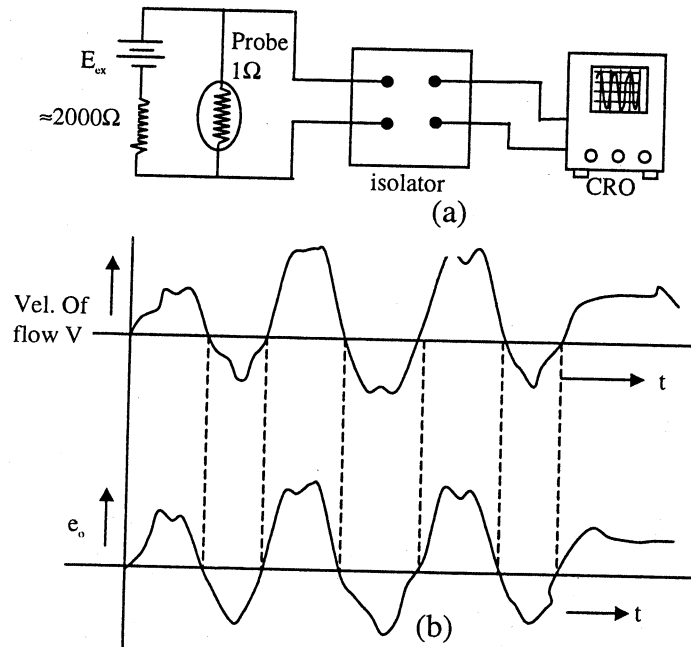


Fig. 4.55

Schematic of a constant current type anemometer with its input and output waveforms

its surface. Here the circuit is designed so that $I^2 R$ heat generated in it is just balanced so that $I^2 R$ heat is essentially constant. Therefore the wire temperature must adjust itself to change the convective heat until equilibrium is reached. This wire temperature is a measure of the flow velocity, which can be measured in terms of its electrical resistance. Fluctuating flow with a frequency of 160 Hz and above cannot be measured using this. A schematic diagram of the constant current type anemometer is shown in fig. 4.55.

4.5.3 Constant Temperature Type (CTT)

In constant temperature form, the current through the wire is adjusted to keep the wire temperature constant. The current required to do this is a measure of the velocity. For measurement of average velocity the constant temperature mode of operation is often used.

As far as the steady flow is concerned a set up shown in fig. 4.56 is alright. But when there is large fluctuation in the velocity of flow, this circuit cannot be used.

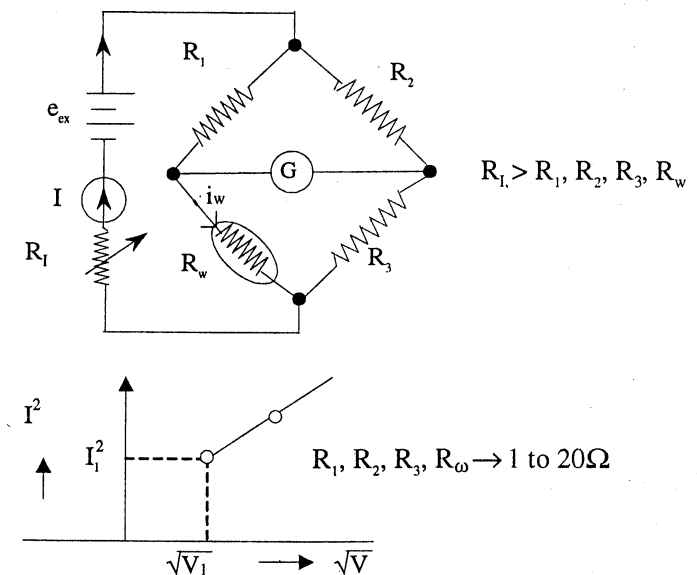


Fig. 4.56

Circuit for constant temperature type anemometer

The constant temperature type of instrument for measuring steady velocities can be extended to measure both average and fluctuating components of velocity by making bridge balancing operation automatic rather than manual through the agency of feedback as shown in fig. 4.57. With zero flow velocity, the bridge excitation is shut off ($i_w=0$), the hot wire assumes the fluid temperature.

The variable resistor R_3 is then manually adjusted so that $R_3 > R_w$, thereby unbalancing the bridge. When the excitation current is turned on, the unbalanced bridge produces an unbalance voltage e_e which is applied to the input of a high gain current amplifier supplying the bridge excitation current.

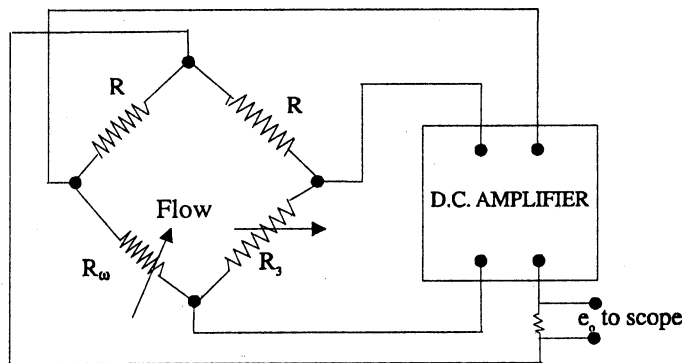


Fig. 4.57

Constant temperature type circuit for fluctuating flow

The current now flowing through R_w increases its temperature thus its resistance. As R_w increases, it approaches R_3 and the bridge unbalance voltage e_e decreases. Since an actual amplifier has limited gain the bridge unbalance voltage cannot become zero.

A typical instrument has flat (within 3 db) frequency response up to 17,000 cps when the average flow velocity is 9 m/s, 30,000 cps for 30 m/s and 50,000 cps at 90 m/s.

4.6 MEASUREMENT OF DIRECTION OF AVERAGE FLOW

A single wire responds essentially to the component of velocity perpendicular to it if the angle between the wire and velocity vector is between 90° and 25° . For this range V is to be replaced by $V \sin \theta$.

For $\theta < 25^\circ$ the heat loss is greater than predicted by $V \sin \theta$ for $\theta = 0$, it is about 55% of that for $\theta = 90^\circ$.

When the probe is mounted on a rotatable base the flow direction angle could be found by determining the probe rotation angle which gives a maximum value of flow velocity. This method is quite inaccurate as θ changes very slowly near 90° . A better procedure (if the flow angle is roughly known), is as follows. The wire is set at about 50° from the flow direction and the current I is measured. The probe is then rotated in the opposite direction until an angle is found at which the same I as before is measured. The bisector angle between the two locations determines the flow direction. This method is more accurate since the rate of change of I with θ is a maximum near 50° .

Fig. 4.58 illustrates a special construction of a rotatable probe. The angle between the two wires is 90° . These two wires are

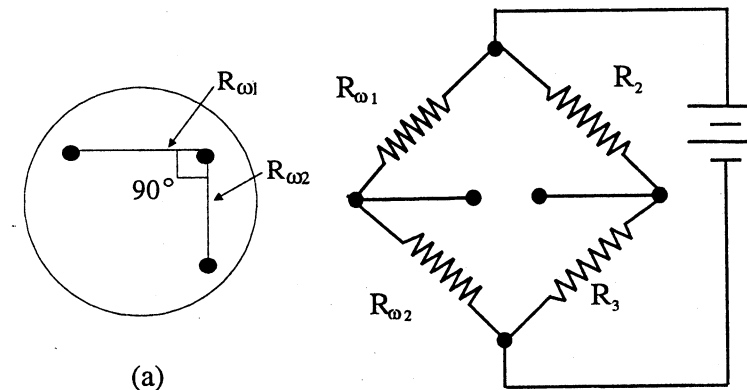


Fig. 4.58

connected in adjacent arms, as shown in fig. 4.57b. The probe is rotated until the bridge is balanced. The bisector of these two wires at the null of the bridge is the direction of flow.

Comparative features of constant current and constant temperature anemometers:

- In constant current type, the current must be set high enough to heat the wire considerably above the fluid temperature for a given average velocity. If the flow should suddenly drop to a much lower velocity or come to rest the hot wire will burn out since the convection loss cannot match the heat generation before the wire temperature reaches the melting point. This drawback is not there in constant temperature type as the temperature of wire is to be kept constant irrespective of the flow variations.
- If the average value of flow velocity changes, the current must be reset in constant current type (CCT) whereas this problem is not there in constant temperature type (CTT) anemometer.
- The error in C.C.T. is more if the fluctuation is more than 5% whereas the fluctuation can be of any size in CTT.
- As the constant temperature type uses a D.C. amplifier, it is usable down to zero frequency (steady velocity) whereas constant current type is good only to about 1 Hz.
- In the C.T.T, the noise level is more and there is difficulty in designing sufficiently high gain d.c. amplifiers without causing instability and drift problem.

So far in the analysis the fluid density ρ was assumed constant. But if P varies, the anemometer actually measures the product ρv i.e. the local mass flow rate.

$$I^2 \propto \sqrt{(\rho v)}$$

The problem encountered in the application of hotwire anemometers are

- Because of the limited strength of fine wire, larger dirt particles striking wire may actually break it.

- Unless the flow is clean, significant calibration changes can occur in a relatively few minutes of operation due to dirt accumulation on the wire.
- At high speeds, wires may vibrate because of aerodynamic loads and flutter effects.

To overcome all the above practical difficulties, a hot film transducer was thought of. Here the resistance element is a thin film of platinum deposited on a glass base, fig. 4.59. The film takes the place of the hot wire. The basic circuitry is similar to the constant temperature hot wire approach. The film transducers have great mechanical strength and may be used at very high temperature by constructing them with internal cooling water passages.

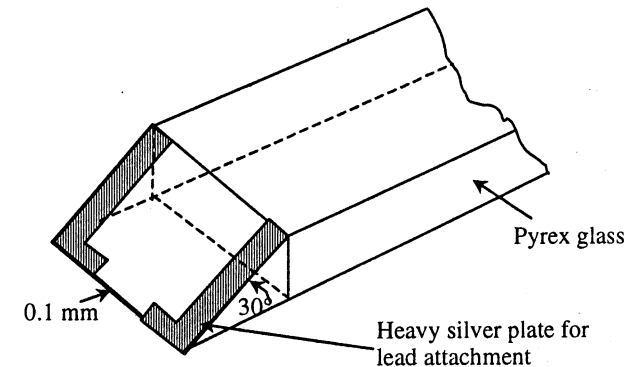


Fig. 4.59

Thin platinum hot film

4.7 OTHER VARIABLE RESISTANCE TRANSDUCER

4.7.1 Contact pressure

The most common transducer based on variable contact pressure between carbon granules is the carbon microphone. The variation in resistance here is due to the variation of the surface area between granules in contact. The main advantages of the carbon microphone is its high output up to 0.5V in ordinary types. Their frequency range is only 30 to 3000 Hz. And their signal noise ratio is low.

As a transducer to measure pressure, we can use thin disks of carbon, graphite or moulded carbonaceous material which are arranged in stacks.

$$R = R_0 + c/p$$

Where R_0 is the 'internal' resistance of the solid disk and c is a constant and p is the pressure on carbon stack with large number of thin disks. R_0 is small in comparison with c/p so that the total resistance R is roughly inversely proportional to the pressure p . The carbon disk material should have a high mechanical strength, low mechanical hysteresis, creep and low temperature coefficient of resistance. In the measurement of pressures above 7 kg/cm^2 , a carbon compound miniature resistor has been used quite successfully.

4.7.2 Humidity Measurement

Lithium Chloride salt in a 3% concentration on poly-vinyl-acetate film was used in the design of an electrolytic hygrometer. This material changes its resistance with humidity change. In order to avoid polarization, alternating current is preferred in the measurement of conductivity in these devices. The conductivity of this material varies to a large extent with temperature. So proper temperature compensation is a must. The materials developed recently as humidity sensing elements are

- Titanium dioxide, a ceramic material.
- A conducting plastics of unspecified composition.

4.7.3 Light

Photo conductive cell is basically a variable resistance transducer. The resistance changes due to photons impingement.

There are basically two varieties

- Bulk photo conductors
- Potentiometer-n junctions or rectifier types.

Photo conductive cells cover the entire spectrum from infra red to ultra violet. Their sensitivity to infra red radiation is of great practical importance.

Lead sulphide can be used up to wave lengths of 3.5 microns. Lead telluride can be used for wave lengths upto 5.8 microns.

Cadmium sulphide can be used up to 0.9 microns.

Selenium sulphide is widely used in indicators of low accuracy, but it is less sensitive to infra red radiation than other materials and there is a longer time lag between illumination and resistance variation.

Semiconductor photo cells have non-linear and temperature sensitive calibration curves and the choice of a suitable type depends very much on the particular job on hand.

4.8 WORKED OUT EXAMPLES

EXAMPLE 4.8.1

- What is the maximum resolution for a 360° single turn wire wound potentiometer with 20V excitation?
- For a 50 turn helical wirewound potentiometer with 20V excitation and 5cm dia, What is the maximum resolution possible?

SOLUTION

- The maximum number of turns per cm for a practical wire wound potentiometer is 150.

For a 360° rotation, the resolution is

$$\frac{360 \times 20}{\pi D \times 150} = 15.3 \text{ volts.}$$

Where D is the diameter in cm of the pot. and $\pi D \times 150$ is the number of turns per 360° movement.

Resolution in terms of degree rotation is

$$\frac{360}{\pi D \times 150} = \frac{.764}{D}$$

- Voltage per 50 turns is 20

Voltage per turn is $\frac{20}{50} \text{ V} = .4 \text{ V}$

$$\begin{aligned} \text{Resolution in terms of voltage} &= \frac{.764}{5} \times 0.4 \\ &= 0.06112 \text{ volts.} \end{aligned}$$

EXAMPLE 4.8.2

A potentiometric transducer with a stroke length of 5 cm, has a total resistance of $10 \text{ K}\Omega$. The voltage across the potentiometer is 7.5 V . When the wiper is 2 cm from one end, what is the output voltage of the potentiometer from that end.

SOLUTION

$$e_o = \frac{2 \times 7.5}{5} = 3 \text{ volts.}$$

EXAMPLE 4.8.3

In an unbonded strain gage circuit shown in fig 4.60, the nominal resistance is 120Ω each. The gage factor is 2.4. Excitation voltage

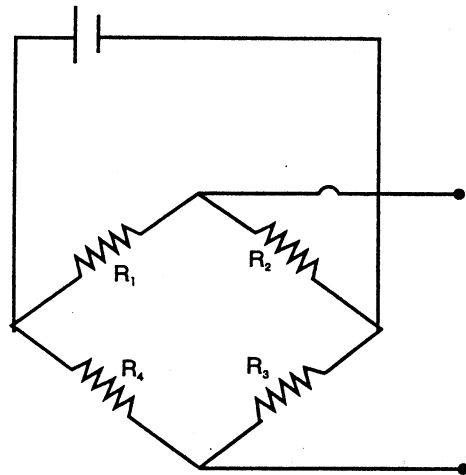


Fig 4.60

Circuit for example 4.8.3.

is 10 V . What is the open circuit output voltage for a tensile strain of 0.001 .

R_1 & R_3 are subjected to tension.

SOLUTION

In an unbonded type of strain gage, when R_1 & R_3 are subjected to tension, R_2 & R_4 will be relieved of a portion of preload tension. Therefore when R_1 & R_3 values increase due to the applied tensile stress, R_2 & R_4 will have a decrease in resistance due to preload release.

The output voltage in a bridge where all the four arms are active is given by eqn. 4.49

$$\begin{aligned} \text{i.e. } e_o &= R_g E_i \cdot F \epsilon \\ &= 120 \times 10 \times 2.4 \times .001 \\ &= 2.88 \text{ V} \end{aligned}$$

EXAMPLE 4.8.4

A platinum thermometer has a resistance of 100Ω at 25° . The temp. coefficient of resistance at 25°C for platinum is $\alpha_{pt} = 0.00392/^\circ \text{C}$

- Find its resistance at 65°C
- If the thermometer has a resistance of 150Ω find the temperature.

SOLUTION

- Resistance at 65°C is

$$\begin{aligned} R_{65} &= R_{25} [1 + \alpha (65 - 25)] \\ &= 100 [1 + 0.00392 \times 40] \\ &= 100 [1 + .1568] \\ &= 115.68 \Omega \end{aligned}$$

- When the resistance is 150Ω , let the temperature be T .

Then $150 = 100 [1 + \alpha (T - 25)]$

$$100 \alpha (T - 25) = 150 - 100 = 50$$

$$100 \times 0.00392 (T - 25) = 50$$

$$T - 25 = \frac{50}{.392} = 127.55$$

$$T = 127.55 + 25 = 152.55^\circ\text{C}$$

EXAMPLE 4.8.5

A thermistor has a resistance temperature coefficient β of $-5\%/^\circ\text{C}$ over a temperature range of 25°C to 50°C . If the resistance of the thermistor is $100\ \Omega$ at 25°C , what is the resistance at 35°C ?

$$R_T = R_0 \left[1 - \frac{\beta}{100} (T - T_0) \right]$$

$$= 100 \left[1 - \frac{5}{100} \times 10 \right]$$

$$= 50\ \Omega$$

EXAMPLE 4.8.6

A thermistor is assumed to have a linear resistance variation with a constant temperature coefficient of resistance of $-0.05\ \Omega/^\circ\text{C}$.

The resistance of the thermistor at 20°C is $1000\ \Omega$

- Calculate the value of its resistance at 25°C
- If this thermistor is connected in series with a copper coil, what is the value of the resistance of the copper coil if the resistance at 20°C and 25°C are the same.

The temperature coefficient of resistance of copper is 0.004°C

SOLUTION

- For a linear range

$$R_T = R_0 [1 - \beta (T - T_0)]$$

$$\therefore R_{25} = R_{20} [1 - 0.05 (25 - 20)]$$

$$= 1000 [1 - 0.05 \times 5]$$

$$= 1000 [.75] = 750\ \Omega$$

b)

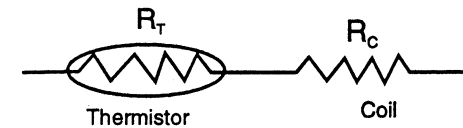


Fig 4.61

It is given that

$$R_T \text{ at } 20^\circ + R_c \text{ at } 20^\circ = R_T \text{ at } 25^\circ + R_c \text{ at } 25^\circ$$

$$1000 + R_c|_{20} = 750 + R_c|_{25}$$

$$R_c|_{25} = R_c|_{20} (1 + \alpha \Delta T) = R_c|_{20} (1 + .004 \times 5)$$

$$= 1.02 R_c|_{20}$$

$$\therefore 1.02 R_c|_{20} - R_c|_{20} = 250 \therefore R_c|_{20} = 12500\ \Omega$$

EXAMPLE 4.8.7

The resistance - temperature relationship is given by

$$R_T = A e^{\frac{b}{T}} \quad (T \text{ in } ^\circ\text{K})$$

Calculate A and b if the resistance of the thermistor at 0°C is $3980\ \Omega$ and $794\ \Omega$ at 50°C

$$3980 = A e^{\frac{b}{273}} \quad (1)$$

$$794 = A e^{\frac{b}{323}} \quad (2)$$

dividing (1) by (2), we get

$$e^{b\left(\frac{1}{273} - \frac{1}{323}\right)} = \frac{3980}{794} = 5.01$$

$$e^{0.005671 b} = 5.01 \quad \therefore b = 2841.53$$

$$A = 3980 e^{-\frac{2841.53}{273}} =$$

EXAMPLE 4.8.8

The resistance variation of a thermistor follows the equation

$$R_T = R_0 e^{\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)}$$

T & T_0 are temp in °K. R_T & R_0 are resistances. β is the characteristic constant of the thermistor material.

It is given that $\beta = 3140$

$$R_{27} = 1050 \Omega.$$

This thermistor is used for temperature measurement and at a particular temperature the resistance is 2330 ohms. What is the temperature?

$$R_T = 2330$$

$$R_{273+27} = 1050$$

$$\text{i. e. } R_{300} = 1050$$

$$\text{Let } \frac{1}{T} - \frac{1}{300} = X$$

$$e^{3140X} = \frac{2330}{1050} = 2.219$$

$$3140X = 0.797$$

$$\therefore X = 2.5382 \times 10^{-4}$$

$$\frac{1}{T} - \frac{1}{300} = 2.5382 \times 10^{-4}$$

$$T = 278.774 \text{ } ^\circ\text{K.}$$

CHAPTER 5

INDUCTANCE TRANSDUCERS

5.1 BASIC PRINCIPLE

Transducers based on the variation of inductance are another group of important devices used in many application. In these transducers self inductance or the mutual inductance of a couple of coils is changed when the quantity to be measured is varied. As a matter of fact inductance, one of the three electrical circuit parameters is related to coil or a pair of coils. Inductance of a coil can be described in terms of number of turns, permeability of the material and the geometric factor. That is

$$L = \mu_0 \mu n^2 G \quad (5.1)$$

where L = inductance of a coil

μ_0 = permeability of air

μ = relative permeability

n = number of turns

G = coil geometry

5.2 LINEAR VARIABLE DIFFERENTIAL TRANSFORMER

The Linear Variable Differential Transformer (LVDT) is the most common mutual inductance element. This can be considered to be a versatile transducer element for most of the electromechanical measuring systems with regards to resolution, hysteresis, dynamic response, temperature characteristics, linearity and life.

LVDT (fig 5.1a) consists of a primary winding, two identical secondary windings and a ferromagnetic movable core. AC carrier

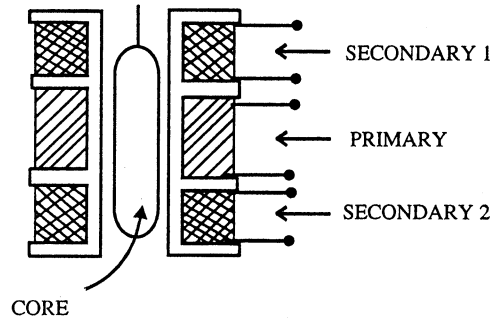


Fig. 5.1a
LVDT cut view

excitation of frequency 60 to 2000 Hz. and magnitude 3 to 15 volts is applied to the primary. The symmetrically spaced, secondary are connected externally in series opposing circuit. Motion of the non-contacting magnetic core varies the mutual inductance of each secondary to the primary which determines the voltage induced from the primary to the secondary. If the core is centered between the secondary windings the voltage induced in each secondary is identical and 180° out of phase, so there is no net output. If the core is moved off the center, the mutual coupling of the primary with one secondary will be greater than with the other and a differential voltage will appear across the secondaries in series. For off center displacements within the range of operation, this voltage is essentially a linear function of displacement. The phase shift between excitation and the output varies with frequency of excitation. The origin of this phase shift can be seen from the following analysis

5.2.1 LVDT Equations

The equivalent circuit of a LVDT is given in fig. 5.1b.

The following equations are in order for the circuit.

$$i_p R_p + L_p \frac{di_p}{dt} - e_{ex} = 0 \quad (5.2)$$

$$i_p = \frac{e_{ex}}{R_p + L_p D} \quad (5.2a)$$

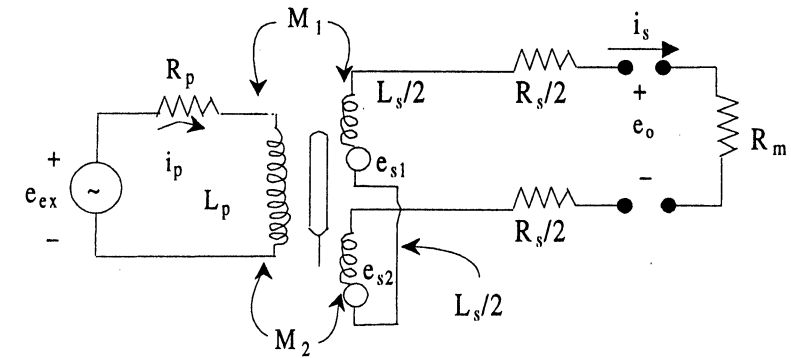


Fig. 5.1b
Equivalent Circuit of LVDT

Where $D \triangleq d/dt$

This symbol $\triangleq$ means by definition that is D is defined by d/dt and there is no history for it.

$$e_{s1} = M_1 \frac{di_p}{dt}, e_{s2} = M_2 \frac{di_p}{dt} \quad (5.3)$$

$$e_s = e_{s1} - e_{s2} = (M_1 - M_2) \frac{di_p}{dt} \quad (5.4)$$

If the output is open circuited for a fixed core position. Using equation (5.2a), the output is

$$e_o = e_s = (M_1 - M_2) \frac{D}{L_p D + R_p} e_{ex} \quad (5.5)$$

$$\frac{e_o}{e_{ex}}(D) = \frac{[(M_1 - M_2)/R_p]}{T_p D + 1} D : T_p \triangleq \frac{L_p}{R_p} \quad (5.6)$$

In terms of frequency response

$$\frac{e_o}{e_{ex}}(j\omega) = \frac{W(M_1 - M_2)/R_p}{\sqrt{(W T_p)^2 + 1}} \angle \Phi \quad (5.7)$$

where $\Phi = 90^\circ - \tan^{-1} W T_p$. So the phase shift between e_o and e_{ex} is a function of the excitation frequency.

5.2.2 LVDT Under Loading Condition

If a voltage measuring device of input resistance R_m is connected to the output terminals, then the situation is different and the equations are

$$\frac{(M_1 - M_2) D}{L_p D + R_p} e_{ex} = (R_s + R_m) i_s + L_s \frac{di_s}{dt} \quad (5.8)$$

$$e_o = i_s R_m \quad (5.9)$$

$$\frac{e_o}{e_{ex}} (D) = \frac{(M_1 - M_2) R_m}{R_p (R_s + R_m)} \frac{D}{(T_p D + 1)(T_s D + 1)} \quad (5.10)$$

where $T_s \triangleq L_s / (R_s + R_m)$ (5.11)

It is obvious from the above expression that the frequency response of $(e_o/e_{ex}) (j\omega)$ has a phase angle of 90° at low frequencies and -90° at high frequencies. Somewhere in between it will be zero. By suitably adjusting the excitation frequency one can achieve a desired phase shift. If for some reason the excitation frequency cannot be adjusted to this value then one of the methods shown in fig. 5.2 can be used to get a desired phase shift.

Now the output voltage at null may not be zero due to harmonics and stray capacitance coupling between the primary and

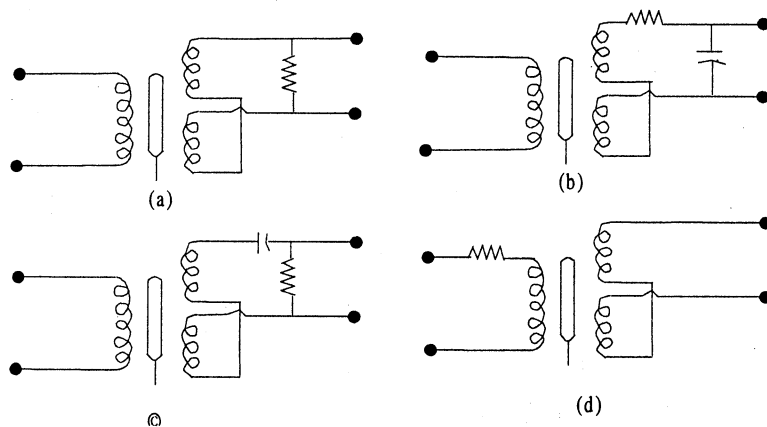


Fig. 5.2

Circuits for phase shift adjustments

secondary. Under normal conditions this is less than 1% of the full scale output voltage and may be quite acceptable. Fig. 5.3 shows some methods for null voltage reduction. The values of R and R_p are not critical but should be as low as possible without loading the excitation source. The core is kept at the middle or centre of travel of the core movement and the R_p wiper is adjusted to get zero or minimum voltage output.

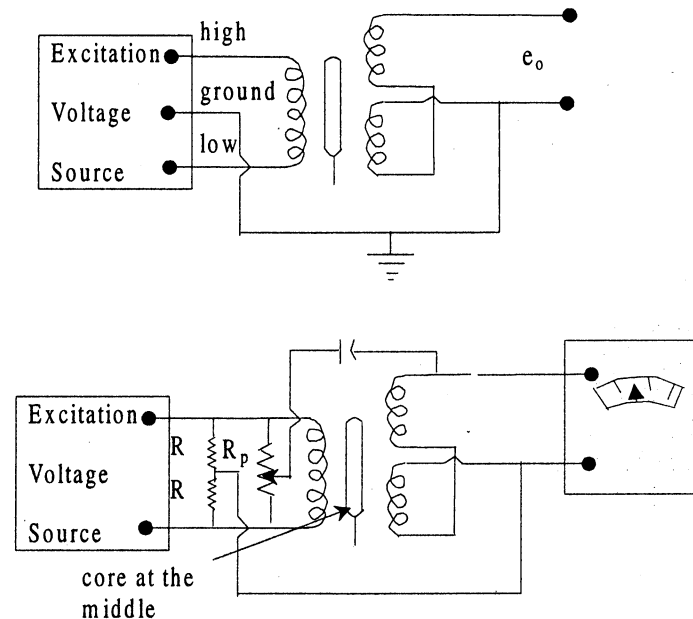


Fig. 5.3

Circuits for zero voltage adjustment at core null position

The output of a differential transformer is a sine wave whose amplitude is proportional to the core motion. An a.c. voltmeter calibrated in linear displacement units is alright for measurement of static or very slowly varying displacements except that the meter will give exactly the same reading for displacements of equal amount on either side. For direction sensitiveness and for rapid core motions, the arrangement shown in fig. 5.4 is suitable.

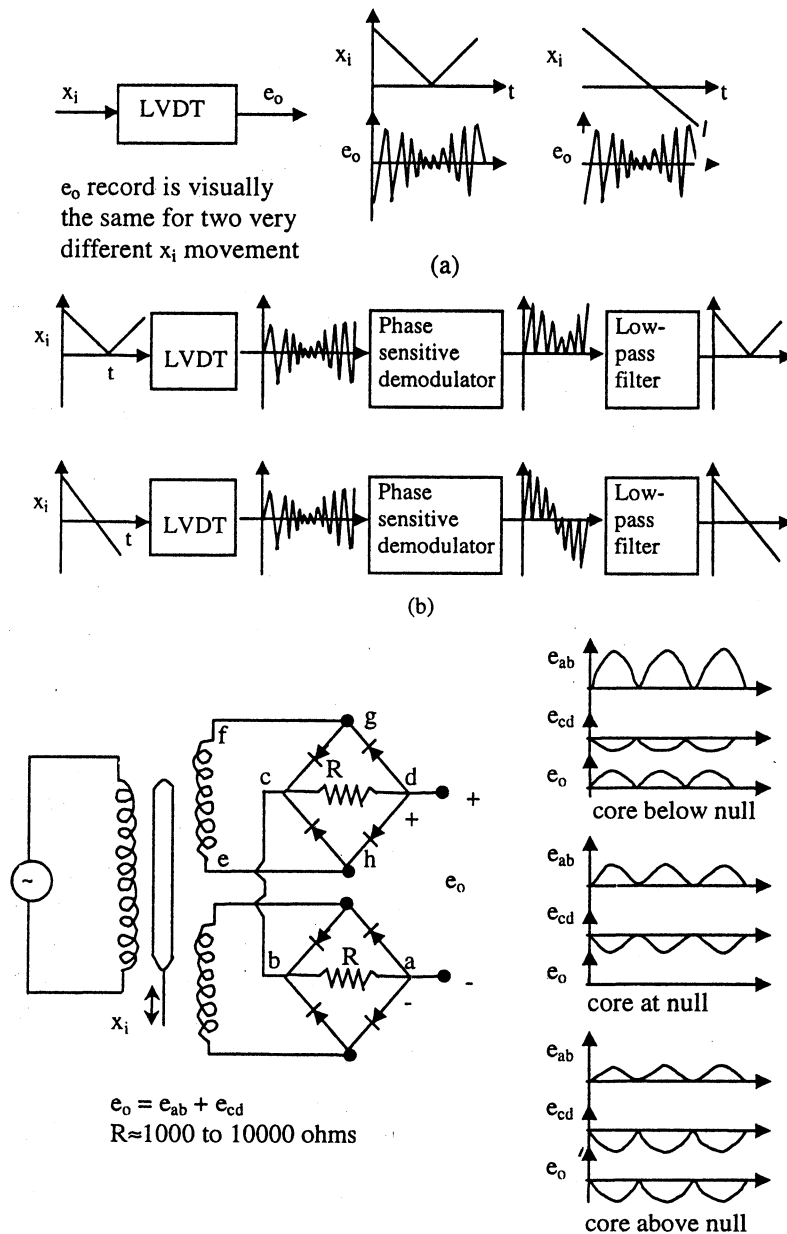


Fig. 5.4
Circuit for phase sensitive demodulation

Fig. 5.4 shows the circuit arrangement for phase sensitive demodulation using semiconductor diodes. If the ratio of the excitation frequency to core motion frequency is 10:1 a simple R_C filter is adequate. The output of this filter then becomes the input to an oscillograph or oscilloscope. For a better response of the LVDT system, a sharp cut off filter may be necessary.

5.2.3 Resonant LVDT

In this type of LVDT, the secondary coils are wound in a manner which markedly increase the distributed capacitance of the coils. The much higher capacity modifies the impedance of the coils and permits partial resonance at a relatively low excitation frequency. The result is exceptionally high output voltage for small core movement.

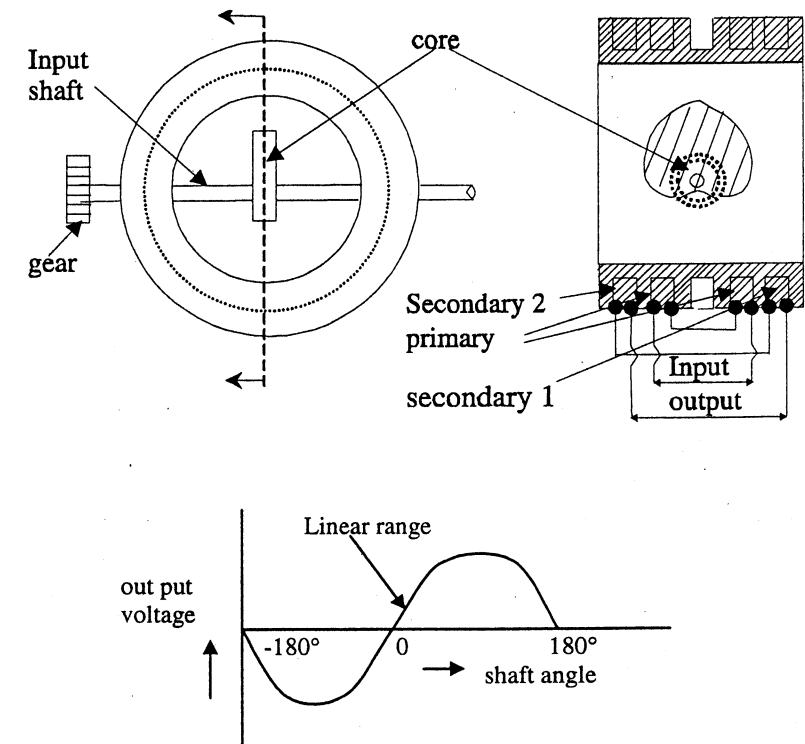


Fig. 5.5
RVDT

5.2.4 Rotary Variable Differential Transformer (RVDT)

The heart of this device is a cordoidal magnetic cam – core pivoted eccentrically on an input shaft across the middle of a coil former. The coil former contains two pairs of identical windings spaced symmetrically above and below the middle. Fig. 5.5 shows the details of this device and its output response characteristics. This device reacts to rotary motion like the LVDT reacts to linear motion. The maximum range of angular position measurement with reasonable linearity is about $\pm 60^\circ$.

5.2.5 DC – LVDT

The concept of a DC - operated transducer element having all the advantages and unique features of the LVDT, has always appealed to meteorologists and instrument users. In practice such a transducer element is achieved by incorporating some form of signal conditioning electronics with a LVDT, fig. 5.6.

The state of the art was dramatically advanced, however, by the development and introduction of the hybrid microcircuit modules. The block diagram of fig. 5.6 shows that the module includes a complete passive demodulator DC – amplifier signal conditioner. DC – LVDTs, incorporating this modular feature, are having high sensitivity low output impedance, excellent reliability and extreme ruggedness.

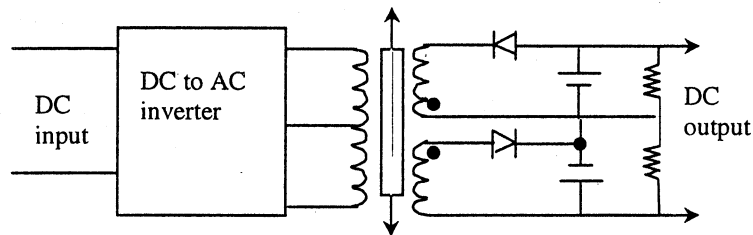


Fig. 5.6
DC - LVDT Circuit

5.3 APPLICATIONS OF LVDT

5.3.1 Displacement Measurement and LVDT Gageheads

One common form of a completely assembled LVDT displacement transducer is illustrated in fig. 5.7. It is commonly known as LVDT gagehead because it is widely used in machine tool inspection and gaging equipment.

The core is connected to a spring loaded probe shaft having a removable tip. The probe shaft is guided in a sleeve bearing that is retained in a case which also encloses the LVDT coil winding. The case is often threaded externally to simplify mounting.

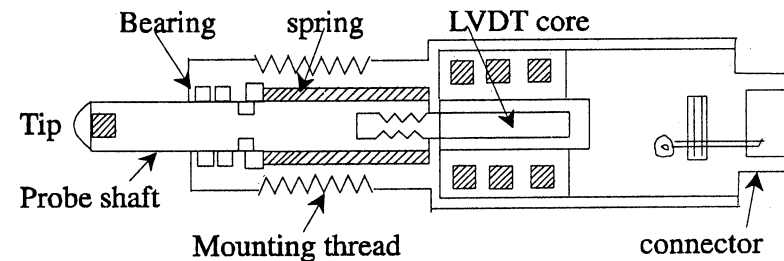


Fig. 5.7
LVDT Gage head

Lever or Finger probe gageheads are useful in places which are inaccessible to the probe of an ordinary LVDT gagehead. Fig. 5.8

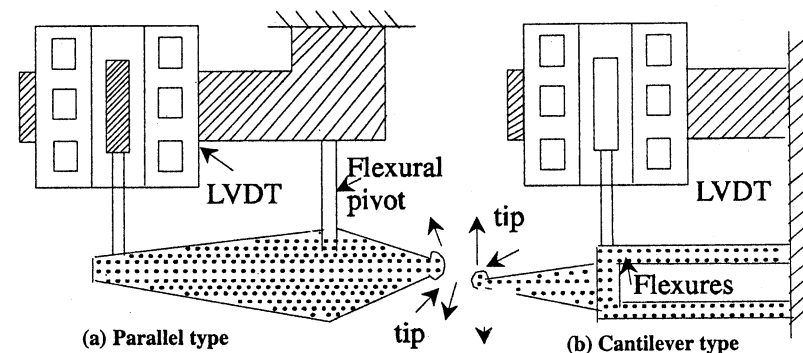


Fig. 5.8
Finger probe gage heads

shows two possible arrangements. Parallel flexure type is preferred as this introduces less error due to core rotation.

Traditional users of LVDT gageheads have been manufacturers of machine tool inspection equipment and control departments. In these applications, the LVDT gagehead and its readout act as more precise substitutes for mechanical dial position indicators. This type of head system can be used to record data electronically, to gather statistical quality control information on manufactured parts or to interface in a closed-loop control system. Another advantage of the LVDT gauging system is that a digital readout can be incorporated, permitting unskilled or semi-skilled personnel to do production line inspection work. Ultra-precise LVDT gauging systems can make measurements that are otherwise difficult or impossible even with precision mechanical dial indicators or other mechanical measuring devices.

5.3.2 LVDT Pneumatic Servofollower

Non contacting position measurement can be done using the pneumatic servo follower shown in fig. 5.9.

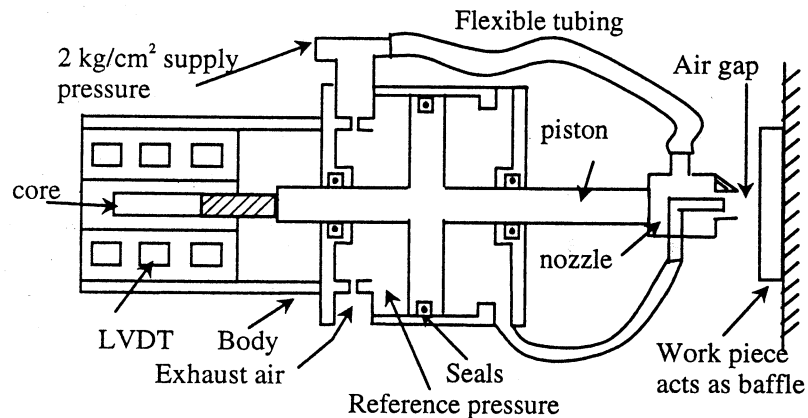


Fig. 5.9
LVDT Servo follower

This pneumatic servofollower consists of a double acting air cylinder having a piston, piston rod, seals and a special nozzle at one end of the piston rod. 2 Kg/cm² pressure air is applied to the

reference pressure chamber through the orifice and then exhausted back to atmosphere through the outlet orifice. Since both orifices are of the same size, the reference pressure developed equals one half of the supply pressure. The supply pressure is also connected to the nozzle, where the baffle is the surface being measured. This gap dependent pressure is fed back to the control pressure chamber. At a particular gap between the surface and nozzle typically 0.075 mm, the control pressure equals one-half supply pressure and the

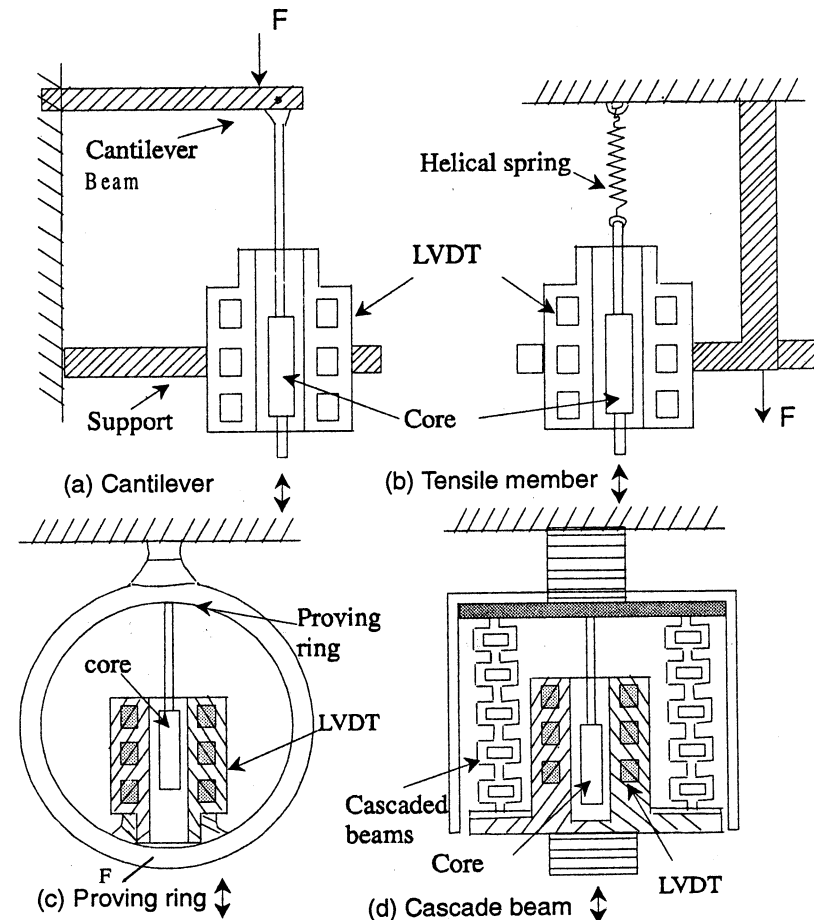


Fig. 5.10
LVDT load cells

piston does not move. If the gap changes because of dimensional or positional change in the gaging surface, the control pressure changes accordingly repositioning the piston and nozzle and restoring the gap. The LVDT coupled to the pneumatic servofollower gives an output either for displacer or to a duplicating system.

5.3.3 LVDT Loadcells

Many elastic members can be loaded in either tension or compression leading to deflection in either direction. The bi-directional nature of the displacement characteristic of a LVDT perfectly complements the bi-directional deflection of an elastic member. Thus LVDT loadcell produces an output voltage linearly proportional to axial force load. Typical Loadcell examples of LVDT's combined with elastic members are illustrated in fig. 5.10.

5.3.4 LVDT Pressure Transducer

This requires an elastically deformable sensing element that responds to omnidirectional fluid pressure. Fig. 5.11 illustrates the arrangement of a typical C-tube LVDT pressure transducer.

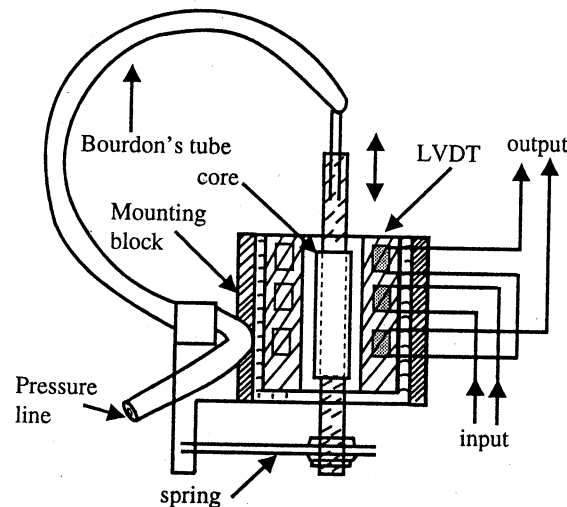


Fig. 5.11
Pressure transducer

The coil is mounted to the block containing the fixed end of the tube and the pressure inlet. A cantilever spring and nonmagnetic core rod maintain the core in the center of the LVDT coil. As the deflection of the tube is small, a voltage output which is a linear function of pressure is obtained. Pressure transducers can also be constructed using helical or twisted Bourdon tubes connected to a RVDT.

For low range pressure measurements axially convoluted bellows, radially convoluted diaphragms, flat diaphragms and welded pressure capsules are alternatives to Bourdon tubes as sensing elements in low range pressure LVDT transducers. Fig. 5.12 shows a capsule type LVDT pressure transducer.

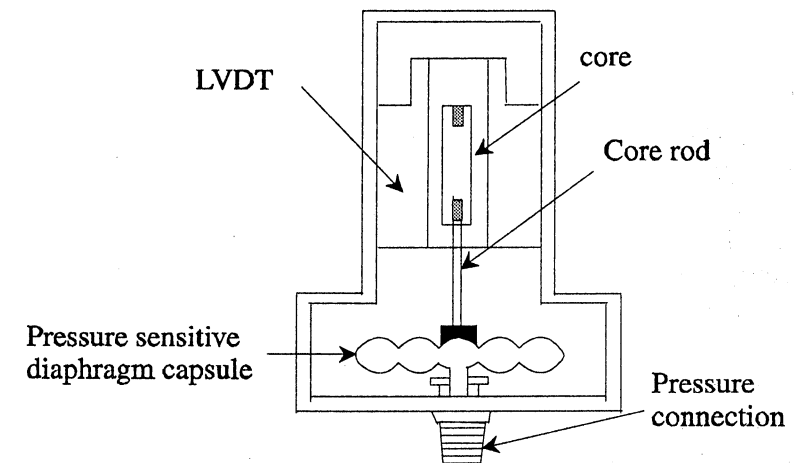


Fig. 5.12
Low pressure LVDT pressure transducer

Thus the LVDT has a wide range of applications. New use is being discovered for this device periodically.

5.4 SYNCHROS

The term synchro means a family of a.c. electromechanical synchronising devices. Synchros perform the functions of angle measurement, angle addition and subtraction, remote angle

transmission. Synchros can also be used for computation of rectangular components of vectors. Synchros for angle measurement are mostly utilised as components of servomechanisms.

The other names for this device are Selsyn, Autosyn which are acronyms for 'self synchronising' and 'automatic synchronising'. Basically synchros are mutual-inductance transducers. The synchro contains a rotor and a stator. The rotor carries a winding as shown in fig. 5.13a and b. The connections to this winding are made through slip rings. The stator is wound exactly like an a.c. motor containing three windings. They are connected in star as shown in fig. 5.13c.

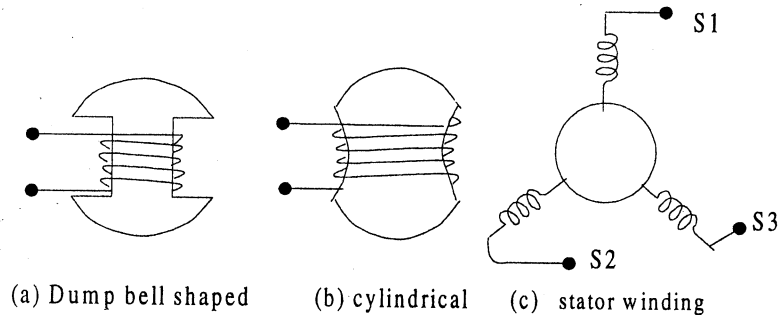


Fig. 5.13
Synchro windings

Rotor has a single winding which is excited with a.c. voltage and it sets up a flux in the stator space. This alternating flux induces voltages in each stator windings. The magnitude of the voltages induced in each stator winding depends on the rotor position with respect to that winding.

For the rotor position shown in fig. 5.14, the voltages in the rotor and stator windings are given by the following equations.

$$\text{Rotor voltage} \quad e_r = A \sin \omega t \quad (5.12)$$

$$\text{Stator voltage} \quad e_1 = e_m \sin \omega t \cos \theta$$

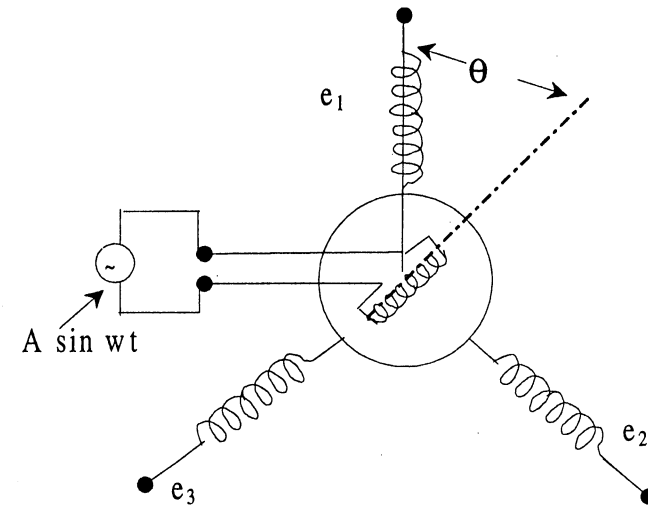


Fig. 5.14
Synchro rotor & stator windings schematic

$$\begin{aligned} e_2 &= e_m \sin \omega t \cos (120 - \theta) \\ e_3 &= e_m \sin \omega t \cos (120 + \theta) \end{aligned} \quad (5.13)$$

The frequency of the stator windings voltages is the same as the frequency of the voltage applied to the rotor. e_m is the peak voltage induced in the stator coil. It may be noticed that all the three-voltages given by equations 5.13 are in same phase. And also for a given set of stator voltages, there is only one corresponding rotor position. The frequency of supply to the rotor is either 50 Hz or 400 Hz. The 400Hz supply is used in aircraft systems for reducing the size of the synchros.

5.4.1 Synchros as Position Transducer

A pair of synchros can be used for transducing angular position to a voltage signal. Consider the system shown in fig. 5.15.

The stator coils are connected as shown (S_1 to S_1 , S_2 to S_2 and S_3 to S_3). The rotor of the Synchro generator (SG) is kept at $\theta = 0$, position. This position is known as Mechanical Zero position. This rotor is supplied with an a.c. voltage.

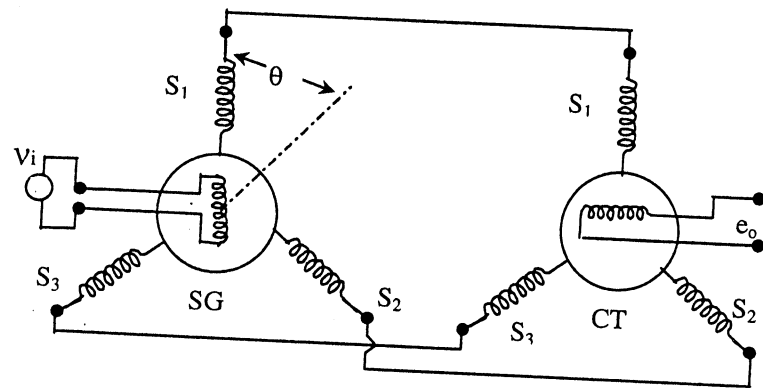
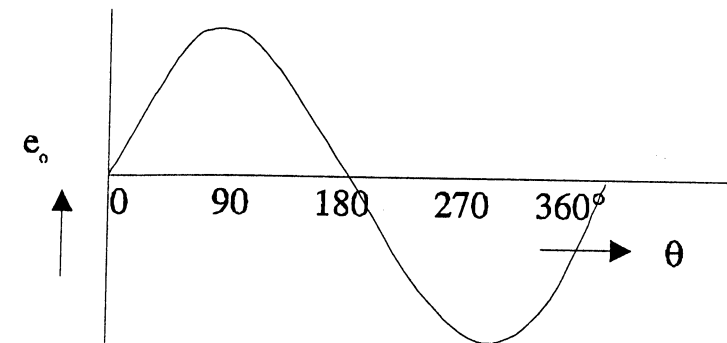


Fig. 5.15
Synchro pair as position transducer

The rotor of the second unit which is known as the control transformer (C.T.) is kept at 90° to the Mechanical zero position of the rotor of the S.G. this rotor is cylindrically shaped in construction to reduce the variation of reluctance with position (fig. 5.13b). The rotor of the S.G. is normally dumbbell shaped as shown in fig. 5.13a. now when $\theta = 0$, the flux is aligned with winding 1 of SG and induces a maximum voltage in winding 1. In other windings also voltages are induced but of smaller magnitudes. These voltages circulate a current in windings S_1 , S_2 and S_3 of C.T. and produce a flux in the stator space of the C.T. This flux pattern will be exactly same as that in SG rotor space in direction and magnitude. This can be shown by considering currents flowing through the stator windings of the C.T. So the flux in the C.T. is aligned with windings 1 of C.T. stator. As the rotor winding of the C.T. is 90° to the winding 1, the voltage in the C.T. rotor is zero. This position of the rotor is known as the electrical zero position. Thus when θ is equal to zero, e_o is also equal to zero. Now assume a 10° displacement is given to the rotor of the S.G. The flux in the C.T. rotor space, is also displaced by the same 10° from the winding 1. The angle between this flux and the C.T. rotor winding is 80° and hence a voltage is induced in the winding. This voltage will be



This negative portion is to indicate 180° phase shift.

Fig. 5.16
Synchro transducer characteristics

maximum if the angle of displacement of the S.G. rotor is 90° . It may be noted that all the time the rotor of the CT is not moved at all. If a graph is plotted between the angular displacement and output voltage, it will be sinusoidal variation as shown in fig. 5.16. the negative half is purposely drawn for θ values ranging from 180° to 360° to stress the point that phase angle between the supply voltage and the output voltage is 180° . This graph gives only the magnitude of the voltage. The frequency will be same as that of the supply voltage. For linear operation, the initial straight line portion of the graph is utilised. In this region the equation is $e_o = K_s \theta$ where K_s is the synchro constant. If the rotor of S.G. is given a sinusoidal motion with 10 oscillations per second, then the actual output will be a modulated wave as shown in fig. 5.17.

This arrangement is used as an error sensing device in position control systems.

5.4.2 Synchros for Angle Telemetry

A pair of synchros can be used for telemetering angular position to a remote location. The connections for this are shown in fig. 5.18.

The first unit is synchrogenerator, and the second unit is known as synchro motor. The motor is equipped with some

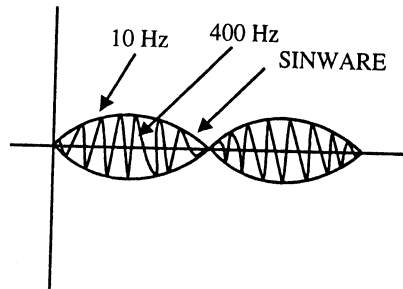


Fig. 5.17
Synchro pair output for a sinusoidal core motion

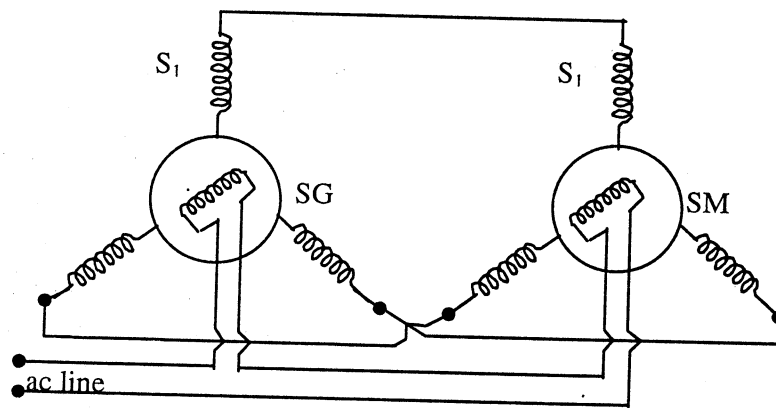


Fig. 5.18
Synchro position transmitter

additional damping units. The angular displacement of the motor follows that of the generator. The SG can be in one room and SM can be in a remote place. There will be five wires connecting them. There are other synchro units known as synchro differential units. These units have three windings in the rotor also. These are used to transmit an intelligence corresponding to the sum of two angular displacements over a distance.

5.5 INDUCTION POTENTIOMETER

In an Induction potentiometer there is one winding on the rotor and one on the stator. (Fig. 5.19).

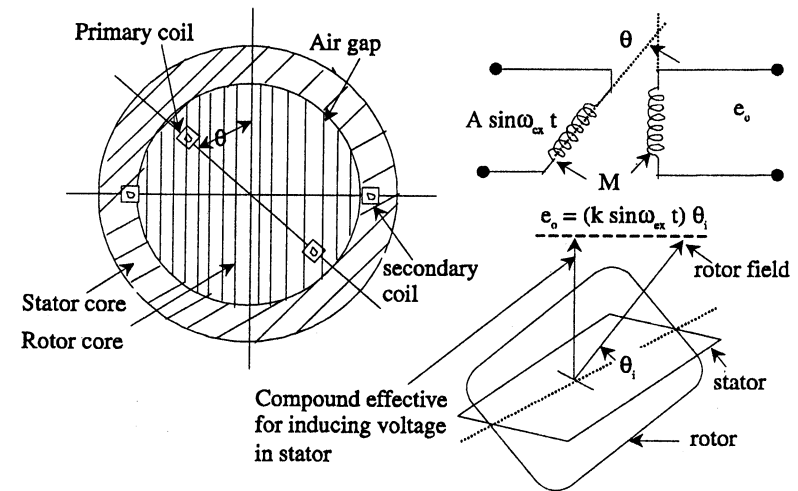


Fig. 5.19
Induction Potentiometer

The primary is excited with alternating current. This induces a voltage into the secondary. The amplitude of this output voltage varies with the mutual inductance between the two coils and this varies with the angle of rotation. For single turn coils the variation with angle would be sinusoidal and only a small linear range around null would be obtained. By carefully distributing the rotor and stator windings a linear relation up to $\pm 90^\circ$ rotation may be obtained.

5.6 VARIABLE RELUCTANCE ACCELEROMETER

Fig. 5.20 shows another common version of the variable reluctance principle. This is an accelerometer for measurement of acceleration in the range $\pm 4g$. Since the force required to accelerate a mass is

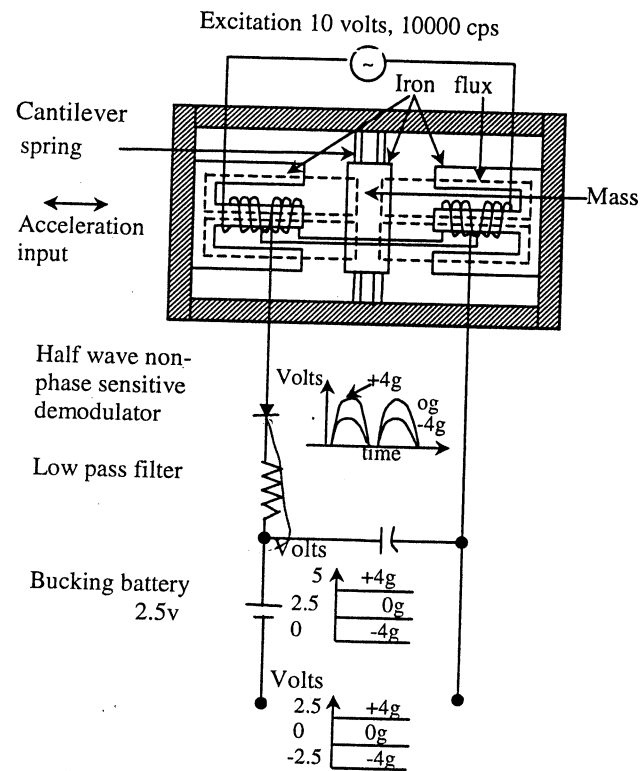


Fig. 5.20
Variable reluctance accelerometer

proportional to the acceleration, the springs supporting the mass deflect in proportion to the acceleration. The mass is of iron and thus serves as both an inertial element for transducing acceleration to force and also as a magnetic circuit element for transducing motion to reluctance. Ordinarily such an instrument would be constructed so that the iron core would be half way between the two E frames when the acceleration was zero, thus giving zero output voltage for zero acceleration. However to detect motion on both sides of zero, a fairly involved phase-sensitive demodulator would be required. To eliminate the demodulator the iron core and springs were adjusted so that core was offset to one side by an

amount equal to the spring deflection corresponding to 4 g acceleration. Thus with no acceleration applied, the output voltage was not zero but some specific value. Then when +4 g of acceleration was applied, the output went to 5.0 volts and when -4 g was applied the output went to zero. With the help of a bucking battery, a simple diode and filtering circuit a phase-sensitive output proportional to the acceleration is obtained. The actual full-scale motion of the mass in this particular instrument is just a few millimeters giving a displacement sensitivity for the variable reluctance element of almost 400 volts /cm.

5.7 MICROSYN

Another variable reluctance element is the Microsyn, a rotary component, shown in fig. 5.21 and widely used in sensitive gyroscopic instruments.

The figure shows the instrument in the null position where the voltages induced in coils 1 and 3 are just balanced by those of coils 2

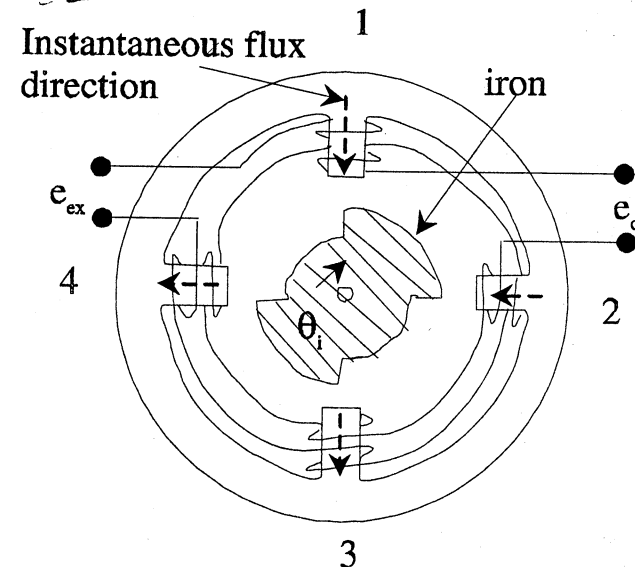


Fig. 5.21

and 4. Motion of the input shaft from the null increases the reluctance of coils 1 and 3 and decreases the reluctance of coils 2 and 4, thus giving a net output voltage e_o . Motion in the opposite direction causes a similar effect except the output voltage has a 180° phase shift. If a direction sensitive dc output is required a phase sensitive demodulator is necessary. The excitation voltage is 5 to 50 volts at 50 to 5000 Hz. Sensitivity is of the order of 0.2 to 5 volts/degree. Nonlinearity is about 0.5% full scale for $\pm 7^\circ$ rotation and 1.0% for $\pm 10^\circ$.

5.8 WORKED OUT EXAMPLES

EXAMPLE 5.8.1

A LVDT has the following specifications. Excitation voltage is 10 V at 400 Hz. The maximum output voltage is 2.5V. Range of core movement is ± 1.5 cm.

- Calculate the output voltage when the core is +1cm away from the null. Sketch the input & output waveform.
- Sketch the output waveform when the core is moved with a 1 Hz sinusoidal motion from the null.

SOLUTION

(a) Output voltage when the core is +1cm away from the null=

$$2.5 \times \frac{1}{1.5} = 1.67 \text{ V}$$

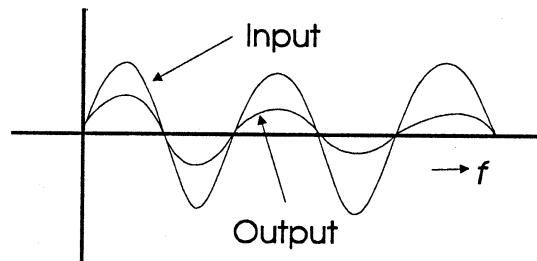


Fig. 5.22a

Input output waveform of the LVDT

(b) The waveform are as given in fig 5.22b

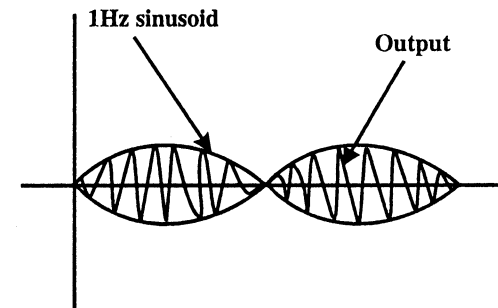


Fig. 5.22b

LVDT output waves for a sinusoidal core motion

EXAMPLE 5.8.2

A LVDT has an output of 6V rms when the displacement is 0.4×10^{-3} m. Determine the sensitivity of this instrument in volt/mm. A 10 volt voltmeter with 100 scale divisions is used to read the output. Two tenths of a division can be estimated with ease. Determine the resolution of the voltmeter. The above arrangement is used in a pressure transducer for measuring the deflection of a diaphragm. The diaphragm is deflected thro' 0.5×10^{-3} m, by a pressure of 1000 N/m^2 . Determine the sensitivity & resolution of this instrument?

SOLUTION

$$\begin{aligned} \text{Sensitivity of the LVDT} &= \frac{6}{0.4 \times 10^{-3}} \text{ Volts/m} \\ &= 15 \text{ Volts/mm} \end{aligned}$$

Resolution of the voltmeter 1 scale divisions = .1 volt

$$\frac{2}{10} \text{ of a division is } \frac{.1 \times 2}{10} = .02 \text{ volt}$$

Resolution of voltmeter alone = .02 volt

$$\text{Resolution of the LVDT and the voltmeter} = \frac{.02}{15} \text{ mm}$$

$$\begin{aligned}
 &= 1.33 \times 10^{-3} \text{ mm} \\
 \text{Pressure sensitivity of the diaphragm} &= \frac{.5 \times \text{mm}}{1000 \text{ N/m}^2} \\
 &= .5 \times 10^{-3} \frac{\text{mm}}{\text{N/m}^2}
 \end{aligned}$$

Voltage sensitivity of the Diaphragm

$$\begin{aligned}
 \text{LVDT system} &= 15 \times .5 \times 10^{-3} = \frac{\text{volt}}{\text{N/m}^2} \\
 &= 7.5 \times 10^{-3} = \frac{\text{volt}}{\text{N/m}^2}
 \end{aligned}$$

Resolution of the Diaphragm

$$\begin{aligned}
 \text{LVDT system} &= \frac{1.33 \times 10^{-3}}{.5 \times 10^{-3}} \text{ N/m}^2 \\
 &= 2.66 \text{ N/m}^2
 \end{aligned}$$

EXAMPLE 5.8.3

A proximity pick up works on the principle of variable reluctance. The inductance coil is 3 mH when the machine part is made out of iron and it is 1 mm away from the pick up. Calculate the maximum

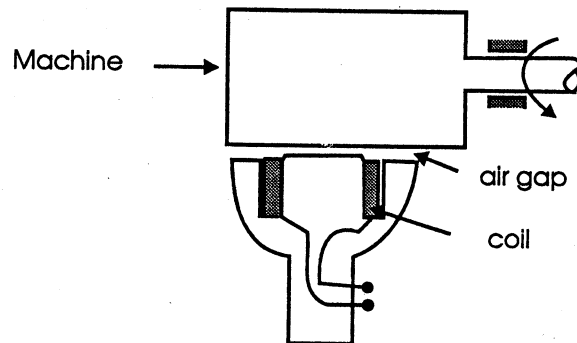


Fig. 5.23
Reluctance pickup

variation of inductance when the moving machine part has 10% surface roughness.

SOLUTION

10% surface roughness means an air gap variation of ± 0.1 mm.

When the air gap is 1mm inductance is 3 mH.

In this type of pick up everything is constant except the reluctance

$$L \propto \frac{1}{\text{Reluctance}} \propto \frac{\mu}{l}$$

Assume the permeability and the area are constant in the neighbourhood

$$L \propto \frac{1}{\text{length}} \text{ of the air gap.}$$

For a maximum gap of 1mm inductance is 3 mH.

$$\therefore \text{For a gap of 1.1m if } 3 \times \frac{1}{1.1} = 2.727 \text{ mH}$$

The maximum variation of Inductance is $\pm .273$ mH

EXAMPLE 5.8.4

A periodic motion with a significant frequency content upto 200 Hz is measured using a LVDT system. The LVDT system consists of LVDT, phase sensitive demodulator (PSD) and an oscillograph galvanometer as shown in fig 5.24. The LVDT is excited with 20 V and 10000 Hz supply

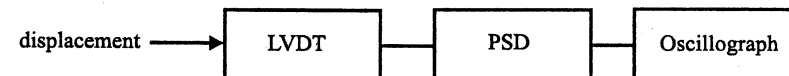


Fig. 5.24
LVDT Measuring system

The oscillograph is a second order system with $\zeta = 0.65$ and a natural frequency of 1000 Hz. What is the error of measurement for frequency components of 10 Hz, 100 Hz & 200 Hz?

Is this arrangement acceptable?

SOULTION

The output of the LVDT will be a modulated wave with carrier 10000 Hz and signal freq. upto 200 Hz.

The output of the PSD is a compound signal consisting of frequencies upto 200 Hz.

The oscillograph being a second order system with $\zeta = .65$ and $w_n = 2\pi \times 1000 = 2000\pi$

The wave form will be distorted due to different level of alternations and phase shifts for different frequency motions of the core.

The second order system dynamics is given by

$$\frac{C(s)}{R(s)} = \frac{K w_n^2}{s^2 + 2\zeta w_n s + w_n^2}$$

k can be assumed to be 1 without loss of generality

$$\therefore M = \left| \frac{C(jw)}{R(jw)} \right| = \frac{w_n^2}{|w_n^2 - w^2 + 2j\zeta w w_n|}$$

$$= \frac{1}{\sqrt{\left[1 - \left(\frac{w}{w_n}\right)^2\right]^2 + \left[2\zeta \frac{w}{w_n}\right]^2}}$$

$$\text{Let } \mu = \frac{w}{w_n}$$

$$\text{Then } M = \frac{1}{\sqrt{(1 - \mu^2)^2 + (2\zeta\mu)^2}}$$

$$\text{When } w = 10 \text{ Hz, } \mu = \frac{2\pi 10}{2000\pi} = \frac{2}{200} = \frac{1}{100}$$

$$M = \frac{1}{\sqrt{(1 - 10^{-4})^2 + (2 \times .65 \times \frac{1}{100})^2}}$$

$$M = 1$$

$$\text{When } w = 100 \text{ Hz, } \mu = \frac{2\pi 100}{2000\pi} = \frac{2}{20} = \frac{1}{10}$$

$$M = \frac{1}{\sqrt{(1 - 10^{-2})^2 + (.13)^2}} = 1$$

when $w = 200 \text{ Hz}$

$$\mu = \frac{1}{5}$$

$$M = \frac{1}{\sqrt{\left(1 - \frac{1}{25}\right)^2 + \left(\frac{1.3}{5}\right)^2}}$$

$$= \frac{1}{\sqrt{1 + 0.676}} = \frac{1}{1.033} = .968$$

The errors in measurement for $w = 10, 100$ & 200 are $0, 0, .032$ respectively.

EXAMPLE 5.8.5

A LVDT output is recorded by a self balancing potentiometric recorder having its natural frequency of 10 Hz and a damping ratio of 0.707. The LVDT is excited by 10V at 50 Hz power supply calculate the maximum frequency of the displacement signal that can be recorded with an error of $\pm 2\%$

SOLUTION

The second order system dynamics of the potentiometric recorder is

$$\frac{C(s)}{R(s)} = \frac{w_n^2}{s^2 + 2\zeta w_n s + w_n^2}$$

Here $\zeta = .707$ & $w_n = 2\pi \times 10$

The frequency response can be obtained by replacing s by jw . The amplitude attenuation for different frequency inputs can be calculated using the following equation

$$M(jw) = \left| \frac{C(jw)}{R(jw)} \right| = \left| \frac{w_n^2}{-w^2 + j2\zeta w_n w + w_n^2} \right|$$

Defining $\mu = \frac{w}{w_n}$

$$M(j\omega) = \left| \frac{1}{1 - \mu^2 + j 2 \zeta \mu} \right|$$

$$M = \frac{1}{\sqrt{(1 - \mu^2)^2 + (\zeta \mu)^2}}$$

The frequency for $\pm 2\%$ error i.e., .98 times the full output can be calculated from the following eqn.

$$\text{i.e., } .98 = \frac{1}{\sqrt{(1 - \mu^2)^2 + (1.4 \mu)^2}}$$

$$(.98)^2 = \frac{1}{(1 - \mu^2)^2 + (1.4 \mu)^2}$$

$$.964 ((1 - \mu^2)^2 + (1.4 \mu)^2) = 1$$

$$[1 - 2\mu^2 + \mu^4 + 1.96 \mu^2] = 1/.964$$

$$\mu^4 - .04 \mu^2 + 1 = 1.037$$

$$\mu^4 - .04 \mu^2 - .037 = 0$$

Applying the quadratic rule to extract the roots, we get

$$\therefore \mu^2 = \frac{.04 \pm \sqrt{(.04)^2 + 4 \times .037}}{2}$$

$$= \frac{.04 \pm .387}{2}$$

$$= .2135 \text{ or } -.1735$$

frequency can not be negative

$$\therefore \mu^2 = .2135$$

$$\mu = .462$$

$$\frac{w}{w_n} = .462$$

$$w = .462 w_n = .462 \times 2 \pi \times 10$$

The freq. at which 2% error occurs is

$$f_c = .462 \times 2 \pi \times 10 / 2 \pi$$

$$= 4.62 \text{ Hz.}$$

CAPACITANCE TRANSDUCERS

6.1 BASIC PRINCIPLE

ONE of the important groups of transducers is the capacitance transducers. Many industrial variables like displacement, pressure, level, moisture, thickness etc. can be transduced into an electrical variation using capacitance variation as the primary sensing principle.

As we know one of the three basic parameters of electrical circuits is capacitance, also known as condenser. This capacitance is a function of area between two plates, the separation between them and the dielectric medium in between

C for parallel plate condenser is

$$C = \frac{A}{x} \epsilon_0 \epsilon_r \quad (6.1)$$

A = common area between plates

ϵ_0 = permittivity of the free space

ϵ_r = relative permittivity (dielectric constant of the medium)

x = distance between them.

So if one of the above three factors changes, the capacitance changes. This change in capacitance can be suitably transduced into a voltage or a frequency signal. For a voltage output, bridge circuits are employed. For frequency variation oscillator circuits are employed. The capacitance is connected to voltage and charge by the equation $Q = VC$ where

Q is charge in coulomb
V is voltage in volts
C is capacitance in farads.

If we differentiate this equation, we get,

$$\frac{dQ}{dt} = V \frac{dC}{dt} + C \frac{dv}{dt} \quad (6.2)$$

Any variable or parameter that affects the capacitance, can be transduced into a voltage or current variation, as per the above equation.

6.2 CAPACITIVE DISPLACEMENT TRANSDUCERS

The most popular form of variable capacitor used in displacement measurement is parallel plate capacitor with a variable air gap. The main problem in this case is the high output impedance.

For example, let us consider a capacitor of area 4 cm^2 , 0.02 cm gap and permittivity of air is $8.854 \times 10^{-12} \text{ farad/m}$. The capacitance of this is

$$C = \frac{A}{x} \epsilon_0 = \frac{4 \times 10^{-4}}{0.02 \times 10^{-2}} \times 8.854 \times 10^{-12} = 17.78 \text{ pf} \quad (6.3)$$

The impedance of this capacitor at a frequency of 10000 Hz .

$$= \frac{1}{2\pi fC} = \frac{1}{2\pi \times 10^4 \times 17.78 \times 10^{-12}} \quad (6.4)$$

$$= 89000 \text{ Ohms}$$

$$= 89 \text{ Kilo ohm}$$

Because of this high impedance level, the noise level is high resulting in complex electronic circuitry. The other problem is that the variation of capacitance with respect to displacement is nonlinear.

$$C = \frac{8.854 A}{x} \text{ pf} \quad (6.5)$$

$$\frac{dC}{dx} = \frac{-8.854 A}{x^2} \text{ pf} \quad (6.6)$$

The sensitivity increases as x decreases in parabolic manner, as shown in fig. 6.1. But percentage change in c is equal to percentage change in x .

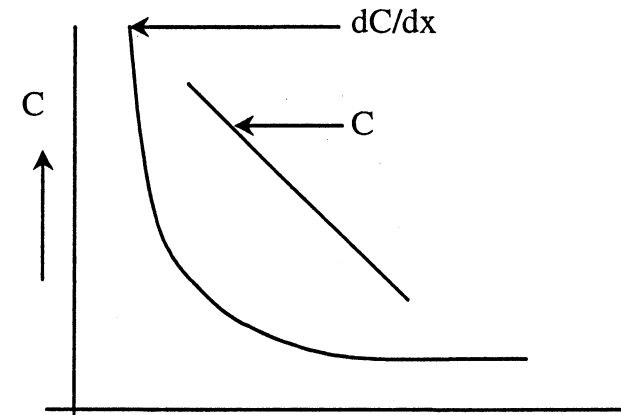


Fig. 6.1
Sensitivity of parallel plate Capacitor

$$\frac{dC}{dx} = \frac{-C}{x} \quad (6.7)$$

$$\frac{dC}{C} = \frac{-dx}{x} \quad (6.8)$$

6.2.1 Single Ended Capacitive Displacement Transducers

The simplest form of displacement transducer is a parallel plate capacitor with plate movable as shown in fig. 6.2.

The capacitance is given by

$$C_x = \frac{A}{\frac{x}{\epsilon_0} + \frac{t}{\epsilon_r \epsilon_0}} = \frac{A \epsilon_0}{x + \frac{t}{\epsilon_r}} \quad (6.9)$$

where A is the common area between the plates
't' is the thickness of the solid dielectric medium
 ϵ_r is the relative permittivity of the solid portion

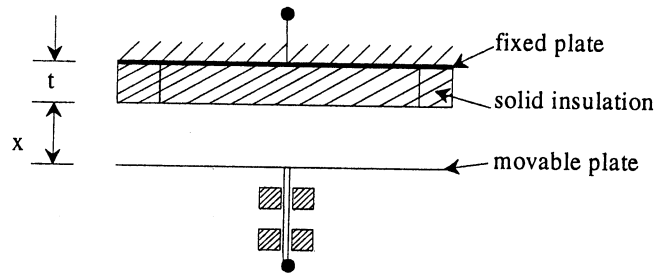


Fig. 6.2

Simple capacitive Displacement Transducer

ϵ_0 is the permittivity of air.

If the air gap is increased by x then the capacitance will get reduced to

$$C_x - \Delta C = \frac{A\epsilon_0}{x + \Delta x + \frac{t}{\epsilon_r}} \quad (6.10)$$

The sensitivity is

$$\begin{aligned} \frac{\Delta C}{C_x} &= \frac{C_x - (C_x - \Delta C)}{C_x} \\ &= 1 - \frac{x + t/\epsilon_r}{x + \Delta x + t/\epsilon_r} = \frac{\Delta x}{x + \Delta x + t/\epsilon_r} \end{aligned} \quad (6.11)$$

$$\frac{\Delta C}{C_x} = \frac{\Delta x}{(x + t/\epsilon_r) + \Delta x} \quad (6.12)$$

If Δx is very small compared to $x + t/\epsilon_r$ it can be deleted, then

$$\frac{\Delta C}{C_x} = \frac{\Delta x}{x + t/\epsilon_r} \quad (6.13)$$

The per unit variation of capacitance is proportional to Δx . thus it is linear over a small range of Δx . The range of linearity can be increased by having another fixed electrode as shown in fig. 6.3a. The circuit connection is shown in fig. 6.3b, which is a unity ratio arm wheatstone bridge or blumlein bridge circuit.

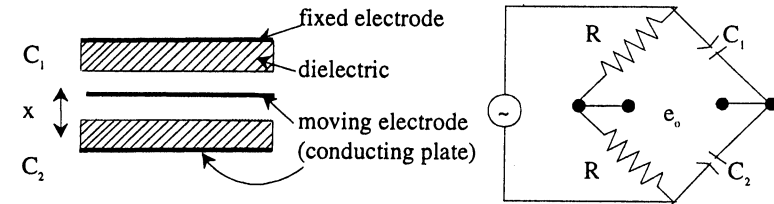


Fig. 6.3a and 6.3b

Two fixed plate capacitive transducer and its circuitry.

For large linear displacements, capacitive transducers where the plates are fixed and the dielectric medium is moved as shown in fig. 6.4, can be used.

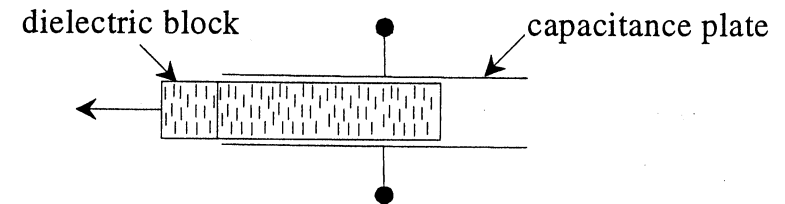


Fig. 6.4

Capacitive transducer for large displacement

6.3 DESIRABLE FEATURES OF CAPACITIVE TRANSDUCERS

Capacitive transducers are having the following advantages:

- Its force requirements are very small
- As the moving plates have very little mass, design of transducers with fast response characteristics is possible
- There is no physical contact between moving and stationary parts
- The capacitance does not depend on the conductivity of the metal electrodes or its variation with temperature.
- The capacitive transducer can be shielded easily against the effect of external electric stray fields.

6.4 PRACTICAL CAPACITOR PICKUPS

6.4.1 Equibar Differential Pressure Transducer

Differential pressure can be transduced by a three terminal capacitor as shown in fig. 6.5.

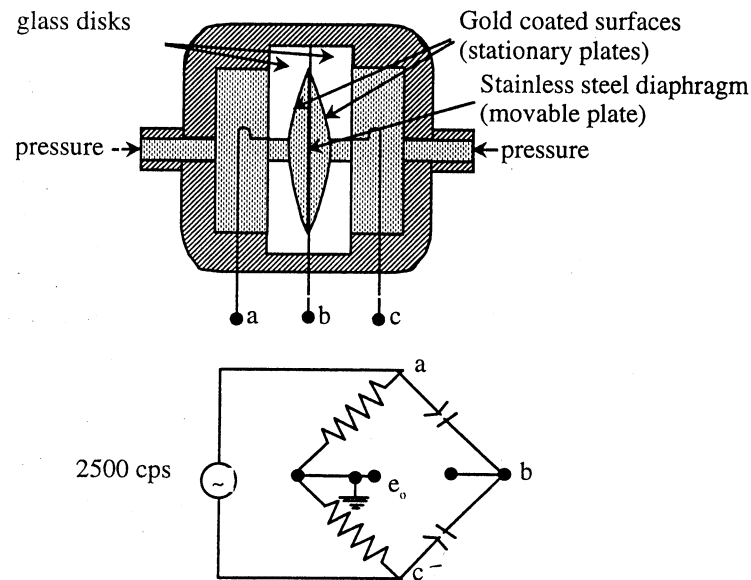


Fig. 6.5

Differential capacitor pressure pick up

Spherical cavity of a depth of about 0.025 mm is ground into the glass disk. These depressions are gold coated to form fixed plates of a differential capacitor. A thin stainless steel diaphragm is clamped between the disks which serves as the movable plate. With equal pressures applied to both ports, the diaphragm is in a neutral position and the bridge is balanced and $e_o=0$.

If one pressure is greater than the other, the diaphragm deflects to the low pressure side, giving an output e_o in proportion to the differential pressure. For the opposite pressure difference e_o exhibits a 180° phase change. The high impedance level requires a cathode follower amplifier at e_o . A direction sensitive d.c. output

can be obtained by conventional phase sensitive demodulation and filtering.

Balance resistors are not shown in fig. 6.5.

6.4.2 Feedback Type Capacitance Proximity Pickup

The diagram and the circuit shown in fig. 6.6a & fig. 6.6b represent a proximity capacitive pickup and a feedback OP-Amp circuit gives a good linearity. This makes use of a high gain operational amplifier, which has

- High input impedance
- High gain of the order of 10^5 to 10^8
- Low output impedance

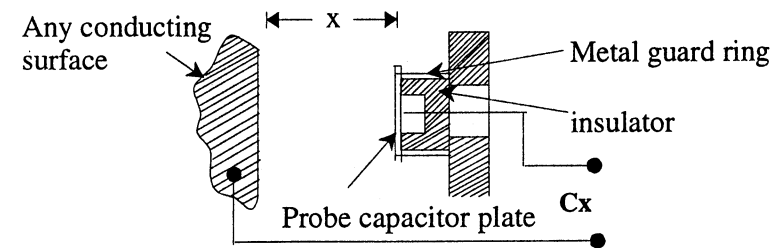


Fig. 6.6a

Proximity capacitive pick up

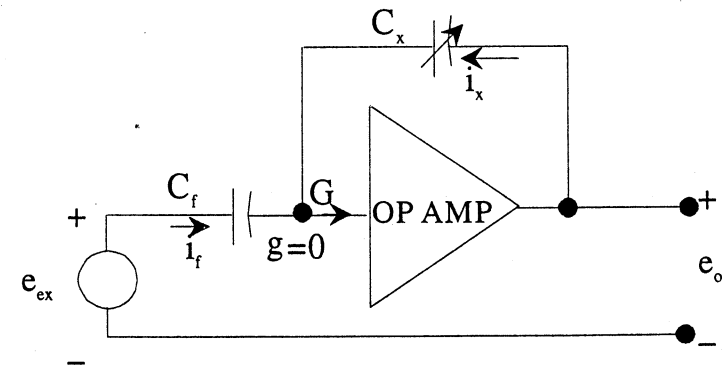


Fig. 6.6b

Feed back circuit

The equations governing its operations are the following:

Applying Kirchhoff's current law for the G junction

$$i_f - i_g + i_x = 0 \quad (6.14)$$

As G is ground potential

$$i_f = C_f \frac{de_{ex}}{dt} \quad (6.15)$$

$$i_x = C_x \frac{de_0}{dt} \quad (6.16)$$

$$\text{As } i_g = 0, i_f = -i_x \quad (6.17)$$

$$C_f \frac{de_{ex}}{dt} = -C_x \frac{de_0}{dt} \quad (6.18)$$

Integrating both sides and substituting $C_x = A\epsilon_0/x$, we get

$$e_0 = -\frac{C_f}{C_x} e_{ex} = -\frac{C_f x}{A\epsilon_0} e_{ex} \quad (6.19)$$

$$= Kx \quad (6.20)$$

Initial conditions or initial capacitor charges are assumed to be zero.

$$\text{where } K = (C_f/A\epsilon_0) e_{ex}$$

Now the output voltage is directly proportional to the plate separation x . Linearity is thus achieved for both large and small motions. In commercial instrument using Op. Amps. e_{ex} is a 50 KHz sinewave of fixed amplitude. The output e_0 is also a 50 KHz sinewave which is rectified and applied to a d.c. voltmeter calibrated directly in distance units.

For vibratory displacement, e_0 will be an amplitude modulated wave as shown in fig. 6.7.

The average value of this wave after rectification is still the mean separation of the plates and can still be read by the same meter as used for static displacements. The vibration amplitude around this mean position is extracted by applying the e_0 signal to a demodulator and a low pass filter with cut off at 10 KHz. The output of this filter is applied to a peak to peak voltmeter directly

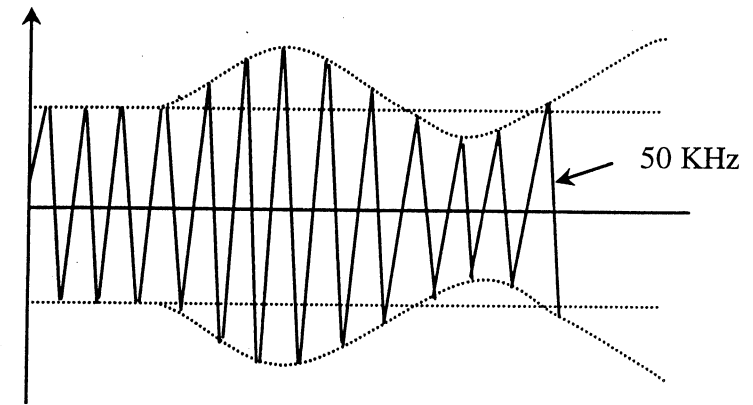


Fig. 6.7
Output for vibration

calibrated in vibration amplitude and is also available at a jack for connection to an oscilloscope for viewing of the vibration waveform. The instrument is provided with six different probes ranging from 0.10 to 25.0 mm in diameter of the capacitance plate and covering the full scale displacement ranges of 0.025, 0.125, 0.25 and 12.5 mm respectively. The overall system accuracy is of the order of 2 percent full scale. Any flat conducting surface may serve as the second plate of the variable capacitor. Thus in vibrating machine parts the parts themselves may often perform this function. The resolution of 0.5 percent of full scale indicates that with the 0.025 mm full scale probe, it is possible to detect motion as small as 125×10^{-6} mm.

Figures 6.8 and 6.9 show the various ways in which this probe can be used for speed measurement.

The capacitance will be maximum for the position shown in fig. 6.8a and minimum for a shaft position 180° to the position shown. Here the shaft has to make an electrical contact through the bearing. The connection through the bearing is unreliable. Therefore a split stator capacitor shown in fig. 6.8b is used which does not require electrical contact with the rotor plate. Many other transducer structures or dielectric configurations can also be used. Some of them are shown in fig. 6.9.

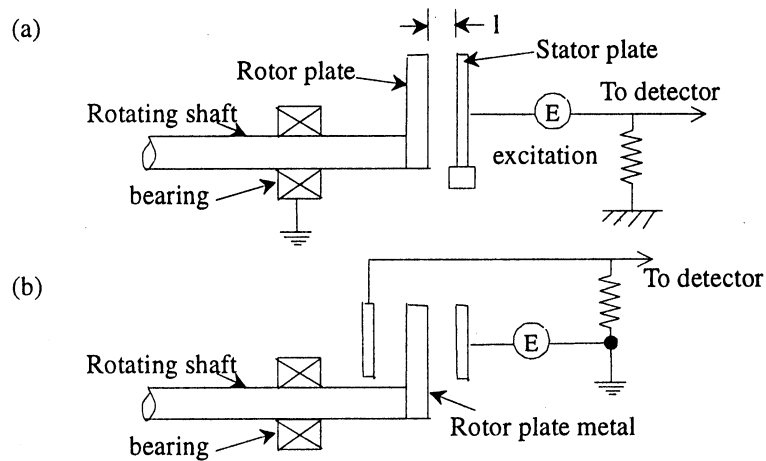


Fig. 6.8
Capacitive probe for speed measurement

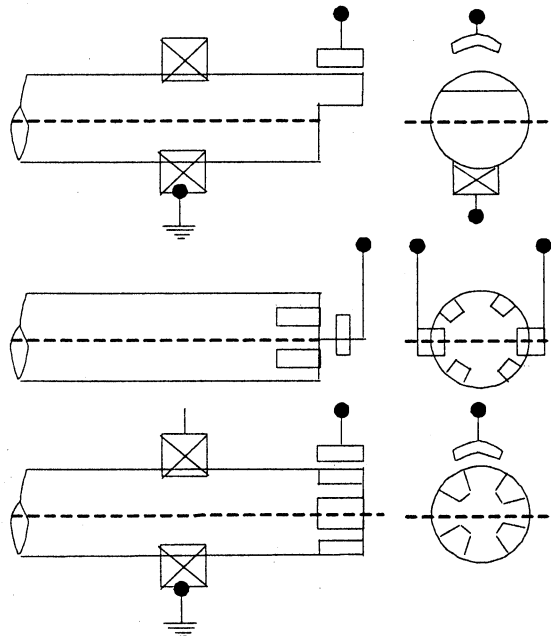


Fig. 6.9
Capacitive probes for speed measurement

6.4.3 Condenser Microphone

Fig. 6.10 is a simplified version of a typical capacitor microphone. The pressure response is found by assuming a uniform pressure p_i to exist all around the microphone at any instant of time. This is actually the case of sufficiently low sound frequencies but reflection and diffraction effects distort this uniform field at higher frequencies. The diaphragm is generally a very thin metal membrane which is stretched by suitable clamping arrangement. Diaphragm thickness ranges from about 0.0025 to 0.050 mm. The diaphragm is deflected by the sound pressure and acts as a moving plate of a capacitance displacement transducer. The other plate of the capacitor is stationary and may contain properly designed damping holes. The damping effect is used to control the resonant peak of the diaphragm response. A capillary air leak is provided to give equalization of steady pressure on both sides of the diaphragm to prevent diaphragm bursting. The variable capacitor is connected into a simple series circuit with a high resistance R and polarized with a d.c. voltage of about 200 volts. This polarizing voltage acts as circuit excitation and also determines the neutral diaphragm position.

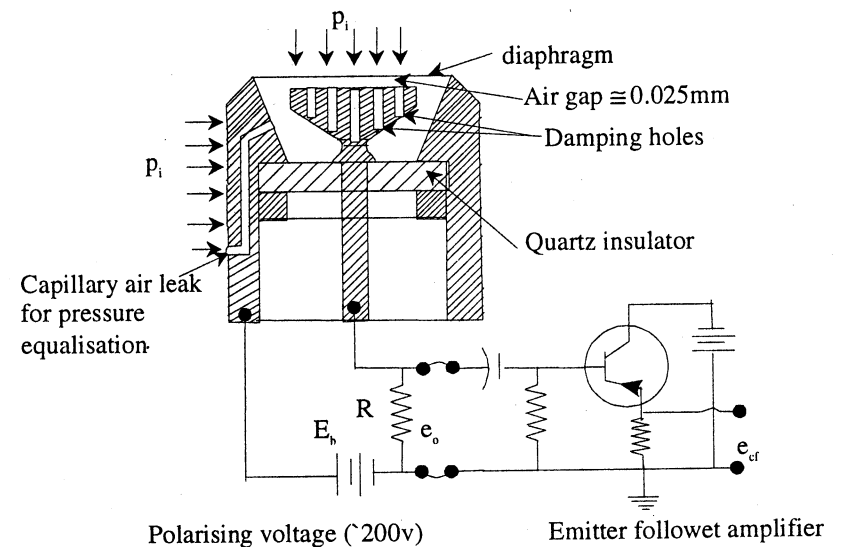


Fig. 6.10
Condenser microphone

6.5 CIRCUITRY FOR CAPACITANCE TRANSDUCERS

The capacitive transducers being active require external power to convert the capacitance change into voltage change which can be precalibrated in terms of the measurand. The various types of circuits utilised for the purpose can be classified into four categories.

- Frequency modulating oscillator circuit
- Circuit employing D.C. excitation
- A.C. bridge for amplitude modulation
- Pulse modulating circuit.

6.5.1 Frequency Modulating Oscillator Circuit

For low frequency variation of the measurand, either Wein bridge oscillator or phase shift oscillator is used. The Wein bridge oscillator shown in fig. 6.11 has many features and hence it has become an universal circuit. The advantages are

- Low distortion less than 0.1 percent
- Amplitude stability against aging, temperature variation and supply voltage variation
- Low output impedance

This has a disadvantage that a ganged dual capacitor is necessary. If a single capacitor is to be employed then Hartley oscillator configuration is to be used which is shown in fig. 6.12. The lowest center frequency is of the order of tens of KHz but 1 MHz is a typical value. This is not suitable for the measurand frequency less than 10Hz as the frequency stability of the low priced oscillator cannot be better than 100 parts per million.

6.5.2 D.C. Circuit For Dynamic Capacitance Variation

When the capacitance variation is dynamic as in the case of condenser microphone, simple RC circuit with DC excitation shown in fig. 6.13 can be used.

The capacitance C is given by $A\epsilon_0/x$ where A the area, ϵ_0 the permittivity of air and x is distance between plates with x_0 as the nominal separation when the capacitance does not change. For

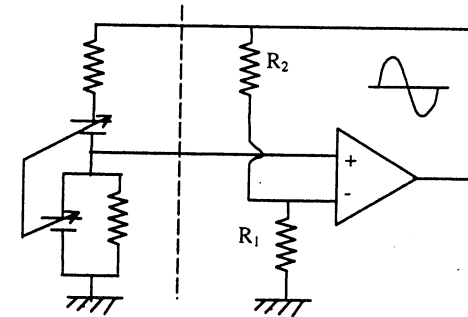


Fig. 6.11
Weins bridge oscillator

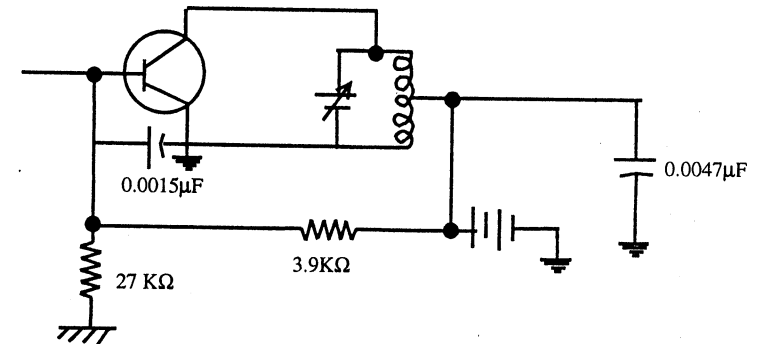


Fig. 6.12
Hartley oscillator

the circuit shown in 6.13, the following equations can be written when q is the electric charge.

$$i = \frac{E - V}{R} = \frac{dq}{dt} = \frac{d(VC)}{dt} \quad (6.21)$$

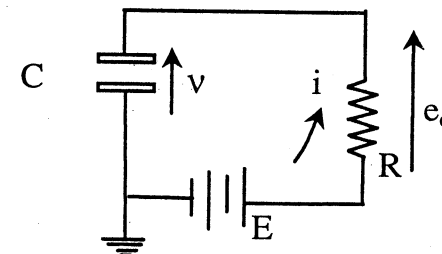


Fig. 6.13
Simple RC circuit for dynamic capacitive transducer

$$= C_0 \frac{dV}{dt} + V_0 \frac{dC}{dt} \quad (6.22)$$

C_0 is the nominal capacitance $= A\epsilon_0/x_0$ & $C = \frac{A\epsilon_0}{x}$

When x is stationary at V_0, x_0 ,
the voltage across the capacitor $= E$

$$\therefore \frac{E - V}{R} = C_0 \frac{dV}{dt} - E \frac{A\epsilon_0}{x^2} \frac{dx}{dt} \quad (6.23)$$

The voltage across the resistor,

$$e_0 = V - E \quad (6.24a)$$

$$\text{and hence, } \frac{de_0}{dt} = \frac{dV}{dt} \text{ as } E \text{ is const} \quad (6.24b)$$

substituting equation 6.24a & 6.24b in 6.23 and assuming $x = x_0 + dx, \cong x_0$

$$-\frac{e_0}{R} = C_0 \frac{de_0}{dt} - E \frac{A\epsilon_0}{x_0^2} \frac{dx}{dt} \quad (6.25)$$

Rearranging equation 6.25 and taking Laplace transform,

$$e_0(s) [1 + RC_0s] = \frac{REC_0s}{x_0} x(s) \quad (6.26)$$

$$\therefore \frac{e_0(s)}{x(s)} = \frac{EC_0Rs}{x_0 [1 + RC_0s]} \quad (6.27)$$

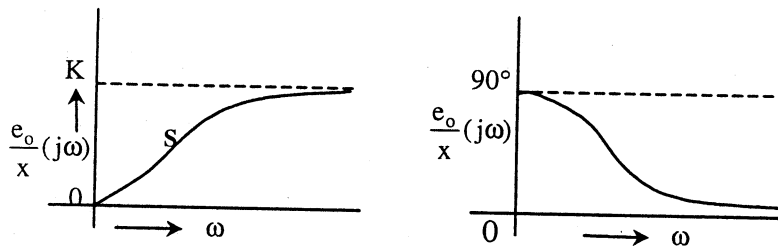


Fig. 6.14

Frequency response of circuit shown in fig. 6.13

Defining $K \triangleq E/x_0$ and $\tau \triangleq RC_0$ we get

$$\frac{e_0(s)}{x(s)} = \frac{K\tau s}{1 + \tau s} \quad (6.28)$$

The frequency response of this circuit is shown in fig. 6.14, which shows that only dynamic displacement can be measured with this circuit.

6.5.3 A.C. Bridges For Amplitude Modulation

Fig. 6.15 shows a simple arrangement for converting capacitance variation into a voltage. The detector may be calibrated to read displacement directly.

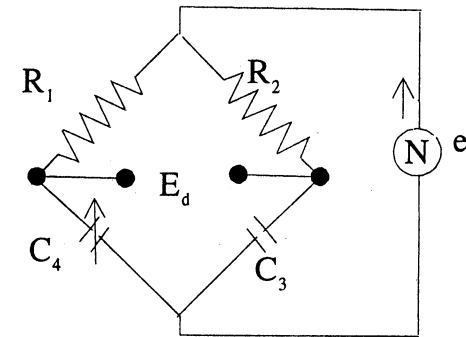


Fig. 6.15

Bridge circuit for capacitance variation

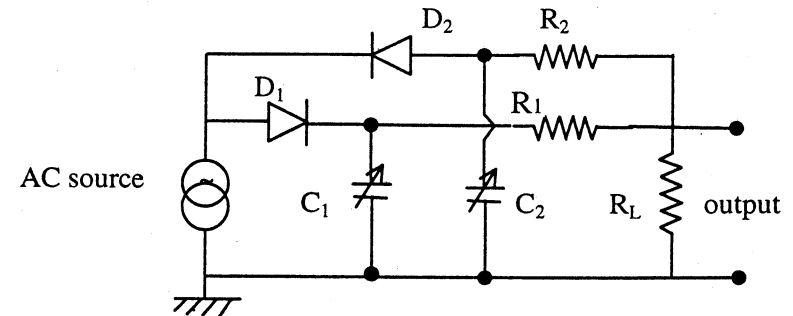


Fig. 6.16

Twin-T network

Twin T network shown in fig. 6.16 is used more often than the other capacitance circuits. This network has got the following advantages.

- The source S, the transducer capacitor C_1 , the comparison capacitor C_2 and the output circuit are grounded.
- The diodes D_1 and D_2 are operated at high voltage level so that they operate in the linear region of their characteristics.
- The output voltage is remarkably high, when operated with a sinusoidal input voltage E_i of 46 V rms at a frequency of 1.3 MHz. A variation of the capacitance difference from -7 to $+7$ pf causes a change of the output voltage from -5 to $+5$ VDC (i.e. 10V swing) into a 1M ohm load. Output voltage from -16 to $+16$ V had been observed, with suitable values of circuit parameters.
- The noise level in the output measured in band width of 7 K Hz, is the order of $10\mu\text{V}$ related to a typical output voltage of 10V. the signal to noise ratio is 10^6 ($S/N = 120\text{db}$).
- The rise time of the output signal depends on the load resistance for a load resistance of 1000 ohms, the rise time is of the order of 20 microseconds so that fast mechanical movements can be measured. The working of this bridge is as follows:

During positive half cycle of the applied voltage, current flows through diode D_1 and C_1 thus charging C_1 .

During the other half of a.c voltage, capacitor C_2 is charged and at the same time C_1 discharges through R_1 and R_L . And C_2 discharges through R_L and R_2 when C_1 is charged.

If C_1 , D_1 , R_1 are identical to C_2 , D_2 , and R_2 , the average current through R_L is zero. This happens when there is no difference between C_1 and C_2 .

A difference in pressure in the differential pressure capacitor pick up, will change C_1 from C_2 and hence a net average current will flow through R_L . & thus producing a voltage output.

This network is capable of producing around ± 5 volts.

6.5.4 Diode Pump

A more versatile circuit to transduce frequency or capacitance variation into a voltage is the diode pump. Fig. 6.17 shows the details of a diode pump circuit used to transduce the capacitance variation into voltage variation. Diode pump works as follows. The transducer is C_1 which is very small compared to C_2 . The time constant RC_2 should be sufficiently small compared to the period of the pulse input. When the pulse goes positive C_1 charges to the peak magnitude V . When the pulse goes to zero, D_2 conducts, so the output voltage jumps to $VC_1 / (C_1 + C_2)$. When pulse goes positive next, D_2 is not conducting and C_2 discharges through R . The wave forms are shown in fig. 6.18. The average value $V_{dc} = V RC_1$. So if V and R are constants the output is directly

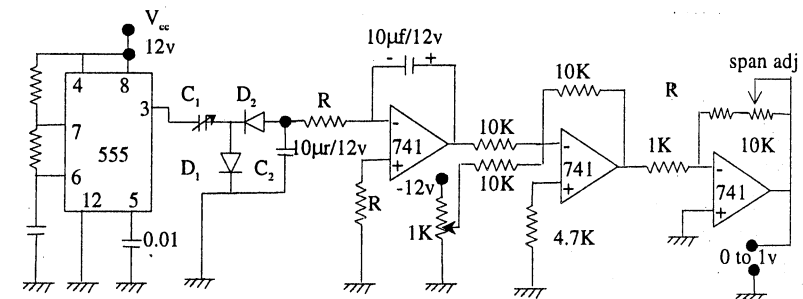


Fig. 6.17
Diode Pump circuit

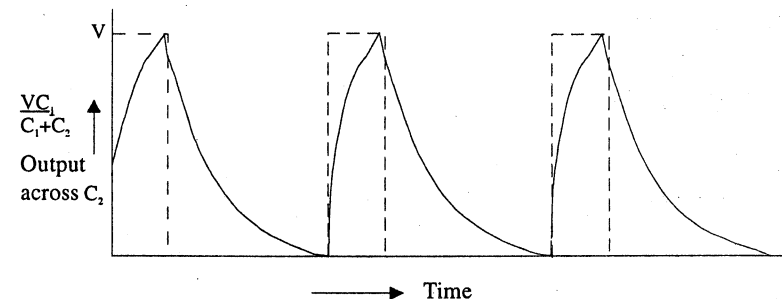


Fig. 6.18
Diode pump voltage waveform

proportional to C_1 . Typical values for the circuit shown in fig. 6.17 is $C_1 = 10\mu\text{f}$, $R = 10\text{K}$, $C_2 = 20$ to 40 pf , output is 1mV/pf .

6.5.5 Pulse Width Modulating Circuit

A differential capacitor variation is transduced into a pulse width by the circuit shown in fig. 6.19. the circuit shown in fig. 6.20 is the IC version of the above principle for transducing the capacitance variation into a digital counter output. When the trigger is applied, the output goes high and the And gate is enabled and the

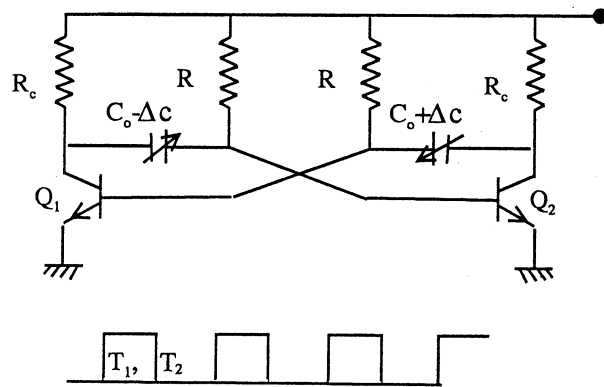


Fig. 6.19

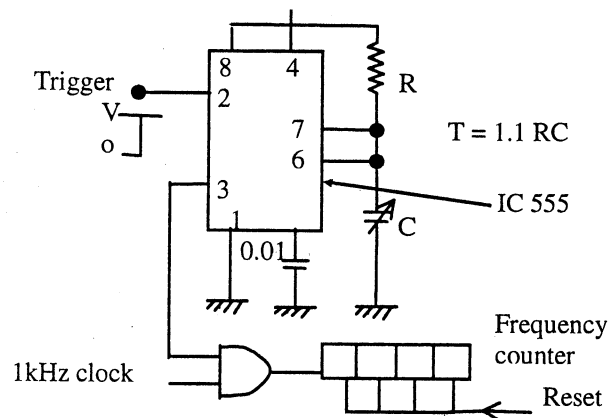


Fig. 6.20

IC version of PWM circuit and a counter

counter starts counting. The counting is inhibited after a time $T = 1.1RC$. For example if we use a 4 digit counter and the clock is 1 KHz, 100 pf can be displayed as 100.0 if $T = 1\text{ sec}$.

$$T = 1.1 R \times C \quad (6.29)$$

$$R = 1 \text{ sec} / (1.1 \times 100 \times 10^{-12}) \quad (6.30)$$

6.8 WORKED OUT EXAMPLES

EXAMPLE 6.8.1

A displacement measuring system consists of 7 parallel plates arranged as shown in Fig. 6.21.

Each plate is of area $5\text{ cm} \times 5\text{ cm}$ and the gap between the plates is 0.25 mm .

Calculate the capacitance when the overlapping distance is 4 cm. The permittivity of air is $8.85 \times 10^{-12}\text{ f/m}$. Calculate also the sensitivity.

SOLUTION

When the overlapping distance is 4 cm, the capacitance between one pair is $C = \frac{A \epsilon}{d}$

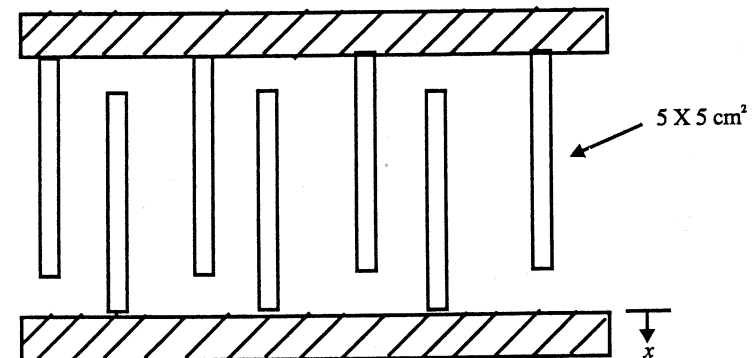


Fig. 6.21

Capacitor bank

$$\begin{aligned}
 &= \frac{4 \times 5 \times 10^{-4} \times 8.85 \times 10^{-12}}{.25 \times 10^{-3}} \\
 &= 708 \times 10^{-1} \times 10^{-12} \\
 &= 70.8 \times 10^{-12} \text{ farads}
 \end{aligned}$$

7 plates form 6 capacitances connected in parallel.

$$\begin{aligned}
 \text{Hence total capacitance} &= 6 \times 70.8 \times 10^{-12} \\
 &= 424.8 \text{ picofarads.}
 \end{aligned}$$

When the distance moved is x cm the capacitance is

$$C_T = \frac{6 \times 5 \times (4 - x) \times 10^{-4} \times 8.85 \times 10^{-12}}{.25 \times 10^{-3}}$$

The sensitivity is

$$\begin{aligned}
 \frac{dC_T}{dx} &= \frac{-6 \times 5 \times 10^{-4} \times 8.85 \times 10^{-12}}{.25 \times 10^{-3}} = 106.2 \text{ pf/cm} \\
 &= -10620 \text{ pf/m} \\
 &= 10.62 \text{ pf/mm}
 \end{aligned}$$

EXAMPLE 6.8.2

A thickness measuring transducer system has a parallel plate capacitive sensors having a pair of plates of area of $10 \text{ cm} \times 20 \text{ cm}$, which are separated by a distance of 0.02 cm . A mica sheet of thickness $0.01 \pm .001 \text{ cm}$ is being passed between the plates calculate the variation of capacitance when the dielectric constant of mica is 8 and the permittivity of air is $8.85 \times 10^{-12} \text{ F/m}$

SOLUTION

Capacitance of the system when the thickness of mica (ref fig. 6.22)

$$\text{is } 0.01 \text{ cm is given by } \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$C = \frac{C_1 C_2}{C_2 + C_1}$$

where C_1 & C_2 capacitances of air & mica respectively

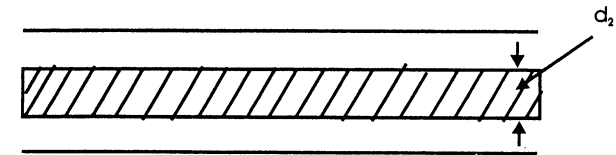


Fig. 6.22

$$C = \frac{A\epsilon_0}{d_1/\epsilon_1 + d_2/\epsilon_2}$$

Where

A is the area of the plates

d_2 is the thickness of mica

d_1 is the remaining air thickness between plates

ϵ_1 is relative permittivity of air

ϵ_2 is relative permittivity of mica

ϵ_0 is the absolute permittivity of air

$$\begin{aligned}
 &= \frac{10 \times 20 \times 10^{-4} (\text{m}^2) \times 8.85 \times 10^{-12} (\text{F/m})}{(.01 \times 10^{-2}/1) + (.01 \times 10^{-2}/8)} \\
 &= \frac{1770 \times 10^{-16}}{.01125 \times 10^{-2}} = 1573.3 \times 10^{-12} \\
 &= 1573.3 \text{ pf}
 \end{aligned}$$

When the mica thickness is $0.01 + .001 \text{ cm}$, the maximum value, then $d_1 = .009$ & $d_2 = .011$

$$\begin{aligned}
 C &= \frac{1770 \times 10^{-16}}{.009 \times 10^{-2} + \left(\frac{.001 \times 10^{-2}}{8} \right)} \\
 &= \frac{1770 \times 10^{-14}}{.010375} \\
 &= 1706.02 \times 10^{-12}
 \end{aligned}$$

when the mica thickness is $.01 - .001 = .009$

$$\begin{aligned}
 C &= \frac{1770 \times 10^{-16}}{.011 \times 10^{-12} + \left(\frac{.009 \times 10^{-2}}{8} \right)} \\
 &= \frac{1770 \times 10^{-14}}{.012125} \\
 &= 1459.79 \times 10^{-12}
 \end{aligned}$$

Therefore when mica is being passed through the plates, the capacitance changes from $1459.79 \text{ pf} \rightarrow 1706 \text{ pf}$ with a nominal value of 1573 pf .

For a mica thickness of $.01 + .001$, $C = 1573 + 133 = 1706 \text{ pf}$
 & for a mica thickness of $.01 - .001$, $C = 1573 - 113.2 = 1459 \text{ pf}$

EXAMPLE 6.8.3

In the variable capacitance velocity pick up shown in fig. 6.23., prove that the current i is directly proportional to the angular velocity $\frac{d\theta}{dt}$

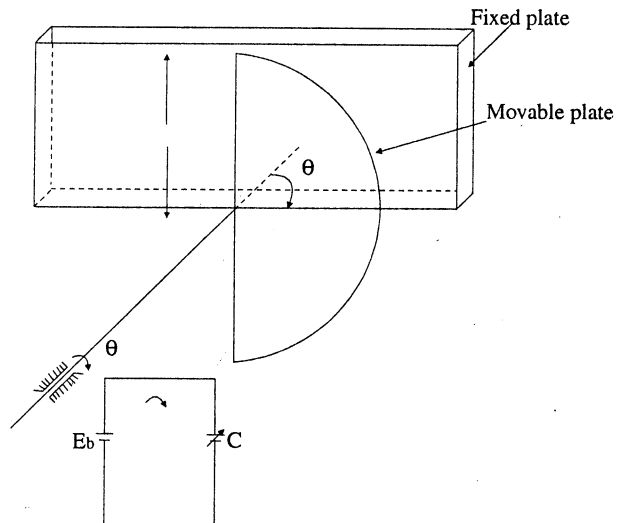


Fig. 6.23
For problem 6.8.3

The overlapping area $= \frac{\theta}{360} \times \pi r^2$

$$C = \frac{\pi r^2}{180} \frac{\theta \epsilon}{d}$$

Where r is the radius of the circular disc ϵ is the permittivity of air & d is the displacement between plates

$= k \theta$ as all the quantities are constant except θ . For the circuit shown,

$$i = \frac{d}{dt} (CE_b) = \frac{dC}{dt} \cdot E_b + C \frac{dE_b}{dt}$$

$$= E_b \frac{dC}{dt} \text{ as } E_b \text{ is constant.}$$

$$i = E_b \frac{d(k\theta)}{dt} = kE_b \frac{d\theta}{dt}$$

Thus the current is proportional to the angular velocity $\frac{d\theta}{dt}$.

EXAMPLE 6.8.4

A capacitive transducer is made up of two concentric cylinder plates as shown in fig 6.24. The inner diameter of the outer cylinder is 5.2 mm and the outer diameter of the inner cylinder is 5 mm . The dielectric medium is air. The length of the electrodes is 25 mm . Calculate the capacitance value.

If the outer cylinder is pulled up by 5 mm , what is the capacitance change.

SOLUTION

The air gap between the cylinders is $(5.2 - 5) / 2 = .1 \text{ mm}$.

Capacitance of a concentric cylinders is given by

$$C = \frac{2\pi \epsilon l}{\ln (D_2/D_1)}$$

Where l is the overlapping length

D_2 is the inner dia of the outer cylinder

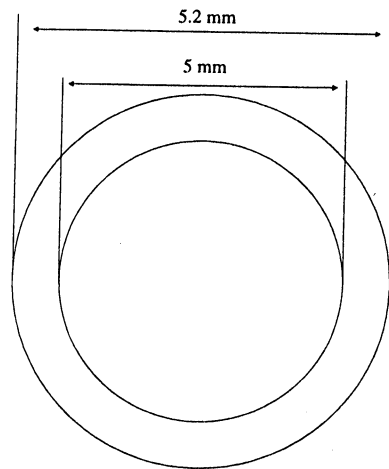


Fig. 6.24
For problem 6.84

D_1 is the outer dia of the inner cylinder
using the above equation to calculate the capacitance for the problem

$$C = \frac{2\pi \times 8.85 \times 10^{-12} \times 25 \times 10^{-3}}{\ln(5.2/5)} \frac{F}{m}$$

$$= 35.445 \text{ pf}$$

If the outer cylinder is pulled up by 5 mm then

$$C = \frac{2\pi \times 8.85 \times 10^{-12} \times 20 \times 10^{-3}}{\ln(5.2/5)}$$

$$= 28.356 \times 10^{-12} \text{ F}$$

$$= 28.356 \text{ pf}$$

The change in capacitance for a 5mm is
 $= 35.445 - 28.356 \text{ pf}$
 $= 7.089 \text{ pf}$

EXAMPLE 6.8.5

A capacitor with flat circular plates of 1.5 cm dia is separated by a parallel gap of 0.02 mm

- What is the sensitivity in farads/mm of the pick up when displacement causes a variation in the gap between the plates.
- If this pick up is connected in the feedback path of an operational amplifier shown in fig 6.25. What is the voltage sensitivity?
- What is the maximum input voltage that can be applied if air exhibits a dielectric breakdown at fields of about 50 000 volts/cm.

SOLUTION

$$(i) \quad C = A \epsilon / d$$

$$= \frac{\pi \times (1.5)^2}{4} \times 10^{-4} \times 8.85 \times 10^{-12}$$

$$= \frac{g}{\text{where } g = \text{gap in m.}}$$

$$\frac{dc}{dg} = - \frac{15.63 \times 10^{-16}}{g^2} = - \frac{15.63 \times 10^{-16}}{(.02 \times 10^{-3})^2} \text{ farads / m.}$$

$$= 39078 \times 10^{-10} \text{ farads / m}$$

$$= 3907.8 \text{ pico farads / mm}$$

$$(ii) \quad C \frac{dV_o}{dt} = \frac{V_i}{R}$$

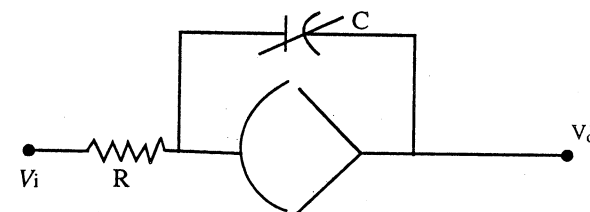


Fig. 6.25
For problem 6.85

$$C V_o = \frac{1}{R} \int V_i dt$$

Let $R = 1M \Omega$

$$V_o = \frac{1}{RC} \int V_i dt$$

To get the voltage sensitivity w.r.to displacement or gap,
replace C by $C = \frac{A\epsilon}{g}$

$$V_o = \frac{g}{RA\epsilon} \int V_i dt$$

Then the voltage sensitivity is

$$\frac{dV_o}{dg} = \frac{1}{RA\epsilon} \int V_i dt$$

Let us calculate

$$\begin{aligned} RA\epsilon &= 1 \times 10^6 \times \frac{\pi \times (1.5)^2 \times 10^{-4}}{4} \times 8.85 \times 10^{-12} \\ &= 15.63 \times 10^{-10} \end{aligned}$$

Then

$$\frac{dV_o}{dg} = \frac{10^{10}}{15.63} \int V_i dt \text{ Volts/m}$$

- (iii) This being an integrater circuit, any input voltage will get integrated and the output voltage of the op-amp. can not be more than the supply voltage. Normal supply voltage being 15 volts, the capacitor will never get punctured.

PIEZOELECTRIC TRANSDUCERS

7.1 INTRODUCTION

An important class of transducers is Piezoelectric transducers. They convert pressure or force into electric charge. These transducers are based upon the natural phenomenon of certain non-metals and dielectric compounds. Materials like Quartz, Rochelle salt, barium titanate develop electrical charge when they are subjected to pressure or force. This effect is reversible meaning thereby if a charge is applied on the surface of these materials, they are deformed. This phenomenon is known as piezoelectric effect. This word piezo is a Greek word meaning force or pressure.

The mechanical deformation producing electrical charge is the basis of many instruments for measuring acceleration, force and pressure. It can also be used as a means of generating high-voltage low current electrical power such as that used in spark-ignition electrode and electrostatic dust filters.

The electrical input / mechanical output direction is applied in small vibration shaker, sonar systems for acoustic detection and location of underwater objects and industrial ultrasonic nondestructive test equipment.

7.2 MATERIALS FOR PIEZOELECTRIC TRANSDUCERS

Twenty of the thirty-two crystallographic elements exhibit piezoelectric properties. Only few materials are useful for piezoelectric transducers. Primarily quartz, Rochelle salt, ammonium dihydrogen phosphate (ADP) and ceramics made with barium titanate, dipotassium tartrate, potassium dihydrogen

phosphate and lithium sulfate are some of the materials used in real applications.

7.2.1 Quartz (SiO_2)

Quartz is mechanically and thermally the most stable among piezoelectric materials. Its internal electric losses are small. Its volume resistivity is higher than 10^{14} ohm-cm. Because of its relatively small piezoelectric effect its application is restricted to such uses where high tensile strength, high mechanical stability or operation at elevated temperature are envisaged. This material can be operated safely up to 550°C . Quartz plates are used primarily for thickness expansion and transverse compression operation.

7.2.2 Rochelle Salt : ($\text{NaKC}_4\text{H}_4\text{O}_6 \cdot \text{H}_2\text{O}$)

$\text{NaKC}_4\text{H}_4\text{O}_6 \cdot \text{H}_2\text{O}$ is stable at room temperature between 35 and 85 percent relative humidity. At higher humidity it will deliquesce and at lower humidity it will dehydrate. Coating with wax is recommended to combat the humidity effects. At a temperature of about 55°C , the crystal will disintegrate. The sensitivity of crystal varies with direction of application of force. Any crystal will have three main axes and the crystal can be cut in different ways choosing X cut or Y cut axes. For Rochelle Salt the dielectric constant in the direction of the Y axis of the crystal varies within wide limits with the temperature and also the resonance frequency of the crystal changes with temperature.

The X cut of the Rochelle salt crystal is primarily for face shear motion, crystals can also be cut for transverse (longitudinal) expansion and the cuts can be combined to form bender or twister bimorph system.

The volume resistivity is of the order of 10^{12} ohm-cm or more. The mechanical strength is considerably lower than that of quartz.

7.2.3 Ammonium Dihydrogen Phosphate : ($\text{NH}_4\text{H}_2\text{PO}_4$)

Ammonium Dihydrogen Phosphate abbreviated as ADP has a higher thermal stability than Rochelle salt. It is stable up to a temperature of 180°C but is used for technical reasons with upper limit of about 125°C . It operates satisfactorily in a range from 0 to 44 percent

relative humidity, if condensation of water on the surface (causing surface leakage) is avoided. The volume resistivity is considerably lower than that of the other substances described above. It is about 10^{10} ohm-cm and decreases with increasing temperature and content of impurities.

The material (primarily the Z cut) is applicable for face shear and longitudinal expansion operation and also for twister bimorphs. The sensitivity is high. A crystal of 1 cm thick gives, for a load of 1 dyne/cm^2 , an output voltage of $1.78 \times 10^{-4} \text{ V}$.

7.2.4 Barium Titanate Ceramics (BaTiO_3)

The single barium titanate crystal is ferroelectric i.e. if exposed to an electric field it becomes polarized and maintains its polarization after the field is removed. Application of the field in the reverse direction causes a reverse polarization. The process follows a hysteresis curve.

In the ceramic polycrystalline form, the individual crystals are randomly oriented. By the application of a polarised electric field (of order of 1000 to several thousand volts per millimeter for about 1 hr. at room temperature) the ferroelectric domains are lined up and the materials becomes piezoelectric. The material loses some of the piezoelectric activity in the first few days and remains fairly constant afterwards. The aging process can be accelerated by subjecting the material to a higher temperature (80°C). The addition of about 4 percent lead titanate increases the long time stability.

Barium titanate can be used for thickness expansion and for transverse expansion operation. They can be combined to form bender bimorphs but they do not operate in shear modes and consequently they cannot be combined to form twister bimorphs.

7.3 EQUIVALENT CIRCUIT OF A PIEZOELECTRIC CRYSTAL

Consider fig. 7.1. The charge generated by the crystal can be expressed as

$$q = K_q X_i$$

where K_q is in coulomb/cm and X_i is the deformation in cm. The resistances and capacitances in the equivalent circuit of fig. 7.1b can be combined as shown in fig. 7.1c. The charge generated can be converted into a current generator as $i = dq/dt$.

For the circuit in fig. 7.1c, the following equations can be written

$$i_{cr} = \frac{dq}{dt} = K_q \frac{dx_i}{dt} = K_q \dot{x}_i \quad (7.1)$$

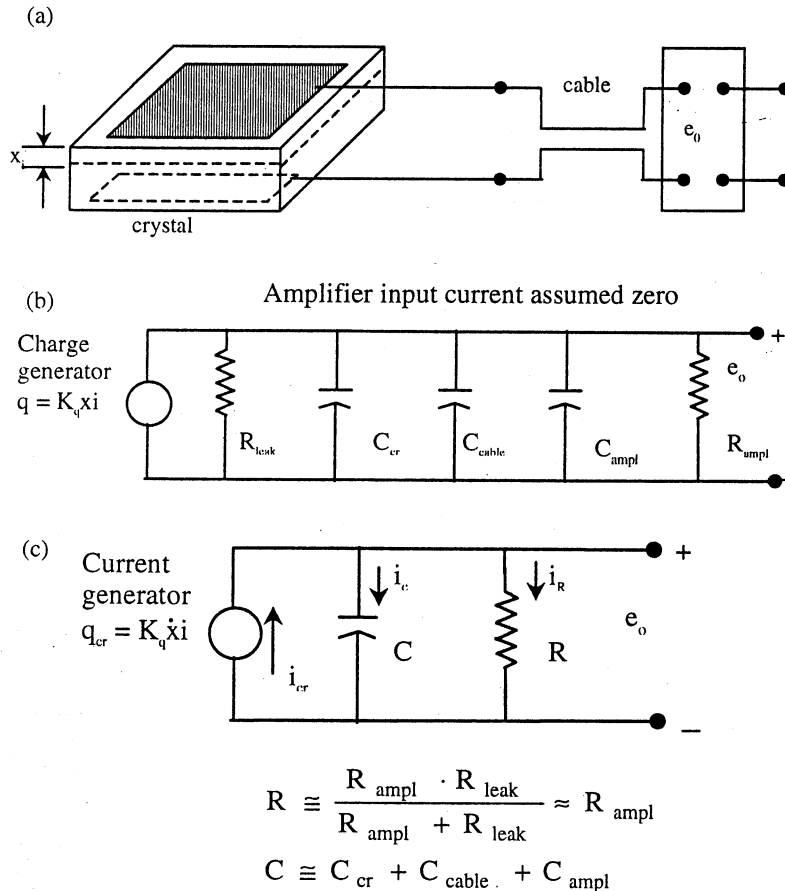


Fig. 7.1

Equivalent circuit of piezoelectric crystal

Then

$$i_{cr} = i_c + i_R \quad (7.2)$$

$$K_q \frac{dx_i}{dt} = C \frac{de_0}{dt} + \frac{e_0}{R} \quad (7.3)$$

Taking the laplace transform on both sides, assuming zero initial conditions,

$$K_q s x_1(s) = C s E_0(s) + \frac{E_0(s)}{R} \quad (7.4)$$

Then

$$\frac{E_0(s)}{X_1(s)} = \frac{K\tau s}{\tau s + 1} \quad (7.5)$$

where $K = K_q/C = \text{sensitivity in volts / cm}$
 $\tau = RC = \text{Time constant in sec.}$

This has a steady state response to a constant x_i is zero just like the capacitance pick up. So static displacement cannot be measured. For a flat amplitude response within say 5 percent, the frequency must exceed ω_1 .

$$\text{Where } (0.95)^2 = (\omega_1 \tau)^2 / [(\omega_1 \tau)^2 + 1] \quad (7.6)$$

$$\text{And } \omega_1 = 3.02/\tau \quad (7.7)$$

To get a better low frequency response, the value of τ should be larger. We can also show that larger the value of τ , the better will be the pulse response and time response of this device.

7.4 PIEZOELECTRIC COEFFICIENTS

A brief discussion of the definitions and relations of the coefficients used to describe piezoelectric properties is given in this section. Double subscripted notation is used to represent the constant. The convention is that the first subscript refers to the direction of the electrical effect and the second to that of mechanical effect using the axis numbering system shown in fig. 7.2.

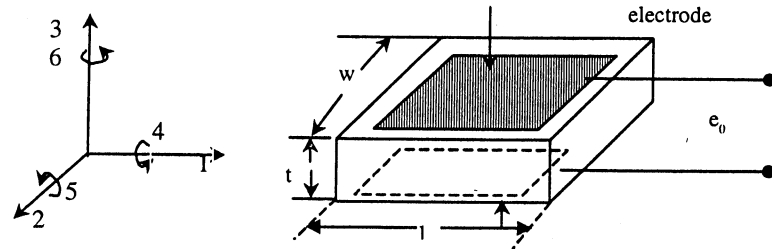


Fig. 7.2
Axis numbering system in crystals

7.4.1 d-coefficient

This is the fundamental piezoelectric coefficient. It is related to the charge generation. The direction of the applied force with respect to the crystal axes, is indicated by two subscripts which are derived from the tensor representation of stress and strain in crystals. Because of crystal symmetry in quartz, only the following coefficients are significant :

$$\begin{aligned} d_{11} &= 2.26 \times 10^{-11} \text{ coulombs/kg} \\ &= 2.30 \times 10^{-12} \text{ coulombs/Newton} \\ d_{14} &= -0.655 \times 10^{-11} \text{ coulombs/kg} \\ &= -0.670 \times 10^{-12} \text{ coulombs/Newton} \end{aligned}$$

$$\text{also, } d_{12} = -d_{11}$$

$$d_{25} = -d_{14}$$

$$d_{26} = -2d_{11}$$

The d-coefficient gives the charge output per unit force input (or charge density per unit pressure) under short-circuit condition, i.e. If the charge is being led away on developing, so that it has no second-order effect on the crystal deformation. In the m.k.s. system it is measured in

$$\frac{\text{Coulombs}}{\text{Newton}} = \frac{\text{Coulombs/m}^2}{\text{Newton/m}^2} \quad (\text{direct effect})$$

In a system of dimensions in which the mechanical and electrical unit of energy is the same, such as the m.k.s. system, the

same numerical coefficient serves the direct and the inverse piezoelectric effect. In the inverse piezoelectric effect the d-coefficient measures the deformation obtained per unit applied voltage under no-load conditions, i.e.

$$\frac{m}{V} = \frac{m/m}{V/m} \quad (\text{inverse effect})$$

7.4.2 g-coefficient

Dividing the d-coefficient by the absolute dielectric constant,

$$\epsilon \epsilon_0 = \epsilon \times 8.85 \times 10^{-12} \text{ (m.k.s)}$$

yields the g-coefficient representing the generated e.m.f. gradient per unit pressure input. It is the most convenient coefficient for computing the output voltage of piezoelectric transducer, if additional shunt capacity is not considered. Its dimension is

$$\frac{V/m}{\text{Newton/m}^2} = \frac{V \cdot m}{\text{Newton}}$$

7.4.3 h-coefficient

While the d-and g-coefficients are related to the applied forces, the h-coefficient is related to a given deformation of the crystal. It is obtained by multiplying the g-coefficient by Young's modulus valid for the appropriate crystal orientation of the material, and thus measures the e.m.f. gradient per unit mechanical deformation, or (V/m)/(m/m).

7.4.4 Coupling Coefficient k

The coupling coefficient can be computed from the square root of the product of h and d. it represents the square root of the ratio of the mechanical energy stored in the crystal, to the electric energy supplied, and is thus a measure of the efficiency of the crystal as an energy converter. It is mainly used in piezoelectric generator-type transducer where the transducer is employed as an electro-mechanical vibrator, and in crystal-type filters. In principle a comparison of the coupling factors of different materials permits the assessment of their relative sensitivities without referring to crystal dimensions. When considering generator-type transducer, however, one must realize that they are usually loaded

mechanically whilst the coupling coefficient refers to short-circuit output conditions, i.e. without mechanical restraint. The required correction, however, is small with materials of low coupling coefficients, such as quartz or even barium titanate, but may approach a value of 2 for Rochelle salt. In instrument transducer work, the coupling coefficient is probably less important than in acoustic and ultrasonic applications.

The piezoelectric sensitivity of quartz varies with temperature. Between 20 and 200°C, the temperature coefficient of d_{11} is about -0.016 percent per degree C. At lower temperatures the variation is less pronounced. If, however, taken above the melting-point, quartz loses its piezoelectric character permanently. While this dielectric constant changes very little at temperatures up to about 250°C, there is an immediate and continuously increasing drop of resistivity, amounting to about six orders

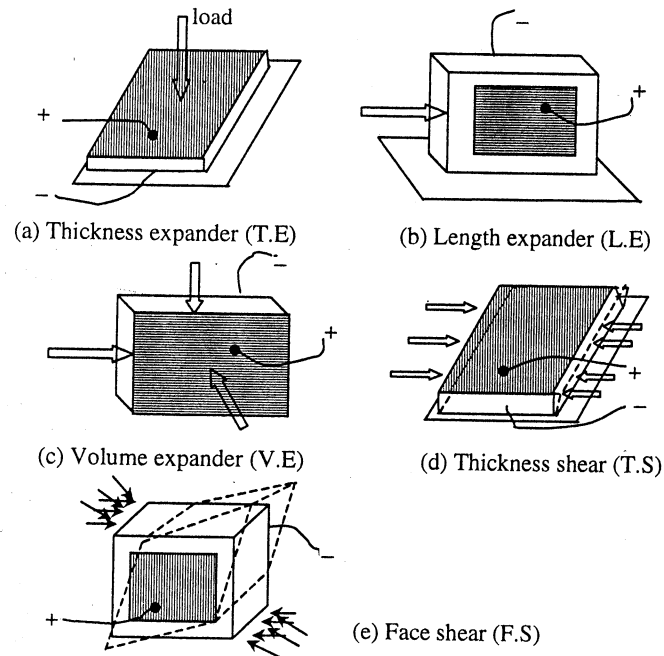


Fig. 7.3

Different ways of deformation of the crystal

between room and Curie-point temperature. Quasi-static measurements with quartz-type transducers at high environmental temperatures are therefore impracticable.

7.5 MODES OF DEFORMATION

There are a number of possible modes of deformation a piezoelectric transducer may suffer under load. They are portrayed in fig. 7.3. Electrodes are placed in axis 1, which are shown as lined face.

7.6 MULTIPLE ARRANGEMENTS (STACKS)

The elements may be connected in series leading to a higher output voltage for the same force. If they are connected in parallel, then a lower output impedance is the result.

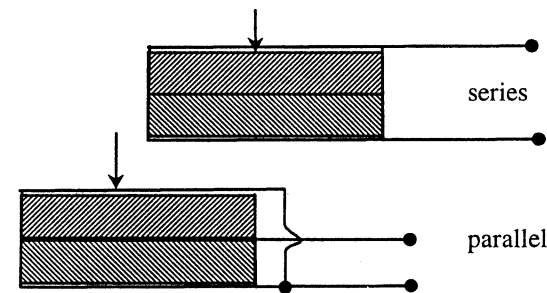


Fig. 7.4

Stacks of crystals

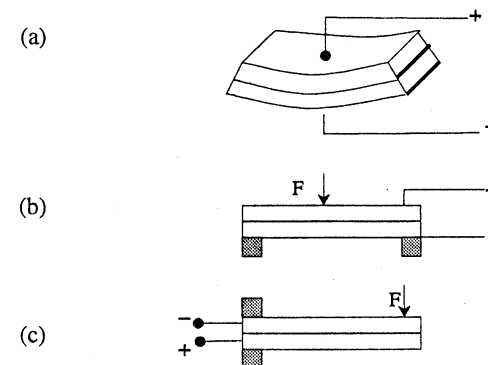


Fig. 7.5

Bimorphs benders

7.6.1 Bimorphs Bender

Bimorphs is the trade name of Brush Electronic Company's Product. Bimorph bender consist of two transverse expanding plates cemented together in such a manner that one plate contracts and the other expands when a voltage is applied.

If the element is free to move, then it will bend as shown (exaggerated) in fig. 7.5a. Fig. 7.5b. and fig. 7.5c show how the Bimorphs can be used to transduce force into a voltage by using as a simply supported beam or cantilever beam. These Bimorph elements has got a higher sensitivity and permits larger deflection than a single solid one.

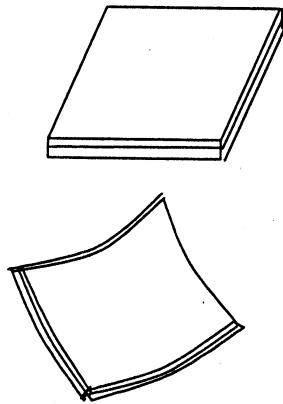


Fig. 7.6
Bimorph twisters

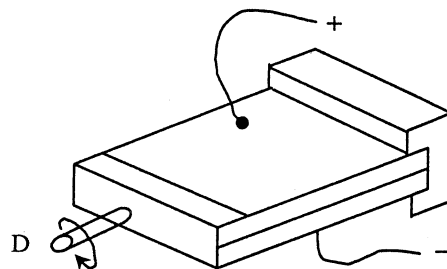


Fig. 7.7
Bimorph torque sensor

7.6.2 Bimorph Twisters

Two face shear plates are cemented together to have a series connection so that their expanding diagonals are perpendicular. If a voltage is applied, and if both plates are free to move then it will bend as shown in fig. 7.6. For transducing torque, the Bimorph twisters can be used as shown in fig. 7.7.

7.7 GENERAL FORM OF PIEZOELECTRIC TRANSDUCERS

Construction details of a piezoelectric accelerometer is as shown in fig. 7.8. The crystal is preloaded to about 600 kg/cm^2 stress by screwing down on the hemispherical spring. The prestressing puts the piezoelectric material at a more linear part of its stress charge curve. Vertical acceleration is converted into a varying force by the

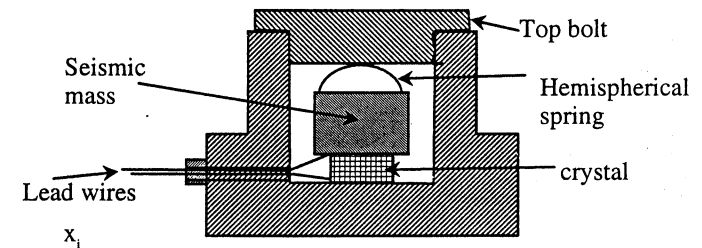


Fig. 7.8
Piezoelectric accelerometer

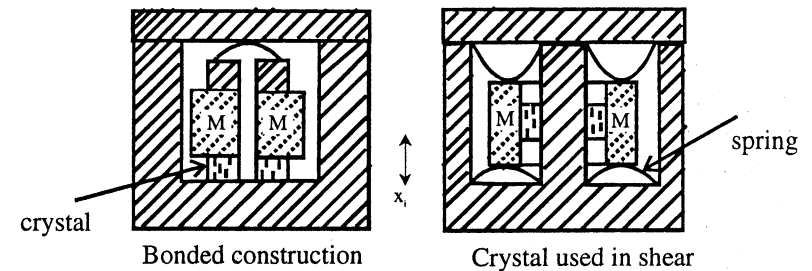


Fig. 7.9
Accelerometer to reduce cross axis sensitivity

seismic mass. It allows measurement of acceleration in both directions without the crystal going into tension. When the preload is applied, a voltage of certain polarity is developed but this soon leaks off to zero. Any further load due to acceleration gives a plus or minus charge depending on the direction of the motion.

7.7.1 Available Typical Piezoelectric Accelerometers

A single low 'g' instrument has a sensitivity of 50 mV/g and will measure acceleration from .03 to 1000 g with a non linearity of 1 percent of full scale. It has a natural frequency of 20 KHz and frequency response of flat ± 5 percent from 20-4000 Hz. When used with 100 mega ohms input impedance capacitance of 600 pf with 1m cable,

cross axis sensitivity is 5 percent

sensitivity drift is ± 10 percent from -30 to $+230^\circ$

size 3 cm cube.

7.7.2 Typical Specification of a Shock Accelerometer

- sensitivity : 5V/g
- range 0 to 1000 g with 1 percent non linearity.
- natural frequency of 35 KHz
- frequency response flat with ± 5 percent from 0.1 to 7000 Hz with charge amplifier.
- pulse response ± 5 percent for pulses shorter than 0.66 sec. No ringing for pulses longer than 0.15 m sec.
- capacitance of 100 pf.
- cross axis sensitivity of 5 percent
- sensitivity drift ± 1 percent from -100 to $\pm 350^\circ\text{C}$

7.8 ENVIRONMENTAL EFFECTS

7.8.1 Effect of Temperature

The temperature dependence of piezoelectric material is far too complex. Very little is known about the temperature dependence of charge sensitivity though it is known from experience that it often roughly follows the tendency of permittivity / temperature characteristics.

7.8.2 Effect of Humidity

Humidity may affect piezoelectric materials in two ways.

- Water soluble crystals if not protected, can stand only a limited range of environmental humidity. Air of lower humidity extracts the water of crystallization and makes it amorphous and higher humidity may dissolve the crystal.
- All piezoelectric materials suffer from loss of insulation resistance when water vapour condenses on their faces.

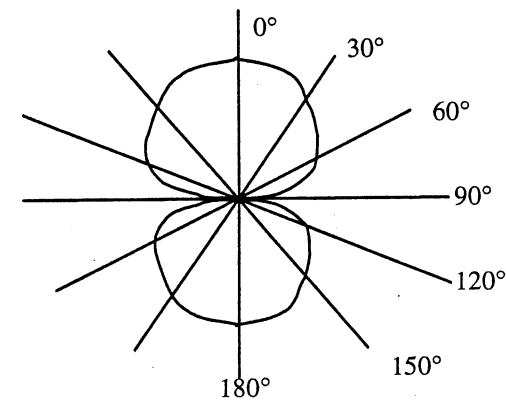


Fig. 7.10
Sensitivity in a radial direction

So water proof coating is a must. Silicone grease is filled into cavities of transducers and connectors.

7.8.3 Transverse Sensitivity

If a linear acceleration transducer is exposed to an acceleration in a radial direction perpendicular to the main axis (direction of polarization) the sensitivity assumes a two lobed shape as shown schematically in fig. 7.10.

7.8.4 Pressure Sensitivity

Pressure sensitivity problem is encountered in piezoelectric acceleration transducers to airborne high intensity noise. Materials with volume expander mode can produce a signal when exposed

to sound pressure. But if a transducer is designed to prevent the exposure of sound pressure to some crystal faces say the cylindrical faces of a disk enclosed in a sealed housing, then a net pressure sensitivity may not be experienced.

7.9 WORKED OUT EXAMPLE

EXAMPLE 7.9.1

A piezo electric transducer has a capacitance of 1000 pf and a charge sensitivity of 4×10^{-6} coulomb/cm.

The connecting cable has a capacitance of 400 pf. The display device has an input impedance of $1 \text{ M}\Omega$ resistance and 50 pf capacitance connected in parallel.

- What is the voltage sensitivity of the piezo electric transducer alone?
- What is the high frequency sensitivity in V/cm of the entire system?
- What is the lowest frequency that can be measured with 2 percent error?

SOLUTION

- The voltage sensitivity of the piezo electric transducer alone is

$$K = \frac{kq}{C} \text{ volts/cm.}$$

$$= \frac{4 \times 10^{-6}}{1000 \times 10^{-2}} = 4 \times 10^3 \text{ volts/cm}$$

- The circuit diagram of the entire system is
This can be reduced to (see fig. 7.12)
The high frequency gain = K

$$= \frac{kq}{C_T} = \frac{4 \times 10^{-6}}{1450 \times 10^{-12}} = 2.7 \times 10^3 \text{ v/cm.}$$

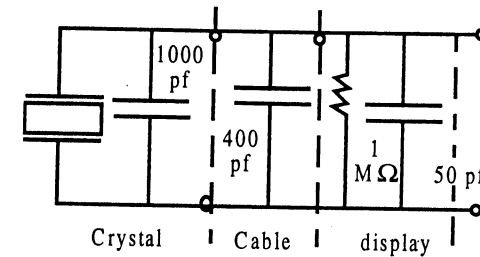


Fig. 7.11
Circuit diagram of the entire system

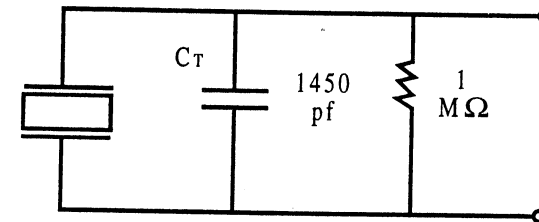


Fig. 7.12
Reduced diagram

- The transfer function of this system is

$$\frac{K \tau s}{1 + s\tau}$$

where

$$K = \frac{kq}{1450 \times 10^{-12}} = \frac{4 \times 10^{-6}}{1450 \times 10^{-12}}$$

$$= 2.7 \times 10^3 \text{ v/cm.}$$

$$\tau = RC = 1 \times 10^6 \times 1450 \times 10^{-12}$$

$$= 1.45 \times 10^{-3} \text{ sec.}$$

The frequency transfer function of the system is = $\frac{K \tau (j\omega)}{1 + j\omega\tau}$

When the error is 2%, the magnitude of the system output is .98K

Therefore, $.98K = \frac{K \omega \tau}{\sqrt{1 + \omega^2 \tau^2}}$

Squaring both sides, we get

$$(.98)^2 = (\omega \tau)^2 / (1 + \omega^2 \tau^2)$$

i.e., $.9604 (1 + \omega^2 \tau^2) = \omega^2 \tau^2$

$$0.0396 \omega^2 \tau^2 = .9604$$

$$\omega^2 \tau^2 = 24.2$$

$$\omega \tau = 4.924$$

$$\omega = 4.924 / \tau = 4.924 / 1.45 \times 10^{-3}$$

$$= 3.396 \times 10^3 \text{ radian/sec.}$$

$$\omega = 3396 \text{ rad/sec.}$$

The lowest frequency f_{low} for 2 percent error

$$f_{\text{low}} = 3396 / 2\pi \text{ cycles/sec}$$

$$= 540.7 \text{ Hz.}$$

EXAMPLE 7.9.2

A piezo electric transducer has a capacitance of 1500 pf and a charge sensitivity of 120×10^{-12} coulombs/Newton. The leakage resistance of the transducer is 10^5 megaohms. The impedance of the measuring system is 1 megaohm parallel with a capacitance of 500 pf.

A pulse force is applied on the transducer with a magnitude of 0.1N and duration 3 millise. Find the response of the transducer.

SOLUTION

The capacitance of the system =

$$C_T = C_{cr} + C_m = 1500 + 500$$

$$= 2000 \text{ pf.}$$

The resistance of the system

$$R_T = \frac{R_L R_M}{R_L + R_M} = R_M \text{ as } R_L \text{ is very large}$$

$$= 10^6 \text{ ohms.}$$

$\therefore$ The time constant of the system

$$\tau = R_T C_T = 10^6 \times 2000 \times 10^{-12}$$

$$= 2 \text{ millise.}$$

The sensitivity of the entire system is

$$K = \frac{kq}{C} = \frac{120 \times 10^{-12}}{2000 \times 10^{-12}} = .06 \text{ V/N.}$$

The transfer function of the system is

$$\frac{\text{Voltage}}{\text{Force}} = \frac{V(s)}{R(s)} = \frac{K T s}{1 + T s}$$

$$= \frac{.06 \times 2 \times 10^{-3} \times s}{1 + 2 \times 10^{-3} s}$$

The pulse input can be considered as two step inputs $u_1(t)$ and $u_2(t)$ as shown in fig. 7.13

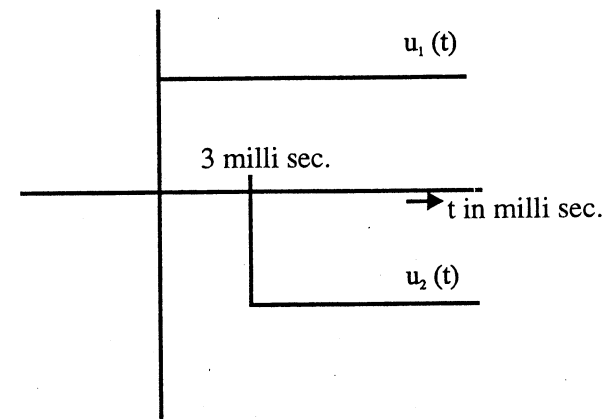


Fig. 7.13
Pulse input as two step inputs.

As far as the response in the initial period upto 3 seconds is considered it is just the step response.

It can be shown that the step response for the piezo electric transducer, whose transfer function is $\frac{K \tau s}{1 + s \tau}$

$V(t) = K R e^{-t/\tau}$, where R is step magnitude

$$\therefore V(t) = 0.06 \times 0.1 \times e^{-t/2 \times 10^{-3}}$$

$$= 6 \times 10^{-3} e^{-t/2 \times 10^{-3}}$$

If t is taken in milli seconds, then $V(t) = 6 \times 10^{-3} e^{-t/2}$,

When the time t becomes larger than 3 milli sec. the duration of the pulse, the response is due one step from 0 time & another negative step starting from 3 msec. for $t > 3$ millisec.

$$V(t) = 6 \times 10^{-3} e^{-t/2} - 6 \times 10^{-3} e^{-\frac{(t-3)}{2}}$$

$$= 6 \times 10^{-3} \left[e^{-t/2} (1 - e^{3/2}) \right]$$

$$= 6 \times 10^{-3} \left[e^{-t/2} (1 - e^{3/2}) \right]$$

The response is shown in fig 7.14

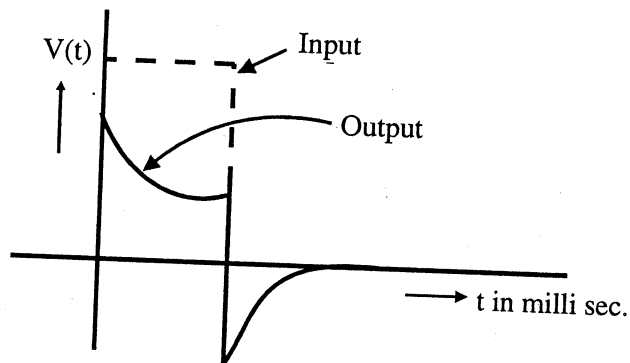


Fig. 7.14

Pulse response of piezoelectric transducer.

EXAMPLE 7.9.3

A piezo electric crystal has a thickness of 2 mm and voltage sensitivity of 0.056 v-m/N. It is subjected to a pressure of 0.5×10^6 N/m² calculate the voltage output..If the permittivity of quartz is 40.6 picofarads/meter, what is the charge sensitivity.

SOLUTION

The output voltage

$$= \text{voltage sensitivity} \times \text{thickness} \times \text{pressure}$$

$$= 0.056 \times 2 \times 10^{-3} \times 0.5 \times 10^6$$

$$= 56 \text{ volts.}$$

Let the charge sensitivity be d coulomb/Newton.

(charge generated per unit force in Newton)

Q = charge developed for a force of F-units

We know $V = \frac{Q}{C}$

$$V = \frac{dF}{A \epsilon_0 \epsilon_r / t}$$

Let the pressure be $p = \frac{F}{A}$

$$V = \frac{d p t}{\epsilon_0 \epsilon_r}$$

where 'A' is the Area of the crystal, ϵ_0 , ϵ_r are the permittivity, t is the thickness of the crystal.

$$d = \frac{V \epsilon_0 \epsilon_r}{p t}$$

We know $\frac{V}{p t} = g$ is the voltage sensitivity of the crystal.

$$\therefore d = g \epsilon_0 \epsilon_r$$

For this problem $\epsilon_0 \epsilon_r$ is given as 40.6 picofarad/meter

$$d = 0.056 \times 40.6 \times 10^{-12}$$

$$= 2.27 \times 10^{-12} \text{ coulomb/Newton.}$$

MISCELLANEOUS TRANSDUCERS

In the previous chapters we have seen a number of transducers based on variable resistance, variable inductance, variable capacitance and piezoelectric property of crystals. There are many other physical properties of materials which can also give rise to a number of transducers. In this chapter, a number of such physical properties and their associated transducers are explained.

8.1 MAGNETOSTRICTIVE TRANSDUCERS

8.1.1 Ferromagnetic materials

Ferromagnetic materials like iron, nickel, 68 permalloy etc., change their magnetic permeability under mechanical stress. This is known as Villari effect.

The permeability can increase or decrease depending upon the material, the type of stress, (compression, tension or torsion) and the magnetic flux density in the sample.

The permeability variation is shown in fig. 8.1 for iron and nickel for different conditions. The ferromagnetic elements which have the above property are known as magnetostrictive or magnetoelastic materials.

This property can be made use of, in constructing a transducer to convert a stress to a variation in induction.

8.1.2 Magnetostrictive Materials

Some of the magnetostrictive materials are

- Nickel
- Permalloy – (nickel alloy with 68% nickel)
- Ferroxcube B

Ferroxcube is highly brittle, so not very much used.

The other two are used for some transducer applications. The B-H curves of nickel and permalloy are shown in fig.8.2. for different values of stress or load. Vibrating stresses and acceleration can be transduced using magnetostrictive material.

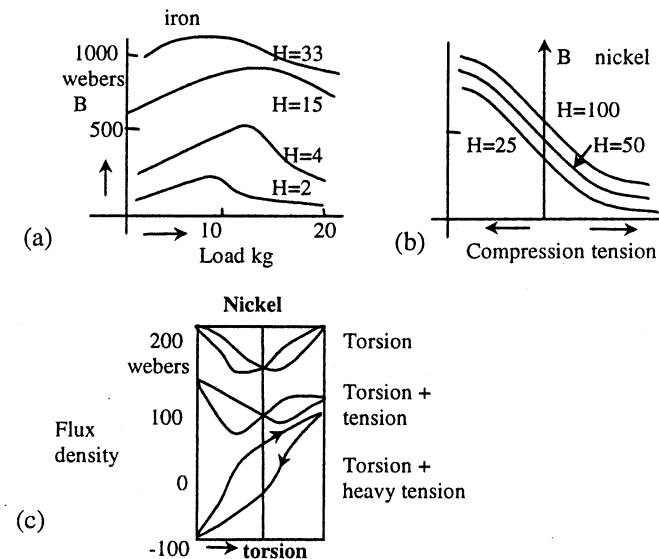


Fig. 8.1
Permeability variation of magnetostrictive elements

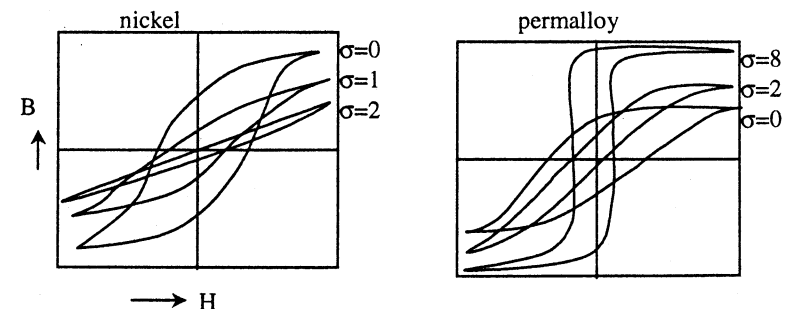


Fig. 8.2
B-H curve for different stress values

8.1.3 Magnetostrictive Load Cell

Load cells can be designed using magnetostrictive element as a basic sensor. Depending upon the cross-sectional area of the probe, forces from several grams up to several tons can be measured directly. The displacement at the input of the transducer is very small-of the order of micro meters.

The yoke Y surrounding the magnetoelastic probe E (fig. 8.3) increases the sensitivity by increasing the magnetic flux because of the permeability change in the probe E for an applied force F and acts as a shield against external stray magnetic fields. The yoke as well as the core should be laminated to reduce eddy currents.

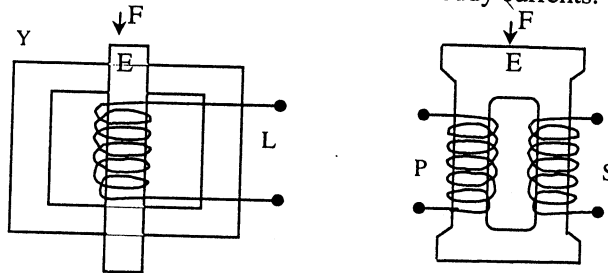


Fig. 8.3

Magnetostrictive loadcells

8.1.4 Magnetoelastic Accelerometer

Magnetoelastic is the other name for magnetostrictive property. It is easy to construct accelerometer using magnetoelastic element as a basic sensing element. One such construction is shown in fig. 8.4.

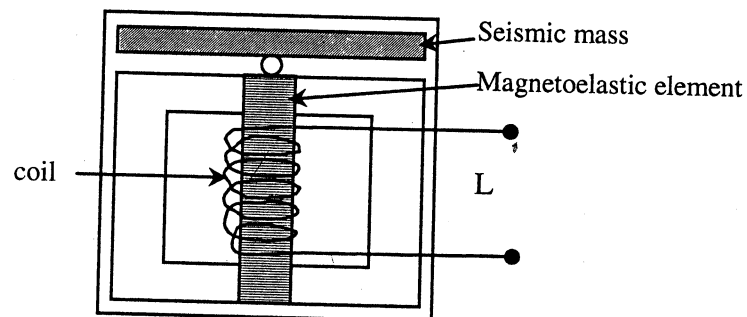


Fig. 8.4

Magnetoelastic accelerometer

8.1.5 Magnetostrictive Phonographic Pickup

Two pick up coils are wound around the two halves of the torsion wire as shown in fig. 8.5. Hysteresis is reduced and linear response is obtained by giving the wire an initial twist.

A stylus displacement will decrease the stress on one side and increase it on the other side by an equal and opposite amount. This increases the magnetic flux in one half the wire and decrease it in the other half. The frequency response is flat over 150 to 15000 cps.

The highest frequency with which the transducer will respond is determined by the mechanical resonance frequency of the magnetoelastic probe. This can be as high as several 10000 Hz.

Eddy currents are likely to introduce limitation at frequency higher than 5000 Hz.

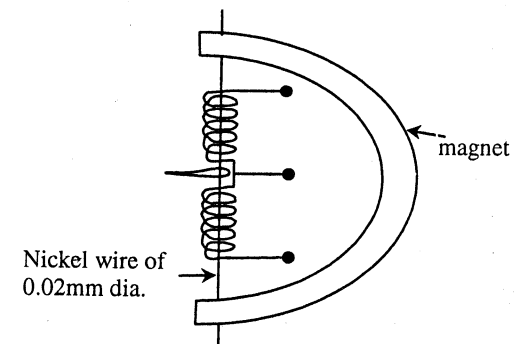


Fig. 8.5

Phonographic pickup

8.1.6 Magnetostrictive Torque Transducers

A schematic diagram of a torque transducer is shown in fig. 8.6a. Thin wall tubes of iron or nickel were magnetised in a direction bisecting the principal planes of stress (either axially or tangentially) with zero applied torque.

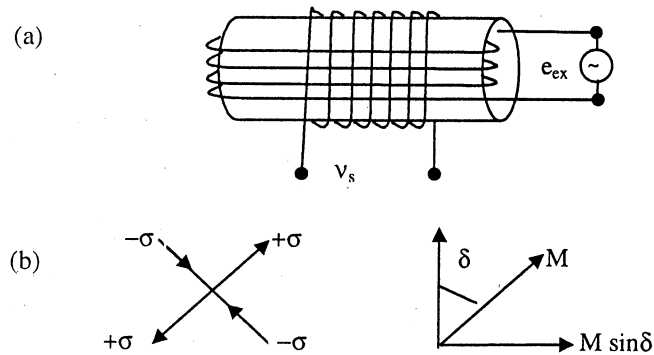


Fig. 8.6
Torque transducer

With zero applied torque, no volts is present in coil V_s – Application of torque generates tensile and compressive stresses of equal magnitude within shaft as shown in fig. 8.6b.

If the material is magnetostrictive this causes a rotation of M towards the direction of tensile stress in which magnetostriction is positive, now the component $M \sin \delta$ induces a voltage in the coil V_s which is proportional to the torque.

8.1.7 Errors in Magnetoelastic Transducers

8.1.7.1 Hysteresis

Hysteresis error can be large, it can be reduced by aging, prestressing, and using the transducer over small ranges of stress variation.

8.1.7.2 Temperature

Increase of temperature causes an increase of inductance at zero stress.

Temperature increase changes the stress sensitivity of the transducer by about 0.2 percent for a variation of 1°C .

8.1.7.3 Variation of supply voltage

Voltage variation causes relatively small variation because inductance variation depends upon the magnetic field strength in

the probe. This first increases with increased field strength and then decreases beyond a certain operating point. Therefore there exists an operating point where small supply-voltage variation have no influence upon the transducer output.

8.1.7.4 Eddy currents

The effects of eddy currents can be reduced by choosing the core material with a high electrical resistivity.

8.1.7.5 Input impedance

The mechanical input impedance can be made larger by choosing the core with a high tensile strength.

8.1.8 Special Features of Magnetostrictive Transducers

Magnetoelastic type of transducers can be built to measure large forces up to several tons and for fast transient phenomena where frequency is of the order of several thousand cycles per sec.

The accelerometers can be built to measure several thousand g. These can be very rugged also.

These instruments need individual calibration as their characteristics depend upon temperature.

The characteristics and their calibration should be checked periodically as they can be changed by environmental factors.

8.2 ELASTIC TRANSDUCERS

The elastic properties of metallic elements are good candidates for constructing force or pressure transducers. In practice bourdon tubes, diaphragms and metallic bellows are used to transduce pressure signal into displacement.

8.2.1 Bourdon Tubes

Some basic forms of bourdon tubes used as sensing elements in pressure transducers are shown in fig. 8.7.

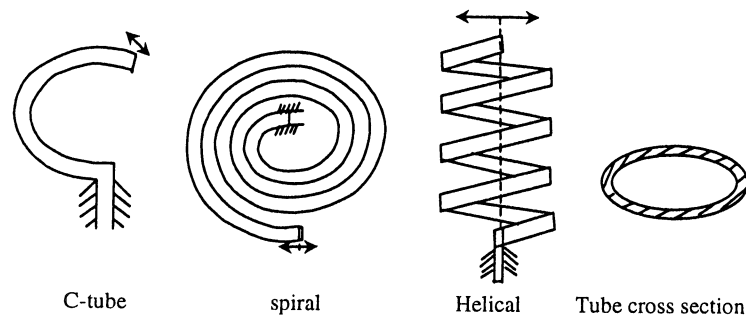


Fig. 8.7
Bourdon tubes

The cross section of the bourdon tube will be elliptical so that any higher pressure inside the tube causes the tube to become circular in cross section. This is because the net force acting on the long diameter of the elliptical cross section F_a (ref fig 8.8) is more than the net force F_b on the short diameter.

The C tube will bend outward when pressure is applied because the outer circumference $k_1 \pi D_2$ of the C tube is more than the inner circumference $k_1 \pi D_1$ as shown in fig. 8.8. D_1 and D_2 are the diameters of the inner surface and outer surface respectively, and k_1 is a constant for the sector angle θ .

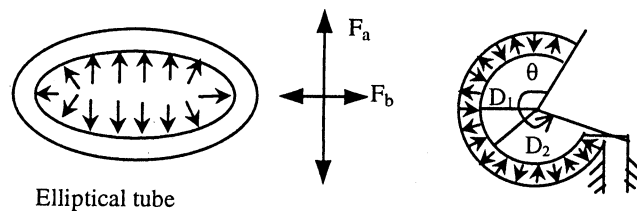


Fig. 8.8
Net force in C bourdon tube.

These bourdon tubes convert pressure into angular displacement which can be converted into a voltage signal by suitable displacement transducers like potentiometer or RVDT.

The C type Bourdon tube can be used for higher pressure ranges of around 5000 kg/cm^2 . For lower pressure values of the order of 50 kg/cm^2 , the spiral and helical configurations are used.

8.2.2 Diaphragm Type Transducers

Some of the diaphragm type elastic elements for transducer applications are shown in fig. 8.9.

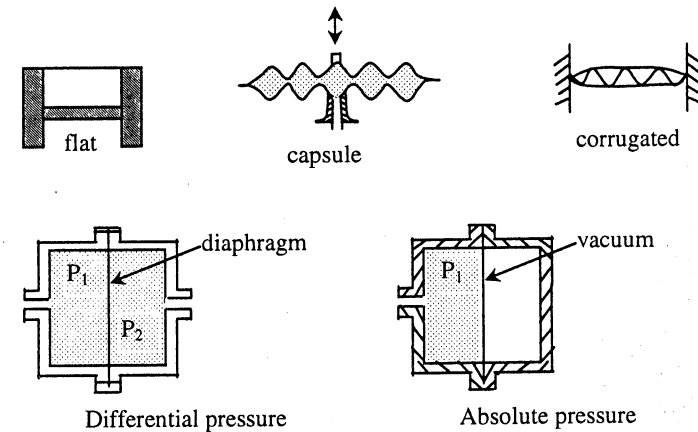


Fig. 8.9
Elastic transducer elements

The mathematical description of the deflection versus pressure characteristics are little involved and hence not attempted here. But the number of practical transducers and transmitters using diaphragm as basic sensors are many.

8.2.3 Bellows

Bellows is another elastic element which is used widely in process instrumentation as basic sensing devices. Basic configurations are shown in fig. 8.10.

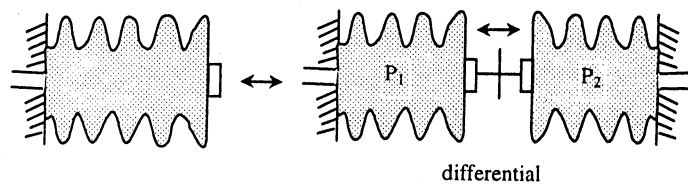


Fig. 8.10
Bellows

8.3 FIBRE OPTIC TRANSDUCERS

Fibre optics is revolutionising the data communication field and it is also finding increasing use in sensing applications. It is known that light can travel only in straight line. But through optical fibre light can travel through bends, curves and corners.

8.3.1 Optical Fibre

An optical fibre is a hair line thin strand of glass or glass like material having two or more layers as shown in fig. 8.11. The refractive indices of the different layers will be reducing from inner core to outer layer. Because of the reducing refractive indices of outer layers, any light passed through the inner core cannot go out of the fibre because of total internal reflection and hence propagates through the fibre which can take any curved shapes.

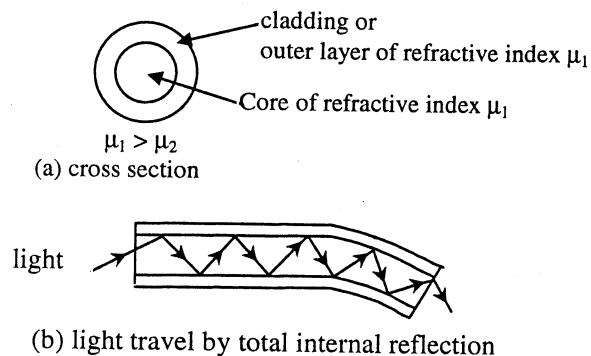


Fig. 8.11
Optical fibre

There are four types of fibre commonly used. They are monomode, multimode, birefringent and coated. Monomode fibre is used in interferometric instruments as the input light should be coherent. In multimode fibre optics, intensity modulating schemes can be employed. Schemes using polarising techniques need birefringent fibre. The coated fibre is useful when compound modulation occurs in situations like magnetic field sensing

Fibre optics can be used to sense variables like force, pressure, temperature, acceleration and displacement.

8.3.2 Microbend Displacement Sensor

When a fibre is deformed into a convoluted shape, part of the light travelling through the fibre is lost to radiation. The loss of light is maximum when the convolutions have a spacing given by $\lambda = 2\pi / \Delta \beta$ where $\Delta \beta$ is the difference in propagation constants between propagating and radiation modes. For the aluminium coated multimode with 120 micron diameter, the optimum spacing was found to be 3 mm. A schematic diagram of a microbend pressure sensor is shown in fig. 8.12.

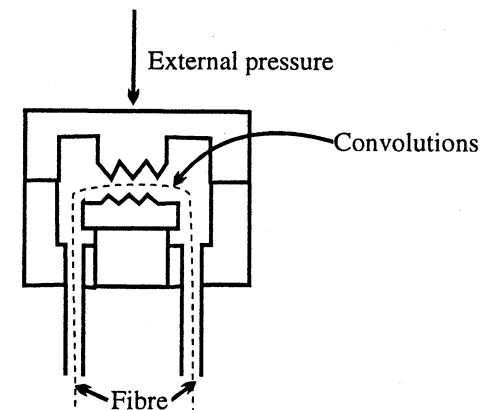


Fig. 8.12
Microbend pressure sensor

In this sensor, the convolution spacing depends on the pressure. The light received by the detector varies according to the convolution spacing which is proportional to the pressure.

8.3.3 Fibre optic displacement sensor

The fibre optic probe used for transducing displacement into electrical signal consists of a bundle of several hundred optical fibers, each with a diameter of 10's of micrometer in diameter. The fiber extends up to 1 meter in length. In the bundle there two sets of fibers, one set for transmitting another set for receiving as shown in fig 8.13

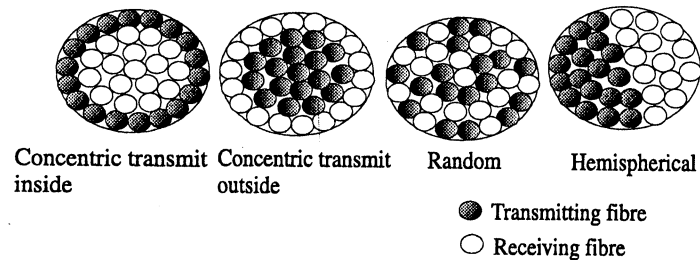


Fig. 8.13
Fiber bundle

The transmitting group is exposed to a light source and hence carries light to the tip of the fibre bundle.

This leaves the bundle at the tip and get reflected by the target surface. The reflected light enters the fibre bundle. Which ever enters the receiving set of a bundle is picked up and processed by the photodetector circuit to produce a d.c. voltage.

If the reflecting surface is nearer more light enters the reflecting group of bundle and hence the d.c output voltage from the photodetector circuit is more. As the reflecting surface is moving farther away the light picked by the bundle is less & hence less output d.c voltage.

Thus the arrangement shown in fig. 8.14, is used as a fibre optic displacement transducer.

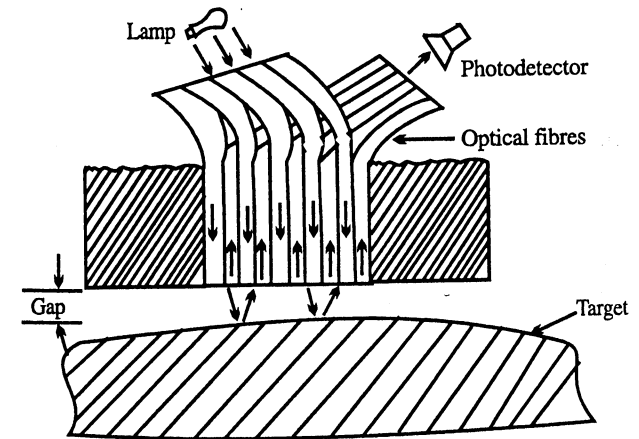


Fig. 8.14
Fiber optic displacement sensor

The amount of light getting reflected depends also on the reflectivity of the surface. The characteristic of a typical fibre optic displacement sensor is shown in fig. 8.15

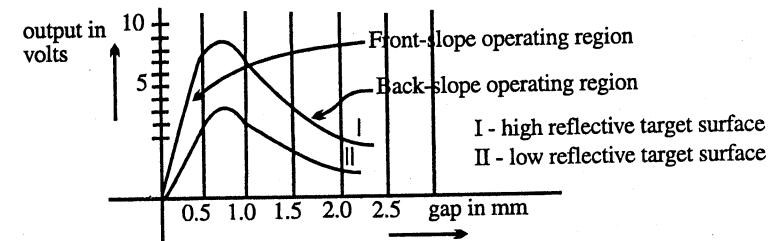


Fig. 8.15
Characteristics of Fibre-optic displacement transducer

This kind of transducers have two useful measuring ranges namely front slope and back slope. At zero gap no light can escape from the transmitting fibre and hence the output of the transducer is zero. As the gap opens up, light comes out of the gap gets

reflected. There is an optimum gap at which max. light is picked up from the reflected light and it gets reduced as the gap increases beyond this value. This kind of characteristics is revealed in the graph. Because of this type of characteristics, there are two useful range for sensor application. One is the front slope region and the second is the back slope region.

Typical fibre optic displacement transducer has a range of .1 to 2.5 mm. with a resolution of .001 mm. The frequency response bandwidth is d.c to 50 Hz. This transducer needs calibration at site as the surface condition may vary from place to place.

8.4. DIGITAL TRANSDUCERS

This term refers to transducers which give digital outputs directly. Many transducers available in the market are analog in nature. Their outputs are continuous functions of time. If these analog transducers are to be interfaced with digital devices, then one has to use analog to digital converters. As many of the instrumentation systems use digital computer as a main component, digital transducers are very useful. Many are under development. Digital transducers directly give output signals in the form of rectangular pulses. If the pulse is present it is taken as 1 and its absence is taken as 0.

A few of them is described in the following sections.

8.4.1 Digital Displacement transducer

Linear or Angular displacements can be converted in digital signals by having digital encoders. Digital displacement encoder is shown in fig. 8.16

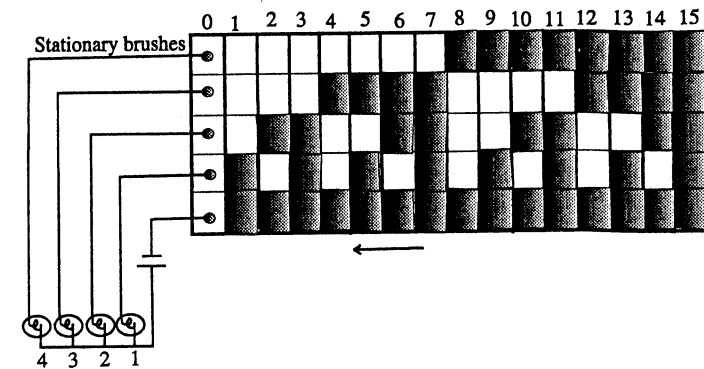


Fig. 8.16

Digital displacement encoder

In this encoder, the brushes or the contactors are kept stationary. The linear displacement is connected to the strip which is having metallic conducting surfaces and nonconducting insulating portions. The linear movement is attached to the coded strip which moves touching the brushes. When the brushes are touching column 1 the bulb 1 glows and all others do not. If we consider glowing bulb as 1 and nonglowing as zeros, then the digital number is 0001. Similarly when the brushes are in column 2, bulb 2 glows and all others do not, which is equivalent to a digital number 0010. In a similar way for all brush positions we can find out the numbers Table 1 shows the digital numbers corresponding to the different brush positions.

Table 8.

column number of brush position	digital numbers
0	0 0 0 0
1	0 0 0 1
2	0 0 1 0
3	0 0 1 1
4	0 1 0 0
5	0 1 0 1
6	0 1 1 0
7	0 1 1 1
8	1 0 0 0
9	1 0 0 1
10	1 0 1 0
11	1 0 1 1
12	1 1 0 0
13	1 1 0 1
14	1 1 1 0
15	1 1 1 1

Thus this encoder converts the linear movement into 16 binary coded digital numbers. If a movement of 7.5 cm of the coded strip, goes through the 16 positions (rows) of the strip, then 7.5 cm motion is converted into 16 binary 4 bit numbers with a resolution of

$$\frac{7.5}{15} = 0.5$$

If a higher resolution is needed, then the number of tracks must be increased and the length of each coded segment should be reduced, which would require finer brushes. The disadvantages of brush pick ups are as usual the brush bounce, brush friction and wear and tear. Arcing of brushes as they move along and

mechanical vibrations are some of the problems associated with this type of encoders.

8.4.2 Shaft angle encoder

Shaft angle encoder can be used for a total angular displacement of 360°. To construct an angular encoder, the linear encoder shown in fig . 8.16 can just be bent to form a disc. as shown in fig 8.17 For the four tracks disc shown in fig 8.17. The resolution is $\frac{360}{15} = 24^\circ$.

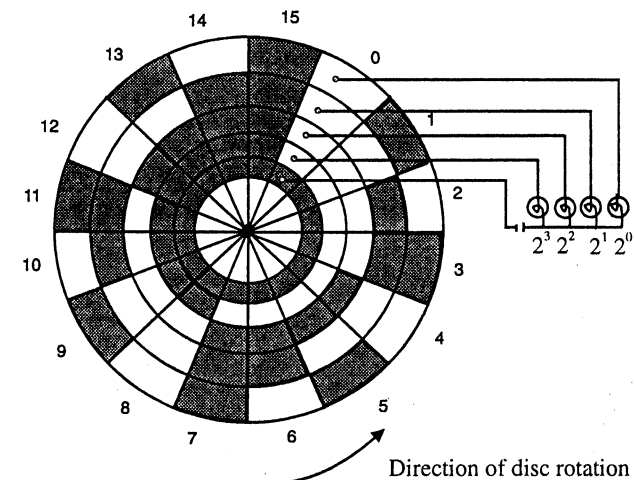


Fig. 8.17
Shaft angle encoder

Discs upto 10 tracks are used to have a resolution of $\frac{360}{1024} = 0.3515^\circ$.

8.4.3 Optical encoders

Optical encoders use optical and photoelectric sensing systems for converting linear or angular movements. These encoders have transparent and opaque areas corresponding to the conducting and non conducting areas instead of conducting and non conducting portions of the mechanical encoders. The sensing system consists of light sources each provided with a focussing lens and an equal

number of photoelectric devices to receive the light from the corresponding sources. The change between transparent and opaque area must be sharp so that the transition between logic '0' and '1' is sure.

8.4.4 Magnetic encoders

In this type of encoders, magnetic tape with magnetised portions and non magnetised portions is moved over sensing heads. The sensing heads have toroidal cores. Each toroidal core has two coils namely reading coil and interrogate coil. The interrogate coil is energised with a constant voltage of 200 kHz signal. The reading coil develops output signal due to transformer action only when

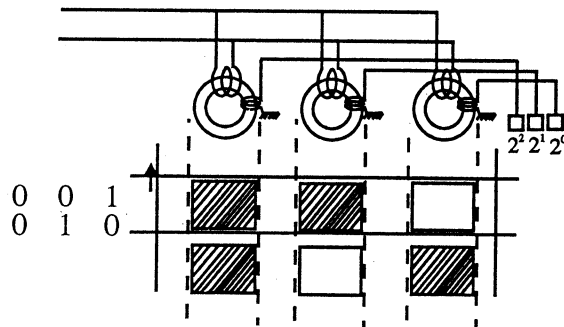


Fig. 8.18
Magnetic encoder

the toroidal core is against the nonmagnetised portion. When the core is against the magnetised portion no voltage is developed because the core is saturated. A schematic diagram of the arrangement is given in fig. 8.18.

8.4.5 Digital speed transducers

8.4.5.1 Variable reluctance type

In variable reluctance type of transducer, a rotating shaft is attached with a toothed rotor, which provides a variable reluctance in a magnetic circuit. One such design is shown in fig. 8.19.

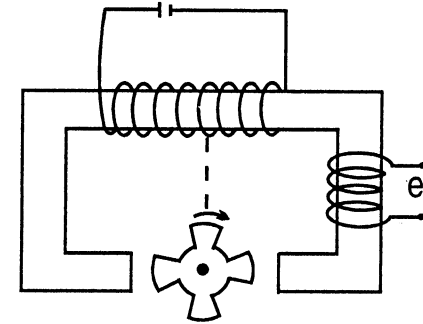


Fig. 8.19
Variable reluctance speed sensor

When the teeth are against the stator poles of the reluctance is less and hence e_0 is more when the slot of the toothed shaft comes against the stator pole, the reluctance is high and hence the voltage induced across e_0 is small.

Thus whenever the teeth cross the pole a voltage pulse appears across e_0 .

By counting the number of pulses per second, we can determine the speed.

The output of the transducer is a series of pulses, this can be interfaced with any digital equipment.

8.4.5.2 Variable capacitance type

The variation of the capacitance between a probe plate and a toothed rotor may be used to generate pulses. The number of pulses per second is equal to the rotor speed and the number of teeth in the serrated rotor. By counting the number of pulses by suitable counters, a digital readout proportional to the speed can be designed.

8.5 HALL EFFECT TRANSDUCERS

8.5.1 Hall element

A number of transducers can be designed using a Hall element. This Hall effect is possessed by certain metals and certain semiconductor elements. When a Hall element is subjected to a magnetic field in one direction, an electric current in another direction which is perpendicular to the direction of the magnetic flux, then an electric potential is generated in a third direction which is mutually perpendicular to the other two directions. The magnitude of the voltage generated depends on the strength of the magnetic field, the strength of the current and the property of the conductor.

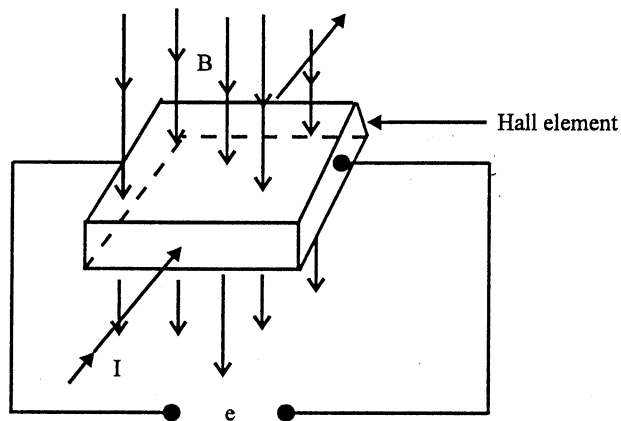


Fig. 8.20
Principle of Hall effect

This is shown in fig. 8.20.

Where e is the hall voltage generated
 I is the current in amperes
 B is the Flux density in weber / m²
 t is the thickness of the Hall element in 'm'
 k_H is the Hall coefficient in $\frac{V - m}{A - Wb\ m^{-2}}$

The nominal values of Hall coefficients for different materials are given in Table 8.1.

Table 8.1 Hall coefficients

Material	Flux density Wb/m ²	Temperature °C	$\frac{K_H}{A \cdot Wb - m^{-2}}$
As	0.4 to 0.8	20	4.52×10^{-9}
Bi	0.113	20	-1×10^{-6}
C	0.4 to 1.8	30	-11.73×10^{-9}
Cu	0.8 to 2.2	20	-52×10^{-12}
Fe	1.7	25	1.1×10^{-9}
Ge	0.001 – 0.8	25	-8×10^{-3}
Si	2	25	4.1×10^{-6}
Sn	0.4	25	-2×10^{-12}
Te	0.3 – 0.9	20	53×10^{-6}

8.5.2 Hall effect transducer for Flux measurement

A semiconductor Hall element can be used to transduce magnetic flux into a voltage. The flat surface of semiconductor plate is kept

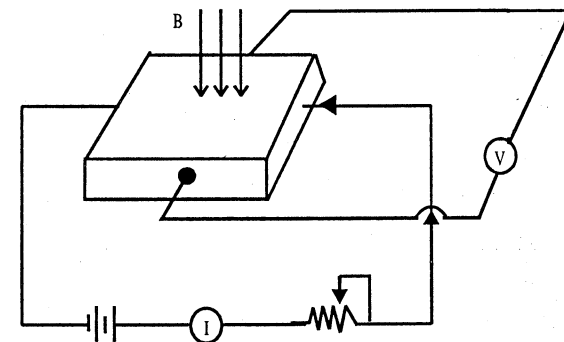


Fig. 8.21
Hall Flux meter

perpendicular to the magnetic field to be transduced. A known fixed current is passed through side of the plate as shown in fig 8.21 Then the voltage generated across the mutually perpendicular surface is proportional to the flux density.

This element gives a continuous output proportional to the magnetic flux variation.

The disadvantages are (i) this element is sensitive to temperature variations. (ii) The Hall coefficient changes from element to element needing individual calibration.

8.5.3 Hall displacement transducer

Hall element can be used for the measurement of displacement of a structural element. Consider the diagram shown in fig. 8.22. A structural member whose displacement is to be measured is fixed with a permanent magnet.

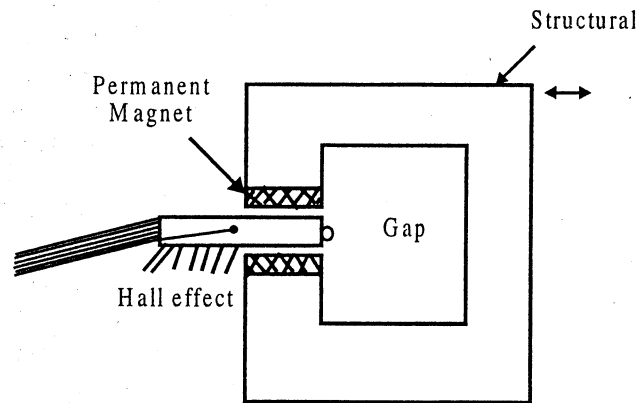


Fig. 8.22
Hall effect displacement sender

In the gap provided in the structural member a Hall element is inserted. The field strength produced by the permanent magnet in the gap varies when the structural member is subjected to a movement or displacement. It might be noted the portion of the Hall element is fixed and it does not move. A fixed current is passed through the Hall element.

The output voltage generated by the Hall element is proportional to the magnetic field lines passing through the Hall element which depends on the relative position of the Hall element and the structural element. The device can be used for displacements ranging from 0.1mm to 10mm.

8.5.4 Hall transducer for current measurement

The diagram shown in fig 8.22 can also be used as an ammeter for measurement of current. Here the portion of the structural member is kept fixed and current I is varied. The voltage generated will be proportional to the current. Once this device is calibrated for different values of current, any unknown current can be measured by passing it through this element.

8.5.5 Hall Wattmeter

Hall element is used to measure power in electrical circuits. Consider the diagram given in fig. 8.23.

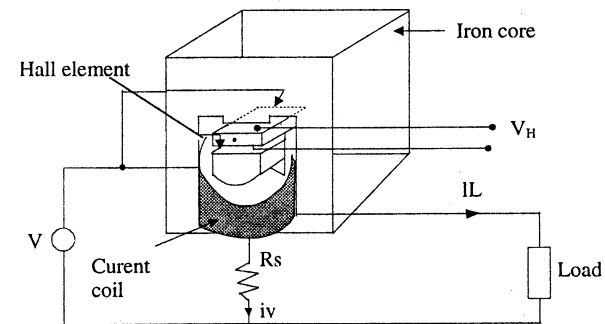


Fig. 8.23
Hall wattmeter

The output voltage of the Hall element V_H is proportional to the instantaneous power consumed by the load. This is explained as follows.

The current I_L drawn by the load flows through the current coil of the Hall wattmeter which produces a magnetic flux. The magnetic flux is passing through one face of the Hall element. A

current i_v which is proportional to the applied voltage v is passed through one side of the Hall element

$$\text{The current } i_v = \frac{v}{R_s}$$

Now the Hall element output $V_H = k_H BI$

$$\begin{aligned} &= k_H (K_B I_L) \left(\frac{V}{R_s} \right) \\ &= K^1 I_L V \end{aligned}$$

$I_L V$ is the power consumed by the load.

Thus the output of the Hall element for the arrangement shown in fig. 8.23 is proportional to the instantaneous power.

8.6 WORKED OUT EXAMPLES

EXAMPLE 8.5.6.1

A Hall element made out of n - Ge with coefficient of $8 \times 10^{-3} \frac{\text{V m}^3}{\text{wb A}}$ is subjected to a magnetic flux density of 0.75 Wb/m². The thickness of this element is 2 mm. When a current of 1.5 Amperes is flowing the element, what is the voltage output.

SOLUTION

$$\begin{aligned} V_H &= \frac{K_H B I}{t} \\ &= \frac{8 \times 10^{-3} \times .75 \times 1.5}{2 \times 10^{-3}} = 4.5 \text{ volts.} \end{aligned}$$

EXAMPLE 8.5.6.2

In Hall wattmeter a voltage applied is 250VDC. This voltage is connected to the Hall element of Bismuth with 2.5 mm thickness with a series resistance of 500 ohms. The current coil produces a flux of .25 Wb/m², when a current of 10 mA is passed through and it is linear upto a current of 2 Ampere. When the Hall voltage is

.25 mv, what is the power consumed by the load circuit. The Hall coefficient for Bismuth is $-1 \times 10^{-6} \frac{\text{V m}^3}{\text{wb Amp.}}$.

SOLUTION

The current through the Hall element caused by the applied voltage is

$$I = \frac{250}{500} = 0.5 \text{ A}$$

Let the flux across the Hall element be B

$$\begin{aligned} \text{we know } V_H &= \frac{K_H B I}{t} \\ &= \frac{1 \times 10^{-6} \times B \times .5}{2.5 \times 10^{-3}} = 2 B \times 10^{-4} \end{aligned}$$

$$\therefore B = \frac{V_H}{2 \times 10^{-4}}$$

$$\text{when } V_H = 25 \text{ mv, } B = \frac{.25 \times 10^{-3}}{2 \times 10^{-4}} = 1.25 \text{ wb/m}^2$$

The current through the current coil for this flux density is

$$\begin{aligned} &\frac{.1}{.25} \times 1.25 \text{ Amperes.} \\ &= 0.5 \text{ Amperes.} \end{aligned}$$

$$\therefore \text{ The power} = 250 \times 0.5 = 125 \text{ watts.}$$

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